



MIDDLE SCHOOL COMPETITIVE EVENTS GUIDE

for the 2026 and 2027
National TSA Conferences

with correlations to
Science, Technology, Engineering, and
Mathematics (STEM) Standards

Technology Student Association

ACKNOWLEDGMENTS



TSA is grateful to many people for their advice and expertise in developing the competitive events program over the years. The organization especially appreciates the dedicated efforts of the Subject Matter Expert Team of the Competition Regulations Committee (CRC), who have refined the event specifications that appear in this guide.

Frank Calfee
Tara Dye
Noah Kelley
Adam Knecht
Rajan Jani
Valerie McCauley
Dale Moll
Angela Quarles
Andy Stephenson

COPYRIGHT 2025—Technology Student Association. Fifteenth Edition

All rights reserved. This guide was developed by the TSA Competition Regulations Committee and sanctioned by the TSA, Inc. Board of Directors. This Guide and its contents are the property of the Technology Student Association. Unless otherwise noted in this guide, no part may be shared or reproduced in any form or by any means without written permission from the executive director of the Technology Student Association.

For more information, please contact:

Technology Student Association
1904 Association Drive, Reston, VA 20191-1540
Phone 703.860.9000 • Toll free 888.860.9010 • Fax 703.738.7486

TSAweb.org • general@tsaweb.org



TABLE OF CONTENTS



ACKNOWLEDGMENTS	2	MIDDLE SCHOOL COMPETITIVE EVENTS	
TSA, THE ORGANIZATION		Audio Podcasting	39
TSA Mission	4	Biotechnology	49
Who Are TSA Members?	4	Career Prep	59
The Role of Competitive Events	4	Challenging Technology Issues	65
About This Guide	4	Chapter Team	73
COMPETITIVE EVENTS PROGRAM		Children's Stories	85
Levels of Competition	5	Coding	95
General Rules and Regulations	5	Community Service Video	101
Competition Regulations Committee	9	Computer-Aided Design (CAD) Foundations	109
Event Coordinator Reminders	9	Construction Challenge	115
Awards	9	Cybersecurity	123
Event Proposal Information	9	Data Science and Analytics	131
DRESS CODE	11	Digital Photography	139
TSA'S LEADERSHIP PROGRAM		Dragster	149
Partnership for 21 st Century Skills (P21)	12	Drone Challenge (UAV)	161
SCIENCE, TECHNOLOGY, ENGINEERING, AND MATHEMATICS (STEM) INTEGRATION	14	Electrical Applications	171
Next Generation Science Standards (Grades 5-8)	15	Flight	179
Standards for Technological and Engineering Literacy (STEL)	20	Forensic Technology	191
Computer Science Teachers Association (CSTA) K-12 Computer Science Standards	24	Inventions and Innovations	197
ISTE Standards for Students	26	Leadership Strategies	205
National Council of Teachers of Mathematics (NCTM) Principles and Standards for School Mathematics	29	Mass Production	213
TSA AND CAREERS		Mechanical Engineering	223
Career Clusters®	32	Medical Technology	233
MIDDLE SCHOOL COMPETITIVE EVENTS ...	35	Microcontroller Design	243
Competitions	36	Off the Grid	251
Competitive Events Eligibility	37	Prepared Speech	259
TSA Competitive Events Rating Form/Rubric	38	Problem Solving	265
		Promotional Marketing	269
		Robotics	277
		Solar Racer	285
		STEM Animation	299
		Structural Engineering	307
		System Control Technology	319
		Tech Bowl	329
		Technical Design	337
		Video Game Design	343
		Website Design	351
		2026 & 2027 FORMS APPENDIX	359

TSA, THE ORGANIZATION



TSA MISSION

The Technology Student Association (TSA) enhances personal development, leadership, 21st century skills, and career opportunities in STEM, whereby members apply and integrate these concepts through intra-curricular activities, competitions, and related programs.

WHO ARE TSA MEMBERS?

TSA is devoted exclusively to the needs of students engaged in science, technology, engineering, and mathematics (STEM). Open to those who are enrolled in or who have completed technology and engineering courses, TSA has 300,000 middle and high school student members across the country. TSA is supported by educators, parents, and business leaders who believe in the need for a technologically literate society. TSA members learn through exciting competitive events, leadership opportunities, and membership activities. It is the intent of TSA to involve as many TSA members as possible in competitive events and provide recognition in a setting of fair play practices using TSA event guidelines.

Explore what TSA has to offer by using this guide and by visiting TSAweb.org for information. With competitive events that range from video game design to structural engineering and much more, there is something to capture the imagination of—and bring out the best in—all students. We hope that with teacher guidance, students will enjoy the challenge of TSA's competitive events at local, state, regional, and national TSA conferences.

The competitions in this guide support a broad spectrum of goals related to STEM curriculum, emphasizing and promoting the development of leadership and 21st century skills, and increasing exposure to future career choices.

THE ROLE OF COMPETITIVE EVENTS

To follow its mission, TSA offers stimulating competitive events. TSA believes that by participating in thoughtfully designed competitions, students learn 21st century skills such as collaboration, perseverance, critical thinking, and problem solving, thereby becoming “winners” irrespective of placement in a competition. Many teachers find that TSA's competitive events provide an excellent motivational tool in the academic environment.

Every two years, TSA's competitive events are reviewed and revised by the Subject Matter Experts (SMEs) of the Competition Regulations Committee (CRC), a standing group of technology and engineering educators with hands-on classroom experience. The *Technology Student Association (TSA) Middle School Competitive Events Guide for the 2026 & 2027 National TSA Conferences* is the result of the collaboration of the CRC SME managers, competitive event coordinators, teachers, proposals of numerous TSA state and chapter advisors, and students who make TSA competitive events current and dynamic. The guide presents rules and regulations for all National TSA Conference competitive events; a view of each event's connection to STEM standards; and suggested careers. Relevant for all levels of competition (state delegations may choose to adopt the national guidelines for state-level competitions), the guide provides an excellent motivational tool for curricular study and activities in the classroom.

ABOUT THIS GUIDE

With the publication of the *2026 & 2027 TSA Middle School Competitive Events Guide*, please note the following:

1. The format of this guide has been streamlined to help familiarize competitors and advisors with the TSA Conference General Rules and Regulations, and the procedures, regulations, and assessment for each event.
2. General rules that apply to all participants across every competitive event are no longer identified in each competition's regulations. Therefore it is critical, and a personal responsibility of each competitive event participant and advisor, to read and fully adhere to the TSA General Rules and Regulations.
3. Every event's guidelines have been revised in some form, whether in content or in format.



COMPETITIVE EVENTS PROGRAM



LEVELS OF COMPETITION

- A. The breakdown of grades noted below is used to designate levels for competition entries. Each level has its own unique competitive events guide.
- Middle School/Junior High School level—Grades 5, 6, 7, 8, 9
 - High School level—Grades 9, 10, 11, 12
 - Ninth graders must compete at the level in which the chapter affiliates.
 - If the configuration of the school includes grades 9-12, ninth grade students must compete in high school events.
 - If the configuration of the school includes grades 6-9 or 7-9, ninth grade students must compete in middle school events.
- B. If the school has a K-12 configuration, or a configuration other than the examples above, national TSA should be contacted for clarification and approval regarding the appropriate school level designation.

GENERAL RULES AND REGULATIONS

NOTE: General rules and regulations apply to *all* competitive events and are *in addition* to each event's specific guidelines.

A. Affiliation and Membership

1. TSA members, advisors, and chapters must be currently affiliated with TSA to enter any competitive event.
2. TSA membership rights extend through the year of graduation.
3. Students who graduate midyear may compete at the national conference that immediately follows their end-of-year graduation.

B. Conference Registration, Attendance, and Participation in Events

1. Individuals who wish to attend the conference must complete conference registration.
2. Students must be registered and be in attendance with an adult chaperone at the National TSA Conference in order to enter and become a semifinalist or finalist in any event.

3. All adult advisors, chaperones, and student participants must be in attendance for the entire conference.
4. National TSA Conference registrants must wear conference identification badges at all times.
5. The TSA competitive event limit is six (6) events per conference participant, individual and team events combined.
6. Team events:
 - a. All team members must be affiliated with the same chapter.
 - b. Registration for any team competitive event requires the identification of all team members. A (one) team captain must be designated by the chapter advisor for any of the events with online submission requirements. The captain is responsible for uploading the competition entry/documentation on behalf of the entire team. Team captain responsibility does not apply for team events that involve a preliminary exam; all team members must take an exam to determine an average team score.
 - c. Unless otherwise designated in a competition's eligibility guideline, the minimum size of a team is two (2) members and the maximum size of a team is six (6) members.

C. Student Responsibilities for Competitions

1. Student members should create their account in the [TSA Student Member Site](#). Affiliated student members may access "member only" resources, including the Student Member Competitive Events Guide, and submit selected competition entries for the national TSA conference.
2. It is the individual responsibility of each participant to obtain all rules and guidelines for competitive events.
3. Lack of knowledge or understanding about a particular event is neither reason nor excuse for an individual to request an accommodating adjustment or change.
4. Students and advisors must routinely check the TSA website, [TSAweb.org](#), for updated information about TSA general rules and competitive event guidelines. One month prior to the conference, no additional updates will be posted to this page.

5. Students who participate in any TSA competitive event are responsible for knowing all updates, changes, and clarifications related to that event.
6. Student competitors are responsible for ensuring that all competition-related websites and internet-based content are accessible from their device (personal or school-issued). TSA assumes no responsibility for a student competitor's inability to access national TSA conference competition platforms and/or web-based content.
7. When a decision is made by a member to compete in an event, it is imperative that the member thoroughly read and understand the rules, regulations, procedures, and evaluation criteria for the chosen event(s). National TSA is responsible for addressing mistakes or discrepancies in the competitive event guides and accomplishes this through competition updates on the TSA website. National TSA is not responsible for interpreting, advising, or suggesting ways that a member might develop a viable response or solution to a competitive event challenge.

D. Competition Entries

1. Entries must be started and completed during the current school year.
2. Entries may be submitted for one (1) year, and one (1) competition only. An infraction of this rule results in disqualification.
3. Each participant/team shall submit only one (1) entry per event.
4. All entries that require a documentation portfolio must be secured in a [clear front report cover](#) unless otherwise indicated in an event's regulations.
5. All entries must be in English.
6. Participants must check in and pick up their event entries at the times and places stated in the conference program, or as announced during the National TSA Conference.

7. If a competition includes a test on-site at the national conference, the test will be administered online only (Wi-Fi service will be provided at no cost only for competitions). Paper tests will not be available unless deemed necessary. Individual participants, or each team member, must bring to the testing site:
 - a. One (1) laptop or tablet (personal or school-issued) capable of networking via Wi-Fi and able to run solely on battery power for up to two (2) consecutive hours. Chrome is the preferred browser.
 - b. Optional: One (1) mouse
8. For any competition that involves the use of a pencil (e.g., for taking a test if necessary, for producing required sketches), participants must provide—and bring to the test site two (2) pencils, either:
 - sharpened standard #2/HB grade with an eraser, or
 - #2 mechanical with an eraser.
9. Entry content:
 - a. National TSA provides guidelines for individual and team entry content but does not bear responsibility for content choices made by participants.
 - b. Entries are evaluated on the basis of an event's official rating form.
10. Projects and/or products:
 - a. Unless otherwise specified, no identifying information—other than a student or team ID#—is to be included on an entry.
 - b. Exceptions to this rule are:
 - i. Middle school competitive events:
 1. Audio Podcasting (student first name[s] only may be used)
 2. Career Prep
 3. Children's Stories
 4. Community Service Video
 5. Construction Challenge
 6. Structural Engineering

- ii. High school competitive events:
 - 1. Children's Stories
 - 2. Digital Video Production
 - 3. STEM Mass Media
 - 4. Structural Design and Engineering
- iii. Events that require submission of a Work Log shall include indication of student initials only.
- c. Unless otherwise noted, for all events that require a display, the size of the display may not exceed 15" deep x 30" wide x 48" high.
- 11. TSA may choose to keep National TSA Conference student entries.
 - a. Such entries may be used by national TSA for promotional purposes. Should that occur, credit for any such entry would be noted by TSA.
 - b. If applicable, the USB flash drive entries become property of TSA and will not be returned.

E. Citations, References, and Copyrighted Material

- 1. For all applicable competitive events, citations/ references must follow a professional citation style of the competitors choosing unless the competitive event specifies a specific formatting style. Failure to use a professional citation style will result in a rules violation of 20% (twenty percent). Some examples of professional citation styles include MLA, APA, Chicago, and IEEE.
- 2. All entries must be the original work of the student participant or student team.
- 3. All ideas, text, images (including those labeled "for reuse"), and sound from other sources must be cited.
- 4. If copyrighted material is used, written permission must be included.
 - a. An Internet search about copyrighted material and copyright fair use is recommended if ideas, text, images, or sound from other sources is incorporated into an event entry.
 - b. Use of the TSA logo is permitted as long as it adheres to the current TSA Branding Guide.
- 5. Failure to follow any of the above procedures results in a 20% rules violation.

- 6. Plagiarized content in any event will result in automatic disqualification.

F. Prohibited Materials, References, and Images

- 1. Hazardous materials, chemicals, lighted or open flames, combustibles, wet cell batteries, and other similar substances are not allowed at the National TSA Conference.
- 2. No liquids, viruses, live plants, or animals may be used as a part of the display.
- 3. Competition entries or presentations by participants must not include racial or ethnic slurs/symbols, reference to gang affiliation, or vulgar, violent, subversive, or sexually suggestive language or images.
- 4. Entries or presentations may not promote products that students may not legally buy, such as tobacco, alcohol, or illegal drugs.
- 5. Images of guns, knives, or other weapons, or items that can be interpreted as weapons, are prohibited.
- 6. The Entertainment Software Rating Board (ESRB) provides industry guidance on the content of video games and/or applications. Please refer to the [ESRB overview](#) for the categories, descriptions and elements of video content that may be included in competition entries. Interpretation of the content of competition entries is at the discretion of the judges.

G. TSA Liability

- 1. TSA is not responsible or liable for any personal property, equipment, competition content, or materials brought to the National TSA Conference for use by a participant or attendee.

H. Event Scheduling Conflicts

- 1. When an event scheduling conflict could prevent an individual from participating in an event, the individual has the right to not compete in an event.

I. Emergencies

- 1. Team member substitution may be allowed should a documented emergency arise in team events that involve written and semifinalist segments. All substitutions must be approved by the event manager and coordinator.

J. Event Judging

1. All events are judged in accordance with the stated event criteria as noted in this competitive events guide.
2. Tier scoring has been implemented in the preliminary round of some events and is intended to streamline the evaluation process used to determine semifinalists.
3. The decisions of judges related to competitive events are final.

K. Procedure for filing a grievance with the Rules Interpretation Panel

The Rules Interpretation Panel (RIP), a group made up of at least three (3) CRC members, monitors and oversees the competitive events during the National TSA Conference. The panel provides a means by which state advisors may express grievances and concerns about conference situations that pertain to events, and it ensures continuity from year to year for competitive event rules and regulations.

1. All concerns must be in writing using the correct form in the guide. The Rules Interpretation Panel Grievance form (see 2026 & 2027 Forms Appendix) must be completed in its entirety.
2. Only state advisors may submit a request to the Rules Interpretation Panel (RIP) at the national conference. Should an individual/team/chapter advisor have a concern about an event, the state advisor shall be the point of contact. National TSA will not accept forms from anyone other than the state advisor.
3. During the conference, the RIP panel will meet to discuss and analyze the advisor's concern.
4. It is the intent of the panel to resolve any grievances at the conference with a written response to the state advisor.
5. Only the state advisor may pick up the written response from the RIP panel.
6. All decisions made by the panel are final. There is no appeal process.

L. Rules Violations and Disqualifications

1. Refer to page 38 for information on the TSA Competitive Event Rating Form/Rubric and Go/No Go Specifications.
2. A rules violation that gives a contestant an unfair advantage will result in a twenty percent (20%) deduction of the total possible points in either a preliminary or semifinal round as applicable.
3. The coordinator or manager of an event has the right to disqualify a contestant when this type of incident occurs.
4. The event coordinator and manager must sign off on both a twenty percent (20%) deduction and a disqualification.

M. Semifinalists

1. Should the competition have a semifinal round, the event will have a minimum of twelve (12) semifinalists.
2. Semifinalists (individuals or teams, as applicable) will compete against one another to determine the top ten (10) finalists in an event.
3. All members of a semifinalist team must participate in the semifinalist portion of an event, unless otherwise noted in the event's regulations

N. Electronic Devices

1. Recording devices are not allowed in certain competitive events.
2. CRC manager and event coordinator approval is required before any event may be recorded.
3. All electronic devices, including but not limited to cell phones, iPads/tablets, electronic readers, smart watches, etc., *must* be turned off unless otherwise noted in specific event regulations.
4. No electronic communication devices of any kind are permitted during competition.

COMPETITION REGULATIONS COMMITTEE

The Competition Regulations Committee (CRC) is composed of dedicated STEM teachers and education professionals from across the country who have made major commitments to create and maintain the high quality of national TSA's competitive events. Some CRC members are charged with reviewing TSA's competitive events, updating them as necessary, and presiding over the competitive events at the annual National TSA Conference.

Ideas and feedback regarding events are always welcome. Guidelines and forms can be found in the Forms Appendix of this guide for proposing a new event and for suggesting revisions to existing event.

EVENT COORDINATOR REMINDERS

TSA appreciates the support of its event coordinators, many of whom are teachers attending the conference with students from their chapters. The busy schedules of these individuals prompt the reminders that follow.

- A. Competitive event coordinators must be present for a mandatory event coordinator's meeting on the first day of the conference.
- B. Competitive event coordinators must be present for conference event check-in and check-out if they are coordinating an event in which these activities take place.
 1. Generally speaking, "check-in" is on the evening of registration day, and "check-out" is held on the day before the awards ceremony.
 2. Tentative schedule information will be available before the conference on the [TSA website](#).
- C. The Competition Regulations Committee, which consists of all the event managers, is available throughout the conference to support coordinators as they supervise competitive events.

AWARDS

At the conference awards ceremony, ten (10) finalists in each event are identified in random order and called to the stage for recognition.

EVENT PROPOSAL INFORMATION

As technology evolves and technology education attempts to keep pace and reflect these changes, new TSA events are added, some are revised, and others are dropped. TSA chapter advisors, state advisors, and others are encouraged to submit proposals for new events.

The following topics reflect potential direction for development:

- 21st century technology
- 3D printing
- Adaptive/Assistive Technology
- Artificial Intelligence
- Cloud computing
- CoDrone
- Cyber Robotics Coding
- Data management
- Economic development
- Electronic publishing
- Engineering
- Environmental technology
- Fluid power technology
- Future technologies
- Green technology
- Innovative power sources
- Lasers/satellites/radar
- Manufacturing technology
- Social media marketing
- Transportation technology

When submitting a proposal for consideration, include these elements:

- Overview (description of the event and participant expectations)
- Eligibility for entry
- Limitations (such as time or entry submission requirements)
- Resource considerations (i.e. are the resources a limiting factor, or are they affordable/readily available to all populations? Can this be executed at the national level?)
- Specific regulations
- Required personnel
- Alignment with STEM standards

Formative ideas are welcome, but the more complete the proposal the less likely it will be misinterpreted. The Competition Regulations Committee (CRC) acknowledges all submissions, and each is given consideration for possible inclusion in a competitive events guide. Once submitted, ideas and events become the property of national TSA. Proposals must be submitted by September 1 of the prior conference year in order to be considered for the next guide.

Find the form in the Forms Appendix of this guide. Proposals must include the submitter's name and complete contact information. Proposals may be mailed to CRC, c/o National TSA, 1904 Association Drive, Reston, VA 20191-1540, or emailed in a Word file attachment to general@tsaweb.org.

DRESS CODE



NATIONAL TSA CONFERENCE DRESS CODE AND OFFICIAL CONFERENCE ATTIRE GUIDELINES

- A. Chapter and state advisors, parents, and chaperones are responsible for seeing that all TSA student members wear TSA competition, general session, or casual attire as occasions may require.
- B. Everyone who is registered for the conference, including parents, guests, and children, must comply with the TSA dress code policy.
- C. TSA attire may be purchased online via the SHOP tab on the [TSA website](#).
- D. Because adults (advisors, parents, and guests) serve as role models at TSA conferences and activities, they are expected to dress appropriately for all TSA occasions they attend.
- E. Students must adhere to the TSA dress code requirements as listed in this section and on the [TSA website](#).
- F. When students compete in any competitive event they must wear competition attire.
- G. Students not in appropriate competition attire when they compete may be allowed to participate in an event, but they will lose twenty percent (20%) of the total possible points per round.

COMPETITION ATTIRE

- 1. **Shirt:** official royal blue TSA shirt
- 2. **Pants, or at least knee-length Skirt:** gray
 - *Unacceptable:* jeans; jeggings; leggings; baggy pants; exterior pocket pants; shorts
- 3. **Shoes:** black dress shoes worn with black or dark blue socks, hosiery (optional):
 - open-toed shoes or sandals are acceptable
 - *Unacceptable:* athletic shoes; flip-flops; military boots; or work boots

Required additional attire for Middle and High School Chapter Team event

- 1. **Blazer:** navy blue with official TSA patch
- 2. **Tie:** official TSA tie (males)
 - Females are not penalized for wearing the official TSA tie to Chapter Team or any other competitive event

Above attire may be worn for other competitions if preferred by event participants.

GENERAL SESSION ATTIRE

- 1. **Shirt:** the official TSA shirt (royal blue) is preferred; button-down shirt or a polo/golf shirt
 - *Unacceptable:* t-shirts; halter tops; tank tops
- 2. **Pants, or at least knee-length Dress or Skirt**
 - *Unacceptable:* jeans; jeggings; leggings; baggy pants; exterior pocket pants; shorts
- 3. **Optional Blazer:** navy blue with official TSA patch
- 4. **Optional Tie:** official TSA tie
- 5. **Shoes:** dress shoes worn with dark socks, hosiery (optional):
 - open-toed shoes or sandals are acceptable
 - *Unacceptable:* athletic shoes; flip-flops; military boots; or work boots

CASUAL ATTIRE

- 1. Appropriate t-shirts, shorts, or jeans.
- 2. Casual attire **may not** be worn at competitions or general sessions.

*Registered parents, guests, and children who are not compliant with General Session Attire and who wish to attend the Awards Ceremony, may be asked to sit in a designated section, if permitted entry.

*Approved by the TSA, Inc. Board of Directors.
The above supersedes any other dress code prior to September 24, 2022.*

TSA'S LEADERSHIP PROGRAM



PARTNERSHIP FOR 21ST CENTURY SKILLS (P21)

In 2002, the Partnership for 21st Century Skills (now the Partnership for 21st Century Learning, or P21) was founded as a non-profit organization by a coalition that included members of the national business community, education leaders, and policymakers.

21st century skills comprise skills, abilities, and learning dispositions that have been identified as being required for success in 21st century society and workplaces by educators, business leaders, academics, and governmental agencies. This is part of a growing international movement focusing on the skills required for students to master in preparation for success in a rapidly changing, digital society. Many of these skills are also associated with deeper learning, which is based on mastering skills such as analytic reasoning, complex problem solving, and teamwork. TSA's competitive events provide a natural platform to highlight the leadership and 21st century capabilities of students.

TSA's leadership program engages participants to be the best member they can be, as they seek knowledge about themselves, the organization, and their community, while developing and demonstrating leadership and 21st century skills. Leadership and 21st century skills components are all specifically tailored for each individual competitive event, and are evaluated based on the official rules and rubrics.

- For example, in one competitive event team members might note the communication, collaboration, and teamwork skills they used to finalize their idea/design in their Work Log. While in another event, a brief discussion of leadership skills and/or 21st century skills that they developed or demonstrated while working on a project might be highlighted as part of an existing presentation/interview. Criteria will be included in the rubric to evaluate the leadership and 21st century skills documented or demonstrated within these components.

TSA's leadership program has recently been revised to incorporate the 21st century skills.

TSA will provide related resources to affiliated chapters through the updated TSA member database. Participation in the TSA competitive events develops leadership and 21st century skills in student members, skills essential for success in the job market.

- There will be other competitive events in which a student/team may naturally demonstrate leadership skills as part of the event. In these events, criteria will be included in the rubric to evaluate the overall leadership and 21st century skills demonstrated.

During the course of preparing for, and participating in a TSA competitive event, participants will study leadership and 21st century skills, and put them into practice. Participants will use the widely accepted leadership and 21st century skills resources, in addition to other resources provided on the TSA website, as they complete the competitive event leadership requirements for all TSA competitions.

TSA Voices: Leadership Lessons from Alumni

A podcast series on the [TSA website](#) featuring TSA alumni who reflect on the TSA experiences that shaped their personal growth and career. In the episodes, alumni (one per episode) share their story by focusing on the 21st Century Leadership Skills they developed through TSA and how those skills played a role in their personal and career journey. These stories offer a meaningful understanding of how the lessons learned in TSA translate into real-world experiences, providing valuable insights for TSA members as they grow as leaders.



TSA's leadership program focuses on the below definitions of leadership and 21st century skills as developed through participation in middle and high school competitions:

Communication: a process by which information is exchanged between individuals through a common system of symbols, signs, or behavior

Collaboration/Social Skills: to work jointly with others, especially in an intellectual endeavor

Initiative: energy or aptitude displayed in initiation of action

Problem Solving/Risk Taking: the process or act of finding a solution to a problem/the act or fact of doing something that involves danger or risk in order to achieve a goal

Critical Thinking (lateral thinking): a method for solving problems by making unusual or unexpected connections between ideas

Perseverance/Grit: continued effort to do or achieve something despite difficulties, failure, or opposition/firmness of mind or spirit--unyielding courage in the face of hardship or danger

Creativity: the quality of being creative

Relationship Building/Teamwork: work done by several associates with each doing a part but all subordinating personal prominence to the efficiency of the whole

Dependability/Integrity: capable of being trusted or depended on/firm adherence to a code of especially moral or artistic values

Flexibility/Adaptability: characterized by a ready capability to adapt to new, different, or changing requirements

SOURCES

en.wikipedia.org/wiki/21st_century_skills

www.edglossary.org/21st-century-skills

www.nea.org/home/34888.htm

SCIENCE, TECHNOLOGY, ENGINEERING, AND MATHEMATICS (STEM) INTEGRATION



In recent years, not only educators, but also political, civic, and industry leaders have pushed for a greater emphasis on STEM education in schools. It is globally recognized that in order for any nation to be competitive, our future generations must develop competency in the 21st century skills afforded through STEM fields. TSA promotes a vision of students literate in these fields and believes competitions within this guide help make that vision a reality.

STEM education is not just an isolated and discreet acquisition of STEM knowledge and skills. Rather, STEM education demands the interdisciplinary application of these academic fields to improve outcomes in comprehension, communication, and problem solving. It is commonly accepted that the correlation between these STEM disciplines is interdependent. In order to develop a deep comprehension of one STEM area, one must simultaneously have an encompassing knowledge of another. For example, to design and engineer with any degree of complexity, one also must be familiar with technology, mathematics, and science. To practice science, one must have a firm knowledge of mathematics and technology.

Beyond necessity, there is another reason for STEM education in our schools and why the TSA program of activities inherently aligns with STEM goals. This reason revolves around teaching, learning, and what motivates our 21st century learners.

When students participate in TSA competitions, they find they must not only embrace the value of design when they compete, but they also must conceptualize, assess, and materialize that vision. Students may choose to work

collaboratively, depending upon the requirements of an event, or they may choose to work independently.

Irrespective of this choice, students develop the essential leadership and critical thinking skills to execute their strategy and align their intention with the STEM objectives set forth in this guide. STEM education is intrinsically exciting, rewarding, and meaningful for instructors and students alike. Through TSA competitive events, instructors challenge students to solve real-world problems through project-based learning and reflective experiences. This rigorous process supplements and complements classroom objectives by asking students to critically evaluate all aspects of their thought processes—from design, to communication, to execution.

Deserving of mention are three other essential areas embedded in most of TSA's competitive events—creativity, innovation, and ethics. Teaching students to think outside the box while considering the ethical consequences provides a global perspective essential to the success of our society. Through TSA competitions, students are asked to design creatively, while assessing the effects and impacts of what they develop.

The competitions found in this guide provide a hands-on venue for learning about STEM. By participating in TSA's competitive events, students gain a broader understanding of these content areas as they experience the satisfaction that comes from applying them to real life, problem-solving situations.

This section of the guide includes commonly accepted national standards for the areas of science, technology, and mathematics, as well as the Accreditation Board for Engineering and Technology (ABET, Inc.) criteria for accrediting higher education engineering programs.



NEXT GENERATION SCIENCE STANDARDS* (GRADES 5-8)**A. Structure and Properties of Matter**

1. **PS1-1:** Develop models to describe the atomic composition of simple molecules and extended structures.
2. **PS1-3:** Gather and make sense of information to describe that synthetic materials come from natural resources and impact society.
3. **PS1-4:** Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed.

B. Chemical Reactions

1. **PS1-2:** Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.
2. **PS1-5:** Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.
3. **PS1-6:** Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.*

C. Forces and Interactions

1. **PS2-1:** Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects.*
2. **PS2-2:** Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.
3. **PS2-3:** Ask questions about data to determine the factors that affect the strength of electric and magnetic forces.
4. **PS2-4:** Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects.
5. **PS2-5:** Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact.

D. Energy

1. **PS3-1:** Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object.
2. **PS3-2:** Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system
3. **PS3-3:** Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.*
4. **PS3-4:** Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.
5. **PS3-5:** Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.

E. Waves and Electromagnetic Radiation

1. **PS4-1:** Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.
2. **PS4-2:** Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials.
3. **PS4-3:** Integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to encode and transmit information than analog signals.

F. Structure, Function, and Information Processing

1. **LS1-1:** Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells.
2. **LS1-2:** Develop and use a model to describe the function of a cell as a whole and ways the parts of cells contribute to the function.

3. **LS1-3:** Use argument supported by evidence for how the body is a system of interacting subsystems composed of groups of cells.
4. **LS1-8:** Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

G. Matter and Energy in Organisms and Ecosystems

1. **LS1-6:** Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms.
2. **LS1-7:** Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.
3. **LS2-1:** Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem.
4. **LS2-3:** Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem.
5. **LS2-4:** Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

H. Interdependent Relationships in Ecosystems

1. **LS2-2:** Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems
2. **LS2-5:** Evaluate competing design solutions for maintaining biodiversity and ecosystem services.

I. Growth, Development, and Reproduction of Organisms

1. **LS1-4:** Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively.

2. **LS1-5:** Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms.
3. **LS3-1:** Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism.
4. **LS3-2:** Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.
5. **LS4-5:** Gather and synthesize information about technologies that have changed the way humans influence the inheritance of desired traits in organisms.

J. Natural Selection and Adaptations

1. **LS4-1:** Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past.
2. **LS4-2:** Apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern organisms and between modern and fossil organisms to infer evolutionary relationships.
3. **LS4-3:** Analyze displays of pictorial data to compare patterns of similarities in the embryological development across multiple species to identify relationships not evident in the fully formed anatomy.
4. **LS4-4:** Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment
5. **LS4-6:** Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

K. Space Systems

1. **ESS1-1:** Develop and use a model of the Earth-sun-moon system to describe the cyclic patterns of lunar phases, eclipses of the sun and moon, and seasons
2. **ESS1-2:** Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system
3. **ESS1-3:** Analyze and interpret data to determine scale properties of objects in the solar system

L. History of Earth

1. **ESS1-4:** Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.
2. **ESS2-2:** Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales
3. **ESS2-3:** Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions

M. Earth's Systems

1. **ESS2-1:** Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.
2. **ESS2-4:** Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity.
3. **ESS3-1:** Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes.

N. Weather and Climate

1. **ESS2-5:** Collect data to provide evidence for how the motions and complex interactions of air masses result in changes in weather conditions.
2. **ESS2-6:** Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates

3. **ESS3-5:** Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

O. Human Impacts

1. **ESS3-2:** Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects.
2. **ESS3-3:** Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment.
3. **ESS3-4:** Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems.

P. Engineering Design

1. **ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.
2. **ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
3. **ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.
4. **ETS1-4:** Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

Although not formally aligned, this standards alignment of TSA competitive events has been developed in accordance with the Next Generation Science Standards (NGSS) model.

*The Next Generation Science Standards (NGSS) were developed by educators, content experts and policymakers, using as a guiding document the Framework for K-12 Science Education from the National Research Council. The Next Generation Science Standards is a registered trademark of Achieve. Neither Achieve nor the lead states and partners that developed the Next Generation Science Standards were involved in the production of this product, and do not endorse it.

NEXT GENERATION SCIENCE STANDARDS																																
Event	Standard Number	A-PS1-1	A-PS1-3	A-PS1-4	B-PS1-2	B-PS1-5	B-PS1-6	C-PS2-1	C-PS2-2	C-PS2-3	C-PS2-4	C-PS2-5	D-PS3-1	D-PS3-2	D-PS3-3	D-PS3-4	D-PS3-5	E-PS4-1	E-PS4-2	E-PS4-3	F-LS1-1	F-LS1-2	F-LS1-3	F-LS1-8	G-LS1-6	G-LS1-7	G-LS2-1	G-LS2-3	G-LS2-4	H-LS2-2	H-LS2-5	
Audio Podcasting																																
Biotechnology		X																				X										
Career Prep																																
Challenging Technology Issues																																
Chapter Team																																
Children's Stories																																
Coding																																
Community Service Video																																
Computer-Aided Design (CAD) Foundations																																
Construction Challenge																																
Cybersecurity																																
Data Science and Analytics																												X	X	X	X	X
Digital Photography																																
Dragster									X																							
Drone Challenge (UAV)														X																		
Electrical Applications																																
Flight									X				X																			
Forensic Technology																																
Inventions and Innovations																																
Leadership Strategies																																
Mass Production																																
Mechanical Engineering																																
Medical Technology		X																						X								
Microcontroller Design																								X								
Off the Grid																																
Prepared Speech																																
Problem Solving																																
Promotional Marketing																																
Robotics										X				X																		
Solar Racer									X					X																		
STEM Animation																																
Structural Engineering																																
System Control Technology																																
Tech Bowl																																
Technical Design																																
Video Game Design																																
Website Design																																

[illegible]

STANDARDS FOR TECHNOLOGICAL AND ENGINEERING LITERACY (STEL)**STEL 1. Nature and Characteristics of Technology and Engineering**

- 1J. Develop innovative products and systems that solve problems and extend capabilities based on individual or collective needs and wants.
- 1K. Compare and contrast the contributions of science, engineering, mathematics, and technology in the development of technological systems.
- 1L. Explain how technology and engineering are closely linked to creativity, which can result in both intended and unintended innovations.
- 1M. Apply creative problem-solving strategies to the improvement of existing devices or processes or the development of new approaches.

STEL 2. Core Concepts of Technology and Engineering

- 2M. Differentiate between inputs, processes, outputs, and feedback in technological systems.
- 2N. Illustrate how systems thinking involves considering relationships between every part, as well as how the system interacts with the environment in which it is used.
- 2O. Create an open-loop system that has no feedback path and requires human intervention.
- 2P. Create a closed-loop system that has a feedback path and requires no human intervention.
- 2Q. Predict outcomes of a future product or system at the beginning of the design process.
- 2R. Compare how different technologies involve different sets of processes.
- 2S. Defend decisions related to a design problem.

STEL 3. Integration of Knowledge, Technologies, and Practices

- 3E. Analyze how different technological systems often interact with economic, environmental, and social systems.
- 3F. Apply a product, system or process developed for one setting to another setting.
- 3G. Explain how knowledge gained from other content areas affects the development of technological products and systems.

STEL 4. Impacts of Technology

- 4K. Examine the ways that technology can have both positive and negative effects at the same time.
- 4L. Analyze how the creation and use of technologies consumes renewable and non-renewable resources and creates waste.
- 4M. Devise strategies for reducing, reusing, and recycling waste caused from the creation and use of technology.
- 4N. Analyze examples of technologies that have changed the way people think, interact, and communicate.
- 4O. Hypothesize what alternative outcomes (individual, cultural, and/or environmental) might have resulted had a different technological solution been selected.

STEL 5. Influence of Society on Technological Development

- 5F. Analyze how an invention or innovation was influenced by its historical context.
- 5G. Evaluate trade-offs based on various perspectives as part of a decision process that recognizes the need for careful compromises among competing factors.

STEL 6. History of Technology

- 6C. Compare various technologies and how they have contributed to human progress.
- 6D. Engage in a research and development process to simulate how inventions and innovations have evolved through systematic tests and refinements.
- 6E. Verify how specialization of function has been at the heart of many technological improvements.

STEL 7. Design in Technology and Engineering Education

- 7P. Illustrate the benefits and opportunities associated with different approaches to design.
- 7Q. Apply the technology and engineering design process.
- 7R. Refine design solutions to address criteria and constraints.
- 7S. Create solutions to problems by identifying and applying human factors in design.
- 7T. Assess design quality based upon established principles and elements of design.
- 7U. Evaluate the strengths and weaknesses of different design solutions.
- 7V. Improve essential skills necessary to successfully design.

STEL 8. Applying, Maintaining, and Assessing Technological Products and Systems

- 8H. Research information from various sources to use and maintain technological products or systems.
- 8I. Use tools, materials, and machines to safely diagnose, adjust, and repair systems.
- 8J. Use devices to control technological systems.
- 8K. Design methods to gather data about technological systems.
- 8L. Interpret the accuracy of information collected.
- 8M. Use instruments to gather data on the performance of everyday products.

STANDARDS FOR TECHNOLOGICAL AND ENGINEERING LITERACY (STEL)		1J	1K	1L	1M	2M	2N	2O	2P	2Q	2R	2S	3E	3F	3G	4K	4L	4M	4N	4O
Event	STEL Benchmark																			
Audio Podcasting							X	X		X	X	X	X				X	X	X	X
Biotechnology			X		X		X				X		X				X	X	X	X
Career Prep			X	X	X								X						X	
Challenging Technology Issues			X	X	X						X		X						X	X
Chapter Team																				
Children's Stories												X								
Coding		X			X			X	X	X		X								
Community Service Video																				
Computer-Aided Design (CAD) Foundations																				
Construction Challenge					X		X		X	X		X	X	X	X	X			X	
Cybersecurity		X	X	X	X		X				X	X	X	X	X	X	X		X	X
Data Science and Analytics					X		X						X						X	
Digital Photography																				
Dragster		X			X		X					X								
Drone Challenge (UAV)				X	X		X	X	X	X	X	X							X	
Electrical Applications		X			X	X	X			X		X								
Flight		X			X		X			X										
Forensic Technology			X													X				X
Inventions and Innovations		X	X		X	X	X	X	X	X	X	X	X	X	X					
Leadership Strategies																				
Mass Production		X			X						X	X			X					
Mechanical Engineering		X		X	X	X	X	X	X	X	X	X			X					X
Medical Technology		X	X	X	X		X					X	X		X				X	X
Microcontroller Design		X			X	X	X	X	X	X		X							X	
Off the Grid		X	X	X	X		X			X		X	X		X		X	X		X
Prepared Speech																				
Problem Solving		X			X		X													
Promotional Marketing					X							X								
Robotics		X			X		X	X		X	X	X								
Solar Racer		X			X							X								
STEM Animation					X							X							X	
Structural Engineering		X			X		X													
System Control Technology		X			X	X		X	X		X	X			X					
Tech Bowl																				
Technical Design					X					X	X	X	X	X						
Video Game Design		X			X			X	X			X								
Website Design		X			X			X	X				X							

STANDARDS FOR TECHNOLOGICAL AND ENGINEERING LITERACY (STEL)															
Event	STEL Benchmark	5F	5G	6C	6D	6E	7P	7Q	7R	7S	7T	7U	7V	8H	8I
Audio Podcasting															
Biotechnology				X	X	X								X	
Career Prep		X		X										X	
Challenging Technology Issues				X	X	X								X	
Chapter Team															
Children's Stories								X	X	X	X	X	X		
Coding								X	X	X	X	X	X		
Community Service Video															
Computer-Aided Design (CAD) Foundations								X	X		X	X	X		
Construction Challenge			X		X			X	X	X	X	X	X	X	
Cybersecurity			X		X		X	X	X	X	X	X	X	X	X
Data Science and Analytics														X	
Digital Photography							X	X	X	X	X	X	X		
Dragster								X	X	X	X	X	X		
Drone Challenge (UAV)							X	X	X	X	X	X	X	X	
Electrical Applications								X	X	X	X	X	X	X	X
Flight					X		X	X	X	X	X	X	X	X	X
Forensic Technology				X										X	
Inventions and Innovations		X	X		X	X	X	X	X	X	X	X	X	X	X
Leadership Strategies															
Mass Production							X	X	X	X	X	X	X	X	X
Mechanical Engineering							X	X	X	X	X	X	X	X	X
Medical Technology					X		X	X	X	X	X	X	X	X	X
Microcontroller Design			X				X	X	X	X	X	X	X	X	X
Off the Grid		X	X		X	X	X	X	X	X	X	X	X	X	X
Prepared Speech															
Problem Solving								X	X	X	X	X	X		
Promotional Marketing							X	X	X	X	X	X	X		
Robotics							X	X	X	X	X	X	X	X	
Junior Solar Sprint							X	X	X	X	X	X	X		
STEM Animation							X	X	X	X	X	X	X		
Structural Engineering								X	X	X	X	X	X		
System Control Technology								X	X	X	X	X	X		
Tech Bowl														X	
Technical Design							X	X	X	X	X	X	X	X	
Video Game Design								X	X	X	X	X	X		
Website Design								X	X	X	X	X	X		

COMPUTER SCIENCE TEACHERS ASSOCIATION (CSTA) K-12 COMPUTER SCIENCE STANDARDS**Level 2: Grades 6-8****CS: Computing Systems**

01. Recommend improvements to the design of computing devices, based on an analysis of how users interact with the devices.
02. Design projects that combine hardware and software components to collect and exchange data.
03. Systematically identify and fix problems with computing devices and their components.

NI: Networks and the Internet

04. Model the role of protocols in transmitting data across networks and the Internet.
05. Explain how physical and digital security measures protect electronic information.
06. Apply multiple methods of encryption to model the secure transmission of information.

DA: Data and Analysis

07. Represent data using multiple encoding schemes.
08. Collect data using computational tools and transform the data to make it more useful and reliable.
09. Refine computational models based on the data they have generated.

AP: Algorithms and Programming

10. Use flowcharts and/or pseudocode to address complex problems as algorithms.
11. Create clearly named variables that represent different data types and perform operations on their values.
12. Design and iteratively develop programs that combine control structures, including nested loops and compound conditionals.
13. Decompose problems and subproblems into parts to facilitate the design, implementation, and review of programs.

14. Create procedures with parameters to organize code and make it easier to reuse.
15. Seek and incorporate feedback from team members and users to refine a solution that meets user needs.
16. Incorporate existing code, media, and libraries into original programs, and give attribution.
17. Systematically test and refine programs using a range of test cases.
18. Distribute tasks and maintain a project timeline when collaboratively developing computational artifacts.
19. Document programs in order to make them easier to follow, test, and debug.

IC: Impacts of Computing

20. Compare tradeoffs associated with computing technologies that affect people's everyday activities and career options.
21. Discuss issues of bias and accessibility in the design of existing technologies.
22. Collaborate with many contributors through strategies such as crowdsourcing or surveys when creating a computational artifact.
23. Describe tradeoffs between allowing information to be public and keeping information private and secure.

CSTA K-12 COMPUTER SCIENCE STANDARDS												
Event	Standard Number	2-CS-02	2-CS-03	2-NI-05	2-NI-06	2-DA-08	2-DA-09	2-AP-11	2-AP-12	2-AP-14	2-AP-14	2-AP-18
Audio Podcasting												
Biotechnology												
Career Prep												
Challenging Technology Issues												
Chapter Team												
Children's Stories												
Coding								X	X	X		
Community Service Video												
Computer-Aided Design (CAD) Foundations												
Construction Challenge												
Cybersecurity				X	X							
Data Science and Analytics						X	X					
Digital Photography												
Dragster												
Drone Challenge (UAV)												
Electrical Applications		X	X									
Flight												
Forensic Technology												
Inventions and Innovations												
Leadership Strategies												
Mass Production												
Mechanical Engineering												
Medical Technology												
Microcontroller Design		X	X					X	X	X	X	X
Off the Grid												
Prepared Speech												
Problem Solving												
Promotional Marketing												
Robotics												
Solar Racer		X	X									
STEM Animation												
Structural Engineering												
System Control Technology		X	X					X	X	X		
Tech Bowl												
Technical Design												
Video Game Design								X	X	X	X	X
Vlogging												
Website Design										X	X	X

**ISTE STANDARDS FOR STUDENTS –
2016 INTERNATIONAL SOCIETY FOR TECHNOLOGY IN EDUCATION****1. Empowered Learner**

Students leverage technology to take an active role in choosing, achieving and demonstrating competency in their learning goals, informed by the learning sciences.

- a. articulate and set personal learning goals, develop strategies leveraging technology to achieve them and reflect on the learning process itself to improve learning outcomes
- b. build networks and customize their learning environments in ways that support the learning process
- c. use technology to seek feedback that informs and improves their practice and to demonstrate their learning in a variety of ways
- d. understand the fundamental concepts of technology operations, demonstrate the ability to choose, use and troubleshoot current technologies and are able to transfer their knowledge to explore emerging technologies

2. Digital Citizen

Students recognize the rights, responsibilities and opportunities of living, learning and working in an interconnected digital world, and they act and model in ways that are safe, legal and ethical.

- a. cultivate and manage their digital identity and reputation and are aware of the permanence of their actions in the digital world
- b. engage in positive, safe, legal and ethical behavior when using technology, including social interactions online or when using networked devices
- c. demonstrate an understanding of and respect for the rights and obligations of using and sharing intellectual property
- d. manage their personal data to maintain digital privacy and security and are aware of data-collection technology used to track their navigation online

3. Knowledge Constructor

Students critically curate a variety of resources using digital tools to construct knowledge, produce creative artifacts and make meaningful learning experiences for themselves and others.

- a. plan and employ effective research strategies to locate information and other resources for their intellectual or creative pursuits
- b. evaluate the accuracy, perspective, credibility and relevance of information, media, data or other resources
- c. curate information from digital resources using a variety of tools and methods to create collections of artifacts that demonstrate meaningful connections or conclusions
- d. build knowledge by actively exploring real-world issues and problems, developing ideas and theories and pursuing answers and solutions

4. Innovative Designer

Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions.

- a. know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems
- b. select and use digital tools to plan and manage a design process that considers design constraints and calculated risks
- c. develop, test and refine prototypes as part of a cyclical design process
- d. exhibit a tolerance for ambiguity, perseverance and the capacity to work with open-ended problems

5. Computational Thinker

Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions.

- a. formulate problem definitions suited for technology-assisted methods such as data analysis, abstract models and algorithmic thinking in exploring and finding solutions
- b. collect data or identify relevant data sets, use digital tools to analyze them, and represent data in various ways to facilitate problem-solving and decision-making
- c. break problems into component parts, extract key information, and develop descriptive models to understand complex systems or facilitate problem-solving
- d. understand how automation works and use algorithmic thinking to develop a sequence of steps to create and test automated solutions.

6. Creative Communicator

Students communicate clearly and express themselves creatively for a variety of purposes using the platforms, tools, styles, formats and digital media appropriate to their goals.

- a. choose the appropriate platforms and tools for meeting the desired objectives of their creation or communication
- b. create original works or responsibly repurpose or remix digital resources into new creations
- c. communicate complex ideas clearly and effectively by creating or using a variety of digital objects such as visualizations, models or simulations
- d. publish or present content that customizes the message and medium for their intended audiences

7. Global Collaborator

Students use digital tools to broaden their perspectives and enrich their learning by collaborating with others and working effectively in teams locally and globally.

- a. use digital tools to connect with learners from a variety of backgrounds and cultures, engaging with them in ways that broaden mutual understanding and learning
- b. use collaborative technologies to work with others, including peers, experts or community members, to examine issues and problems from multiple viewpoints
- c. contribute constructively to project teams, assuming various roles and responsibilities to work effectively toward a common goal
- d. explore local and global issues and use collaborative technologies to work with others to investigate solutions

Although not formally aligned, this standards alignment of TSA competitive events has been developed in accordance with the ISTE Standards for Students framework. The ISTE Standards for Students are a framework for teaching and learning in the digital age and are adopted by schools, districts, states locally, nationally and internationally. The ISTE Standards for Students are a registered trademark of International Society for Technology in Education (ISTE). ISTE was not involved in the production of this product and does not endorse, support, or sponsor it.

ISTE STANDARDS FOR STUDENTS																													
Event	Standard Number	1a	1b	1c	1d	2a	2b	2c	2d	3a	3b	3c	3d	4a	4b	4c	4d	5a	5b	5c	5d	6a	6b	6c	6d	7a	7b	7c	7d
Audio Podcasting		X			X	X	X	X	X	X	X	X	X	X	X	X	X					X	X	X	X	X	X	X	X
Biotechnology		X								X	X	X	X	X	X			X	X										
Career Prep																													
Challenging Technology Issues																													
Chapter Team																													
Children's Stories																													
Coding				X	X																						X		
Community Service Video				X	X					X		X	X				X					X	X	X	X	X	X	X	X
Computer-Aided Design (CAD) Foundations				X	X									X	X							X	X	X	X				
Construction Challenge																													
Cybersecurity			X			X	X	X	X				X				X										X	X	X
Data Science and Analytics		X						X		X	X	X	X				X	X	X	X			X	X			X	X	X
Digital Photography					X					X																			
Dragster																													
Drone Challenge (UAV)		X		X	X									X	X	X	X				X	X	X	X	X		X	X	X
Electrical Applications																													
Flight																													
Forensic Technology																													
Inventions and Innovations																													
Leadership Strategies																													
Mass Production																													
Mechanical Engineering																													
Medical Technology		X		X	X					X	X	X	X	X	X	X	X	X	X				X	X			X	X	X
Microcontroller Design		X	X		X					X		X	X	X	X	X	X	X	X	X	X	X	X	X			X	X	X
Off the Grid																													
Prepared Speech																													
Problem Solving																													
Promotional Marketing							X															X	X	X					
Robotics		X		X	X									X	X	X	X	X			X	X	X	X	X		X	X	X
Solar Racer																													
STEM Animation		X		X	X		X							X								X	X	X	X		X	X	X
Structural Engineering																													
System Control Technology		X		X	X					X		X	X	X	X	X	X	X	X	X	X	X	X	X					X
Tech Bowl																													
Technical Design														X			X												
Video Game Design		X		X	X			X																				X	X
Website Design		X	X		X	X		X	X													X	X	X	X	X		X	X

**NATIONAL COUNCIL OF TEACHERS OF MATHEMATICS (NCTM)
PRINCIPLES AND STANDARDS FOR SCHOOL MATHEMATICS****A. Numbers and operations**

1. Understand numbers, ways of representing numbers, relationships among numbers, and number systems
2. Understand meanings of operations and how they relate to one another
3. Compute fluently and make reasonable estimates

B. Algebra

1. Understand patterns, relations, and functions
2. Represent and analyze mathematical situations and structures using algebraic symbols
3. Use mathematical models to represent and understand quantitative relationships
4. Analyze change in various contexts

C. Geometry

1. Analyze characteristics and properties of two- and three-dimensional geometric shapes, and develop mathematical arguments about geometric relationships
2. Specify locations and describe spatial relationships using coordinate geometry and other representational systems
3. Apply transformations and use symmetry to analyze mathematical situations
4. Use visualization, spatial reasoning, and geometric modeling to solve problems

D. Measurement

1. Understand measurable attributes of objects and the units, systems, and processes of measurement
2. Apply appropriate techniques, tools, and formulas to determine measurements

E. Data analysis and probability

1. Formulate questions that can be addressed with data, and collect, organize, and display relevant data to answer them
2. Select and use appropriate statistical methods to analyze data
3. Develop and evaluate inferences and predictions that are based on data
4. Understand and apply basic concepts of probability

F. Problem solving

1. Build new mathematical knowledge through problem solving
2. Solve problems that arise in mathematics and in other contexts
3. Apply and adapt a variety of appropriate strategies to solve problems
4. Monitor and reflect on the process of mathematical problem solving

G. Reasoning and proof

1. Recognize reasoning and proof as fundamental aspects of mathematics
2. Make and investigate mathematical conjectures
3. Develop and evaluate mathematical arguments and proofs
4. Select and use various types of reasoning and methods of proof

H. Communication

1. Organize and consolidate mathematical thinking through communication
2. Communicate mathematical thinking coherently and clearly to peers, teachers, and others
3. Analyze and evaluate the mathematical thinking and strategies of others
4. Use the language of mathematics to express mathematical ideas precisely

I. **Connections**

1. Recognize and use connections among mathematical ideas
2. Understand how mathematical ideas interconnect and build on one another to produce a coherent whole
3. Recognize and apply mathematics in contexts outside of mathematics

J. **Representation**

1. Create and use representations to organize, record, and communicate mathematical ideas
2. Select, apply, and translate among mathematical representations to solve problems
3. Use representations to model and interpret physical, social, and mathematical phenomena

Reprinted with permission from *Principles and Standards for School Mathematics*, copyright 2000 by the National Council of Teachers of Mathematics (NCTM). All rights reserved.

National Council of Teachers of Mathematics (NCTM) Principles and Standards for School Mathematics																																					
Event	Standard Number	A1	A2	A3	B1	B2	B3	B4	C1	C2	C3	C4	D1	D2	E1	E2	E3	E4	F1	F2	F3	F4	G1	G2	G3	G4	H1	H2	H3	H4	I1	I2	I3	J1	J2	J3	
Audio Podcasting																				X																	
Biotechnology																											X										
Career Prep																																					
Challenging Technology Issues																											X										
Chapter Team																											X										
Children's Stories																																					
Coding																																					
Community Service Video			X																																		
Computer-Aided Design (CAD) Foundations									X	X		X	X																								
Construction Challenge													X	X							X																
Cybersecurity															X	X	X	X			X																
Data Science and Analytics	X		X	X	X		X								X	X		X							X		X		X								
Digital Photography																																					
Dragster								X	X	X		X	X							X							X										
Drone Challenge (UAV)												X		X						X																	
Electrical Applications													X	X	X	X			X																		
Flight			X	X	X				X			X	X					X	X	X	X					X		X									
Forensic Technology															X						X																
Inventions and Innovations																																					
Leadership Strategies																																					
Mass Production													X							X																	
Mechanical Engineering								X	X											X																	
Medical Technology	X	X	X	X	X		X	X					X					X	X	X	X					X		X									
Microcontroller Design																																					
Off the Grid													X							X																	
Prepared Speech																																					
Problem Solving													X						X																		
Promotional Marketing													X																								
Robotics												X	X							X																	
Junior Solar Sprint	X	X	X						X	X		X	X					X	X	X	X					X		X									
STEM Animation	X	X	X			X														X						X		X									
Structural Engineering			X	X				X	X			X	X						X		X					X		X									
System Control Technology	X	X	X	X	X	X	X		X	X	X	X	X					X	X	X	X				X		X		X								
Tech Bowl									X	X		X	X							X																	
Technical Design									X	X		X	X							X																	
Video Game Design									X	X	X	X	X																								
Website Design													X							X																	

All learners deserve to participate in meaningful career exploration and preparation experiences through Career Technical Education (CTE) that prepare them for the changing and interconnected world of work. This requires systems and structures that are accessible, responsive to evolving industry needs, and flexible for the needs of each state and community. Since 2002, The National Career Clusters® Framework has provided a shared structure and language for CTE program design across the United States. Advance CTE serves as the steward of the Framework.

In fall 2024 following 2 years of input and validation, Advance CTE released a modernized Framework designed to serve as a bridge between education and work and a central building block for consistently designed and high-quality CTE programs.

The modern Framework consists of 14 Clusters and 72 Sub-Clusters that serve as the primary organizing structures for CTE programs. These structures are supported by five Cluster Groupings aligned to purpose and impact of included careers, and three Cross-Cutting Clusters that provide both skills and careers that can stand alone and intersect with all other Clusters. Twelve Career-Ready Practices to ensure that every program includes the skills that are essential to every career and life.

The TSA Competitions and The National Career Clusters Framework® grid illustrates the interconnectedness between individual TSA competitions and the Framework®. Use these together as a starting point to help your students become informed about careers and develop a plan to reach their life goals. Please note that several TSA competitions incorporate annual themes and challenges that may align with additional clusters, cluster groupings and subclusters.



© 2024. The National Career Clusters Framework®. All rights reserved. More information on the Career Clusters®, Pathways to College and Career Readiness can be found at careertech.org/career-clusters.

CROSS CUTTING CLUSTERS

DIGITAL TECHNOLOGY

- Data Science & Artificial Intelligence
- Information Technology (IT) Support & Services
- Network Systems & Cybersecurity
- Software Solutions
- Unmanned Vehicle Technology
- Web & Cloud

MARKETING & SALES

- Market Research, Analytics, & Ethics
- Marketing & Advertising
- Retail & Customer Experience
- Strategic Sales

MANAGEMENT & ENTREPRENEURSHIP

- Business Information Management
- Entrepreneurship & Small Business
- Leadership & Operations
- Project Management
- Regulation

CLUSTER GROUPINGS/SUBCLUSTERS

A. CARING FOR COMMUNITIES

1. Education

- Early Childhood Development
- Education Administration & Leadership
- Learner Support & Community Engagement
- Teaching, Training & Facilitation

2. Healthcare & Human Services

- Behavioral & Mental Health
- Biotechnology Research & Development
- Community & Social Services
- Health Data & Administration
- Personal Care Services
- Physical Health

3. Public Service & Safety

- Emergency Response
- Judicial Systems
- Local, State & Federal Services
- Military & National Security
- Public Safety

B. BUILDING & MOVING**1. Advanced Manufacturing**

- Engineering
- Industrial Machinery
- Production & Automation
- Robotics
- Safety & Quality Assurance

2. Construction

- Architecture & Civil Engineering
- Construction Planning & Development
- Equipment Operation & Maintenance
- Skilled Trades

3. Supply Chain & Transportation

- Air & Space Transportation
- Ground & Rail Transportation
- Maintenance & Repair
- Marine Transportation
- Planning & Logistics
- Purchasing & Warehousing

C. CREATING & EXPERIENCING**1. Arts, Entertainment, & Design**

- Design & Digital Arts
- Fashion & Interiors
- Fine Arts
- Lighting & Sound Technology
- Media Production & Broadcasting
- Performing Arts

2. Hospitality, Events, & Tourism

- Accommodations
- Conferences & Events
- Culinary & Food Services
- Travel & Leisure

D. CULTIVATING RESOURCES**1. Agriculture**

- Agribusiness
- Agricultural Technology & Automation
- Animal Systems
- Food Science & Processing
- Plant Systems
- Water Systems

2. Energy & Natural Resources

- Clean & Alternative Energy
- Conservation & Land Management
- Ecological Research & Development
- Environmental Protection
- Resource Extraction
- Utilities

E. INVESTING IN THE FUTURE**1. Financial Services**

- Accounting
- Banking & Credit
- Financial Strategy & Investments
- Insurance
- Real Estate

TSA AND CAREERS																* Several competitions incorporate annual themes and challenges that may align with additional clusters, cluster groupings and subclusters.															
Event		Standard	CROSS-CUTTING CLUSTERS				CLUSTER GROUPINGS/SUBCLUSTERS																								
			DIGITAL TECHNOLOGY	MARKETING AND SALES	MANAGEMENT & ENTREPRENEURSHIP	A1	A2	A3	B1	B2	B3	C1	C2*	D1	D2*	E															
Audio Podcasting			X	X								X				E1*															
Biotechnology			X				X							X	X																
Career Prep					X																										
Challenging Technology Issues			X			X	X					X																			
Chapter Team			X		X	X						X																			
Children's Stories				X	X	X						X																			
Coding			X								X	X																			
Community Service Video			X	X	X	X	X	X				X		X																	
Computer-Aided Design (CAD) Foundations									X																						
Construction Challenge							X		X	X	X	X		X																	
Cybersecurity			X				X	X			X			X																	
Data Science and Analytics			X	X			X	X						X																	
Digital Photography			X					X				X																			
Dragster									X	X	X																				
Drone Challenge (UAV)			X					X	X	X	X	X		X																	
Electrical Applications								X	X	X				X																	
Flight			X						X	X	X																				
Forensic Technology					X			X																							
Inventions and Innovations					X			X	X																						
Leadership Strategies				X				X																							
Mass Production			X		X				X	X	X			X																	
Mechanical Engineering					X			X	X	X	X			X																	
Medical Technology						X	X	X						X																	
Microcontroller Design					X				X	X	X																				
Off the Grid			X					X		X	X			X	X																
Prepared Speech																															
Problem Solving						X	X	X						X																	
Promotional Marketing			X	X			X	X				X																			
Robotics			X				X	X	X	X	X	X		X																	
Solar Racer			X						X		X				X																
STEM Animation			X						X	X		X																			
Structural Engineering						X	X	X	X	X	X			X																	
System Control Technology									X	X	X																				
Tech Bowl			X			X						X																			
Technical Design									X	X	X			X																	
Video Game Design			X		X	X		X				X																			
Website Design			X		X	X		X				X																			

MIDDLE SCHOOL COMPETITIVE EVENTS

NEW EVENTS

- Audio Podcasting
- Drone Challenge (UAV)
- Robotics
- Solar Racer

OF NOTE

- Some event's guidelines have been revised in some form, whether in content or in format.
- In addition to specific event guidelines, all advisors and participants must read the General Rules and Regulations in this guide.
- Every two years the specifics of many events are changed, keeping the competitions dynamic.

Audio Podcasting
Biotechnology
Career Prep
Challenging Technology Issues
Chapter Team
Children's Stories
Coding
Community Service Video
Computer-Aided Design (CAD) Foundations
Construction Challenge
Cybersecurity
Data Science and Analytics
Digital Photography
Dragster
Drone Challenge (UAV)
Electrical Applications
Flight
Forensic Technology
Inventions and Innovations
Leadership Strategies
Mass Production
Mechanical Engineering
Medical Technology
Microcontroller Design
Off the Grid
Prepared Speech
Problem Solving
Promotional Marketing
Robotics
Solar Racer
STEM Animation
Structural Engineering
System Control Technology
Tech Bowl
Technical Design
Video Game Design
Website Design

COMPETITIVE EVENTS ELIGIBILITY



2026 & 2027 MIDDLE SCHOOL COMPETITIONS	ELIGIBILITY
Each participant/team shall submit only one [1] entry for the following competitive events.	
Audio Podcasting	three (3) teams per state; individual entries are permitted
Biotechnology	five (5) teams per state, consisting of at least three (3) individuals
Career Prep	one (1) individual per chapter
Challenging Technology Issues	three (3) teams of two (2) individuals per state
Chapter Team	one (1) team of six (6) individuals per chapter
Children's Stories	three (3) teams per state; individual entries are permitted
Coding	one (1) team of two (2) individuals per chapter
Community Service Video	one (1) team per chapter; individual entries are permitted
Computer-Aided Design (CAD) Foundations	two (2) individuals per state
Construction Challenge	one (1) team of at least two (2) individuals per chapter
Cybersecurity	three (3) individuals per chapter
Data Science and Analytics	three (3) teams of two to three (2-3) individuals per state
Digital Photography	three (3) individuals per state
Dragster	two (2) individuals per chapter
Drone Challenge (UAV)	three (3) teams of two (2) individuals per state
Electrical Applications	one (1) team of two (2) individuals per chapter
Flight	two (2) individuals per chapter
Forensic Technology	one (1) team of two (2) individuals per chapter
Inventions and Innovations	one (1) team of two (2) to four (4) individuals per chapter
Leadership Strategies	three (3) teams of three (3) individuals per state
Mass Production	one (1) team of at least three (3) individuals per chapter
Mechanical Engineering	one (1) team of two (2) individuals per chapter
Medical Technology	three (3) teams of two to three (2-3) individuals per state
Microcontroller Design	one (1) team per chapter; individual entries are permitted
Off the Grid	three (3) teams per state; individual entries are permitted
Prepared Speech	three (3) individuals per state
Problem Solving	one (1) team of two (2) individuals per chapter
Promotional Marketing	one (1) individual per chapter
Robotics	three (3) teams of two to six (2-6) team members per state
Solar Racer	one (1) team of two to four (2-4) individuals per chapter; one (1) entry per team
STEM Animation	three (3) teams of at least two (2) individuals per state
Structural Engineering	one (1) team of two (2) individuals per chapter
System Control Technology	one (1) team of three (3) individuals per state
Tech Bowl	one (1) team of three (3) individuals per chapter
Technical Design	one (1) team of two (2) individuals per chapter
Video Game Design	one (1) team of two (2) to six (6) individuals per chapter
Website Design	one (1) team of at least three (3) and a maximum of six (6) individuals per chapter

TSA COMPETITIVE EVENTS RATING FORM/RUBRIC



The Technology Student Association (TSA) Middle School Competitive Events Guide for the 2026 & 2027 National TSA Conferences contains a rating form (rubric) for each competition. Rubrics are embraced by STEM educators because they provide a way to evaluate performance. The use of descriptors for each criterion being measured in a rubric increases consistency and a greater understanding of the evaluation process. The TSA rating form/rubric provides a way for TSA members to better prepare for competitions, for advisors to carefully assist them in the process, and for judges to effectively evaluate participants and their entries.

GO/NO GO SPECIFICATIONS

- Each competitive event has a Go/No Go Specifications checklist placed at the beginning of the official event rating form/rubric.
- Specifications in the checklist are required and must be met, or the individual or team will not be allowed to compete in the event.
- Refer to each competitive event's official rating form/rubric for details.



AUDIO PODCASTING



OVERVIEW

Applying leadership and 21st century skills, participants use digital audio technology to create original content around a pre-determined technology theme. Podcasting encourages good storytelling, voice acting, and folly sound effects to create a coherent creative work. The theme will be posted on the [TSA website](#) under *Themes & Problems*. Semifinalists complete an on-site challenge to produce additional podcast(s). Required criteria, such as lines of dialog, topics, etc. will be revealed at the semifinalists orientation meeting.

ELIGIBILITY

Three (3) teams per state may participate. Teamwork is strongly encouraged, but an individual may participate solo in this team event.

TIME LIMITS

PRELIMINARY ROUND

- A. All components of the chapter's entry (audio file URL and documentation PDF) must be finished, submitted, and accessible via the Internet by 11:59pm ET on a designated date mid-May.
- B. The audio piece must be more than one (1) minute and less than five (5) minutes in length.
- C. A deduction of five (5) points total will be incurred for each fifteen (15) seconds under the one (1) minute minimum and for each fifteen (15) seconds over the five (5) minute maximum length.
- D. The timing starts with the first sound and continues until the last sound ends.

SEMIFINAL ROUND

- A. The number of podcasts and timing will be revealed at the semifinalist orientation meeting.
- B. Participants have thirty-six (36) hours, beginning at the semifinalists meeting, to complete and submit the entire production.
- C. The time limits for the on-site audio podcast will be provided at the semifinalist orientation meeting.

- D. A deduction of five (5) points will be incurred for each fifteen (15) seconds under the minimum and for each fifteen (15) seconds over the maximum length.
- E. The timing starts with the first sound and continues until the last sound ends.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants design an original audio podcast piece.
- C. Participants record their design process within a documentation portfolio.
- D. Participants submit a URL to the audio file (in MP3 or suitable format) and a documentation PDF of the required documentation by 11:59pm ET on a designated date in mid-May.
- E. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. Judges independently assess the entries using the following procedure:
 - 1. Judges score the Audio Podcast criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the Documentation Portfolio criteria of those top twenty-four contestants to determine the top twelve (12) semifinalist teams.
- B. A list of the twelve (12) semifinalists (in random order) will be posted on the first full day of the conference.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to receive the on-site challenge information. TSA competition attire is required to receive the on-site challenge.

- B. The event coordinator distributes the materials, information, directions, and deadlines to each team.
- C. Each team supplies its own audio production and editing equipment to complete its entry.
- D. Teams are responsible for submitting a hyperlink of their audio solution and a PDF of a completed Student Copyright Checklist (see 2026 & 2027 Forms Appendix) using the submission procedures provided by the event coordinator.
- E. Entries are reviewed independently by judges.
- F. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. The audio podcast and required documentation must be uploaded or located online and accessible for evaluation by the posted deadline.
- B. The entry must point directly to the team's entry. Entries that require a software download or a request that access be granted will not be judged.
- C. Entries received, or changes made to submitted entries after this deadline will not be judged.
- D. The audio podcast may include the first name of the students in the podcast and is exempt from the general rule that no identifying information may be used. However, students will not be allowed to use last names, their school name, or the state they are from.
- E. Audio Podcast:
 - 1. The URL must point directly to the audio file and not require any permissions or installation of software for evaluation.
 - 2. Music may accompany the audio piece but is not required.

- 3. The audio piece must be greater than one (1) minute and less than five (5) minutes in length.
- 4. There will be a five (5)-point deduction for:
 - a. each five (5) seconds under the one (1)-minute minimum
 - b. each 15 seconds over the five (5)-minute maximum length.
- 5. All audio pieces must be the original work of the team and must have been completed within the current school year.
- 6. Free, non-copyrighted sounds, loops, or other audio elements may be incorporated into audio pieces. The sources of these elements and the way in which they are used in the audio piece must be described in the portfolio.
- 7. Each actual and/or synthesized voice used in the final piece must be illustrated in a timeline format in the portfolio.
- 8. Where applicable, all ideas, sounds, and loops from other sources must be cited. If copyrighted material is used, proper written permission must be included.

F. Podcast Cover Art

- 1. All entries must include a podcast cover art graphic (JPEG or PNG).
- 2. The graphic must fit a square ratio, set to 1400 x 1400 pixels.
- 3. The podcast cover art must be visible (attached) to the audio clip and visible when listening to the podcast.

PRELIMINARY ROUND

A. Documentation Portfolio:

- 1. The documentation portfolio should be complete, well written, and professional in organization and appearance.
- 2. Documentation materials (comprising a "portfolio") are required and must be submitted as a multi-page PDF document with pages in this order:
 - a. Title page with the title of the audio piece, the event title, Chapter ID, the conference city and state, and the year; one (1) page

- b. Table of contents; pages as needed
- c. Podcast Cover Art. The graphic must fit a square ratio, set to 1400 x 1400 pixels.; one (1) page
- d. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
- e. Self-evaluation of the piece using criteria from the official rating form; pages as needed.
- f. Audio composition. Each actual and/or synthesized voice used in the final audio must be illustrated graphically using a timeline format; pages as needed
- g. When audio elements are used that were NOT created by the team, the source, effects applied, the way each element was incorporated into the audio piece and how each element corresponds to the audio must be included; pages as needed.
- h. List of hardware, software, and instruments used in the development of the audio piece; one (1) page
- i. List of references that includes sources for materials (non-copyrighted); pages as needed
- j. Completed Student Copyright Checklist (see 2026 & 2027 Forms Appendix); one (1) page

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to receive the on-site challenge information. TSA competition attire is required to receive the on-site challenge.
- B. The event coordinator explains any recording restrictions for the specific property.
- C. All participant voice audio must be the original work of the team and must have been completed during the event timeline.
- D. On-site Submission Information:
 - 1. Participants may choose any audio hosting site (such as an UNLISTED YouTube URL), or a sharable link in cloud storage, as long as the audio podcast is located online and accessible for evaluation.

- 2. The URL must point directly to the participant's entry. Entries that require a software download or request access be granted will not be judged.
- 3. Entries received or changes made to submitted entries after the deadline will not be judged.
- 4. Students must complete the Student Copyright Checklist (see 2026 & 2027 Forms Appendix) and save it as a PDF to be submitted electronically with the entry online.
- 5. When applicable, all ideas and sounds from other sources must be cited. Copyright materials may NOT be used. If including a citation page, the page must be included as the second page of the Student Copyright Checklist in the PDF in D.4.
- 6. National TSA will not provide wireless Internet.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. The audio podcast

Tier 2

- B. The documentation portfolio

SEMIFINAL ROUND

- A. The completed on-site audio podcast(s)

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Artist
- Audio designer or engineer
- Audio operator or technician
- Broadcast technician
- Music composer

AUDIO PODCASTING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Audio composition was submitted in proper format and accessible
- ☐ TIER 2 – PDF of the documentation portfolio was submitted
- ☐ ENTRY NOT EVALUATED

TIER 1 – AUDIO PODCAST (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Creativity and Uniqueness (X1)	The podcast feels like a standard piece; it does not make strides to distinguish itself from any other podcast.	The work involves some original aspects or manipulations of podcast ideas and form; it explores and includes at least one (1) or more podcast elements.	The episodes includes highly original, unusual, or imaginative episode ideas; it explores and includes at least two (2) or more podcast elements.	
Artisanship (X1)	The work gives no sense of a completed podcast episode idea; there is no clear beginning, middle, or end section; the form appears random, rather than organized.	One (1) podcast element has been used to organize the episode's ideas, content, and overall form, which are somewhat coherent.	The episode(s) presents at least one (1) complete episode idea; the piece has a coherent and organized form with a clear beginning, middle, and end; additional elements are used to organize the episode's ideas and form.	
Energy and Style (X2)	The piece lacks liveliness, vitality, and vigor; there is no flair, elegance, or grace to the form.	The episodes generates an initial level of energy that appeals to the listener; the style and content is somewhat distinctive.	The liveliness and forcefulness of the episodes excite the listener; the style is truly unique and electrifying.	
Appropriateness (X1)	The podcast episodes, ideas or concepts are not appropriate and acceptable for use in the event.	The podcast idea or concept presented is acceptable and somewhat fitting.	The podcast episodes' ideas or concepts presented are fitting and serve as an excellent example of the type of work expected.	
Overall Appeal (X2)	The work does not present an effective general impression; the podcast content does not hold the listener's interest.	The podcast includes some interesting content and form ideas; the general impression is pleasant and moderately effective.	There is strong, interesting, and effective audio appeal; the work is designed to be enjoyed by the listeners.	
TIER 1 – AUDIO PODCAST SUBTOTAL (70 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				

TIME DEDUCTIONS					
There will be a five (5) point deduction for each fifteen (15) seconds under the minimum time or each fifteen (15) seconds over the maximum time allowed for the audio podcast.					
Total time under		Fifteen (15)-second intervals under		Under time deduction	
Total time over		Fifteen (15)-second intervals over		Over time deduction	
TOTAL TIME DEDUCTION					

TIER 1 – SUBTOTAL (70 points)	
-------------------------------	--

TIER 2 – DOCUMENTATION PORTFOLIO (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Podcast Cover Art (X2)	The cover art is present, but not creative and doesn't reflect the podcast.	The cover art is present and the correct size; the cover art is of good quality and creates the desire to enjoy the podcast.	The cover art is present and the correct size; the cover art is exemplary and truly promotes the podcast; it is evident that the team connects the cover art to the podcast.	
Work Log and Self-Evaluation (X1)	The Work Log and/or self-evaluation are incomplete, and/or missing key components.	The Work Log and/or self-evaluation are somewhat complete and incorporate reflections and efforts of the team.	A complete and concisely written Work Log and self-evaluation are provided and incorporate the efforts and reflections of the team.	
Track Timeline (X1)	The track timeline is incomplete and/or not created correctly; the timeline does not correlate with the actual music production.	The track timeline is largely complete and attempts to correlate with the actual podcast production.	The track timeline is of exemplary quality; it correlates completely with the podcast production and is easy to follow.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

PRELIMINARY SUBTOTAL (120 points)	
-----------------------------------	--

SEMIFINAL ON-SITE PODCAST (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Creativity and Uniqueness (X1)	The podcast feels like a standard piece; it does not make strides to distinguish itself from any other podcast.	The work involves some original aspects or manipulations of podcast ideas and form; it explores and includes at least one (1) or more podcast elements.	The episodes include highly original, unusual, or imaginative episode ideas; it explores and includes at least two (2) or more podcast elements.	
Artisanship (X1)	The work gives no sense of a completed podcast episode idea; there is no clear beginning, middle, or end section; the form appears random, rather than organized.	One (1) podcast element has been used to organize the episode's ideas, content, and overall form, which are somewhat coherent.	The episode(s) presents at least one (1) complete episode idea; the piece has a coherent and organized form with a clear beginning, middle, and end; additional elements are used to organize the episode's ideas and form.	
Energy and Style (X2)	The piece lacks liveliness, vitality, and vigor; there is no flair, elegance, or grace to the form.	The episodes generate an initial level of energy that appeals to the listener; the style and content is somewhat distinctive.	The liveliness and forcefulness of the episodes excite the listener; the style is truly unique and electrifying.	
Appropriateness (X1)	The podcast episodes, ideas or concepts are not appropriate and acceptable for use in the event.	The podcast idea or concept presented is acceptable and somewhat fitting.	The podcast episodes' ideas or concepts presented are fitting and serve as an excellent example of the type of work expected.	
Overall Appeal (X2)	The work does not present an effective general impression; the podcast content does not hold the listener's interest.	The podcast includes some interesting content and form ideas; the general impression is pleasant and moderately effective.	There is strong, interesting, and effective audio appeal; the work is designed to be enjoyed by the listeners.	
SEMIFINAL ON-SITE PODCAST SUBTOTAL (70 points)				

TIME DEDUCTIONS					
There will be a five (5) point deduction for each fifteen (15) seconds under the minimum time or each fifteen (15) seconds over the maximum time allowed for the audio podcast.					
Total time under		Fifteen (15)-second intervals under		Under time deduction	
Total time over		Fifteen (15)-second intervals over		Over time deduction	
TOTAL TIME DEDUCTION					

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

SEMIFINAL SUBTOTAL (70 points)	
---------------------------------------	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (190 points)	
--	---------------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

AUDIO PODCASTING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Assistant for check-in, one (1)
- C. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Copies of an on-site challenge for semifinalist teams and judges.
- C. One (1) computer with monitor as needed for judges
- D. One (1) extension cord and one (1) power strip as needed for judges
- E. Tables and chairs for event coordinator, judges, and participants

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Judges determine the twelve (12) semifinalists and discuss and break any ties.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the evaluation of entries is to begin, meet with your judges and check in, personnel to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the CRC event manager before the evaluation begins.
- F. Ensure the judges have access to the online judging system.

SEMIFINAL ORIENTATION

- A. Prior to the semifinalist orientation meeting, the coordinator should tour the facilities and develop a list of restricted areas and/or specific restrictions for the event. This event should be shared with the event manager prior to the event meeting. The coordinator should mention at the event meeting that teams must be courteous to all guests in common areas or designated filming areas.
- B. Meet with all participants at the designated time and place to deliver the specific criteria, including any required props and dialog.
 - 1. Ensure that all participants understand regulations regarding equipment allowed, behavior, deadlines, and submission requirements.
 - 2. Share the online submission link with the participants.
 - 3. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond the participant's control.
 - 4. In order to compete, participants must be on the entry list, or must have approval of the CRC.

SEMIFINAL SCORING

- A. Close the online submission link.
- B. Ensure judges have access to the entries.
- C. Judges independently evaluate the entries.
- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
 - 1. To deduct 20% of the total possible points or
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants conduct research on a contemporary biotechnology issue as it relates to a chosen theme on the [TSA website](#) under *Themes & Problems*. Participants will document their research and make a display. The information gathered may be student performed and researched, or a re-creation or simulation of research performed by the scientific community. Semifinalist teams present and are interviewed about their topic.

ELIGIBILITY

Five (5) teams per state, consisting of at least three (3) individuals, may participate.

TIME LIMITS

Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:

- A. One (1) minute for set-up
- B. Seven (7) minutes for the presentation
- C. Two (2) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- A. Participants access the annual theme on the [TSA website](#) under *Themes & Problems*.
- B. Participants research the issue using credible sources.
- C. Participants prepare their display and documentation portfolio according to the regulations and requirements.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to set up the display.
- B. Entries are independently reviewed by judges, with neither students nor advisors present.
 - 1. Judges score the display criteria to determine the top twenty-four (24) preliminary teams, which will not be posted.
 - 2. Judges score the documentation criteria of those top twenty-four (24) teams to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. One (1) team representative reports at the time and place stated in the conference program to sign-up for a presentation/interview time.
- B. Three (3) team representatives report at least ten (10) minutes prior to their assigned time to the designated place for their presentation/interview.
- C. Semifinalist team representatives participate in the on-site presentation and interview.
- D. Team representatives pick-up their entry from the display area at the time and place stated in the conference program.
- E. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Students must understand the fundamental concepts and principles of the contemporary biotechnology issue they research. Research about the issue should focus on significant impacts (opportunities and risks) on the environment, economy, and society, as well as any important ethical considerations.
- B. Display guidelines:
 1. The size of the display may not exceed 15" deep x 30" wide x 48" high.
 2. A/C electricity may not be used.
 3. If the display requires power, they must be powered by dry-cell batteries or photo-voltaic cells.
 - a. The power supply must physically fit within the display dimensions.
 - b. All power must be switched off once the team has completed set-up.
 - c. If teams want judges to activate any electronic device in their display, complete instructions must be provided to judges on how to power up the display.
 4. If operating instructions are necessary, they must be clearly displayed and should include information about turning on devices, logging in, etc.
- C. Documentation Portfolio:
 1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Definition and explanation of the issue as it relates to the theme; one (1) page
 - d. Research base: an explanation of the importance of the issue in contemporary biotechnology today, including possible solutions; maximum three (3) pages
 - e. Support materials such as logs, graphs, sketches, drawings, illustrations, photos, etc.; maximum four (4) pages
 - f. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
 - g. A list of references and credible sources; a minimum of three (3) different types of references must be used; pages as needed
 - h. Work must be original or cited, using a professional citation style of the competitors choosing. Failure to use a professional citation style will result in a rules violation of 20% (twenty percent). Some examples of professional citation styles include MLA, APA, Chicago, and IEEE; pages as needed
 - i. Photo/Film/Video Consent and Release forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed
- D. Once the display set-up time frame has closed, participants may not re-enter the event area.
- E. Violation of the following will result in disqualification:
 1. No liquids, viruses, live plants, or animals may be used as a part of the display.
 2. No harmful or illegal substances may be displayed.

SEMIFINAL ROUND

- A. The presentation is limited to three (3) team representatives. Each member should be an active participant in the presentation.
- B. Representatives should reference their display and documentation during the presentation and interview.
- C. Teams may bring and use audio/visual materials or a laptop for their presentation. Projectors and screens are not permitted. Access to power will not be provided.

D. Time limits:

1. One (1) minute for set-up
2. Seven (7) minutes for presentation
3. Two (2) minutes to respond to questions from judges.
4. A five-(5) point deduction will be incurred for any time infraction.

EVALUATION**PRELIMINARY ROUND****Tier 1**

- A. The display

Tier 2

- B. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation
- B. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Botanist
- Chemical engineer
- Food scientist
- Molecular biologist
- Plant geneticist

BIOTECHNOLOGY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Display is present
- ☐ TIER 2 – Documentation portfolio is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – DISPLAY (90 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Definition and Explanation of the Issue and Solution (X1)	Unclear definition and explanation of the issue is evident; it is difficult to understand the solution being communicated; an illogical explanation is presented.	Issue is defined and explained adequately; the solution is acceptable.	There is evidence of a clear and concise definition and explanation of the issue; explanation is presented and communicated in an organized and concise manner.	
Explanation of Impacts (X2)	Explanation is missing a discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	Explanation addresses some of the issue's relevance to environmental, economic, social, and/or ethical considerations.	Explanation includes a full discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	
Supporting Information (X1)	Supporting information does not help to clarify documentation and/or it is of little significance to the issue.	Supporting information is appropriate and helps supplement the documentation by providing clarity to the issue.	Supporting information is highly effective and of excellent quality.	
Research, References, and Resources (X1)	Documentation lacks adequate research base, and/or very few credible sources are referenced.	Research is generally appropriate, with adequate credible sources.	A comprehensive research base that includes credible sources is evident.	
Communication of Issue (X1)	It is difficult to understand the issue being communicated; an illogical explanation is presented.	The issue is communicated and thoughts are somewhat organized.	The issue is communicated in an organized, clear, and concise manner.	
Communication of Solution (X1)	It is difficult to understand the solution being communicated; an illogical explanation is presented.	The solution is communicated and thoughts are somewhat organized.	The solution is communicated in an organized, clear, and concise manner.	
Creativity (X1)	The display lacks creativity; no, or very few, design principles are integrated in the display.	Some elements of creativity exist in the display, and essential design principles are generally evident.	The display exudes creativity; essential design principles and elements are well integrated.	
Aesthetics and Artanship (X1)	Display is unorganized and sloppy; display seems to be an afterthought or thrown together.	Display is somewhat organized and aesthetically pleasing.	Display is logical, organized, cohesive, and aesthetically pleasing.	
TIER 1 – DISPLAY SUBTOTAL (90 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (90 points)

TIER 2 – DOCUMENTATION PORTFOLIO (60 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Definition and Explanation of Issue (X1)	Definition and explanation of the issue is unclear.	Issue is generally defined and explained, but some points could use additional clarification.	Clear and concise definition and explanation of the issue is evident.	
Research Base (X1)	Research is inadequate, and/or very few credible sources are referenced.	Research has been conducted appropriately, with some credible sources included.	Research indicates evidence of a comprehensive assortment of materials that are credible sources.	
Support Materials (X1)	Support materials do not help clarify the documentation and/or are of very little significance to the issue.	Support materials are somewhat appropriate and lend some clarity to the issue.	Support materials are of excellent quality; if not original, they are cited; support materials clarify the issue.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
Quality and Effectiveness (X1)	Portfolio appears to have been thrown together; distracting errors in punctuation, grammar, and spelling are evident in the documentation.	Portfolio is generally organized; punctuation, grammar, and spelling are correct for the most part, with few errors.	Work is of exceptional quality and well organized; punctuation, grammar, and spelling are correct, with no errors.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (60 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (150 points)

SEMIFINAL PRESENTATION/INTERVIEW (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participants seem unorganized and unprepared for the presentation, with an illogical explanation of the problem and solution.	Participants are generally prepared for the presentation; explanation of the problem and solution are somewhat communicated and organized.	The presentation is logical, well-organized, and easy to follow; the problem and solution are communicated in an organized and concise manner.	
Articulation (X1)	Communication of the solution is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the solution is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the solution is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Delivery (X1)	The team is verbose and/or uncertain in its presentation/interview; participants' posture, gestures, and lack of eye contact diminish the delivery; little to no references are made to the display and/or documentation portfolio.	The team is generally well-spoken and clear in its presentation/interview; participants' posture, gestures, and eye contact are acceptable in the delivery; some references are made to the display and/or documentation portfolio.	The team is well-spoken and distinct in its presentation/interview; participants' posture, gestures, and eye contact result in a polished, natural, and effective delivery; multiple appropriate and effective references are made to the display and/or documentation portfolio to enhance the delivery.	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit an understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the project	
Team Participation (X1)	Majority of the presentation/interview is made by one member of the team; the other team members may be disengaged.	Team members generally are engaged in the process, though one member may take on more responsibility than the other(s).	All team members are actively involved in the presentation/interview.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (60 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
TIME DEDUCTIONS				
A five-(5) point deduction will be incurred for any time infraction.				
SEMIFINAL SUBTOTAL (60 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				
TOTAL (210 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

BIOTECHNOLOGY

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinalist round, two (2) or more (preferably the same judges from the preliminary round)
- C. Assistant, one (1)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Display tables for entries
- C. Table and chairs for team presentations, as needed
- D. Stopwatch
- E. Measuring tape
- F. Display tables for entries
- G. Table and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Check in the entries at the time stated in the conference program.
- B. No more than two (2) team representatives may check in:
 - 1. The display
 - 2. The documentation portfolio
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Requirements for attire do NOT apply during check-in, only on the first full day of the conference.
- F. Each entry must include the team's identification number in the upper right-hand corner of the display and somewhere visible on the documentation portfolio

PRELIMINARY ROUND

- A. Judges independently score the display criteria which will determine the top twenty-four (24) teams, which will not be posted.
- B. Judges score the documentation criteria of those top twenty-four (24) teams to determine the top twelve (12) semifinalists.

- C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must all initial either of the violations on the rating form.
- D. Create a sign-up sheet for semifinalist presentations/ interviews and communicate any instructions to the semifinalist participants.

SEMIFINAL ROUND

- A. Prepare a list of five to ten (5-10) standard interview questions to be asked of all participants.
- B. Semifinalists report to sign-up for the presentation at the time and place stated in the conference program.
- C. Check in semifinalists and confirm with the teams their order of presentation and the procedure. Team representatives should report at least ten (10) minutes prior to their assigned time to the designated place for their presentation.
- D. Manage the presentations and interviews.
- E. If judges ask questions, they should be the same questions for each team for the purpose of comparing one team to another.
- F. Discuss rule violations and have all relevant parties initial the rating form.
- G. Judges independently assess the semifinal presentation and interview.
- H. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants conduct research on a technology-related career according to a theme posted on the [TSA website](#) under *Themes & Problems* and prepare a cover letter and a job-specific resume. Semifinalists participate in a mock interview.

ELIGIBILITY

One (1) individual per chapter may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. All components of the chapter's entry (PDF) must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.
- B. Entries received, or changes made to submitted entries after this deadline will not be judged.

SEMIFINAL ROUND

- A. Ten (10) minutes is allowed for the mock interview.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants concentrate their efforts conducting research on the selected technology-related career.
- D. Participants prepare a cover letter and a job-specific resume.
- E. Participants submit the PDF by 11:59 p.m. ET on a designated date in mid-May.
- F. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalists (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to sign up for a mock interview time.
- B. Semifinalists report at the assigned time and place to participate in the mock interview.
- C. Simulating a real interview, semifinalists bring with them an additional printed copy of their job-specific resume to the judges at the time of the interview.
- D. The printed copy of the job-specific resume provided to the judges during the interview will not be returned.
- E. Judges evaluate the interviews.
- F. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Participants enter this event with the following scenario in mind:
 - 1. You have graduated from high school and have the appropriate level of education and training (i.e., college degree, technical school, certification and training, etc.) for an entry-level position that is required for successful employment in your selected career.
 - 2. Your training, education, and other qualifications for the entry-level position are realistic for successful employment in your chosen career and are reflected in your resume and cover letter.

- B. Participants should use fictitious home address and telephone number information when completing the documents, however, correct participant names must be used.
- C. Each participant submits a PDF of (no cover page, table of contents, or release forms):
 - 1. Cover letter:
 - a. Must be typed
 - b. Is limited to one (1) single-sided, 8½" x 11" page
 - c. Must include an opening, body, and conclusion
 - d. Headshots or photos are not permissible.
 - 2. The job-specific resume:
 - a. Must be typed
 - b. Is limited to two (2) single-sided, 8½" x 11" pages
 - c. Headshots or photos are not permissible.

EVALUATION

PRELIMINARY ROUND

- A. The cover letter and job-specific resume

SEMIFINAL ROUND

- A. The mock interview

Refer to the official rating form for more information

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the career areas featured in the TSA AND CAREERS section of this guide.

CAREER PREP

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Cover letter and job-specific resume submitted
- ☐ ENTRY NOT EVALUATED

COVER LETTER (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Introduction (X1)	Introduction fails to describe the participant, and/or does not clearly identify the position or reason for contact, and/or does not indicate how the position was discovered, and/or does not indicate interest, and/or does not grab the employer's attention.	Introduction describes many of the elements (e.g., participant description, how the position was identified, reason for applying, etc.) and briefly addresses others (e.g., how the position was discovered, interest level, etc.), resulting in an adequate introduction.	Introduction incorporates all elements—the participant, the position or reason for contact, how the position was discovered, and genuine interest in the position—and ultimately grabs the employer's attention.	
Body (identification of skills) (X1)	Participant fails to identify any skills or qualifications, does not explain interest, and does not indicate how his/her skills would provide benefit to the company.	Participant indicates his/her skills, with a general explanation of how the skills relate to the position at hand; participant conveys interest, and briefly connects the skills to benefits for the company.	Participant provides one or two strong qualifications and clearly relates these skills to the job at hand; participant clearly explains how his/her interest and skills can benefit the company.	
Conclusion (X1)	The conclusion does not include a thank-you to the employer and/or, does not note contact information, is not assertive, and/or lacks mention of follow-up after a given period of time.	The conclusion may or may not include a thank-you and follow-up to the employer with contact information; overall, it is adequate.	The conclusion includes a thank-you to the employer for his/her time and the applicant's contact information; it is assertive and mentions a method of follow-up within a given period of time.	
Overall Writing Quality and Grammar (X1)	Writing does not make sense; participant has written too much or too little; there are multiple spelling and grammatical errors.	Writing is average and includes relevant information and content; it is somewhat convincing to an employer; there are minor spelling or grammatical errors.	Writing flows well, is clear, concise, and gets right to the point; it is convincing and contains no spelling or grammatical errors.	
COVER LETTER SUBTOTAL (40 points)				

JOB-SPECIFIC RESUME (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Audience and Purpose (X1)	Resume does not address a particular audience; the purpose is not clearly stated; revision is needed.	Audience and purpose are generally implied; the resume is somewhat tailored to the employer.	Audience is clearly addressed and the resume is tailored to the employer; the purpose is clearly stated.	
Presentation and Format (X1)	Resume does not have a clear design format; headers are not used or are used incorrectly or inappropriately; resume does not use reverse chronological format.	Resume attempts (and partially succeeds) to use a reverse chronological format; headers are used for a somewhat professional and concise presentation.	Resume follows a reverse chronological format; it uses clear and appropriate headers to organize information; it has a professional appearance.	
Language and Style (X1)	Participant fails to use action words; phrasing is wordy and lacks focus.	Participant uses some action words and some concise and clear words in most of the resume.	Participant consistently uses strong, clear, and concise words throughout the resume.	
Overall Writing Quality and Grammar (X1)	Writing does not make sense; participant has written too much or too little; there are multiple spelling and grammatical errors.	Writing is average and includes relevant information and content; it is somewhat convincing to an employer; there are minor spelling or grammatical errors.	Writing flows well, is clear, concise, and gets right to the point; it is convincing and contains no spelling or grammatical errors.	
JOB-SPECIFIC RESUME SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
PRELIMINARY SUBTOTAL (80 points)				

SEMIFINAL MOCK INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participant is unorganized and unprepared for the interview; an illogical explanation of the career choice is presented.	Participant is generally prepared for the interview; an explanation of the career choice is somewhat communicated and organized.	Interview responses are clear, well organized, and easy to follow; the responses reflect knowledge of the career choice and are communicated in a concise manner.	
Knowledge (X2)	Participant seems to have little understanding of the concepts in the career choice; answers to questions may be vague.	Participant exhibits a general understanding of the career choice.	Participant shows clear evidence of a thorough understanding of the career choice.	
Delivery (X1)	Participant is verbose and/or uncertain in the interview; participant's posture, gestures, and lack of eye contact diminish the interview.	Participant is somewhat well-spoken and clear in the interview; participant's posture, gestures, and eye contact are acceptable in the interview.	Participant is well-spoken and clear in the interview; participant's posture, gestures, and eye contact result in a polished, natural, and effective interview.	

SEMIFINAL MOCK INTERVIEW (50 points) – continued				
Articulation (X1)	Communication is full of illogical thoughts that lack clarity, and/or there is insufficient information provided about the career choice; leadership and/or 21 st century skills are not evident.	Communication is somewhat logical, easy-to-follow, and/or there is sufficient information provided describing the career choice; leadership and/or 21 st century skills are somewhat evident.	Communication is clear and concise, and there is ample information provided about the career choice; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL MOCK INTERVIEW SUBTOTAL (50 points)				
Rules violations (a deduction of 20% of the total possible points in the semifinalist sections above) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (50 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (130 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CAREER PREP

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more (preferably the same judges who reviewed the written entries)
- C. Assistants, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Interview questions appropriate to the annual selection of technology-related careers
- C. Manila folder for semifinal resumes
- D. Tables and chairs for event coordinator, judges, and participants

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Judges determine the twelve (12) semifinalists.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.

- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. On the first full day of the conference, post a list of the twelve (12) semifinalists in random order.
- B. Create a sign-up sheet for semifinal mock interviews.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to sign-up for the mock interview.
- B. Semifinalists report at the assigned time and place for the mock interview.
- C. Manage the interviews.
- D. Judges use the same official rating form for both the preliminary and semifinal round of judging.
- E. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- F. If necessary, manage security and the removal of materials from the event area.

CHALLENGING TECHNOLOGY ISSUES



OVERVIEW

Applying leadership and 21st century skills, participants collaborate to prepare and deliver a debate-style presentation with participants explaining opposing views of a current technology issue. The annual topics will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) teams of two (2) individuals per state may participate.

TIME LIMITS

- A. Fifteen (15) minutes is allowed to prepare.
- B. A minimum of two (2) minutes and a maximum of four (4) minutes is allowed to present.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Both members of the team must report at the time and place stated in the conference program to receive an assigned presentation time.
- B. Both members must report with their materials to the preparation room at the assigned time.
- C. Topic selection for each round:
 - 1. A member from each team will randomly select one of the annual topics.
 - 2. After each selection, the topic will be returned to the pile and shuffled for the next team.
- D. At the end of the fifteen (15)-minute preparation time, each team is escorted to the presentation room where the team presents opposing views of the selected issue.

- E. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. A representative from each semifinalist team reports at the time and place stated in the conference program to sign up for a presentation time.
- B. Semifinalist teams report to the preparation room at their assigned time for the same procedure used in the preliminary round.
- C. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants report to the preparation room at the assigned presentation time.
- B. Materials:
 - 1. Team members may bring non-electronic reference materials for use while in the preparation room.
 - 2. No other form of assistance is allowed.
 - 3. Participants must provide their own pencils or pens for this event.
 - 4. Participants may provide note cards, if desired.
- C. Note cards:
 - 1. Note cards may be used during the presentation.
 - 2. The use of note cards may result in score deductions if they detract from the effectiveness of the presentation.

D. Event debate:

1. Using leadership and/or 21st century skills, the team presents on one (1) topic from the topics posted on the [TSA website](#) under *Themes & Problems*
2. Team may use first names ONLY and no other identifying information during their presentation.
3. One team member states the topic prior to the first speaker speaking in the presentation.
4. After stating the selected topic, the presentation time begins once a team member begins to speak.
5. Teams present opposing views of the selected issue in debate style.
6. Teams may use the following suggested format to present opposing views:
 - Introduction (pro) Introduction (con)
 - Position (pro) Position (con)
 - Conclusion (pro) Conclusion (con)
7. A timekeeper notifies a presenting team thirty (30) seconds before the end of the four (4)-minute time limit by holding up a “30-seconds remaining” card.
8. Time is called at four (4) minutes, at which time the presentation must stop. No overtime speaking is allowed.

SEMIFINAL ROUND

- A. Participants report to the event area at the time and place stated in the conference program to receive an assigned debate time.
- B. Participants report to the preparation room at the assigned time.
- C. Preliminary round procedures are used for the semifinal round.

EVALUATION

A. The debate

Note: Scores are reset for the semifinal round and are not added to the preliminary scores.

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Lobbyist
- Management executive
- Motivational speaker
- Politician
- Public policy specialist

CHALLENGING TECHNOLOGY ISSUES

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ A team of two (2) individuals is present

☐ ENTRY NOT EVALUATED

PRO SIDE (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Introduction (X1)	Introduction does not show an attempt to get the attention of the audience, and/or it does not outline points clearly and distinctly.	Introduction shows an effort to grab the attention of the audience; previewing points are somewhat organized in a logical manner.	Introduction employs an attention-getter, clearly states the thesis, and previews main points of the issue.	
Organization (X1)	Ideas may not be focused or developed; the main purpose is not clear; there are no transitions.	The main idea is evident, and the organizational structure is generally strong; ideas are relevant and flow smoothly.	Ideas are clearly organized, developed, and supported, with smooth transitions.	
Topic Knowledge (X1)	Team does not have a grasp of the topic; inaccurate, generalized, or inappropriate support material is used; over dependence on notes is evident.	Team demonstrates a simple grasp of the topic; support material is stated; the team is at ease when it elaborates on the topic.	Team exhibits full knowledge and a clear grasp of the topic; support material is introduced and referenced accurately.	
Conclusion (X1)	Conclusion ends abruptly (no logical and/or organized conclusion is presented); it does not restate main points of the position.	Conclusion is somewhat engaging; it restates main points and wraps up the position on the topic.	Conclusion is engaging and restates main points of the topic in a comprehensive, competent, and articulate way, reinforcing the team's understanding of the subject; the position is well stated and closes with a connection to the introduction.	
Voice and Language (X1)	Vocabulary choice may be limited; speech is peppered with slang or jargon; voice is dull; language is questionable or inappropriate for the setting.	Presentation is respectful; language is appropriate for the setting and word choices are generally precise; voice is appropriate.	Language is appropriate for the setting and voice tone and pitch energize the presentation; vocabulary choices are vivid and precise.	
Delivery (X2)	Delivery detracts from the message; eye contact may be limited; presenters may tend to look at the floor, mumble, speak inaudibly, fidget, or read from cards; gestures and movements may be excessive.	Delivery is appealing, though some hesitancy may be observed; posture, eye contact, gestures, facial expressions, volume, pace, etc., do not detract from the message.	Delivery is natural, confident, and enhances the message; posture, eye contact, gestures, facial expressions, volume, pace, etc., indicate confidence.	
PRO SIDE SUBTOTAL (70 points)				

CON SIDE (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Introduction (X1)	Introduction does not show an attempt to get the attention of the audience and/or does not outline points clearly and distinctly.	Introduction shows the effort to grab the attention of the audience; previewing points are somewhat organized in a logical manner.	Introduction employs an attention-getter, clearly states the thesis, and previews main points of the issue.	
Organization (X1)	Ideas may not be focused or developed; the main purpose is not clear; main points are difficult to identify; there are no transitions.	The main idea is evident, and the organizational structure is generally strong; ideas are relevant and flow smoothly.	Ideas are clearly organized, developed, and supported, with smooth transitions.	
Topic Knowledge (X1)	Team does not have a grasp of the topic; inaccurate, generalized, or inappropriate support material is used; over dependence on notes is evident.	Team has a simple grasp of the topic; support material is stated; the team is at ease when it elaborates on the topic.	Team exhibits full knowledge and a clear grasp of the topic; support material is introduced and referenced accurately.	
Conclusion (X1)	Conclusion ends abruptly (no logical and/or organized conclusion is presented); it does not restate main points of the position.	Conclusion is somewhat engaging; it restates main points and wraps up the position on the topic.	Conclusion is engaging and restates main points of the topic in a comprehensive, competent, and articulate way, reinforcing the team's understanding of the subject; the position is well stated and closes with a connection to the introduction.	
Voice and Language (X1)	Vocabulary choice may be limited; speech is peppered with slang or jargon; voice is dull; language is questionable or inappropriate for the setting.	Presentation is respectful; language is appropriate for the setting and word choices are generally precise; voice is appropriate.	Language is appropriate for the setting and voice tone and pitch energize the presentation; vocabulary choices are vivid and precise.	
Delivery (X2)	Delivery detracts from the message; eye contact may be limited; presenters may tend to look at the floor, mumble, speak inaudibly, fidget, or read from note cards; gestures and movements may be excessive.	Delivery is appealing, though some hesitancy may be observed; posture, eye contact, gestures, facial expressions, volume, pace, etc. do not detract from the message.	Delivery is natural, confident, and enhances the message; posture, eye contact, gestures, facial expressions, volume, pace, etc. indicate confidence.	
CON SIDE SUBTOTAL (70 points)				

PARTICIPATION (10 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Team Member Participation (X1)	One team member does the majority of the speaking and/or debating on the topic; the other member seems disengaged from the presentation; leadership and/or 21 st century skills are not evident.	Both team members are engaged in the debate, but one clearly takes the lead; the other member only replies to or refutes statements; leadership and/or 21 st century skills are somewhat evident.	Both team members are actively involved in the presentation, debate, and rebuttals of the topic; shared responsibility is evident throughout; leadership and/or 21 st century skills are clearly evident.	
PARTICIPATION SUBTOTAL (10 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
--	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.

TOTAL (150 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CHALLENGING TECHNOLOGY ISSUES

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more per heat
 - 2. Semifinal round, two (2) or more per heat
- C. Assistants, two (2) per preparation room
- D. Timekeeper, one (1) per presentation room

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. A time sign-up sheet
- C. Lined paper and 3" x 5" note cards
- D. Stopwatch, one (1) per preparation room and one (1) for the presentation room per heat; one (1) set for the semifinal round
- E. Note cards with each topic
- F. A card with "30 seconds remaining" printed clearly, one (1) per presentation room
- G. Tables and chairs for event coordinator, judges, and participants

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, etc. Notify the event manager of any potential problems.

- E. One (1) hour before the event is scheduled to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRESENTATION TIME SIGN-UP

- A. Check in the teams at the time stated in the conference program.
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- E. When the teams report to the event area:
 - 1. Assign times for their presentations. Times should be scheduled at ten (10)-minute intervals.
 - 2. Depending upon the number of entries, heats may be necessary to determine semifinalists.

PRELIMINARY ROUND

- A. Preparation:
 - 1. After each team reports to the preparation room at the assigned time and participants have been seated, review the time limits and distribute paper and note cards.
 - 2. Allow each team to randomly select one of the annual topics from the shuffled pile of note cards.
 - 3. Start the time and inform participants of the end of the preparation time.
 - 4. Seven (7) minutes after the first team enters the preparation room, the second team enters the preparation room, goes to a different section of the room, and begins its presentation preparation, again with fifteen (15) minutes allowed for preparation; and so on for all the teams.

5. Each team in turn is allowed to enter the presentation room at six (6)-minute intervals, thus enabling a constant flow of participants to speak before the evaluators in a timely fashion. (This allows for one [1] minute to enter the presentation room and announce the entry number, up to five [4] minutes for the presentation, and one [1] minute to exit the room.)
 6. Monitor students during the preparation time and supply extra paper if needed.
 7. At the end of the fifteen (15)-minute preparation time, collect the topic and escort the team to the presentation room.
- B. Presentation:
1. After the team is in position, introduce the team by entry number only. Please note that team members may use first names only and no other identifying information during their presentation.
 2. Judges tell the team to start, and timing the event begins.
 3. The timekeeper notifies the team thirty (30) seconds before the end of the four (4)-minute time limit by holding up a “30 seconds remaining” card.
 4. Time is called at four (4) minutes, at which point the speakers must stop.
- C. Judges independently assess the entries.
- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
1. To deduct twenty percent (20%) of the total possible points in this round
 2. To disqualify the entry
- The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. Judges determine the twelve (12) semifinalists.
- F. Create a semifinalist sign-up sheet for final presentations.

SEMIFINAL ROUND

- A. At least one (1) hour before the event begins, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- B. Semifinalist teams report at the time and place stated in the conference program to sign up for a presentation time.
- C. Oversee semifinalist presentations, following the same procedure used in the preliminary round.
- D. Use the same official rating form for both the preliminary and semifinal rounds of evaluation.
- E. Semifinalist scores are used independently and are not aggregated with the preliminary scores
- F. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
 1. To deduct twenty percent (20%) of the total possible points in this round or
 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- G. If necessary, manage security and the removal of materials from the area.

CHAPTER TEAM



OVERVIEW

Participants take a written parliamentary procedures test in order to qualify for the semifinals. Applying leadership and 21st century skills, semifinalist teams are challenged to complete an opening ceremony, items of business, parliamentary actions, and a closing ceremony within a specified time period.

ELIGIBILITY

- A. Participants are limited to one (1) team of six (6) individuals per chapter.
- B. Team members do not have to be elected officers of the local chapter.
- C. Teams that take the written test and advance to the semifinalist portion of the event must be comprised of the same six (6) members.

TIME LIMITS

PRELIMINARY ROUND

- A. One (1) hour is allowed for the test. Six (6) members must be present at the same time and take the test individually.

SEMIFINAL ROUND

- A. Teams have fifteen (15) minutes with no penalty, and up to seventeen (17) minutes with penalty (see Time over chart) are permitted to complete a presentation that incorporates the set-up time, items of business, and required parliamentary actions.
- B. The time begins when the team is handed the event materials; the time ends when the gavel is rapped to close the meeting, or at seventeen (17) minutes (at that point all team members other than the secretary must leave the room). The secretary may then be taken to another room to complete the minutes.
- C. The secretary has five (5) additional minutes to complete the minutes of the meeting.

- D. Teams are penalized five (5) points per thirty (30) seconds for going over the allotted time, based on the following scale:

Time over fifteen (15) minutes	Penalty
15:01 to 15:30	five (5) points per evaluator
15:31 to 16:00	ten (10) points per evaluator
16:01 to 16:30	fifteen (15) points per evaluator
16:31 to 17:00	twenty (20) points per evaluator

ATTIRE

TSA competition attire is required. Note additional specific attire requirements for Chapter Team found in the General Rules and Regulations and on the [TSA website](#). The additional specific attire is required for both the preliminary and semifinal rounds of this event.

PROCEDURE

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants report for the test at the time and place stated in the conference program.
- B. A parliamentary procedures test is administered at the same time to all team members.
- C. Twelve (12) teams with the highest averaged scores are selected as semifinalists for the oral presentation. A semifinalist list in random order is posted.

SEMIFINAL ROUND

- A. A semifinalist team representative reports at the time and place stated in the conference program to sign up for an oral presentation time.
- B. Semifinalist teams report at the assigned time and place for the oral presentation.

- C. Each team follows the procedure for opening and closing a local chapter meeting:
 - 1. A list of three (3) parliamentary actions are provided by the event coordinator and given to each team for the oral demonstration.
 - 2. Each team then closes the meeting according to the prescribed procedure.
 - 3. Concerning the reading of the TSA Creed by the secretary during the closing ceremony, a chapter has the option to recite the creed using one (1) or more of its team's members.
 - 4. The secretary has five (5) additional minutes to complete the minutes of the meeting.
 - 5. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Team members take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 - 1. Student identification numbers must be entered on the scantron form in the space indicated.
 - 2. Participants must stop work immediately when time is called.

- 3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.
- 4. All tests and scantrons must be turned in before leaving the test area.
- E. The average of the test scores of the six (6) team members will be divided by five (5) for the score that determines team ranking.
- F. These same six (6) team members compete in the semifinal round of the event, should the team qualify.
- G. Teams consist of a president, vice president, secretary, treasurer, reporter, and sergeant-at-arms.

SEMIFINAL ROUND

- A. Any team that fails to appear at the designated time is placed at the end of the list and allowed to participate at the discretion of the evaluators and event coordinator if time permits.
- B. Written materials, other than those provided by National TSA, may not be taken in the event room.
- C. Materials provided to teams:
 - 1. A set consisting of secretary's minutes, copy of Chapter Team Official Minutes form for secretary's minutes, a treasurer's report, a copy of the TSA creed, and a list of parliamentary actions.
 - 2. Paper, six (6) pens, a calculator, and six (6) 3" x 5" note cards.
 - 3. Gavel
- D. Optional materials: A timepiece and/or a non-programmable calculator may be used by a team.
- E. Room setup:
 - 1. Officer symbols and gavel (only) are placed on a long table with the United States flag positioned to the right of the president's rostrum (if available) and the host state flag to the left.
 - 2. The president's rostrum (if available) should be centered between the two (2) flags.

3. The symbols of the officers should be placed in front of the respective officers.
 4. The host state banners are optional and do not add to or subtract from a team's score.
- F. The semifinal portion of the event includes:
1. The call to order
 2. The pledge to the flag
 3. Roll call
 4. Order of business
 5. Closing ceremony
- G. Semifinal time limits and rules:
1. Teams have a time limit of fifteen (15) minutes to complete a presentation that incorporates the set-up time, items of business, and required parliamentary actions.
 2. Official timing will begin when the materials are given to the president and will stop at the team's final gavel to end the meeting.
 3. Five (5) points will be deducted for every thirty (30)-second interval over the allotted time (see TIME LIMITS).
 4. Bonus points will be awarded for additional motions and parliamentary actions by the officers, other than the president.
 5. At the conclusion of the oral presentation, each team secretary has five (5) minutes to write a copy of chapter minutes that will be submitted to an evaluator. The coordinator will begin timing the five (5) minutes when the secretary is seated at the area designated for writing of the minutes.
 6. All materials given to team members, and the chapter minutes recorded during the presentation, must be handed to the evaluators before the team leaves the room.
 7. No reference should be made to a team's school, chapter name, city, or state. Exception: The state name on a TSA patch is acceptable.
- H. Teams are welcome to use desired pronouns during the oral demonstration.

EVALUATION

PRELIMINARY ROUND

- A. The team's average test score.

SEMIFINAL ROUND

- A. The demonstration of a chapter business meeting

Refer to the official rating form for more information.

NOTES

There are a number of ways to learn about parliamentary procedure. The standard reference is *Robert's Rules of Order, Newly Revised*. Information about the latest edition of parliamentary procedure may be found online at robertsrules.com. Additional information may be found online at www.parliamentarians.org.

For writing proper minutes, also refer to *Robert's Rules of Order, Newly Revised*.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Parliamentarian
- Recording Clerk
- Government Official
- Lawyer
- Legal Aide

CHAPTER OPENING AND CLOSING CEREMONIES

OPENING CEREMONY

- A. At the prescribed time for meetings, the president assumes his/her position behind the rostrum in the front center of the room.
- B. Other officers are seated to the left and right of the president. They are seated in this order from stage left to right: vice president, treasurer, secretary, president, reporter, and sergeant-at-arms.

HOST STATE BANNER (OPTIONAL)

U.S. FLAG SGT.-AT-ARMS REPORTER PRESIDENT SECRETARY TREASURER VICE PRES. STATE FLAG

(OFFICERS FACING AUDIENCE)

AUDIENCE

- President: (raps gavel twice) The meeting will please come to order. Mr./Ms. Sergeant-at-Arms, are all the officers in their places?
- Sergeant-at-Arms: They are, Mr./Ms. President.
- President: (raps three [3] times for assembly to rise) Mr./Ms. Sergeant-at-Arms, please lead the assembly in the Pledge to the Flag of the United States of America.
- Sergeant-at-Arms: (leads Pledge to the Flag)
- President: (raps once and assembly is seated) Mr./Ms. Secretary, will you please call the roll.
- Secretary: Mr./Ms. Sergeant-at-Arms.
- Sergeant-at-Arms: Present. The symbol of my office is the “hearty handshake” (officer points to symbol), and it is my responsibility to see that the assembly is comfortable and properly welcomed. It is also my duty to serve as doorkeeper for this organization.
- Secretary: Mr./Ms. Reporter.
- Reporter: Present. The symbol of my office is the beacon tower (officer points to symbol), and it is my duty to see that our school, community, and national association have a complete report of our organization’s activities.
- Secretary: Mr./Ms. President.
- President: Present. The symbol of my office is the gavel (officer points to symbol). The duties vested in me by my office are to preside at all regular and special meetings of this organization and to promote cooperation in carrying out the activities and work of our organization. Mr./Ms. Secretary.
- Secretary: Present. The symbol of my office is the pen (officer points to symbol), and it is my responsibility to see that accurate and proper records are kept of all business and correspondence of this association. Mr./Ms. Treasurer.

- Treasurer: Present. The symbol of my office is a balanced budget (officer points to symbol), and it is the duty of my office to keep accurate records of all funds and see that our financial obligations are met promptly.
- Secretary: Mr./Ms. Vice-President.
- Vice President: Present. The symbol of my office is a star (officer points to symbol), and it is the duty of my office to see that we always have a strong membership, a good work program, and are alert to the welfare of our chapter.
- Secretary: Mr./Ms. President, all officers are present and in their place.
- President: Mr./Ms. Sergeant-at-Arms, do we have guests present?
- Sergeant-at-Arms: If so, introduce guest(s); if not, state the following: No, Mr./Ms. President.
- President: Mr./Ms. Secretary, we are ready to transact our business.
- Teams dispose of the assigned business following the suggested order of business.

CLOSING CEREMONY

- President: (raps three [3] times; assembly rises) Mr./Ms. Secretary, will you please (read) or (lead us in) the TSA Creed.
- Secretary: (recites the TSA Creed; when presented at state and national competitions, the creed may be presented in some more original method.)
- President: Will the assembly repeat the TSA Motto after me. (Motto is spoken.) Does anyone know of any reason why this assembly should not adjourn? If not, I will entertain a motion to adjourn. (following motion to adjourn, a second, and a vote) I now declare this meeting adjourned until a special meeting is called or until our next regular meeting (raps once with gavel).

SUGGESTED ORDER OF BUSINESS FOR CHAPTER MEETINGS

1. The president calls the meeting to order with opening ceremonies.
2. Roll call is taken and a quorum is established.
3. The secretary reads the minutes of the preceding meeting. Any necessary corrections and/or additions are made and the minutes are approved as read or corrected.
4. The treasurer's report is received as read and placed on file, subject to audit.
5. The chairperson calls for committee and officer reports as necessary. If a committee has no report, let the committee so state.
6. Unfinished business is addressed.
7. New business is addressed.
8. The program, if any, is held. The chairperson presides with the assistance of the program chairperson or the committee chairperson.
9. Make announcements.
10. Adjournment with closing ceremonies.

CHAPTER TEAM OFFICIAL MINUTES

Team ID number _____

Date _____

Location of conference _____

Participants may use the back of this page, if necessary.

Secretary's signature _____ Date _____

CHAPTER TEAM

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Team is present for demonstration

☐ ENTRY NOT EVALUATED

TEAM TEST (10 points)

Record the scores of the six (6) team members in the boxes below. Calculate the average of their scores. Divide the average by five (5) for the score that the team will receive out of ten (10) points. Record the score in the column space to the right.

#1	#2	#3	#4	#5	#6	
TEAM TEST SUBTOTAL (10 points)						

Record scores
in the column
spaces below.

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (10 points)

BUSINESS MEETING DEMONSTRATION (190 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
PREPARATION FOR MEETING (30 points)			
Official Attire/Poise (X2)	Appearance is untidy; grooming is lacking; clothing is not consistent in coloration and visual appearance; shoes are the wrong color; poise and confidence are missing.	Overall appearance is neat and consistent; grooming is good, and professional appearance is adequate.	Overall appearance is cohesive, polished, and business professional.
Placement of Flags and Officer Symbols; Officer Seating (X1)	Flags are not placed in the correct order; and/or officer symbols are not in the correct order; and/or are not aligned properly on the table; and/or not all officers are seated in the proper arrangement, resulting in a sloppy and haphazard appearance.	Placements and seating are generally businesslike and professional, with some inconsistencies (e.g., flags are in the correct order but are not aligned with other aspects of the officer gear; and/or several of the symbols are in proper order but are some are misaligned; and/or officers are seated in the proper arrangement, but some chairs are misaligned, etc.).	Flags are completely aligned and in proper order and placement; officer gear is placed in the correct order and in proper alignment on the table; the seating arrangement is precise, businesslike, and professional.

Record scores
in the column
spaces below.

BUSINESS MEETING DEMONSTRATION (190 points) – continued			
KNOWLEDGE OF TSA (20 points)			
Opening Ceremony (X1)	Many items of sequence and order are incorrect; officers make several mistakes.	Officers make few, if any, sequence and order mistakes, resulting in a fairly smooth opening ceremony.	The opening is smooth and efficient.
Closing Ceremony (X1)	Officers make several mistakes; the creed recitation is sloppy and the overall effort is unpolished.	Appropriate procedures are followed, with some mistakes made (e.g., creed recitation).	The closing is outstanding, with no mistakes; the presentation is highly polished.
KNOWLEDGE OF PARLIAMENTARY PROCEDURE (140 points)			
Voting Procedures (X1)	Several significant mistakes are made in voting procedures.	Only a few mistakes are made in voting procedures.	All voting procedures are correct, smooth, and efficient.
Debate (exclude president) (X3)	Only a few officers participate effectively in the debate, which is loosely presented.	Most officers participate in the debate process and are somewhat convincing.	All officers participate in and present a highly cohesive debate.
Parliamentary Actions (X5)	Only one (1) of the required actions is completed correctly.	At least two (2) of the actions are completed correctly, with adequate effort.	All three (3) actions are completed correctly, with notable and inspiring effort.
Communication (X2)	Communication is unclear; some mumbling occurs and/or voices are too loud or too soft; and/or problems occur with verbal expression (e.g., grammar, sentence structure).	Communication is generally clear, with appropriate volume of voices and only minor problems with articulation or verbal expression.	Communication is clear, concise, and easy to understand; voices are well modulated and speakers are articulate.
Treasurer's Report (X1)	The report is incorrect or not complete; math and spelling errors are evident.	The report generally is correct and complete, with only one (1) or two (2) math or spelling errors.	The report is correct and complete, with no math or spelling errors.
Chapter Minutes (X2)	The format of the minutes is incorrect or not complete; poor grammar and spelling errors are evident.	The format of the minutes is generally correct and complete, with few grammar and/or spelling errors.	The minutes are formatted correctly, are complete, and have no grammar or spelling errors.
BUSINESS MEETING DEMONSTRATION SUBTOTAL (190 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
SEMIFINAL SUBTOTAL (200 points)			
TIME DEDUCTIONS (NO TEAM MAY GO BEYOND 17 MINUTES)			
A five-(5) point deduction will be incurred for every thirty (30)-second interval over the allotted time. Multiply the number of intervals by five (5) and record the total deduction in the column to the right. # of intervals X 5 = _____ (total deduction)			

BONUS (20 points)					
For Additional Motions and Parliamentary Actions (by officers other than the president) (X2)	Only one (1) or two (2) of the additional actions is/are completed correctly; the effort is uninspiring.	Three (3) or four (4) of the actions are completed correctly.	All five (5) of the supplementary actions are completed correctly in an efficient and effective manner.		
BONUS SUBTOTAL (20 points)					

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (200 points)	
---	---------------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CHAPTER TEAM

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Assistants, two (2) or more for the written test and two (2) or more for the oral presentations
- D. Timekeeper

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Opening and closing ceremonies script
- C. List of parliamentary actions; one (1) copy for each semifinalist team and copies for the judges.
- D. Copies of secretary's minutes
- E. Copies of treasurer's report
- F. Copies of the Chapter Team Official Minutes document
- G. Paper, six (6) pens, one (1) nonprogrammable calculator, and six (6) 3" x 5" note cards, per team
- H. Officer's symbols and gavel
- I. United States flag
- J. State flag (optional)
- K. Stopwatches
- L. Table rostrum, if available
- M. One (1) long table or two (2) tables and six (6) chairs for team members
- N. One (1) table and three (3) chairs for judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the [National TSA Conference Competition Requirements](#)).
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. All participants and evaluators should be in the room at this time.
- D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. Administer the test.
- G. Determine individual and team scores.

- H. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

SEMIFINAL ROUND

- A. Check in semifinalists at the time stated in the conference program. Inform the teams of their order of performance and review the procedure to be followed.
- B. When a team enters the oral performance room, the materials (secretary's minutes, copy of Chapter Team Official Minutes form for secretary's minutes, a treasurer's report, a copy of the TSA creed, and a list of parliamentary actions) are provided to the president. At this point, begin the allotted time.
- C. The event coordinator or an assistant is responsible for introducing each team by entry number only when the previous team has finished its presentation.
- D. Teams may take chapter paraphernalia (officer symbols and gavel only) into the performance room if desired, or they may use what is provided by the event coordinator.
- E. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
1. To deduct twenty percent (20%) of the total possible points in this round or
 2. To disqualify the entry
- The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- F. If necessary, manage security and the removal of materials from the area.

OVERVIEW

Applying leadership and 21st century skills, participants create an illustrated children's story of high artistic, instructional, and social value. The narrative may be written in prose or poetry and take the form of a fable, adventure story, or other structure. The physical story book should be of high quality and designed to meet the annual theme, which will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) teams per state may participate; individual entries are permitted.

TIME LIMITS

SEMIFINAL ROUND

- A. Five (5) minutes are allowed to answer questions from the judges.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual design challenge on the [TSA website](#) under *Themes & Problems*.
- C. Participants concentrate their efforts in researching, writing, and illustrating children's books and literature, as well as the creation of storybooks similar to the annual design challenge (e.g., pop-ups, interactive books, etc.).
- D. Participants develop and construct a high-quality children's storybook with illustrations.

- E. Participants record the design process in a documentation portfolio.
- F. Participants "field test" their storybook and document outcomes and findings.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report to the time and place stated in the conference program to check in:
 - 1. The storybook
 - 2. The documentation portfolio
- B. Entries are reviewed by judges with neither students nor advisors present.
 - 1. Judges score the Storybook criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the Documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. One (1) team representative reports to the event area at the time and place stated in the conference program to sign up for an interview time.
- B. No more than two (2) representatives report at the assigned time and place for the interview.
 - 1. Both representatives participate in the interview process. The interview process lasts up to five (5) minutes.
- C. Judges score the interview criteria.
- D. No more than two (2) team representatives pick up the team's entry from the display area at the time and place stated in the conference program.
- E. Ten (10) finalists are announced during the conference award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

A. Storybook:

1. Participants design and create an entirely original storybook, complete with narrative and illustrations to meet the annual design challenge.
2. The physical storybook should be of high quality, designed to meet the age group for which it is intended. For the purposes of this event, children are defined as those twelve (12) years or younger.
3. Together with the storybook, the narrative and accompanying illustrations should result in an experience that delights, enlightens, and contributes to the wholesome development of a child.
4. The storybook and narrative with accompanying illustrations should take no more than ten (10) minutes to read and view.
5. The physical storybook must not exceed 12" x 12" when closed.
6. There is no limit on the number of inside pages (may be one or two-sided).
7. The team must determine which format best presents the team's narrative and illustrations.
8. There must be a minimum of seven (7) illustrations that enhance the story and deepen the child's understanding and enjoyment of the reading experience.
 - a. An illustration on the book's cover may count as one (1) of the required seven (7) illustrations. The team may use the cover illustration within the story as well
 - b. All illustrations MUST be original, freehand, and/or computer-generated drawings made by the team member(s).
 - c. All computer-generated work MUST be developed from primitive lines and shapes and be the sole work of the team members.
 - d. Physical or computer templates, previously existing drawings, characters, backgrounds, etc., are NOT PERMITTED. The storybook may include the name of the author(s) and illustrator(s) on the cover. This inclusion of the name of the author(s) and illustrator(s) is an exception to the General Rules.
9. Copyrighted material is NOT PERMITTED.
10. The physical storybook must be the original work of the team members.
11. If narrative or illustrations appear in the story and they are not authored by one of the team members, the team will be disqualified.
12. The book must be designed, engineered, created, and bound together solely by the team; no professional binding is allowed.
13. Photographic verification of the book construction process must be included in the portfolio.
14. The story must be no more than fifteen hundred (1500) words.
 - a. There will be a five (5)-point deduction for every hundred (100) words over the fifteen hundred (1500) word limit.
 - b. Stories containing two thousand (2000) or more words will be disqualified.
 - c. There is no minimum number of words required.
 - d. The word count (number of words comprising the story's narrative) must be listed on the inside of the front cover of the book.
15. Publishing rights remain with the authors and illustrators.

B. Documentation Portfolio:

1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:

- a. Title page with the title of the story, the event title, the conference city and state, the year, the team/individual chapter ID number, and the word count (number of words comprising the story's narrative); one (1) page
- b. Table of contents; pages as needed
- c. Purpose of story; one (1) page, to include:
 - i. Story's intent
 - ii. Summary of storyline and theme
 - iii. Intended audience (age, gender, demographics, and special disabilities, if any)
 - iv. Word count – Number of words comprising the story's narrative
- d. Photographic verification of book construction and binding; pages as needed
- e. Field Tests Summary. A "field test" is a reading of the storybook to a group of children in the intended target age range; pages as needed
 - i. A minimum of two (2) field tests must be conducted.
 - ii. Participants must document each field test with a summary paragraph that details the outcomes and findings.
 - iii. Each summary paragraph must include the date, time, and location of the field test.
 - iv. Each "field test" must be signed off by the chapter advisor.
- f. Research summary: A written summary of the research, writing strategies, problems encountered, and solutions developed in the writing and illustrating of the story; one (1) page.
- g. Project summary: A written summary of the research into the creation of storybooks similar to the annual design challenge (e.g., paper folding, interactive features in books). The summary must include the process and challenges the team encountered and the solutions developed in overcoming them; one (1) page.
- h. A list of tools, software (if any), and techniques used in the creation of the physical storybook and illustrations; pages as needed.
- i. References/research sources; one (1) page.
- j. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
- k. Photo/Film/Video Consent and Release forms. If the entry contains images of people (minors require parental consent), proof of consent must be included (see 2026 & 2027 Forms Appendix); pages as needed

PRELIMINARY ROUND

- A. No more than two (2) team representatives drop off the storybook and documentation portfolio at the time and place stated in the conference program. NOTE: The documentation portfolio and storybook must be submitted together, in order for the entry to be judged.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. The physical storybook

Tier 2

- B. The documentation portfolio

SEMIFINAL ROUND

- A. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

Depending upon the subject of the story this event may align with one (1) or more of the STEM (Science, Technology, Engineering, and Mathematics) educational standards. Refer to the STEM Integration section of this guide for more information.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Writer
- Illustrator
- Educator
- Editor
- Publisher
- Graphic artist

CHILDREN'S STORIES

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Storybook is present
- ☐ TIER 2 – Documentation portfolio is present and the story is 1999 words or less (verified on cover page and in portfolio)
- ☐ ENTRY NOT EVALUATED

TIER 1 – STORYBOOK (130 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Story Narrative (X3)	Narrative is poorly written; there is little apparent purpose; it is lacking a coherent theme and storyline.	Narrative's purpose is clear, with a focused theme and storyline; the narrative has good pacing and development of characters and events.	Narrative is extremely well written with a clear purpose; storyline is fast paced and exciting; the details are rich and enchanting.	
Illustrations (X3)	Artisanship of the illustrations reflects little technical skill; illustrations add little value to the story's narrative, storyline, and/or theme.	Artisanship of most illustrations reflects good technical skill; illustrations add to the story's narrative, storyline, and theme.	Artisanship of illustrations is excellent, reflecting sophisticated technical skills; illustrations enhance the story's narrative, storyline, and theme, and they are of high esthetic quality.	
Book Construction and Concept (X3)	Book construction demonstrates little or no creativity or innovation; minimal consideration is given to basic design principles and book construction; book is poorly constructed or is not bound.	Construction of the book is of good quality and demonstrates some degree of creativity and innovation and an understanding of basic design principles; adequate choice of materials was used in its construction.	Book is designed with attention to detail; construction is of high quality and demonstrates a thorough understanding of design principles; an excellent choice of materials was used in the construction of the book.	
Impact (X3)	Story (narrative, with the illustrations) is lacking in purpose and coherence; it is not very interesting; it lacks artistic, and/or instructional, and/or social value.	Story (narrative, with the illustrations) reflects a purpose and incorporates artistic, instructional, and social value; it is somewhat compelling and entertaining.	Story (narrative, with the illustrations) is beautifully told; it is compelling, entertaining, purposeful, and it reflects high artistic, instructional, and social value.	
Theme (X1)	The annual theme is not addressed.	The annual theme is somewhat addressed but doesn't contribute to the effectiveness of the overall design.	The annual theme is addressed and contributes to the effectiveness of the overall design.	
TIER 1 – STORYBOOK SUBTOTAL (130 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (130 points)

TIER 2 – DOCUMENTATION PORTFOLIO (80 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Purpose of Story (X1)	Story's intent, storyline, and theme are poorly explained and/or the intended audience is not identified.	Story's intent, storyline, theme, and intended audience are adequately explained.	Story's intent, storyline, theme, and intended audience are complete and well explained.	
Photographic Verification (X1)	Photographic verification is very unorganized or is missing.	Photographic verification is somewhat disorganized and is missing a few components; the process is somewhat outlined.	Photographic verification has all components and is well organized; the process is clearly outlined; there is photographic verification of book construction and binding process.	
Field Test Summaries (X1)	Summaries are poorly organized and/or incomplete.	Summaries are adequately detailed and organized and contains most of the required components, including a record of the field tests.	Summaries are well documented and contains all the required components, with special attention given to field tests.	
Research and Project Summary (X2)	Summary of the research, design, and writing process is poorly done and/or is incomplete.	Summary of the research, design, and writing process is generally clear and complete.	Summary of the research, design, and writing process is very well written, detailed, clear, and complete.	
Tools and Software (X1)	Lists are missing several components, and/or they are unorganized.	Most components are included; the lists are generally organized and complete.	All components are included and organized in the lists.	
Research Base (X1)	There are few references listed, and/or the references listed show little relevance to the project's goal; sources lack credibility.	There are a sufficient number of references listed, with some credible references.	Many credible and high-quality references are listed, reflecting research in writing and illustrating for children, and in child development.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (80 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

Story length violation: For stories exceeding 1500 words, a deduction of 5 points will be incurred for every 100 words more than 1500 and up to 2000. Stories of 2000 words or greater will be disqualified. Example: 1600 – 1699 words, 5 points; 1700 – 1799 words, 10 points; 1800 – 1899 words, 15 points; 1900 – 1999 words, 20 points; 2000 words and above, disqualified.

PRELIMINARY SUBTOTAL (210 points)

SEMIFINAL INTERVIEW (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit a general understanding of the concepts in their project.	Participant shows clear evidence of a thorough understanding of their project.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL INTERVIEW SUBTOTAL (30 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (30 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (240 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CHILDREN'S STORIES EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants for check-in, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Stopwatch
- C. Display tables for entries (minimum width 18")
- D. Table and chairs for judges and two (2) semifinalist team representatives.

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with the judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Check in the entries at the time stated in the conference program.
- B. Participants check in:
 - 1. The storybook
 - 2. The documentation portfolio
- C. Portfolios should be held by the coordinator and not placed with the storybook.
- D. Each entry must include the participant's identification number in the upper right-hand corner of the entry.
- E. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- F. In order to compete, participants must be on the entry list or must have approval of the CRC.
- G. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- H. Position the entries for evaluation and viewing.
- I. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Judges independently review the entries based on the following criteria:

Tier 1

- 1. Judges score the storybook criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

- 2. Judges score the documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

- B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

- 1. To deduct twenty percent (20%) of the total possible points in this round
- 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

- C. Create a semifinalist sign-up sheet for each team's final presentation.

SEMIFINAL ROUND

- A. Two (2) team representatives report to the event area at the time and place stated in the conference program to sign up for an interview time.
- B. Two (2) team representatives report at the assigned time and place for the seminal interview.
- C. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
- D. Manage the interviews:
 - 1. Team members must be prepared to discuss illustrations included in the story.
 - 2. Both team members participate in the interview process (lasting up to five [5] minutes).
- E. Judges score the reading and interview criteria.
- F. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Participants will demonstrate their knowledge of computer science and coding concepts (e.g., language syntax, data structures, control flow, object oriented programming, etc.) by taking a test. Applying leadership and 21st century skills, semifinalists participate in an on-site programming challenge. Specific requirements, such as the programming language, are posted on the [TSA website](#) under *Themes & Problems*. Completed solutions are objectively measured to determine the best and most effective solution for the stated problem.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. One (1) hour is allowed for the test. Both members must be present at the same time and take the test individually.

SEMIFINAL ROUND

- A. Two (2) hours are allowed for the on-site challenge.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants report at the time and place stated in the conference program for the test.
- B. Both team members take the test simultaneously.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. Participants bring their own computer systems to the event area at the time and place stated in the conference program.
- B. Participants are given a problem, evaluation criteria, materials, and allotted two (2) hours for the design and development of the solution.
- C. Each solution is tested and presented to the judges as soon as completed.
- D. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Team members take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 - 1. Student identification numbers must be entered on the scantron form in the space indicated.
 - 2. Participants must stop work immediately when time is called.
 - 3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.

4. All tests and scantrons must be turned in before leaving the test area.
- E. The average of the test scores of the two (2) team members determines team ranking.
- F. These same two (2) team members compete in the semifinal round, should the team qualify.

SEMIFINAL ROUND

- A. Participants respond to a design challenge consisting of a series of coding problems that must be completed on-site at the conference within a two (2)-hour time frame.
- B. All work must be completed in the event area during the time specified for the event.
- C. Participants are required to bring:
 1. One (1) laptop. Laptop must be fully charged with enough battery for the duration of the semifinal round.
 2. One (1) computer mouse (if desired).
 3. Pencils and/or pens.
 4. Notebook paper or copier paper.
 5. Portable power bank (if desired).
- D. Participants do NOT have access to electrical power/ outlets during the event.
- E. Participants do NOT have access to the Internet during the event. The use of hot spots is not permitted.
- F. Participants must have all software development tools needed for the competition downloaded and accessible on their computers.
- G. All solutions must be tested, demonstrated and presented by participants in front of the judges.
- H. Participants may NOT use any generative artificial intelligence (AI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.) in developing their solution(s).

EVALUATION

PRELIMINARY ROUND

- A. Each team's averaged test score

SEMIFINAL ROUND

- A. The solution to the on-site problem
 1. Each problem in the programming challenge will have an objective correct answer.

STEM INTEGRATION

Depending upon the subject of the problem, this event has connections to one (1) or more of the STEM areas of Science, Technology, Engineering, and Mathematics.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer software engineer
- Mathematician

Participant/Team ID# _____

CODING

2026 & 2027 OFFICIAL RATING FORM
MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Computer hardware is present☐ ENTRY NOT EVALUATED

TEST SCORE (50 points)

Average of the two (2) team member's test scores.

TEST SCORE SUBTOTAL (50 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (50 points)

ON-SITE PROBLEM – OBJECTIVE (80 points)

Evaluation: The ranking will be determined based on the submission time of the solution(s) to the on-site problem.

1st: 80 Points

2nd: 75 Points

3rd: 70 Points

4th: 65 Points

5th: 60 Points

6th: 55 Points

7th: 50 Points

8th: 45 Points

9th: 40 Points

10th: 35 Points

ON-SITE PROBLEM (OBJECTIVE) SUBTOTAL (80 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (80 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (130 points)	
--	---------------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CODING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Assistants, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Tables and chairs for participants
- C. Tables and chairs for judges
- D. Twelve (12) copies of a well-written, technologically appropriate problem for each semifinalist team that can be objectively measured; one (1) copy per team
- E. Adequate conditions, tools, materials, monitoring, and testing devices for the problem
- F. Stopwatch or clock for timekeeper

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the [National TSA Conference Competition Requirements](#)).
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. All participants and judges should be in the room at this time.
- D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. Distribute any necessary materials.
- G. Monitor the one (1)-hour test.
- H. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

SEMIFINAL ROUND

- A. At least one (1) hour before the event is scheduled to begin, meet with semifinalist judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- B. Check in the semifinalist teams and equipment. Teams must bring:
 - 1. One (1) laptop
 - 2. Extra charged laptop battery or extra charged laptop as backup (but only one laptop may be used at any time)
 - 3. One (1) computer mouse
 - 4. Teams may also bring pencils and paper.
 - 5. Portable power bank (if desired)

- C. Participants do NOT have access to electrical power/ outlets during the event.
- D. Participants do NOT have access to the Internet during the event.
- E. Participants must have all software development tools needed for the competition downloaded and accessible on their computers.
- F. Once teams are seated and general announcements have been given, distribute and review the problem and start the time.
- G. All solutions must be tested, demonstrated and presented by participants in front of the judges. Judges and assistants observe, with judges evaluating solutions as soon as appropriate.
- H. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- I. Manage security and the removal of materials from the area.

COMMUNITY SERVICE VIDEO



OVERVIEW

Applying leadership and 21st century skills, participants create and submit a video that depicts their local TSA chapter's involvement with a community service project (e.g., American Cancer Society) of their choice.

ELIGIBILITY

One (1) team per chapter may participate. Individual entries of one (1) are permitted.

TIME LIMITS

PRE-CONFERENCE

- A. All components of the chapter's entry (community service video URL and documentation portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59 p.m. EST on a designated date in mid-May.
- B. The video cannot exceed duration of more than two (2) minutes.
- C. A deduction of five (5) points will be applied to videos exceeding the time limit.
- D. There is no minimum length restriction.
- E. The video will be timed from the first sound or picture to the final sound or picture.

SEMIFINAL ROUND

- A. Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:
 - 1. Seven (7) minutes for the presentation
 - 2. Three (3) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Teams identify a community service project.
- C. Teams prepare a video observing the requirements set forth under the Regulations and Requirements section of this event.
- D. Participants submit the community service video URL and the documentation portfolio PDF by 11:59 pm EST on a designated date in mid-May.
- E. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalist teams (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Semifinalist teams sign up for a presentation/interview time at the date and time noted in the conference program.
- B. The semifinalist team or individual may report to the event area for the presentation/interview.
- C. Following the presentation, semifinalist teams will have a chance to answer questions about their entry, specifically about the community service project and the video's design and development process.
- D. Ten (10) finalists will be announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Participants may choose to submit their entry electronically via a HYPERLINK to the designated submission file as an UNLISTED YouTube URL of the video, or a shareable link in cloud storage. The video must be located online and accessible for evaluation.
- B. The URL must point directly to the participant's entry. Entries that require a software download or request that access be granted will not be judged.
- C. Entries received, or changes made to submitted entries after the deadline will not be judged.
- D. The year must be clearly illustrated at the beginning of the video.
- E. This entry is exempt from General Rule D.9., which states that a chapter name must not appear in an entry.
- F. All video footage, graphics, special effects, and audio clips must be originally created/filmed by the participants.
- G. The source for any graphics, music, or sounds not created by the members must be properly cited on the references/resources page of the documentation portfolio.
- H. Documentation Portfolio:
 - 1. The documentation portfolio must be submitted as a multi-page PDF document with pages in this order:
 - a. Title page with the event title, conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Community Service Project Overview – introduce the topic/subject, detail the chapter's involvement in the project, describe why the chapter selected the project, and describe how the chapter/recipients benefited; one (1) page
 - d. References/Resources; pages and needed
 - e. Student Copyright Checklist (see 2026 & 2027 Forms Appendix); one (1) page

- f. Photo/Film/Video Consent and Release forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed
- g. Work Log (see 2026 & 2027 Forms Appendix); pages as needed

SEMIFINAL ROUND

- A. One (1) team representative reports at the time and place stated in the conference program to sign-up for a presentation/interview time.
- B. Semifinalist teams report at least ten (10) minutes prior to their assigned time to the designated place for their presentation/interview.
- C. Semifinalist teams will have a chance to make a presentation and answer questions about their entry, specifically about the community service project and the video's design and development process.
- D. Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:
 - 1. Seven (7) minutes for the presentation
 - 2. Three (3) minutes to respond to questions from the judges

EVALUATION

PRELIMINARY ROUND

- A. The video
- B. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation and interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Videographer
- Director
- Actor
- Screenplay writer
- Audiovisual technician

COMMUNITY SERVICE VIDEO

2026 & 2027 OFFICIAL RATING FORM MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Video entry was submitted
- ☐ Documentation PDF was submitted
- ☐ ENTRY NOT EVALUATED

VIDEO PRODUCTION (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Planning (X1)	Video shows little or no evidence of planning, analysis, or research.	Video shows good analysis of the project; concepts make sense.	Video shows a complete and insightful analysis; content is outlined properly.	
Camera (X1)	Problems are evident with camera focus, steadiness, and framing.	Camera work is clearly focused and framed; creative and close-up shots are somewhat used.	Steady and creative shots that enhance the video are evident in the camera work; there is good use of close-ups.	
Lighting (X1)	Poor ambient lighting choices and/or heavy back-lighting are evident.	Adequate lighting on subjects and proper lighting techniques are evident.	Excellent and creative use of lighting to propel the story emotionally is evident.	
Continuity and Pacing (X2)	Show sequencing in the video is incomprehensible; shots are left too long; edit points have glitches.	Pace and timing of the video are well structured; clips move along, telling the story; moderate use of transitions is evident.	Shots logically pace the story in an interesting way; excellent and purposeful use of transitions is evident.	
VIDEO PRODUCTION SUBTOTAL (50 points)				

VIDEO EFFECTIVENESS (100 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Topic and Context (X3)	Video does not show clear evidence of where and when the event(s) happened and why.	Video shows where and when the event(s) happened and why they occurred.	Video shows where and when the event(s) happened, why they occurred, and what factors contributed to their development.	
Theme (X2)	The theme is not clear, or the project has a loose-fitting, vaguely stated connection.	Video conveys the theme and implies the connection.	Video shows a thorough understanding of the theme and clearly uses the theme as a basis throughout.	
Significance of Topic (X2)	Video shows little evidence of the significance/importance of the topic.	Video states the topic's significance adequately.	Video clearly states the significance of the topic.	
Creativity and Originality (X1)	Little original thought or creativity are evident in the design and production of the video.	Original thought and creative elements are expressed and generally highlighted in the video.	Originality and creativity are at the forefront of the video.	
Video Efficacy (X2)	The video does not meet project goals; an unclear, sloppy message is evident.	The video topic is presented with insights; viewer can see that the video adequately meets the objective.	The video is highly focused, with a rich variety of supporting material.	
VIDEO EFFECTIVENESS SUBTOTAL (100 points)				

TIME DEDUCTION	
Five (5) points is deducted for the video duration exceeding two (2) minutes. The video timing commences from the first sound or picture to the final sound or picture.	

DOCUMENTATION PORTFOLIO (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Community Service Project Description (X2)	The overview is missing details and content and appears to be lacking information; minimum effort appears to have been given.	The overview provides adequate information and includes descriptions and details to help support the video.	The overview is clear and concise and includes the introduction, involvement, descriptions, and benefits of the project.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
DOCUMENTATION PORTFOLIO SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (190 points)

SEMIFINAL PRESENTATION/INTERVIEW (50 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions are vague.	Participants exhibit a general understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the concepts in the project.	
Articulation (X2)	The presentation/interview is full of illogical thoughts that lack clarity, and/or there is insufficient information provided that describes the project.	The presentation/interview is somewhat logical, easy-to-follow, and/or there is sufficient information provided that describes the project.	The presentation/interview is clear, concise, and there is ample information provided that describes the project.	
Delivery (X1)	The team/individual is verbose and/or uncertain; participants' posture, gestures, and lack of eye contact diminish the interview.	The team/individual is somewhat well-spoken and clear; participants' posture, gestures, and eye contact are adequate.	The team/individual is well-spoken and distinct; participants' posture, gestures and eye contact result in a polished, natural, and effective interview.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (50 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL (240 points)**

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

COMMUNITY SERVICE VIDEO

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 1. Preliminary round, two (2) or more (judging takes place online).
 2. Semifinalist round, two (2) or more (preferably the same judges as the preliminary round).

MATERIALS

- A. Coordinator's packet, containing:
 1. Event guidelines, one (1) copy for the coordinator and for each judge
 2. TSA Event Coordinator Report
- B. One (1) stopwatch
- C. Table and chairs for judges

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. EST on a designated date in mid-May. The results will be shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.

- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details related to the event.
- F. Ensure the judges have access to the online judging system.

SEMIFINAL ROUND

- A. At least one (1) hour before the event is scheduled to begin, meet with judges and review the time limits, procedures, and regulations.
- B. Semifinalist teams report to the event area at the time and place noted in the conference program to sign up for a presentation/interview.
- C. Manage completion of the on-site presentation/interviews.
- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 1. To deduct twenty percent (20%) of the total possible points in this round
 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. If necessary, manage security and the removal of materials from the event area.

COMPUTER-AIDED DESIGN (CAD) FOUNDATIONS



OVERVIEW

Applying leadership and 21st century skills, participants demonstrate their understanding of CAD fundamentals as they create a two-dimensional (2D) graphic representation of an engineering part or object. For example, participants may be given an isometric drawing and would be expected to generate the required 2D views, complete with dimensions.

ELIGIBILITY

Two (2) individuals per state may participate.

TIME LIMITS

- A. Thirty (30) minutes to set-up.
- B. Two (2) hours to develop the drawing(s).
- C. One (1) hour for final evaluation process.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants bring their own computer systems to the event area at the time and place stated in the conference program.
- B. Each participant, with one (1) assistant (an instructor, fellow student, or adult chaperone), is allowed thirty (30) minutes to set up and test equipment. At the end of the thirty (30)-minute set-up period, assistants are required to leave the area.
- C. Participants are given a design problem to solve during a two (2)-hour work session.
- D. Participants work independently, without assistance from judges, teachers, or fellow participants.

- E. At the end of the session, participants save their work on their hard drives and back up saved work on a USB flash drive.
- F. At the end of the session, participants will submit their final projects as a .PDF file on the USB flash drive. All USB flash drives become the property of TSA and will not be returned.
- G. Judges circulate to evaluate the entries and ask questions of the participants.
- H. Participants shall reserve one (1) additional hour for the final evaluation process.
- I. Participants report to the event area at the time and place stated by in the conference program to pick up their equipment.
- J. The top ten (10) finalists are announced during the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

- A. Participants provide their own computer hardware:
 - 1. Only one (1) computer and one (1) monitor are allowed per student. Laptop computers are recommended.
 - 2. Computers must be equipped with a USB port.
 - 3. Software needed for the challenge; preference is that the software is downloaded; however, if the software is online, National TSA will not provide access to the internet. Students may opt to provide their own hotspots, but National TSA WILL NOT guarantee reception quality or functionality.
 - 4. One blank USB flash drive labeled with the registered student identification number, for submission.
 - 5. Power strip/surge protector and 20' extension cord (optional and not provided by TSA)

6. Reference materials, which must be printed resources; no online resources will be allowed.
 7. Pencils
- B. Participants are provided with sketching paper and electricity.
- C. Participants are advised to save their work onto their hard drives every fifteen (15) minutes.
- D. Participants identify their work using only their student identification number.
- E. Participants are not permitted to:
1. Leave the event room without permission from the event coordinator. If a participant must use the restroom he/she is accompanied by an escort.
 2. Share solutions to problems, reference materials, hardware, or software.
- F. Breakdown of equipment is permitted only after the work of all participants has been evaluated.

EVALUATION

- A. A PDF of the design solutions submitted on a USB. Participants must be able to save their drawing as a PDF without assistance.
- B. In the event of a tie, the participant's completion time for the problem will be used as the tiebreaker.

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Engineer
- Automobile designer
- CAD professional
- Machine designer

CAD FOUNDATIONS

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Computer hardware is present
- ☐ USB drive is present
- ☐ CAD Software is present
- ☐ ENTRY NOT EVALUATED

LAYOUT (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Geometry and Drawing Orientation (X2)	The correct views, constructive geometry, and orientation have not been selected or used throughout the drawing process and final layout.	Most of the views, constructive geometry, and orientation selected and used are correct and in the proper layout.	All of the views, constructive geometry, and orientation that have been selected and used are correct and in the proper layout.	
Line Conventions (X2)	Proper line types have not been used throughout the drawing process.	Most of the proper line conventions have been used, with one or two mistakes.	All of the correct line types are used in the correct locations.	
Dimensioning (X2)	Many of the necessary dimensions are missing or placed incorrectly.	Most of the required dimensions are included and placed correctly.	All of the necessary dimensions are included and correctly placed; this includes hole & thread notes (if applicable).	
Title Block (X1)	The title block is missing or is missing scale, title of drawing, student ID, or date.	The title block is present, but is missing either scale, title of drawing, student ID, or date.	The title block is present and contains all required elements: scale, title of drawing, student ID, and date.	
LAYOUT SUBTOTAL (70 points)				

SOFTWARE UTILIZATION (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Application of Drawing Knowledge (X1)	There is little evidence that the student is able to read and interpret mechanical drawings.	There is some evidence that the student is able to read and interpret mechanical drawings.	There is overwhelming evidence that the student can read and interpret mechanical drawings.	
Use of CAD Features and Functions (X2)	There is little evidence of an understanding and application of CAD functions.	There is evidence of a general understanding and effective application of CAD functions.	A complete and effective understanding and application of CAD functions is evident.	
SOFTWARE UTILIZATION SUBTOTAL (30 points)				
Rules violations (a deduction of 20% of the total possible points in the semifinalist sections above) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				
TOTAL (100 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

COMPUTER-AIDED DESIGN (CAD) FOUNDATIONS

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Assistants, one (1)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. One (1) ream of 8½" x 11" white copier paper
- C. Statement of problem as a hard-copy sketch; copies, as needed
- D. Tables and chairs for event coordinator, judges, and participants
- E. Access to AC electricity for each member.

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

ON-SITE CHALLENGE

- A. Check the registration list and assign participants to work stations.
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. All participants and judges should be in the room at this time. Participants not present may be disqualified.
- E. Allow thirty (30) minutes for participants and their assistants (no more than one [1] per participant) to set up equipment.
- F. At the end of the thirty (30)-minute set-up time, non-participants are required to leave the event area.
- G. Review the time limits, procedures, regulations, and protocol of the event with the participants.
- H. Distribute copies of the CAD problem.
- I. Remind participants to save their work at regular time intervals.
- J. Answer any appropriate questions concerning the CAD problem.
- K. Begin the event and announce the ending time.
- L. Announce the time remaining to work at one (1) hour, thirty (30) minutes, fifteen (15) minutes, and five (5) minutes before time is called.
- M. When time is called, participants stop, save their work on their hard drives, and back-up on their USB flash drives.
- N. Each entry must include the student's identification number.
- O. Participants are required to remain in the area while their entries are being judged, but may be asked to leave the room while judges discuss entries.
- P. Record the time that each participants finished the CAD challenge. In the event of a tie, the participant's completion time for the problem will be used as the tiebreaker.
- Q. Judges circulate and evaluate entries as they are completed.

R. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

S. Breakdown of equipment is permitted only after the work of ALL participants has been evaluated.

T. If necessary, manage security and the removal of materials from the event area.

CONSTRUCTION CHALLENGE



OVERVIEW

Applying leadership and 21st century skills, participants submit a scale model display, and portfolio that documents their design idea and plan for their construction solution for the annual theme. Semifinalists demonstrate expertise in construction design through a presentation and an interview. The annual theme will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

One (1) team of at least two (2) individuals per chapter may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. The teams design and construction plan to address the annual theme must be completed during the current school year.

SEMIFINAL ROUND

- A. Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:
 - 1. Seven (7) minutes for the presentation
 - 2. Three (3) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants design a solution to address the construction problem outlined in the annual theme. The annual theme will be posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants create a scale model/prototype, display, and documentation portfolio while observing the regulations.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in:
 - 1. The scale model/prototype
 - 2. A display
 - 3. A documentation portfolio
- B. Entries are reviewed by judges with neither students nor advisors present based on the following criteria:
 - 1. Judges score the Model/Prototype and Display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the Documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. A representative from each semifinalist team reports at the time and place stated in the conference program to schedule a presentation/interview time.
- B. Two to four (2-4) representatives from each team report at the assigned time and place to participate in a brief presentation, discuss their chapter's community project, and respond to questions.
- C. Participants pick up their entries from the display area at the time specified in the conference program.
- D. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. The chapter must plan and prototype a construction project to address the annual theme.
- B. Chapters must be involved in the majority of stages of development of the project, from planning to construction of the prototype (where appropriate), by experiencing as many facets of the undertaking as possible.
- C. Model/Prototype and Display:
 1. The materials that make up the scale model/prototype must be student-constructed and assembled.
 2. The display may occupy a space no more than 15" deep x 30" wide x 48" high.
 3. A/C electricity may not be used.
 4. If the display and/or model/prototype requires power, they must be powered by dry-cell batteries or photo-voltaic cells.
 - a. The power supply must physically fit within the display dimensions.
 - b. All power must be switched off once the team has completed set-up.
 - c. If teams want judges to activate any electronic device in their model/prototype/display, complete instructions must be provided to judges on how to power up the model/display.
 5. If operating instructions are necessary, they must be clearly displayed and should include information about turning on devices, logging in, etc.
- D. Once the display set-up time frame has closed, participants may not re-enter the event area.
- E. Identifying information must not be placed on the scale model/prototype, display, or in the portfolio for the purpose of revealing the TSA chapter, however, the entry is excluded from General Rule D.9, and the chapter name or other identifying information may appear in the entry documentation portfolio but only in media clippings, photographs, etc.
- F. Documentation Portfolio:
 1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. A design brief describing your team's proposal for addressing the theme's construction project. Your design brief should include:
 - i. Your inspiration, proposed solution, and highlight any major features or elements of the project; maximum two (2) pages.
 - ii. Sketches and initial design ideas for the construction project; pages as needed.
 - iii. A description of why major materials were selected for the project; one (1) page.
 - iv. A list of materials to be used in the construction of the proposed solution; maximum two (2) pages.
 - v. A set of drawings outlined in your proposed design for the construction project; pages as needed
 - vi. A description of the regulations or design standards being utilized for the project and how they impacted the design and construction; maximum two (2) pages.
 - d. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
 - e. References must follow a professional citation style of the competitors choosing. Failure to use a professional citation style will result in a rules violation of 20% (twenty percent). Some examples of professional citation styles include MLA, APA, Chicago, and IEEE; pages as needed.
 - f. Photo/Film/Video Consent and Release Forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed

PRELIMINARY ROUND

- A. No more than two (2) team representatives set up the entry and submit the portfolio for judging.

SEMIFINAL ROUND

- A. Using leadership and/or 21st century skills, two to four (2-4) representatives are allowed a maximum of ten (10) minutes to present and respond to interview question.

EVALUATION**PRELIMINARY ROUND****Tier 1**

- A. The model/prototype
B. The display

Tier 2

- C. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation and interview

Refer to the official rating form for more information.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Architect
- Community planner
- Construction manager
- General contractor
- Product designer

CONSTRUCTION CHALLENGE

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Model/prototype and display are present
- ☐ TIER 2 – Documentation portfolio is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – MODEL/PROTOTYPE/DISPLAY (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Model/Prototype (X2)	Model/prototype is sloppy and does not represent the project.	Model/prototype has some design flaws, but it represents the project.	Model/prototype has a high quality of craftsmanship and clearly depicts the project.	
Display (X2)	Display is sloppy and does not represent the project.	Display has some design flaws, but it represents the project.	Display has a high quality of craftsmanship and clearly depicts the project.	
TIER 1 – MODEL/PROTOTYPE/DISPLAY SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
TIER 1 – SUBTOTAL (40 points)				

TIER 2 – DOCUMENTATION PORTFOLIO (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Contents (X1)	Portfolio is unorganized and/or is missing three (3) components.	Portfolio is generally organized but is missing two (2) components.	Portfolio has all required components in order and is well organized.	
Design Brief (X1)	The design brief is incomplete, lacks a clear explanation of inspiration or solution, and does not highlight significant features or elements.	The design brief provides a basic explanation of inspiration and solution, with a mention of major features, but lacks depth or organization.	The design brief is comprehensive, clearly articulating inspiration, a well-thought-out solution, and highlighting all major features with professional presentation and organization.	
Sketches and Initial Design Ideas (X1)	Sketches and initial ideas are unclear, disorganized, or fail to communicate the proposed design effectively.	Sketches and design ideas are moderately detailed, organized, and convey the proposed design with some room for improvement in clarity or depth.	Sketches and initial ideas are detailed, professionally rendered, well-organized, and clearly convey the creativity and functionality of the proposed design.	
Description of Material Selection (X1)	The description is vague, lacks justification for material choices, or does not connect materials to project goals.	The description provides reasonable justifications for material selection, with some connection to project goals, but may lack thoroughness or specific details.	The description thoroughly justifies material choices, demonstrating a deep understanding of their alignment with project goals and design constraints.	
List of Materials (X1)	The materials list is incomplete, poorly organized, or lacks essential details such as quantities or specifications.	The materials list is complete and organized, providing necessary details but may miss opportunities for improved clarity or precision.	The materials list is comprehensive, meticulously organized, and provides clear, precise details about quantities, specifications, and their relevance to the project.	
Drawings (X1)	The drawings are incomplete, unclear, or fail to meet basic standards of technical drawing.	The drawings are complete and meet basic technical standards, but may lack precision, clarity, or advanced features.	The drawings are detailed, precise, professionally presented, and fully adhere to technical drawing standards, effectively communicating the design.	
Description of Regulations or Standards (X1)	The description of regulations or design standards is superficial, lacks detail, or fails to explain their impact on the project.	The description addresses relevant regulations or standards and provides a general explanation of their impact on the project.	The description is thorough, clearly identifies all relevant regulations or standards, and provides an insightful analysis of their impact on design and construction.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed, generally organized, and contains all of the required components.	Log is well documented and contains all required components.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (80 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
PRELIMINARY SUBTOTAL (120 points)				

SEMIFINAL PRESENTATION/INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participants seem unorganized and unprepared for the presentation/ interview; an illogical explanation of the need and solution is presented.	Participants are generally prepared for the presentation/interview; an acceptable explanation of the need and solution are communicated.	Presentation/interview is logical, well organized, and easy to follow; the need and solution are communicated in an organized and concise manner.	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit a general understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the project.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Team Participation (X1)	The majority of the delivery is made by one member of the team; the partner(s) may be disengaged from the presentation/interview.	Team members generally are engaged in the presentation/ interview, though one (1) member may take on more responsibility than the other(s).	All team members are actively involved in the presentation/ interview and responses to questions; there is shared responsibility between team members.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (50 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				TOTAL (170 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

CONSTRUCTION CHALLENGE

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants:
 - 1. Two (2) or more individuals assigned to check in, receive entries, and direct students to the display set-up area
 - 2. One (1) individual assigned for security during the set-up time

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Display tables for entries (needed at set-up time through pick-up time prior to the close of the conference)
- C. Stopwatch to monitor semifinalist presentations and interviews
- D. Tables and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is scheduled to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. No more than two (2) team representatives check in:
 - 1. The scale model/prototype
 - 2. The display
 - 3. The documentation portfolio
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- E. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Judges independently review the entries based on the following criteria:

Tier 1

- 1. Judges score the model/prototype and display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

- 2. Judges score the documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

CONSTRUCTION CHALLENGE

B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

C. Create a sign-up sheet for the semifinal interviews.

SEMIFINAL ROUND

- A. One (1) semifinalist team representative reports to the time and place stated in the conference program to sign-up for a presentation/interview time.
- B. Two to four (2-4) team representatives report at the assigned time and place for the presentation/interview.
- C. Manage completion of the presentation/interviews.
- D. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
- E. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Participants complete a cybersecurity test covering general cybersecurity vocabulary and knowledge needed to execute tasks commonly performed by all levels of cybersecurity professionals. Applying leadership and 21st century skills, participants prepare a presentation addressing a specific cybersecurity issue and present to a group of hypothetical corporate board members (i.e., judges). Participants must explain the importance of cybersecurity and why it is essential that the organization invest in such measures. Participants will answer questions from the judges following the presentation. The problem statement will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) individuals per chapter may participate.

TIME LIMITS

- A. One (1) hour to complete the test.
- B. Five (5) minutes are allowed for the presentation and interview broken down as follows:
 - 1. Three (3) minutes for the presentation
 - 2. Two (2) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual problem on the [TSA website](#) under *Themes & Problems*.
- C. Participants prepare a presentation according to the regulations.

PRELIMINARY ROUND

- A. Participants report at the time and place stated in the conference program for the test.
- B. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program to schedule a time for the semifinal presentation/interview.
- B. Semifinalists report at the assigned time and place stated in the conference program for the presentation/interview.
- C. Semifinalists present their presentation to the judges and answer questions.
- D. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No. 2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 - 1. Student identification numbers must be entered on the scantron form in the space indicated.
 - 2. Participants must stop work immediately when time is called.

3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.
4. All tests and scantrons must be turned in before leaving the test area.

SEMIFINAL ROUND

A. The Presentation/Interview

1. Any participant that is late or fails to arrive to present at the designated time results in an automatic disqualification.
2. Participants enter this event with the current year's problem which can be found on the *Competition Updates* page on the [TSA website](#).
3. No identifying information other than the participant's identification number and conference title and year are permitted.
4. Participants shall focus on the following in their presentation:
 - a. Identify and explain the security problem.
 - b. Explain how physical and digital security measures protect electronic information.
 - c. Use real-world data metrics and situations to support the claim.
 - d. Explain trade-offs when selecting and implementing cybersecurity recommendations.
5. Participants must use fictitious company names if needed for both the company in which the participant is "employed," as well as the fictitious company offering cybersecurity services. Real names may not be used.
6. Participants may use the presentation platform of their choice, such as PowerPoint, Prezi or Google Slides, however, audio may not be used.
 - a. Participants need to provide their own laptop. AC power will not be provided.
 - b. Each laptop device needs an HDMI connection or HDMI adapter.
 - c. Participant must be prepared to use the laptop screen if a digital projector and/or white background are not available.
7. The participant must orally present key points to the corporate board members.
8. No commercial or copyrighted material may be used.
9. Highlight your proposal with creative visual elements (e.g., graphics, photos, titles, transitions) to actively engage the audience. Participants may use images "labeled for reuse," but must properly cite the source (refer to the General Rules).
10. Photo/Film/Video Consent and Release Forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix).
11. Staple or paperclip the citation page (if necessary) and any Photo/Film/Video Consent and Release Forms (if necessary) together and present them to the corporate board members (judges) prior to the presentation.
12. Timing:
 - a. Participants are allowed five (5) minutes for the presentation/interview, broken down as follows:
 - b. Three (3) minutes to present solution to the theme/problem
 - c. Two (2) minutes to answer questions
 - d. A timepiece may be used by the participants, if desired
 - e. One (1) point per ten (10)-second interval is deducted for speaking over three (3) minutes. Time commences when the presenter begins speaking.

B. Judges score the presentation.

- C. The top ten (10) finalists are announced at the TSA conference awards ceremony.

EVALUATION

PRELIMINARY ROUND

A. The test

SEMIFINAL ROUND

A. The presentation/interview

Refer to the official rating form for more information

RESOURCES

- NICSS Cybersecurity for Students: niccs.cisa.gov/education-training/cybersecurity-students
- NICSS CISA Learning: niccs.cisa.gov/education-training/cisa-learning
- Center for Internet Security - Insights: www.cisecurity.org/insights/all
- Cyber.org: cyber.org
- Hack the Box: www.hackthebox.com
- TryHackMe: tryhackme.com
- IBM Skillbuilder – Cybersecurity: skillsbuild.org/students/course-catalog/cybersecurity
- SANS: www.sans.org

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Information support & services
- Network systems
- Programming & software development
- Web & digital communications
- Technical support specialist
- Computer software engineer
- Cybersecurity engineer
- Cryptographer
- Cyber Crime Investigator
- Cyber defense incident responder
- Cyber forensics expert
- Cyber legal advisor
- Cyber operator
- Vulnerability assessor

CYBERSECURITY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Participant is present for presentation/interview
- ☐ ENTRY NOT EVALUATED

TEST (50 points)	
TEST SUBTOTAL (50 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (50 points)	
---	--

SEMIFINAL PRESENTATION/INTERVIEW (130 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
PRESENTATION/INTERVIEW (100 points)				
Identification and Explanation of Issue (X2)	Identification and explanation of the issue is unclear.	Issue is defined and explained appropriately, however, some points need clarification.	A clear and concise definition and explanation of the issue is evident.	
Explanation of Importance (X2)	There is little evidence of research; there is a lack of understanding of the issues cited.	There is some evidence of research; an adequate understanding of the issues is present.	Thorough research is clearly evident with a firm understanding of the issues established.	
Creativity, Aesthetics, and Artisanship (X1)	The presentation lacks creativity; the work is unorganized and sloppy.	Some visual elements of creativity exist in the work; presentation is generally organized in its explanation of the issue and the visual elements somewhat enhance the presentation.	The presentation exudes creativity; essential design principles and elements are well integrated; presentation logically communicates an important idea and is engaging.	

SEMIFINAL PRESENTATION/INTERVIEW (130 points) – continued					
PRESENTATION/INTERVIEW (100 points) – continued					
Articulation (X2)	Communication of the proposal is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the proposal is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the proposal is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.		
Overall Impact (X2)	The presentation does not detail or enhance the essential components of the participant's problem identification and proposal.	The presentation somewhat enhances the essential components of the participants problem identification and solution.	The presentation greatly details and enhances the essential components of the participant's problem identification and solution.		
Interview (X1)	Responses to questions are not clear and show evidence that the member is not knowledgeable of the subject.	Responses to questions are clear and organized and demonstrate knowledge of the subject.	Responses to the questions are highly clear and well organized and demonstrate high knowledge of the subject.		
STAGE PRESENCE (30 points)					
Delivery (X1)	Participant is verbose and/or uncertain in his/her presentation; participant's posture, gestures, and lack of eye contact diminish the presentation.	Participant is somewhat well-spoken and clear in his/her presentation; participants' posture, gestures, and eye contact are acceptable in the presentation.	Participant is well-spoken and distinct in its presentation; the participant's posture, gestures, and eye contact result in a polished, natural, and effective presentation.		
Appearance (X1)	Participant's appearance is unprofessional, sloppy, and/or inappropriate.	Participant's appearance is adequate and appropriate.	Participant's appearance is clearly exceptional, appropriate, and professional.		
Confidence (X1)	Participant appears nervous during the presentation; poor posture, poor eye contact, and lack of confidence are evident; participant reads the presentation from notes.	Participant is generally poised, displays eye contact, and is confident, with little sign of nervousness; participant reads from the presentation or notes for key points, but presents clearly.	Participant "commands" the room, and is exceptionally poised, confident, and positive; participant does not use rely heavily on the presentation or reference materials.		
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (130 points)					

TIME DEDUCTIONS	
One (1) point per ten (10)-second interval is deducted for speaking over three (3) minutes. Time commences when the competitor begins speaking.	
TOTAL TIME FOR PRESENTATION	
TIME DEDUCTION	
TOTAL TIME DEDUCTIONS	

Record scores in the column spaces below.

SEMIFINAL SUBTOTAL (130 points)	
--	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (180 points)	
--	---------------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____



CYBERSECURITY

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Assistants for the set-up, monitoring, and clean-up of the preliminary and the semifinal rounds, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Tables and chairs for participants
- C. Tables and chairs for judges
- D. A projector (if available) with HDMI input for computers to project the presentation if desired.
- E. Presentation screen or wall to project the presentation (if available)
- F. Stopwatch or clock for the timekeeper

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the [National TSA Conference Competition Requirements](#)).
- B. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. Begin the event at the scheduled time by closing the doors and checking the entry list.
- E. All participants and judges should be in the room at this time.
- F. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- G. In order to compete, participants must be on the entry list or must have approval of the CRC.
- H. Distribute any necessary materials.
- I. Monitor the one (1)-hour test.
- J. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges and CRC manager must all initial either of these actions on the rating form.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to sign up for a presentation time.
- B. Semifinalists report at the assigned time and place for the on-site challenge with:
 - 1. One (1) laptop, or other presentation hardware (e.g. iPad)
- C. Participants do NOT have access to electrical power/outlets during the event.

- D. Participants can have access to the Internet during the event.
- E. Participants must have the presentation downloaded and accessible on their computers.
- F. Participants may project their presentation via the projector (if available).
- G. Participants are given eight (8) minutes to present, broken down as follows:
 - 1. Two (2) minutes to set up
 - 2. Three (3) minutes for the presentation
 - 3. Two (2) minutes to respond to questions
 - 4. One (1) minute to clear material
- H. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- I. Manage security and the removal of materials from the area.

OVERVIEW

Data science is a broad term for implementing a variety of methods to find connections between data to gain insightful knowledge about a particular issue. Data analytics is an even more focused way of sorting through collected data to make predictions.

Applying leadership and 21st century skills, participants conduct research on the annual topic and analyze the provided data set posted on the [TSA website](#) under *Themes & Problems*, document their research in a supporting portfolio and create a display. Semifinalist teams participate in an on-site presentation highlighting their pre-conference submission.

ELIGIBILITY

Three (3) teams of two to three (2-3) individuals per state are permitted.

TIME LIMITS

SEMIFINAL ROUND

- A. Five (5) minutes are allowed for the presentation
- B. Five (5) minutes are allowed for a question and answer session

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual topic and provided data set on the [TSA website](#) under *Themes & Problems*.
- C. The provided data set must be used.
- D. Participants research the issue using credible sources, analyze the provided data set, and draw conclusions.
- E. Participants prepare their documentation and display according to the regulations.

PRELIMINARY ROUND

- A. Participants report at the time and place stated in the conference program to:
 - 1. Set up the static entry display
 - 2. Submit the documentation portfolio
- B. Entries are reviewed by judges and based on the following criteria:
 - 1. Judges score the display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. A representative from each semifinalist team reports at the time and place stated in the conference program to schedule a time for the on-site presentation/interview.
- B. Semifinalist teams report at the assigned time and place stated in the conference program for the on-site presentation/interview.
- C. Semifinalists are given no more than five (5) minutes to present their presentation to the judges, followed by a judge-led question and answer session that will last no more than five (5) minutes.
- D. Teams pick up their entry from the display area at the time and place stated in the conference program.
- E. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants must understand the fundamental concepts and principles of the contemporary issue researched. Research about the issue shall focus on:
1. Analysis of the provided data.
 2. Representation of that data in statistical graphs.
 3. Synthesis of the provided data in terms of factors influencing the issue, societal impacts, and ethical considerations.
- B. Display:
1. The display shall include, at a minimum, the following major key points:
 - a. The team identification number and the event name in the upper-right hand corner of the display. No other identifying information shall be included.
 - b. The definition and explanation of the issue.
 - c. An explanation of the importance of the issue including problems, impact, and possible solutions or recommendations (if applicable).
 - d. Emphasis shall be placed on the graphs, which should depict not only the issue, but any causal factors.
 2. Highlight your display with creative visual elements (e.g., photos, titles) to actively engage the audience. Participants may use images "labeled for reuse," but must properly cite the source(s) on the display (refer to the General Rules).
 3. No commercial or copyrighted material may be used. If the entry contains images of people, proof of consent must be provided for each person in the presentation.
 - a. Minors require parental consent
 - b. Use the Photo/Film/Video Consent and Release form (see 2026 & 2027 Forms Appendix) for any individuals included in the presentation.
 - c. Participants must attach the consent forms within the documentation portfolio
 4. May be three-dimensional as long as it does not exceed the size limitations of 15" deep x 30" wide x 48" high
 5. Any digital display (e.g. iPad) must fit within the allotted display space. All power must be switched off once the team has completed set-up. If teams want judges to activate any electronic device in their display, complete instructions must be clearly displayed and should include information about turning on devices, logging in, etc.
 6. A/C electricity may not be used.
 7. If the display requires power, it must be powered by dry-cell batteries or photo-voltaic cells.
 8. Any power source used must fit within the display dimensions.
 9. No liquids, viruses, live plants, or animals may be used as a part of the display.
 10. No harmful or illegal substances may be displayed.
 11. Once the display set-up time frame has closed, participants may not re-enter the event area.
- C. Documentation Portfolio:
1. The documentation portfolio shall include all supporting material such as research notes, links to articles, sketches, illustrations, etc.
 2. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page that includes the event title, the conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Supporting materials; maximum ten (10) pages
 - d. A summary of the issue, one (1) page
 - e. Analysis of the provided data; pages as needed
 - f. A summary of findings and recommendation, one (1) page
 - g. A list of software used, references, and credible resources; a minimum of three (3) different types of resources must be used; pages as needed
 - h. Work Log (see 2026 & 2027 Forms Appendix); pages as needed

- i. Photo/Film/Video Consent and Release Forms. If the entry contains images of people on the display or in the documentation portfolio (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed

SEMIFINAL ROUND

- A. The on-site presentation must include all team members.
- B. A semifinalist representative reports at the time and location in the conference program to sign-up for a presentation/interview time.
- C. Semifinalists report at the assigned time and place stated in the conference program to give their presentation/interview.
- D. Participants are permitted to present for up to five (5) minutes on their analysis, conclusions, and/or recommendations.
- E. The question and answer session is limited to five (5) minutes.
- F. Participants are NOT permitted to:
 1. Use their phones during the presentation or the interview.
 2. Contact outside sources for assistance.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. The display

Tier 2

- B. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Data scientist
- Data analyst
- Actuary
- Economist
- Epidemiologist
- Forensic accountant
- Market researcher
- Meteorologist
- Operations research analyst
- Quality engineer

DATA SCIENCE & ANALYTICS

2026 & 2027 OFFICIAL RATING FORM MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Display is present
- ☐ TIER 2 – Documentation portfolio is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – DISPLAY (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Explanation of Impacts (X1)	Explanation is missing a discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	Explanation addresses some of the issues relevant to environmental, economic, social, and/or ethical considerations.	Explanation includes a full discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	
Data Charts and Graphs (X2)	The data is not represented in charts and graphs.	The data is represented in charts and graphs and somewhat supports the analysis of the team.	The data is represented in charts and graphs and supports the analysis of the team.	
Communication of Issue (X2)	It is difficult to understand the issue being communicated in the display; display organization does not aid in conveying the points.	The issue is somewhat communicated by the data presented, imagery, and organization of the display.	The issue is communicated in an organized, clear, and concise manner through effective use of imagery and the display organization.	
Creativity (X2)	The display lacks creativity; no, or very few, design principles are integrated in the display.	Some elements of creativity exist in the display, and essential design principles are generally evident.	The display exudes creativity; essential design principles and elements are well integrated.	
Aesthetics and Artisanship (X1)	Work is unorganized and sloppy; display seems to be an afterthought or thrown together.	Display shows an organized presentation of the issue; design and layout considerations are evident in some aspects.	Display is exemplary in logically communicating important data; design and aesthetics are considered and add to the effectiveness of the display.	
TIER 1 – DISPLAY SUBTOTAL (80 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
TIER 1 – SUBTOTAL (80 points)				

TIER 2 – DOCUMENTATION PORTFOLIO (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Support Materials (X2)	Support materials do not help clarify the documentation or are of little significance to the issue.	Support materials are appropriate and help supplement documentation by providing clarity to the issue.	Support materials are of excellent quality; if not original, they are cited; support materials clarify the issue.	
Research Base (X2)	Research is inadequate, and/or very few credible sources are referenced.	Research has been conducted appropriately, with some credible sources included.	Research indicates evidence of a comprehensive assortment of materials that are credible sources.	
Quality, Effectiveness, and Mechanics (X1)	Portfolio appears to have been thrown together; distracting errors in punctuation, grammar, and spelling are evident in the documentation.	Portfolio is generally organized; punctuation, grammar, and spelling are generally correct, with few errors.	Work is of exceptional quality and well organized; punctuation, grammar, and spelling are correct, with no errors.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (60 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				

PRELIMINARY SUBTOTAL (140 points)	
--	--

SEMIFINAL PRESENTATION/INTERVIEW (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Articulation (X2)	The analysis is full of illogical thoughts that lack clarity, and/or there is insufficient supporting information provided; leadership and/or 21 st century skills are not evident.	The analysis is somewhat logical, generally easy to follow, and/or there is sufficient supporting information provided; leadership and/or 21 st century skills are somewhat evident.	The analysis is clear, concise, and there is ample supporting information provided; leadership and/or 21 st century skills are clearly evident.	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their analysis; responses to questions may be vague.	Participants exhibit a general understanding of the concepts in their analysis.	Participants show clear evidence of a thorough understanding of data analysis; the analysis is accurate and supported by evidence.	
Delivery (X1)	The team is verbose and/or uncertain in their analysis; the participants' posture, gestures, and lack of eye contact diminish the delivery; participants rely too heavily on the notes.	The team is somewhat well-spoken and clear in their analysis; participants' posture, gestures, and eye contact result in an acceptable delivery; participants rely on the notes.	The team is well-spoken and distinct in its presentation/interview; participants' posture, gestures, and eye contact result in a polished, natural, and effective delivery.	
Team Participation (X1)	The majority of the presentation is made by one member of the team; the partner(s) may be disengaged.	Team members are generally engaged in the process, though one member may take on more responsibility than the other(s).	All team members are actively involved in the presentation/interview.	

SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (60 points)		
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____		
SEMIFINAL SUBTOTAL (60 points)		
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.		
TOTAL (200 points)		

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

DATA SCIENCE AND ANALYTICS

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistant, one (1)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. USB with provided data set for judge reference.
- C. Measuring tape for judges
- D. Display tables for entries
- E. Table and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough judges have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Check in the entries at the time stated in the conference program. Participants:
 - 1. Set up the static entry display
 - 2. Submit the documentation portfolio
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have CRC approval.
- D. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- E. Each entry must include the team's identification number in the upper right-hand corner of the entry.
- F. Instruct participants to position the entries for viewing.
- G. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Entries are reviewed by judges based on the following criteria:

Tier 1

- 1. Judges score the display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

- 2. Judges score the documentation portfolio criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

- B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

- 1. To deduct twenty percent (20%) of the total possible points in this round
- 2. To disqualify the entry

The event coordinator, judges and CRC manager must all initial either of these actions on the rating form.

- C. Judges determine the twelve (12) semifinalists.

SEMIFINAL ROUND

- A. A representative from each semifinalist team reports to the time and place stated in the conference program to sign-up for the on-site presentation/interview.
- B. Semifinalist teams report at the assigned time and place stated in the conference program for the on-site presentation/interview.
- C. Semifinalists are not permitted to:
 - 1. Use their phones during the presentation or during the interview
 - 2. Contact outside sources for assistance
- D. Semifinalists are given no more than five (5) minutes to present their presentation to the judges.
- E. Semifinalists are given no more than five (5) minutes to answer questions from the judges.
- F. Judges use the same official rating form for both the preliminary and semifinal round of evaluation.
- G. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- H. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants produce a digital photographic portfolio that represents or relates to a chosen theme posted on the [TSA website](#) under *Themes & Problems*. Semifinalists demonstrate competency in the production of a series of digital photographs taken at the conference site within the time limit specified and complete a presentation/interview.

ELIGIBILITY

Three (3) individuals per state may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. All components of the individual's entry (digital photographic portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.

SEMIFINAL ROUND

- A. One-half (½) hour is allowed to set up equipment and receive the on-site challenge and procedures.
- B. Two (2) hours are allowed to complete the on-site challenge.
- C. The presentation/interview will be three (3) minutes with two (2) minutes for the presentation and one (1) minute for the interview.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants review the annual theme posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants produce a digital photographic portfolio while observing the regulations.

- D. Participants submit the digital photographic portfolio PDF by 11:59 p.m. ET on a designated date in mid-May.
- E. Submission information will be provided on [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalists (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program for the on-site challenge with all of the required equipment and software noted in the Regulations and Requirements section.
- B. Participants are allowed one-half (½) hour to set up and test their equipment.
- C. At the end of the set-up time, the event coordinator presents the on-site challenge and related procedure to the semifinalists.
- D. The challenge involves students taking photographs addressing a specific theme, or documenting a current aspect of the conference, such as a competitive event, a special focus of the site, a general session, etc.
- E. Semifinalists are given two (2) hours to complete the challenge, including taking their pictures, and editing or enhancing them. Editing must take place in the event room.
- F. The semifinalists save the final photos in a multimedia presentation on their laptop.
- G. Members can power down the laptop once finished and then power back up at the presentation/interview to save battery life.
- H. The laptop with the semifinal solution must remain in the room for judging and the presentation/interview.
- I. The participant signs up for a presentation/interview time once the two (2) hours has concluded.
- J. Participants report at the assigned time and place stated in the conference program for the presentation/interview.
- K. Judges assess the entries and the interview responses.
- L. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Photographs included in the portfolio must be unified.
- B. Participants are solely responsible for all aspects of the competition, including taking the photographs, editing, and completing the portfolio.
- C. Digital Photographic Portfolio:
 1. The digital photographic portfolio must be submitted as a multi-page PDF document with pages in this order:
 - a. Cover page – must include the event title, the conference city and state, the year, and the participant's ID number; one (1) page.
 - b. Summary of the theme – must follow the annual challenge, including a detailed description of how the theme was interpreted by the participant, why the particular subjects were chosen, as well as the challenges that were faced in the selection of the subjects, in taking the photos, and in selecting and editing the final images; pages as needed.
 - c. Photos:
 - i. Submission: the original photo and the edited version should appear together on the same page and must be labeled "ORIGINAL" and "EDITED." The description must NOT be on the page with the original and edited photos.
 - ii. On the page immediately following the Original and Edited photos should be a description of the two photos. The description should include the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length) and include a synopsis of how the original photo was edited/changed, and how the theme is addressed by both the style and content of the edited photo.
 - iii. Enhancements and/or editing should be done ethically and in a way that makes a better photograph.
 - iv. The order of the photos in the portfolio should be as follows: Original photo and edited photo on the same page, description on the following page, then repeat for the remaining photos.
 - d. Resource page – must include a list of resources used to complete the portfolio, including camera, software, hardware, etc.; pages as needed
 - e. Consent and Release Forms – Recognizable individuals pictured in the images must give their written consent before the images can be used in this event (see Photo/Film/Video Consent and Release form in 2026 & 2027 Forms Appendix). All necessary consent forms must be included at the end of the multi page PDF album. Students may note in their photo description when a consent form has been deemed necessary and is included in the portfolio; pages as needed (if needed). NOTE: If consent forms are deemed necessary by the judges and have not been included by the competitor, the entry will not be judged.
- D. The entry must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.
- E. Entries received, or changes made to submitted entries after this deadline will not be judged.

SEMIFINAL ROUND

- A. Participants are required to provide their own equipment, including:
 1. One (1) laptop. Laptop must be fully charged with enough battery for the duration of the semifinal round
 2. One (1) computer mouse (if desired)
 3. Portable power bank (if desired)
 4. A method to transfer the images to the computer (such as a media reader) for editing

5. Tripod
 6. DSLR or dedicated “point-and shoot” camera, with a timer feature and/or remote trigger as well as the ability to be mounted to a tripod; cell phones are permitted as cameras
 7. Semifinalists may bring optional related accessories (i.e., filters, reflectors, extra batteries, etc.) to use in the on-site challenge.
- B. Participants do NOT have access to electrical power/ outlets during the event.
- C. No internet access will be provided during the competition.
- D. When preparing their multimedia presentation, semifinalists must include the following information for each photograph:
1. Camera make (e.g., Nikon, Canon, Panasonic, Sony, etc.)
 2. Camera model (e.g., CoolPix, Rebel, 5D, etc.)
 3. F-stop at which each photograph was taken
 4. Exposure time of each photograph
 5. ISO value, aperture (f-stop number), and shutter speed
 6. How the photo relates to the on-site challenge
 7. Any software or apps used to edit the photograph
- E. All multimedia presentations of images must be saved to the laptop and will be used by the member and the judges during the presentation/interview.

EVALUATION

PRELIMINARY ROUND

- A. The digital photographic portfolio

SEMIFINAL ROUND

- A. The multimedia presentation
B. The presentation/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Animator
- Photojournalist
- Research and development scientist
- Reporter
- Website designer

DIGITAL PHOTOGRAPHY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Digital portfolio was submitted
- ☐ Consent/Release forms are present (if required)
- ☐ ENTRY NOT EVALUATED

DIGITAL PHOTOGRAPHIC PORTFOLIO (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Principles and Elements of Design and Composition and Creativity (X2)	There is little or no evidence of an understanding or creative use of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	Images demonstrate some knowledge and creative use of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	Images clearly demonstrate excellent and creative use of a variety of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	
Technical Quality (X1)	Photos are out of focus; blurriness is unintentional and does not contribute to the overall composition; photos display incorrect exposure values, white balance, and/or range of tones; there is little or no consideration given to lighting and/or special effects.	Photos are in focus; photos display correct exposure values, white balance, and/or range of tones; images exhibit some attention to lighting and/or the use of special effects.	Photo subject(s) are in sharp focus; blurriness is used effectively to enhance the composition; photos display correct exposure values, white balance, and/or range of tones; images are enhanced by attention to lighting and/or the use of special effects.	
Conveyance of the Theme (X1)	The images are flat and lack a clear connection to the theme; viewers are not drawn into the scene; images do not clearly convey the theme/challenge.	The images show some emotional depth; viewers make an emotional connection with the images/pictures and are drawn into the scene or learn something from the subject(s); images convey the competition theme/challenge.	The images show emotional depth; viewers make an instant emotional connection with the pictures and are drawn into the scene or learn something from the subject(s); images clearly convey the competition theme/challenge.	
Written Statements (X1)	A written statement may or may not accompany each photo, and/or the statement includes only some of the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length), and/or an explanation of the relevance of the image to the challenge is not present and/or is written poorly.	A written statement accompanies each photo; the statement includes most of the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length); an explanation of the relevance of the image to the challenge is included.	A written statement accompanies each photo; the statement includes the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length); an explanation of the relevance of the image to the challenge is presented in a clear and concise manner.	

DIGITAL PHOTOGRAPHIC PORTFOLIO (70 points) – continued				
Resources (X1)	A resources list is present, but graphic and/or software packages used are not mentioned.	A resources list is present and the primary software packages used are included; contains an adequate list of resources.	Detailed and concise resources are provided; all software packages used are included.	
DIGITAL PHOTOGRAPHIC PORTFOLIO SUBTOTAL (70 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				

PRELIMINARY SUBTOTAL (70 points)

SEMIFINAL CHALLENGE (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Principles and Elements of Design/Composition and Creativity (X2)	There is little or no evidence of an understanding or creative use of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	Images demonstrate some knowledge and creative use of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	Images clearly demonstrate excellent and creative use of a variety of compositional elements (line, shape, form, value, space, texture, color, framing, emphasis, balance, unity, contrast, movement/rhythm and pattern/repetition).	
Technical Skill (X2)	Participants do not demonstrate knowledge of and understanding of how to utilize photographic equipment and techniques; participants show little or no creativity in staging and photographing subjects.	Participants demonstrate some knowledge of and an understanding of how to utilize photographic equipment and techniques, or use it only minimally in the on-site challenge; participants show some creativity in staging and photographing subjects.	Participants clearly demonstrate knowledge and understanding of how to utilize photographic equipment and techniques; participants show creativity in staging and photographing subjects; participants show a high level of skill in the use of the tools to create high quality images.	
Technical Quality (X1)	Photos are out of focus; blurriness is unintentional and does not contribute to the overall composition; photos display incorrect exposure values, white balance, and/or range of tones; there is little or no consideration given to lighting and/or special effects.	Photos are in focus; photos display correct exposure values, white balance, and/or range of tones; images exhibit some attention to lighting and/or the use of special effects.	Photo subject(s) are in sharp focus; blurriness is used effectively to enhance the composition; photos display correct exposure values, white balance, and/or range of tones; images are enhanced by attention to lighting and/or the use of special effects.	
Impact and Theme (X1)	The images are flat and lack emotional depth; viewers are not drawn into the scene; images do not clearly convey the theme/challenge.	The images show some emotional depth; viewers make an emotional connection with the images/pictures and are drawn into the scene or learn something from the subject(s); images convey the competition theme/challenge.	The images show emotional depth; viewers make an instant emotional connection with the pictures and are drawn into the scene or learn something from the subject(s); images clearly convey the competition theme/challenge.	

SEMIFINAL CHALLENGE (70 points) – continued				
Written Statements (X1)	A written statement may or may not accompany each photo, and/or the statement includes only some of the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length), and/or an explanation of the relevance of the image to the challenge is not present and/or is written poorly.	A written statement accompanies each photo; the statement includes most of the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length); an explanation of the relevance of the image to the challenge is included.	A written statement accompanies each photo; the statement includes the photo's meta data (camera make/model, f/stop, exposure time, ISO speed, focal length); an explanation of the relevance of the image to the challenge is presented in a clear and concise manner.	
SEMIFINAL CHALLENGE SUBTOTAL (70 points)				

SEMIFINAL PRESENTATION/INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization Use of visual materials (X1)	Participant seems unorganized and unprepared for the presentation; illogical explanation of the photographs and/or stylistic elements is presented.	Participant is generally prepared for the presentation; explanation of photographs and/or stylistic elements is communicated and somewhat organized.	The presentation is logical, well organized, and easy to follow; explanation of the photographs and/or stylistic elements are communicated in an organized and concise manner.	
Knowledge (X1)	Participant seems to have little understanding of the stylistic photographic elements used.	Participant exhibits an understanding of the stylistic photographic elements used in their project; the presentation is for the most part, logical and/or clear.	Participants show clear evidence of a thorough understanding of the stylistic photographic elements; the presentation is concise and logical.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Integration of Theme (X1)	The current theme is not demonstrated in the product.	The current theme is adequately demonstrated in the product.	The current theme is exceptionally well demonstrated in the product.	
Delivery (X1)	The participant is verbose and/or uncertain in his/her presentation/interview; participants' posture, gestures, and lack of eye contact diminish the presentation/interview.	The participant is somewhat well-spoken and clear in his/her presentation/interview; participants' posture, gestures, and eye contact are acceptable in the presentation/interview.	The participant is well-spoken and distinct in its presentation/interview; the participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation/interview.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (120 points)	
--	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (190 points)	
--	--------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

DIGITAL PHOTOGRAPHY

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more (online)
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Semifinal round materials
 - 1. Evaluation schedule
 - 2. Challenge and guidelines
 - 3. Interview questions

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Collect the rating forms electronically and bring them to conference on a flash drive.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size,

chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. On the first full day of the conference, post a list of the twelve (12) semifinalists in random order.

SEMIFINAL ROUND

- A. Before the on-site challenge session, check the room set-up. There should be enough seating and table space for twelve (12) semifinalists.
- B. Participants must provide their own equipment, including computers and cameras.
- C. Begin the event at the scheduled time by closing the doors and checking the entry list.
- D. Late entries are considered on a case-by-case basis and only when the lateness is caused by circumstances beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. All participants should be in the room at this time. Participants registered but not present may be disqualified.
- G. Review the on-site challenge and procedures with the participants.
- H. Manage the semifinalist session with judges, who may observe participants taking pictures or editing.

- I. As participants complete the on-site challenge, keep all laptops in the room for judging and presentation/ interviews, and manage the removal of participants' equipment.
- J. As participants depart, ensure they sign up for the presentation/interview.
- K. Judges independently score the multimedia presentation.
- L. Participants report at the time and place stated in the conference program for the presentation/interview
- M. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- N. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, Participants design and produce a race-worthy CO₂-powered dragster according to stated specifications, using only specified materials. The current year's theme/problem will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Two (2) individuals per chapter may participate.

TIME LIMITS

- A. The dragster and drawing are submitted at the time and place stated in the conference program.
- B. All raceable cars will make one (1) qualifying time run, then judges check for spec compliance, leaving the top sixteen (16) legal cars as semifinalists for the semifinal double elimination bracket.
- C. Sixteen (16) qualifying car builders participate in a five (5)-minute interview.
- D. Drawings and cars must be picked up at the specified time and place stated in the conference program.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants prepare their documentation and dragster model according to regulations.

PRELIMINARY ROUND

- A. Participants check in the following at the time and place stated in the conference program:
 - 1. The dragster
 - 2. One (1) full-size metric drawing of the completed vehicle
 - 3. One (1) letter-sized printed document listing all parts and materials

- B. Entries are reviewed by judges to determine specification adherence and safety on the track.
- C. Safe dragsters race for qualifying time on the same lane of a raceway.
- D. The top sixteen (16) qualifying cars, based on time trials, are evaluated against the required specifications for this event.
- E. Dragsters that do not meet event regulations are disqualified and lower qualifying cars are moved up until sixteen (16) dragsters meeting specifications are determined.
- F. Judges score the dragster body production quality, paint finish, assembly, the technical drawing scale and dimensioning as well as drawing completion and precision quality.

SEMIFINAL ROUND

- A. The top sixteen (16) car builders report at the time and location stated in the conference program to sign up for an interview.
- B. The top sixteen (16) car builders report to the track at the posted time for a five (5)-minute interview.
- C. The top sixteen (16) cars race in a double-elimination format to earn points for the race portion of the event.
- D. Drawing, design, materials list, and body finish points are combined with race points to determine the final standings.
- E. Following the end of the race, participants pick up their entries from the display area at the time and place stated in the conference program.
- F. Ten (10) finalists are announced during the conference award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants must check for a theme/problem on the [TSA website](#) under *Themes & Problems*.
- B. Drawings:
 1. The two (2)-view (top and side) drawing with metric dimensions is made on one piece of drawing paper no larger than 11" x 17" in size.
 2. Drawings are developed using standard engineering practices and procedures.
 3. The drawing may be produced using traditional drafting methods or CAD.
 4. A one (1) letter sized page with the Materials List must be printed on or attached to the back of the technical drawing.
 5. The title block includes only the participants identification number, which is assigned at registration time and is placed on the entry and drawing during check-in.
- C. Dragsters that do not meet all specifications/ tolerances are disqualified and not permitted to progress to the semifinal round.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place in the conference program to sign up for an interview time.
- B. Semifinalists arrive at the time and place in the conference program for the interview.
- C. The Race:
 1. The official distance between the start line and the finish line on the race track is twenty (20) meters.
 2. No repair or maintenance is allowed after the entries have been submitted.
 3. In the event that the vehicle is damaged by conference personnel, the event coordinator rules as to whether or not the vehicle may be repaired by the student entering the vehicle. This is the only reason a student is allowed to touch his/her vehicle after registration.
 4. All CO₂ cartridges for the race are provided by national TSA.

5. Cars that lose wheels, bearings, screw-eyes will not continue to race.
6. Damaged wheels may not be replaced.

EVALUATION

PRELIMINARY ROUND

- A. Sixteen (16) qualifying entries

SEMIFINAL ROUND

- A. Dragster construction, interview, and race points
- Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Aeronautical engineer
- Automotive designer
- Automotive modeler
- Industrial designer
- Industrial engineer
- Mechanical engineer
- Race car engineer

Dragster body		
	MINIMUM	MAXIMUM
1. The Dragster body must be one (1) piece of any species of wood or can be from plastics, including urethane modeling foam. It may be produced using traditional hand tools, power tool, CNC Milling and 3D printing, following applicable safety rules. a. Two (2) or more like or unlike pieces of material glued together are not considered one (1)-piece b. Any type of lamination will result in disqualification. c. No add-ons, such as body strengtheners, fenders, plastic canopy, exhausts, or air foils may be attached to or enclosed within the vehicle. Hydro dipping technique is permitted. d. Fiberglass, vinyl wrap, and shrink wrap are considered body strengtheners and cannot be used on the car body for any reason. e. Decals may be used for decoration only; they may not be used to gain an aerodynamic advantage, i.e., decals cannot cover the exterior axle holes or be used to cover open areas of the body.		
2. Body length	*(2026) – 280mm	290mm
* denotes specific school year requirement	*(2027) – 230mm	240mm
3. Body height with wheels		75mm
4. Body mass (completed car without CO ₂)	*(2026) – 40g	55g
* denotes specific school year requirement	*(2027) – 55g	65g
5. Body width at the point the axles pass through the body, front and back	35mm	42mm
6. Vehicle total width (including wheels).		90mm

Axles/axle holes/wheelbase		
	MINIMUM	MAXIMUM
1. Dragsters must have two (2) axles per car, no more.		
2. Bottom of axle hole or bearing above bottom of car body. (NOTE: This will only be measured at the lowest point of the side surfaces of the wood car body at the axle hole.)	5mm	10mm
3. Axle hole from front and rear of car	10mm	100mm
4. Wheelbase (axle distance apart at farthest points)	105mm	Not specified
5. Bearings, bushings and lubricants may be used.		
6. Glue may only be used to secure bearings to body.		

Spacer washers/clips		
	MINIMUM	MAXIMUM
1. Spacer washers		10
2. Axle clips		8
3. Silicone or any other type of glue/adhesive may not be used in place of wheel clips to hold wheels or axles in place.		

Power plant (CO ₂ cartridge hole)		
	MINIMUM	MAXIMUM
1. The power plant hole must be at the farthest point at the rear of the car and must be drilled parallel to the racing surface to assure proper puncture of the CO ₂ cartridge. a. A minimum of 5mm thickness around the entire power plant hole must be maintained on the dragster for safety. b. There should be no paint inside the CO₂ cartridge hole.		
2. Hole depth	45mm	55mm
3. Safety zone thickness	5mm	
4. Chamber diameter	19mm	20mm
5. Lowest point of chamber diameter to race surface (with wheels)	26mm	40mm

Eye screws		
	MINIMUM	MAXIMUM
1. Dragsters must have no more than two (2) eye screws per car that meet tolerances. a. Eye screws must not make contact with the racing surface. b. The track string must pass through both eye screws, which are located on the center line of the bottom of the car. c. Eye screws may be glued in place with CA glue or epoxy. d. It is the responsibility of the car designer/engineer to see that the eye screw holes are tightly closed to prevent the track string from slipping out. e. Any adjustments must be done prior to event check-in.		
2. Inside diameter	3mm	5mm
3. Distance apart (at farthest points)	150mm	Not Specified

Wheels		
	MINIMUM	MAXIMUM
1. A dragster must have exactly four (4) wheels, each of which separately must meet regulations in items 2 and 3 below. a. All four (4) wheels must touch the racing surface at the same time. b. All wheels must roll. c. Wheels must be made entirely from plastic. d. Dimensions must be consistent for the full circumference of the wheel.		
2. Wheel diameter	30mm	40mm
3. Wheel width*	2mm	18mm
* Width is determined by the continuous point of contact between the wheel and track or flat surface.		

DRAGSTER

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Car is present
☐ ENTRY NOT EVALUATED

PRELIMINARY DRAGSTER QUALIFYING RACE / POST RACE INSPECTION		YES	NO
Qualified Top 16 in Qualifying Race Speed			
Passed Top 16 Regulations Inspection			
Qualified Top 16 Legal (still safe to race) Cars for Interview and Final Race			
Please mark an "X" for each criteria. If all three are marked YES, place an "X" in the final box.			

SEMIFINAL DRAGSTER CONSTRUCTION (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Dragster Body Production Quality (X1)	Dragster exhibits poor production quality, with a crude and rough surface and little or no attention to detail.	Dragster shows evidence of proper production techniques; it is adequate but may need improvement.	Dragster displays excellent production techniques, with obvious attention to detail and quality.	
Body Paint/Finish (X1)	Surface preparation is inadequate; the body is unprimed, with poorly applied final finish.	Dragster body is painted and finished adequately.	Dragster body finish is exemplary; body is smooth, shiny, and exhibits quality.	
Vehicle Assembly (X1)	Dragster exhibits poor or sloppy assembly of parts (loose wheels, eye screws are not level, and/or they are loose, etc.).	Dragster is well assembled, and adequately meets standards.	Dragster is properly assembled, with obvious evidence of attention to detail.	
Drawing Scale and Dimensioning (X1)	The drawing is present, but is not to scale; dimensions are missing, or dimensioning is poorly done.	The drawing is acceptable and to scale; it is a close representation of the vehicle, but some dimensions may be missing.	The drawing is exemplary, exact, and includes all pertinent dimensions.	
Drawing Completion and Quality (X1)	The drawing is sloppy, missing parts, and lacking quality.	The drawing is complete, and the quality is adequate.	The drawing is complete and precise, and of exceptional quality.	

SEMIFINAL DRAGSTER CONSTRUCTION (60 points) – continued				
Materials List (X1)	Materials List is present, but has very few items listed and lacks organization; has the appearance of being thrown together at the last second.	Materials list is present and lists the majority of parts.	Materials list is complete and detailed.	
SEMIFINAL DRAGSTER CONSTRUCTION SUBTOTAL (60 points)				

SEMIFINAL INTERVIEW (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Car Builder Interview (X2)	The participant shows very limited knowledge of (and has difficulty articulating) how the car was produced or decisions made during the production; the participant exhibits a basic understanding of design elements and functionality, and the rationale is inconsistent or absent.; leadership and/or 21 st century skills are not evident.	The participant demonstrates some knowledge of the dragster production and has adequate knowledge of some processes or reasoning behind the vehicle design; leadership and/or 21 st century skills are somewhat evident.	The participant demonstrates competence and knowledge related to the design and production of the dragster and articulates the "reasoning" behind the decisions made; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL INTERVIEW SUBTOTAL (20 points)				

SEMIFINAL RACE (55 points)								
1st	2nd	3rd	4th	5th & 6th	7th & 8th	9th-12th	13th – 16th	
55 Points	50 Points	45 Points	40 Points	35 Points	30 Points	25 Points	15 Points	
SEMIFINAL RACE SUBTOTAL (55 points)								

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

SUBTOTAL (135 points)	
------------------------------	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (135 points)
---	---------------------------

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____



DRAGSTER

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Recorder for double elimination chart, one (1)
- D. Assistants, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries (three [3] per entry; one [1] each for the car, drawing, and materials list)
- B. Time trial record sheet
- C. Qualifier Interview Time Slot sheet
- D. Double elimination bracket chart
- E. CO₂ cartridges
- F. Go/No-Go gauges for all judges
- G. Metric scientific scales (triple beam balance or digital)
- H. Mono-filament fishing line (50lb) for track (4 pre-tied, 2 on track, and 2 reserve)
- I. Race track set, including a starting gate and a finish gate with digital timer and winning lane indicator
- J. Padding for the finish gate
- K. One (1) or more test cars
- L. Tables for the display of cars and for evaluation
- M. Table at the starting line for arranging and holding cars prior to the races
- N. Table at the finish gate for the placement of cars after the races and to hold eliminated cars
- O. Table for the official time keeper

- P. When using a computer controlled track, provide the proper computer for the software being used, all necessary connections, and a printer (placed on the official time keeper's table)
- Q. A method for displaying the time trial and race brackets

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. Ensure the judges have access to the online judging system.

CHECK-IN

- A. Participants report to the time and place stated in the conference program and check in:
 - 1. The dragster entry
 - 2. Full-size metric drawing of the completed vehicle
 - 3. A letter-sized, printed materials list fixed to the back of the technical drawing
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- E. Check to see that each entry drawing includes the participant's identification number in the upper right-hand corner of the paper.

- F. Position each entry (dragster and drawing) for evaluation and viewing.
- G. Secure the entries in the designated area.

PRELIMINARY ROUND

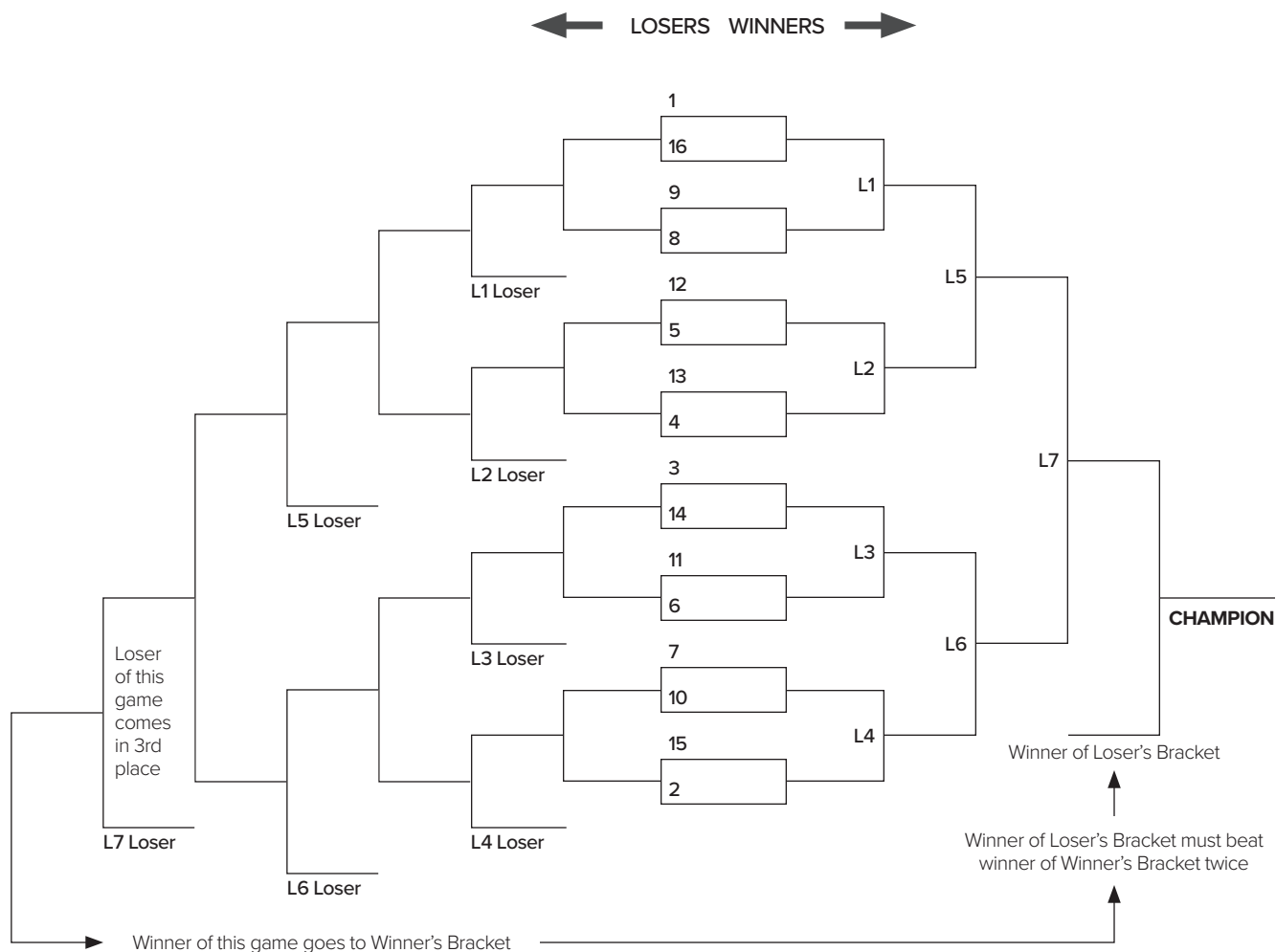
- A. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- B. Begin the time trials at the scheduled time.
 - 1. Every race-worthy car should be tested.
 - 2. Students do not have to be present.
 - 3. Public viewing is allowed.
- C. Position a judge at the starting gate to ensure that all cars are positioned in the starting gate correctly.
- D. Position another judge at the finish line.
- E. If there is a misfire or if a time is not properly recorded, a rerun may be ordered at the discretion of the event coordinator.
- F. Record preliminary times on a time trial record sheet.
- G. Assist judges with the inspection of the cars to specifications/tolerances and then the evaluation of the design, drawing, and construction categories.
- H. Judges verify that the top sixteen (16) qualifying cars meet Regulation C specifications.
- I. Entries that do not meet specifications are removed.
- J. Cars that are damaged or broken during the qualifying round are deemed non-raceable and also are removed.
- K. Only raceable cars, as determined by the judges, are allowed to compete for the semifinalist category.
- L. Lower qualifying cars are moved up until there are sixteen (16) legal semifinalists.
- M. Submit the semifinalist results to the designated location.
- N. Place each car in the double elimination race bracket (see next page for sample) according to the rank of its qualifying time.
- O. Post Semifinalist List and at a designated time and place have semifinalists sign up for interview times.

SEMIFINAL ROUND

- A. Begin the semifinals at the scheduled time.
- B. Car builders report to the track at the posted time for a five (5)-minute interview.
- C. Only the sixteen (16) qualifying cars are raced.
- D. Students do not have to be present.
- E. Public viewing is allowed.
- F. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- G. If necessary, manage the security and removal of materials from the event area.
- H. Checking for ID, allow competitors to claim their entries at a specified time at the track location.

RACE BRACKET FOR 16-CAR DOUBLE ELIMINATION

Double Elimination Tournament Chart Seeded 16 player Field



DRONE CHALLENGE (UAV)



OVERVIEW

Applying leadership and 21st century skills, participants will demonstrate their knowledge of drones by participating in manual flight missions related to a specific theme area. Teams will also study the principles of flight and research the use of drones in a specified theme area to create a documentation portfolio. Semifinalists will participate in an on-site interview.

The annual theme will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) teams of two (2) individuals per state may participate.

TIME LIMITS

- A. Ten (10) minutes prior to assigned times teams can set up their assigned pit area.
- B. Thirty (30) minutes session to test and correct any problems. During this time judges will also perform a safety check.
- C. Ten (10) minutes running clock to complete the challenge.
- D. Ten (10) minutes for the semifinalist interview.

ATTIRE

TSA competition attire, safety glasses, and safety vests are required for this event.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme on the [TSA website](#) under *Themes & Problems*.
- C. Participants complete their documentation portfolio and test their UAV drone.

PRELIMINARY ROUND

- A. While wearing safety glasses and safety vests, participants will submit their drone, submit documentation portfolios, and sign up for a setup and testing time at the time and place stated in the conference program.
- B. While wearing safety glasses and safety vests, participants attend a fifteen (15)-minute orientation meeting at the time and place stated in the conference program.
- C. Students will arrive at the assigned place and time to:
 - 1. Set up their pit areas
 - 2. Set up their drone
- D. Entries are reviewed by judges to determine safety.
- E. Safe drones will be given opportunity to test.
- F. Top sixteen (16) scores on drone testing will have their documentation portfolios evaluated.
- G. A list of twelve (12) semifinalists (in random order) is posted.

SEMINFINAL ROUND

- A. Participants report at the time and place stated in the conference program to sign up for an interview time.
- B. Participants report at the assigned time and place for the ten (10) minutes interview.
- C. Portfolio, challenge, and interviews scores are combined with race points to determine the final standings.
- D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants must check for the current year's design challenge specifications on the [TSA website](#) under *Themes & Problems*.

B. Documentation Portfolio:

1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team identification number; one (1) page
 - b. Table of Contents; one (1) page
 - c. Explanation of the problem(s); one (1) page
 - d. Communication of solution(s) - illustrating and explaining up to three (3) ways drones may be used related to the given theme; maximum three (3) pages
 - e. Resources/Bibliography; pages as needed
 - f. Work Log (see 2026 & 2027 Forms Appendix); pages as needed

UAV Drone Specifications

- A. Drones can be purchased commercially or built by the team members. Drones can be RC controlled or Wi-Fi.
- B. With all components attached, drone must not exceed the following dimensions:
 1. 9" width (at the furthest point)
 2. 9" length (at the furthest point)
 3. 5" height (as measured from the surface the drone is resting upon to the highest point of the drone with all its components attached)
- C. Drone weight (with all components attached) must not exceed 3.5oz (100g).
- D. Propeller guards/cage may be used if, once attached, the overall drone size does not exceed the dimensions listed above.
- E. A gamepad controller or smartphone/tablet may be used for manual flight.
- F. First Person View (FPV) Goggles are not allowed.
- G. If using Wi-Fi, teams must change their drone Wi-Fi name (e.g. TELLO-A9BBB3) to their official TSA Team or Chapter ID number (e.g. T9999).

Manual Flight Overview

- A. Participants must wear safety glasses and safety vests at all time during the event (including check-in, submission, and orientation meeting).
- B. Participants choose which missions to complete manually to score the most points.
- C. Various points will be assigned for each mission accomplished.
- D. Missions do not need to be navigated in any specific order.
- E. The manual flight attempt lasts one (1) minute and the timer never pauses.
- F. Participants will be given up to three (3) attempts to achieve their best score by completing the flight missions.
- G. Each member of the team must complete at least one (1) attempt as the pilot.
- H. The lowest scoring flight attempt will be dropped. The top two (2) manual flight attempts will be added together to determine the total flight score.

DURING THE COMPETITION

- A. Participants must wear safety glasses and safety vests at all times during the event.
- B. At the hot table, participants power up their drone and connect to their Drone Control System.
- C. Only one (1) team at a time competes within the airspace.
- D. The spotter places the drone on the launch pad. An attempt begins with a countdown.
- E. The pilot and spotter must remain outside the airspace and controls the drone from outside the perimeter (protective netting).
- F. Participants will have one (1) minute to complete the flight missions manually. The time never pauses.
- G. Participants will be given up to three (3) attempts to achieve their best score. Each member of the team must complete at least one (1) attempt as the pilot. The lowest scoring flight attempt will be dropped.

Drone Flight Rules

A. The manual flight attempt specifications are as follows:

1. Only one (1) team will be allowed in the competition area.
2. A hot table will be set up for teams to power up the drone and connect to the Drone Control System.
 - a. One a team that is next to fly should be at the hot table.
 - b. Teams may have spare parts (e.g., propellers, batteries) available, but must supply their own parts.
3. During an attempt, one (1) team member will serve as the pilot and one (1) team member will serve as the spotter.
4. At flight time, the drone will be placed on the launch pad inside the protective netting and then both members must be outside of the protective netting prior to the start and/or power to the propellers.
 - a. Power cannot be applied to the propellers until the attempt begins.
 - b. The attempt will begin with a countdown ("3, 2, 1, GO") by the timekeeper. At that time, the drone must take off.
 - c. The drone must be in contact with the launch pad when the word "go" is announced. Takeoff before the official start will result in a score of 0 for the attempt.
5. During the attempt, the following applies:
 - a. If the drone unintentionally stops during the attempt and no power is applied to the drone propellers, the spotter may enter the airspace, pick up the drone, and move it to the launch pad to continue the attempt.
 - b. If a part of the drone becomes detached during the attempt, it may be retrieved after the attempt is scored and there is not penalty. If the spotter retrieves the detached part during the attempt, a 20% rules violation will be applied in Tier 1.
 - c. The spotter may not touch any part of the missions. A violation of this rule will result in a score of 0 for the attempt.
6. Missions do not need to be navigated in any specific order. Points are awarded based on successfully completing individual missions.
7. Missions may be attempted multiple times, but only the highest-scoring, single attempt will be recorded.
8. Participants will not be penalized for accidental contact with mission and/or airspace elements.
9. Intentional damage to missions will result in a score of 0 for the attempt.
10. Participants will be asked to crash land or ground their drone if it poses a threat to any individuals or contacts the protective netting excessively.
11. At the end of the attempt, everything must be preserved as-is.
 - a. If the drone is moving, it must be landed, and the propellers powered down.
 - b. The drone must be left in place.
 - c. Missions completed after the end of the attempt will not be scored.
12. The judge will discuss what happened with the participants, mission by mission.
 - a. If the participants agree with the score sheet, the participants initials the score sheet and the score is final.
 - b. Any challenges must be made at this time by one of the participating team members to the coordinator or lead judge.
 - c. The coordinator or lead judge will make the final decision.
13. The team will be given three attempts. Each member of the team must serve as the pilot on at least one (1) of the attempts. The lowest scoring flight attempt will be dropped. The two (2) highest scoring flight attempts will be added together for the manual flight score.
14. Only competing participants and event officials may be in the event area.

15. Judges may inspect the drone at any time before, during, or after an attempt.
16. Any additional rules, regulations, or guidelines established by the event coordinator must be followed.

Drone Safety

- A. Power cannot be applied to the propellers unless the drone is within the airspace.
- B. Participants may not fly in an intentionally dangerous manner.
- C. Participants must not fly their drones over or near other individuals.
- D. Participants may only fly their drone when instructed to do so by a judge.
- E. Participants will be asked to emergency stop or crash their drone if its flight course poses a threat to any individuals.
- F. The Drone Control System must remain untouched while a participant is placing the drone on the launch pad.
- G. Participants will adhere to all safety rules and directions of event officials.

Battery Safety

- A. Participants may only connect a battery to the drone when the drone is on the hot table and told to do so by an official.
- B. Participants should always be present during the charging of a lithium polymer (LiPo) battery.
- C. Never charge a battery that is puffy or punctured.
- D. Stop charging immediately if a battery heats up.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. Drone testing

Tier 2

- B. Documentation portfolio

SEMINFINAL ROUND

- A. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

DRONE CHALLENGE (UAV)

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Drone, safety vests, and safety glasses
- ☐ TIER 2 – Documentation Portfolio
- ☐ ENTRY NOT EVALUATED

TIER 1 – MANUAL DRONE FLIGHT (60 points)		
Evaluation: Each team has three attempts. Drop the lowest score (do not record) and enter the two (2) highest scores.		
Highest Attempt Score (60 points)		
Second Highest Attempt Score (60 points)		
TIER 1 – MANUAL DRONE FLIGHT SUBTOTAL (120 points)		
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____		
TIER 1 – SUBTOTAL (120 points)		

TIER 2 – DOCUMENTATION PORTFOLIO (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Explanation of the Problem(s) (X2)	Problem(s) to be solved by drone is/are difficult to understand or presented in an illogical manner.	Problem(s) to be solved by the drone is/are defined and communicated adequately.	Evidence of a clear definition and explanation of problem(s) to be solved by drone; explanation is presented in an organized manner.	
Communication of Solution(s) (X2)	It is difficult to understand the solution(s) being presented; there is little or no evidence of understanding drone capabilities.	The solution(s) is/are communicated; there is some evidence of research and understanding of drone capabilities and use.	The solutions(s) is/are communicated in a clear concise manner; there is solid evidence of in-depth research and understanding of drone capabilities and use.	
Research/Bibliography (X1)	The research is inadequate, and/or very few credible sources are referenced.	The research is adequate, and it includes a few credible sources.	The research is comprehensive, and credible resources are included.	
Work Log (X1)	The Work Log is not complete.	The Work Log is included and mostly complete.	The Work Log is complete and fully documents project work.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (70 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
PRELIMINARY SUBTOTAL (190 points)				

SEMIFINAL INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	There is little or no evidence of understanding drone capabilities.	There is some evidence of research and understanding of drone capabilities.	There is solid evidence of in-depth research and understanding of drone capabilities and use.	
Application (X1)	Use of drone solution to problem(s) presented in an illogical manner.	Use of drone as solution to problem(s) communicated adequately.	Use of drone as solution to problem(s) presented clearly and concisely.	
Team Participation (X1)	Majority of the interview responses are made by one member of the team; a partner may be disengaged.	Both team members are generally engaged in the interview, though one member may take on more responsibility.	Both team members are actively involved in the interview and show understanding of the drone solution.	
SEMIFINAL INTERVIEW SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (40 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				
TOTAL (230 points)				

Comments: I certify these results to be true and accurate to the best of my knowledge. JUDGE Printed name: _____ Signature: _____

DRONE CHALLENGE (UAV)

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Pit Area Judge/Inspector (1)
- D. Assistants, two (2)

MATERIALS

- A. Coordinator's packet and box, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for entries, as needed
- B. Time trial record score sheet
- C. Qualifier interview time slot sheet
- D. Interview questions
- E. Testing Arena – two (2) 10' x 10' tents frame only. Covered with bird netting.
- F. Course materials based on theme
- G. Table with power for the Pit Area for the competitors
- H. Table for inspection and tabulation
- I. 2-Step ladder with platform for judging and setup
- J. Stations or caution tape to separate spectators and participants
- K. Safety glasses and vests for judges and coordinators
- L. Blue and red tamper proof tape for indicating frequency of controllers
- M. Fire extinguisher

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.

- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Participants report to the time and place stated in the conference program and check in:
 - 1. Portfolio
 - 2. Sign up for setup/testing time
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. Safety vests and safety glasses must be worn at all times during the event in addition to the requirements for attire.
- E. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Complete the fifteen (15)-minutes team orientation meeting at the time and place stated in the conference program.
- A. Assist judges with the check in/setup of pit area.
- B. Assist judges with the drone portion of the event and then the judging of the portfolios of the top sixteen (16) entries following the drone portion.

C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round or
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

D. Begin the time trials at the scheduled time.

1. Every Drone that is compliant with rules should have the opportunity to be tested.
2. Public viewing is allowed.
3. Announce starting time stop if there is an issue teams do not get additional testing time if they need to complete a repair of adjustment.

E. Position a judge on either side of the testing area to view the trial.

F. If a time is not properly recorded, a rerun may be ordered at the discretion of the event coordinator.

G. Record preliminary times on a time trial record sheet.

H. The testing of the drone times will determine the top sixteen (16) results.

I. Evaluate the top sixteen (16) trials portfolios.

J. Create the semifinalist sign-up sheet.

SEMIFINAL ROUND

A. Post the top twelve (12) teams with interviews times.

B. Drone builders report to the designated area posted time for a ten (10)-minute Drone Team interview.

C. Conduct interviews with the qualifying top twelve (12) Drone Teams.

D. Begin the semifinals at the scheduled time.

E. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.

F. If necessary, manage the security and removal of materials from the event area.

ELECTRICAL APPLICATIONS



OVERVIEW

Participants take a test of basic electrical and electronic theory to qualify as semifinalists. Applying leadership and 21st century skills, semifinalists assemble a specific circuit from a schematic diagram using their own kit and make required electrical measurements. Semifinalists explain their solution during an interview.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate. No individuals will be allowed to compete.

TIME LIMITS

PRELIMINARY ROUND

- A. One (1) hour is allowed for the test. Both members must be present at the same time and take the test individually.

SEMIFINAL ROUND

- A. Semifinalists are allotted one (1) hour to solve the circuit problem.
- B. Upon completion of the circuit, or at the end of the time limit, semifinalists are questioned about their solution in an interview.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants report to the event area at the time and place stated in the conference program for the test.
- B. Participants follow the specific regulations and adhere to the directions provided on-site by the event coordinator.

- C. Team members take the test individually within the time limit.
- D. The top twelve (12) teams qualify as semifinalists.
- E. The preliminary round score will be the average test score for the two (2) members.
- F. A semifinalist list (in random order) is posted.

SEMIFINAL ROUND

- A. Participants report to the event area at the time stated in the conference program.
- B. Participants build a circuit from the provided schematic diagram and make electronic measurements with their multimeter at the designated positions in the circuit, within the time limit, using their own kit.
- C. The semifinalist teams are interviewed before they leave.
- D. Ten (10) finalists are announced during the conference award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Team members take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 - 1. Student identification numbers must be entered on the scantron form in the space indicated.
 - 2. Participants must stop work immediately when time is called.

3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.
 4. All tests and scantrons must be turned in before leaving the test area.
- E. The average of the scores of the two (2) team members determines team ranking.
- F. These same two (2) team members compete in the semifinal round, should the team qualify.

SEMIFINAL ROUND

- A. All work must be completed in the event area during the time specified for the event.
- B. Semifinalists provide:
1. A toolkit with identification (school name, address, and advisor cell phone number). Each toolkit must contain all required materials to fabricate the solution. The following is a suggested list of materials:
 - a. (Minimum) 1.375" x 3.25" solderless circuit breadboard 10 x 30 pin positions
 - b. One (1) 9-volt battery with snap-on battery connector
 - c. One (1) speaker (wires pre-soldered)
 - d. Two (2) LEDs
 - e. Twelve (12) connector wires
 - f. Pushbutton switch (wires pre-soldered)
 - g. One (1) photocell
 - h. One (1) potentiometer
 - i. One (1) IN4003 diode
 - j. One (1) IC555 integrated circuit
 - k. One (1) 2N3906 transistor
 - l. One (1) 2N3904 transistor
 - m. Resistors (minimum of one [1] each, ohms): 10, 10K, 47, 100, 220, 1K, 2.2K, 3.3K, 6.8K, 16K, 33K, 120K, 330, 470K

- n. Capacitors (in microfarads): .01, .1, 10, 100, 1000
- o. S106B1 SCR
- p. Wire strippers
- q. Standard 4-function calculator (scientific calculators will not be permitted)
- r. Digital multimeter

- C. Paper to complete on-site calculations is provided by TSA.
- D. All other equipment necessary to solve the on-site problem is provided by the coordinator.
- E. Semifinalists remain with their circuit solution until the judges have completed the interview.

EVALUATION

PRELIMINARY ROUND

- A. The average of the team members' scores is factored into the final total score.

SEMIFINAL ROUND

- A. The accuracy and degree of completion of the circuit problem in the allotted time, and the exit interview, determine the final standings.

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Electrical engineer
- Electrical technician
- Electrician
- Electronic analyst
- Electronic designer
- Research assistant

ELECTRICAL APPLICATIONS

2026 & 2027 OFFICIAL RATING FORM MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Toolkit is present
☐ ENTRY NOT EVALUATED

WRITTEN TEST SCORE (50 points)

AVERAGE OF THE TWO (2) TEAM MEMBER'S WRITTEN TEST SCORE SUBTOTAL (50 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (50 points)

SEMIFINAL SOLUTION TO ON-SITE PROBLEM (90 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Solution Accuracy (X2)	Solution attempt is evident but the solution is not complete, and/or there is no final solution.	Solution is not complete, though some measurements can be taken.	Solution is accurate and complete.	
Proper Use of Components (X1)	Components are not used properly, and/or they are placed in the wrong sequence.	Components are used correctly, however, they may be placed in the improper sequence.	Components are used correctly and they are in the proper sequence and arrangement.	
Accuracy of Measurements (X1)	Measurements taken and calculated are 0-49% accurate.	Measurements taken and calculated are 50-89% accurate.	Measurements taken and calculated are 90-100% accurate.	
Articulation (X1)	The interview is full of illogical thoughts that lack clarity, and/or there is insufficient information provided describing the solution; leadership and/or 21 st century skills are not evident.	The interview is somewhat logical, easy-to-follow, and/or there is sufficient information provided that describes the solution; leadership and/or 21 st century skills are somewhat evident.	The interview is clear, concise, and there is ample information provided that describes the solution; leadership and/or 21 st century skills are somewhat evident.	

SEMIFINAL SOLUTION TO ON-SITE PROBLEM (90 points) – continued			
Delivery (X1)	Participants are verbose and/or uncertain in the interview; participant's posture, gestures, and lack of eye contact diminish the interview.	Participants are somewhat well spoken and clear in the interview; participant's posture, gestures, and eye contact are acceptable in the interview.	Participants are well-spoken and distinct in the interview; participant's posture, gestures, and eye contact result in a polished, natural, and effective interview.
Organization (X1)	The team seems unorganized and unprepared for the interview; an illogical explanation of the solution is presented.	The team is generally prepared for the interview; an explanation of the solution is communicated adequately.	The interview is logical and easy to follow; the solution is communicated in an organized and concise manner.
Knowledge (X2)	The team seems to have little understanding of the necessary concepts; answers to questions may be vague.	The team exhibits understanding of the concepts involved in the solution.	The team shows clear evidence of a thorough understanding of the concepts involved in the solution.
SEMIFINAL SOLUTION TO ON-SITE PROBLEM SUBTOTAL (90 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
SEMIFINAL SUBTOTAL (90 points)			
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (140 points)			

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ELECTRICAL APPLICATIONS

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, semifinal round, two (2) or more
- C. Proctors (preliminary round to administer the written test), two (2) or more
- D. Assistants, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Stopwatch for the assistant/timekeeper
- C. On-site problem:
 - 1. Twelve (12) copies of the on-site circuit diagram problem, one (1) for each team
 - 2. Twelve (12) copies of the calculation worksheet, one (1) for each team, if necessary
 - 3. Any additional electrical components needed for the on-site problem
- D. Adequate conditions (inside or outside) for on-site testing devices as needed for the designated circuit
- E. Tables and chairs for the participants, event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the [National TSA Conference Competition Requirements](#)).
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. All participants and judges should be in the room at this time.
- F. Manage the one (1)-hour test.
- G. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program for the semifinal challenge.
- B. Provide the on-site circuit problem, any additional electrical components, and the worksheet to the semifinalists.
- C. Supervise the one (1)-hour on-site circuit problem.
- D. Judges conduct semifinalist interviews at a location separate from the other semifinalists
- E. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- F. Manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants study the principles of flight and design in order to fabricate a glider that stays in flight for the greatest elapsed time. The glider must be designed to be launched from a catapult that is provided on-site. The design process is documented in a portfolio that is submitted for evaluation.

ELIGIBILITY

Two (2) individuals per chapter may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. Participants test their pre-built glider for four (4) flights with no additional trim time.

SEMIFINAL ROUND

- A. Forty-five (45) minutes are allowed for the construction of a glider.
- B. Ten (10) minutes are allowed for trimming (test flights) of their glider.

ATTIRE

TSA competition attire is required.

SAFETY

- A. Participants are required to provide and wear safety-approved eyewear during all phases of this event.
- B. Prescription eyewear needs to have side shields to be considered safety eyewear.
- C. Should a participant remove his/her eyewear during the event, he/she will be reminded once to replace it. If there is a second infraction, the participant will be disqualified and asked to leave the competition.
- D. TSA will not supply safety eyewear.
- E. Participants must be instructed by their advisors on the proper use of cyanoacrylate (CA) glue.

PROCEDURE

PRE-CONFERENCE.

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. The solution must generate aerodynamic lift associated with gliding. The solution must attempt to maintain altitude using the force of lift. This requirement applies to both the preliminary and semifinal rounds.
- B. A solution that only relies on the force of drag for a slow decent is not permitted.

On-site Testing of Pre-Built and Trimmed Gliders

- A. Participants report to the time and place stated in the conference program with:
 - 1. The completed glider
 - 2. The documentation portfolio
 - 3. Safety eyewear
- B. On-site Testing:
 - 1. During the testing participants must provide and wear safety eyewear.
 - 2. No trim time is allotted during the preliminary round.
 - 3. After check-in, participants will test their pre-built glider for four (4) flights.
 - 4. Launch Procedures:
 - a. Participants are called by their group timer to the designated launch area.
 - i. Each participant receives a turn to fly his/her glider.
 - ii. Participants must do all four (4) flights consecutively during their turn.

- iii. The glider is hooked to the rubber loop of the catapult provided by TSA, and the participant pulls the glider's shark tooth point back to the wooden stop in front of the 350mm stop block or less on the catapult. The altitude and angle of the catapult (with the glider on it) are determined by participants as the glider is launched.
- iv. The participant releases the glider after getting the OK from the official timer.
- 5. Flight time begins when the glider is released.
- 6. Flight time ends when the glider hits the floor or ground, or when it comes to rest on an obstruction.
- 7. One repair is allowed after the individual time trials have begun.
 - a. The repair must be made in three (3) minutes or less.
 - b. No additional trimming is allowed after the repair.
- C. Scoring: After the fourth flight, the top three (3) flight times are combined to obtain the total flight time; if a plane is unable to be tested the time is marked as a zero (0).
- D. Twenty (20) semifinalists are determined by the top twenty (20) total test flight times.
- E. Ties are broken by determining the longest single flight time.
- F. Participants pick up their entries at the time and place stated in the conference program.

SEMIFINAL ROUND

On-site Reconstruction of Glider and Flight Testing

- A. Twenty (20) semifinalist members report to the event area at the time and place stated in the conference program to demonstrate the leadership and/or 21st century skills acquired by participating in an on-site reconstruction of the glider.
- B. Participants provide their own tool box and building materials for the on-site construction portion (see regulations).
- C. Participants must provide and wear safety eyewear.
- D. Participants use their metric technical drawing to fabricate a glider.
- E. Participants have ten (10) minutes to trim their glider in the designated area.
- F. Documentation portfolios are evaluated.
- G. Participants are allowed four (4) opportunities to fly their gliders for official times. The semifinal launch procedures are the same as the preliminary round.
- H. The combined flight time of the best three (3) of the four (4) flights is used to determine the ten (10) finalists.
- I. Ties are broken by determining the longest single flight time.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Documentation Portfolio:
 - 1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with event title conference city and state, the current year, and the participant's ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Full-size metric technical drawing of the glider including dimensions on an 11" x 17" paper, which may be folded to fit into the sheet protector. The technical drawing must:
 - i. Be created using CAD, or be hand-drawn with traditional mechanical drawing instruments
 - ii. NOT be a freehand sketch
 - iii. Depict all parts that make up the glider
 - iv. Be drawn to full scale
 - v. Be drawn on a single sheet of paper that does not exceed 11" x 17"

- vi. Participants are not allowed to enter the semifinal on-site glider construction round without a completed technical drawing included
- d. Pictures of two (2) test gliders are included in the portfolio (one [1] picture of each test glider, for a total of two [2] pictures)
- e. A flight log for each pictured test glider (see Flight Log sample) must be included; pages as needed
- f. A detailed illustration demonstrating compliance with the rules and features, including design principles used in building and adjusting gliders, must be included on 11" x 17" paper (may be folded to fit in the sheet protector)
- g. A technical review of one flight log detailing launch, trim, and flights of the glider must be included; pages as needed

B. For pre-built glider and home-testing catapult regulations and specifications, refer to the "Glider Reconstruction Materials" section in the semifinal round of this guide.

PRELIMINARY ROUND

The Flight Test

- A. Participants are required to provide and wear safety eyewear for this event.
- B. Catapults for timed flights at the National TSA Conference are supplied by TSA.
- C. During time trial flights, ONLY catapults provided by TSA may be used.
- D. No trim time is allotted.

SEMIFINAL ROUND

Reconstruction of the Glider

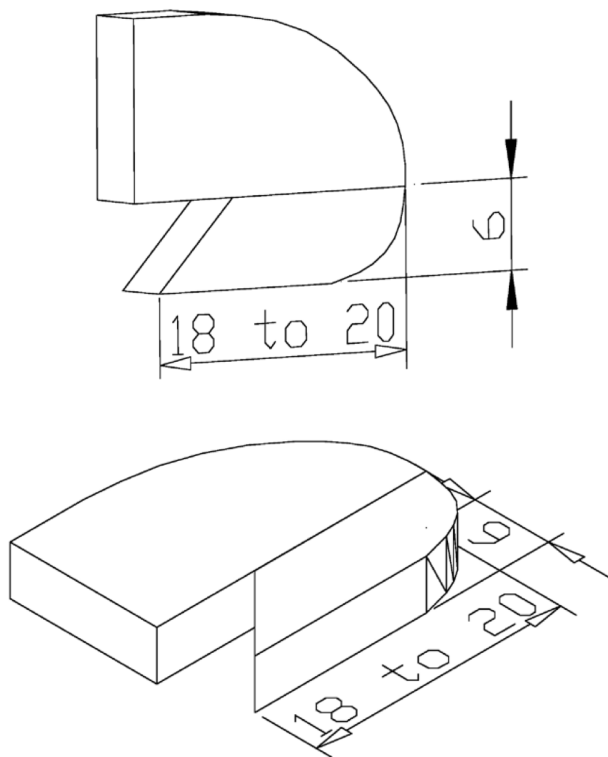
- A. Participants demonstrate their expertise by reconstructing the glider.

B. Participants are required to provide their own tool box for use in the semifinal on-site construction challenge. Participants should bring only the tools needed and leave the rest behind. Transporting and checking in will be made simpler with a smaller and lighter tool box.

1. Each tool box must:
 - a. Include identification (school name, address, and advisor cell phone number)
 - b. Not exceed twenty (20) inches (508 mm) length x ten (10) inches (254 mm) width x ten (10) inches (254 mm) height
 - c. Contain all items needed to fabricate the solution
 - d. Participants are not permitted to share tool boxes
 - e. The following is a suggested list of tools:
 - i. Cutting devices – none may be electric
 - ii. Adhesives – This event requires the use of cyanoacrylate glue (best known as Super/Krazy glue) instead of aliphatic resin glue. Participants should practice with this material before the conference.
 - iii. Aerosol and electric applicators are not allowed
 - iv. A bottle of Uncure or Debonder is recommended
 - v. A single two (2)-ounce bottle of accelerant (pump or drip) is permitted
 - vi. Temporary fastening devices
 - vii. Straight pins
 - viii. Clamps
 - ix. Tape
 - x. A cutting surface that prevents table-top marring (required)
 - xi. Rulers, straightedges, and/or measuring scales
 - xii. Abrasives sheets, sponges, boards
 - xiii. Marking devices (pens, pencils, etc.) and sharpener
 - xiv. Sheet of wax paper, as large as is needed for the competition (required)

BLANKS	Length	Width	Thickness
Fuselage	MAX 300mm or 11 $\frac{7}{8}$ "	MIN 13mm or $\frac{1}{2}$ "	MIN 3mm or $\frac{1}{8}$ "
Wing	MAX 300mm or 11 $\frac{7}{8}$ "	MAX 76mm or 3"	MIN 1.5mm or $\frac{1}{16}$ "
Stabilizer	MAX 150mm or 5 $\frac{7}{8}$ "	MAX 50mm or 2"	MIN .75mm or $\frac{1}{32}$ "
Fin	MAX 76mm or 3"	MAX 25mm or 1"	MIN .75mm or $\frac{1}{32}$ "
Shark Tooth	MAX 20mm or $\frac{3}{4}$ "	MAX 6mm or $\frac{1}{4}$ "	MIN 3mm or $\frac{1}{8}$ "

FINISHED GLIDER SIZE	Length	Width	Notes
Fuselage Measured without ballast (clay, etc)	MAX 285.75mm or 11 $\frac{1}{4}$ "		No extra length allowed for grip
Wing Span	MAX 285.75mm or 11 $\frac{1}{4}$ "		
Wing Chord		MAX 76mm or 3"	Measured parallel to fuselage at widest point
Shark Tooth			Can be part of the fuselage or added on after. Dimensions per the illustration below



2. Glider reconstruction materials (supplied by the participant)
 - a. Participants are not permitted to share.
 - b. Moldable ballast material, i.e., clay
 - c. Any materials are allowed to be used to create the glider. The material blanks must fit the regulations below.
 - i. Templates, jigs, and fixtures MAY be used in constructing gliders (these are to help facilitate fast and accurate construction), however, these templates, jigs, and fixtures must be developed and built by students, and must not be the same material as the blanks to eliminate confusion with pre-made parts.
 - ii. Blank specifications:
 1. Only one (1) piece is allowed for each glider part listed. Choose materials carefully.

2. A blank is the starting size of material before cutting or sanding.
3. The semifinal glider individual parts must be cut from the blanks on-site during the semifinal round.
3. Catapult specifications (to be used for trim and testing at home, school, and during preparation prior to time trial flights):
 - a. Catapults for timed flights at the National TSA Conference are supplied by TSA.
 - b. During time trial flights, ONLY catapults and rubber bands provided by TSA may be used.
 - c. Catapults are made from hardwood or plywood.
 - d. Participants who prefer to do so may use their own catapults during trim flights.
 - e. Catapult wooden stick dimensions:
 - i. Laminate a piece of wood (10mm thick x 45mm wide x 700mm long) to a second piece of wood (6mm thick x 45mm wide x 350mm long), aligning the pieces at the handle end and gluing them face-to-face (see drawing).
 - ii. The handle is 20mm thick x 30mm wide x 150mm long and is attached by screws to a 15mm thick x 30mm wide x 75mm long block using a middle-lap joint. The 75mm long block then is screwed to the laminated main catapult stick beginning at 400mm from the muzzle end.
 - iii. The rubber loop is a #19 rubber band $3\frac{1}{2}$ " x $\frac{1}{16}$ " threaded through the screw eye of the launcher.
 - iv. The screw eye is attached to the center of the 15mm thick x 15mm wide x 45mm long wooden block connected to the underside of the muzzle end of the catapult.
4. Storage container—All student-made items and fixtures must fit in the tool box, which is not to exceed 254mm high x 254mm wide x 508mm long.
5. Student made fixtures may include:
 - a. Traction plate with sandpaper (150mm x 300mm maximum) attached to a thin piece of rigid material, i.e., plywood, foam core board, press board, cardboard, plastic, etc.
 - b. Dihedral fixture — this is an all-wood apparatus that assists in sanding the critical dihedral joints and secures the model as the glue dries to ensure a precise prototype.

EVALUATION

PRELIMINARY ROUND

- A. The combined flight time of the best three (3) of the four (4) flights determine the top twenty (20) semifinalists. Scores will be reset to zero (0) for the semifinalist round.

SEMIFINALIST ROUND

- A. The documentation portfolio
- B. Points earned for the combined flight time of the best three (3) of the four (4) flights

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

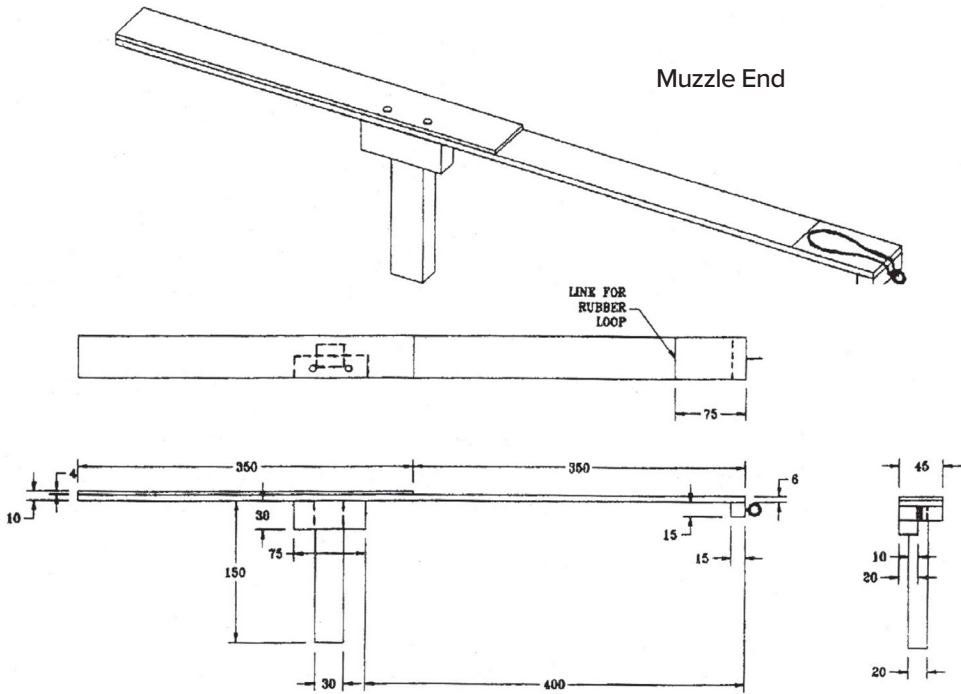
This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Aeronautical engineer
- Aircraft systems engineer
- Physics instructor

CATAPULT DRAWING



FLIGHT LOG SAMPLE

Glider #1 or Glider #2 (circle one)			Dates:	
Flight #	Time aloft	Flight pattern	Trim adjustment	Advisor sign off
#1				
#2				
#3				
#4				
#5				
#6				
#7				
#8				
#9				
#10				

FLIGHT

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Glider is present for flight times

☐ ENTRY NOT EVALUATED

PRELIMINARY ROUND – FLIGHT TIMES	
Flight times recorded to the nearest tenth (.1) of a second. Enter the top three (3) flight times	
Duration of Flight #1	Seconds
Duration of Flight #2	Seconds
Duration of Flight #3	Seconds
PRELIMINARY ROUND – FLIGHT TIMES (combine flight #1, flight #2, and flight #3) TOTAL SECONDS	

Rules violations (a deduction of 20% of the total combined flight times for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.
Indicate the rule violated: _____

PRELIMINARY ROUND SUBTOTAL (SECONDS)

SEMIFINAL ROUND – DOCUMENTATION PORTFOLIO (70 points)			
CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.
Full Scale Technical Drawing (X1)	Technical drawing is missing two (2) or more components; parts of the glider are not shown; non-metric dimensioning is used; technical drawing is not drawn to full scale and/or is on paper larger than 11" x 17", and/or it is sloppy.	Technical drawing may be missing one (1) component; the technical drawing is largely correct and neatly completed.	All components are included in the technical drawing and the drawing is correctly and neatly completed.

Record scores
in the column
spaces below.

SEMIFINAL ROUND – DOCUMENTATION PORTFOLIO (70 points) – continued					
Technical Drawing/ Built Glider Correlation (X1)	Glider built for the competition does not match the technical drawing in dimensions or appearance; glider is not designed/built properly for the event.	Glider is similar to the technical drawing within a tolerance of 5mm; glider is designed correctly to fly in the competition.	Glider is within a tolerance of 2mm of the technical drawing; glider is constructed exactly as the technical drawing illustrates.		
Test Glider Pictures (X1)	One test glider photo is missing, and/or pictures are not clearly visible, and/or they lack definition/detail of each glider.	Pictures of both test gliders are included; each picture is clearly visible, but pictures provide only adequate definition and/or detail.	Both test glider pictures include significant details and annotations about each glider; clearly visible pictures are defined.		
Detail Drawing (X1)	Detail drawing is unclear, non-compliant with the regulations and design feature specifications; adjustments are not addressed or are missing.	Detail drawing illustrates compliance with most rules and design features used in building; adequate details for adjustments are provided.	Detail drawing illustrates compliance with all rules and design features used in building and adjusting the glider.		
Flight Logs (X1)	One flight log is missing, and/or the logs are incomplete, and/or advisor signature is not included.	Both logs are included and they are generally complete.	Both logs are included and are complete, with a thorough understanding of a flight log's purpose as a flight aid.		
Technical Review of Flight Log (X1)	Review of flight logs are missing many details of launching; trimming and flying of one glider are not clear; leadership and/or 21 st century skills are not evident.	Review of flight logs provide adequate details of launching, trimming and flying of one glider; leadership and/or 21 st century skills are somewhat evident.	Review of flight logs are complete, with a thorough understanding of launching, trimming and flying of one glider; leadership and/or 21 st century skills are clearly evident.		
SEMIFINAL ROUND – DOCUMENTATION PORTFOLIO SUBTOTAL (70 points)					

SEMIFINAL FLIGHT TIMES (70 points)							
Flight times recorded to the nearest one hundredth [.01] of a second. Enter the top three (3) flight times.							
Duration of Flight #1						Seconds	
Duration of Flight #2						Seconds	
Duration of Flight #3						Seconds	
SEMIFINAL FLIGHT TIMES (combine flight #1, flight #2, and flight #3) TOTAL SECONDS							
1st	2nd	3rd	4th	5th	6th	7th	8th
70 Points	67 Points	64 Points	61 Points	58 Points	55 Points	52 Points	49 Points
9th	10th	11th & 12th	13th & 14th	15th & 16th	17th & 18th	19th – 20th	
46 Points	43 Points	36 Points	30 Points	24 Points	18 Points	12 Points	
SEMIFINAL FLIGHT SCORE SUBTOTAL (70 points)							

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (140 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (140 points)	
--	--------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

FLIGHT

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants, two (2) or more
- D. Timekeepers, two (2) or more

MATERIALS

- A. Coordinator's packet, containing
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge/assistant
 - 2. TSA Event Coordinator Report
- B. Stopwatches, two (2) or more
- C. Measuring scales
- D. First aid kit with strip bandages and debonder
- E. Catapults, five (5)
- F. #19 rubber bands
- G. Metric rulers

SAFETY

- A. Participants are required to provide and wear safety-approved eyewear during all phases of this event.
- B. Prescription eyewear needs to have side shields to be considered safety eyewear.
- C. Should a participant remove their eyewear during the event, they will be reminded once to replace it. If there is a second infraction, the participant will be disqualified and asked to leave the competition.
- D. TSA will not supply safety eyewear.
- E. Participants must be instructed by their advisors on the proper use of cyanoacrylate (CA) glue.

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Check in participants at the time stated in the conference program.
- B. Participants check in:
 - 1. The completed glider
 - 2. The documentation portfolio
 - 3. Safety eyewear
- C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Distribute the list of entrants assigned to each designated judge/timer.

- F. Timed flight procedure:
 1. Each flight time is recorded to the nearest one hundredth (.01) of a second.
 2. After the fourth flight, the top three (3) flight times are added together, then divided by three (3) to obtain the average flight score; each glider is placed with its documentation portfolio.
 3. Three (3) groups may fly simultaneously in the assigned area for the event, with consideration for the safety of gliders and participants.
 4. Each participant receives a new rubber band for each of the test flights.
- G. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 1. To deduct twenty percent (20%) of the total possible points in this round
 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- H. After the test flight, combine the scores and determine the top twenty (20) semifinalists. Evaluate the gliders to make sure they meet the rules criteria.
- I. After the gliders have been tested and evaluated, secure the holding area so that the gliders and documentation portfolios remain safe until the scheduled time for pickup.
- G. Designate times for groups to make four (4) official flights for time.
- H. Timed flight procedure:
 1. Each flight time is recorded to the nearest one hundredth (.01) of a second.
 2. After the fourth flight, the top three (3) flight times are averaged to obtain a score; each glider is placed with its documentation portfolio.
 3. Three (3) groups may fly simultaneously in the assigned area for the event, with consideration for the safety of gliders and participants.
 4. Each participant receives a new rubber band for each of their test flights.
 5. Judges calculate scores based on the results of the test flight.
 6. Gliders are evaluated.
- I. Documentation portfolios are judged.
- J. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- K. If necessary, manage security and the removal of materials from the event area.

SEMIFINAL ROUND

- A. Check in semifinalists at the time stated in the conference program.
- B. Check to verify that the tool boxes and building materials are within the specifications outlined.
- C. Announce any specific rules and regulations pertaining to the on-site construction challenge.
- D. Manage the on-site construction of gliders.
- E. After the gliders have been constructed, secure the holding area so that the gliders and documentation portfolios remain safe until the scheduled time for trimming.
- F. Designate times for test flying/trimming and communicate the thirty (30)-minute segments scheduled for each group of participants.

OVERVIEW

Participants take a test of basic forensic science theory to qualify as semifinalists. Applying leadership and 21st century skills, semifinalists demonstrate expertise of forensic science concepts through the application of techniques within a limited time frame. Each year, three (3) skills or techniques will be posted on the [TSA website](#) under *Themes & Problems*. Students will need to be familiar with and able to demonstrate all three (3) skills. During the semifinalist portion, the students will demonstrate one (1) skill to the judges. Students will be notified of the presentation topic upon completion of the test.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate. No individuals will be allowed to compete.

TIME LIMITS

PRELIMINARY ROUND

- A. One (1) hour is allowed for the test. Both members must be present at the same time and take the test individually.

SEMIFINAL ROUND

- A. Five (5) minutes is allowed to set up supplies and prepare the demonstration. Semifinalists are required to bring their own supplies.
- B. Ten (10) minutes is allowed to demonstrate the selected skill and answer questions from the judges.
 - 1. Seven (7) minutes for the demonstration
 - 2. Three (3) minutes for the interview.
- C. Three (3) minutes is allowed to clean the demonstration area.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants report for the test at the time and place stated in the conference program.
- B. A forensic science test is administered to both team members at the same time.
- C. Participants will be given a copy of the skill they are required to demonstrate upon completion of their test.
- D. Twelve (12) teams with the highest averaged scores are selected as semifinalists for the on-site problem.
- E. A list of semifinalists in random order is posted.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program to sign-up for the on-site skills challenge.
- B. Participants report at the assigned time and place for their on-site skills challenge. The specific skill will have been provided at the completion of the test.
 - 1. Participants use the allotted time to set up necessary materials and prepare the demonstration.
 - 2. Participants demonstrate the selected skill in the application of using proper forensic science techniques, and to respond to questions.
 - 3. Participants clean the demonstration area and pack necessary supplies for the next competitors.
- C. The top ten (10) finalists are announced during the conference award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Team members take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 1. Student identification numbers must be entered on the scantron form in the space indicated.
 2. Participants must stop work immediately when time is called.
 3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.
 4. All tests and scantrons must be turned in before leaving the test area.
- E. The average of the scores of the two (2) team members determines team ranking.
- F. These same two (2) team members compete in the semifinal round, should the team qualify.

SEMIFINAL ROUND

- A. No reference may be made concerning the name of the team, the team members, or their school. Team members write their team identification number on any material used to demonstrate their skill.
- B. No reference materials may be used during this event.
- C. No observers are allowed in the event or preparation rooms during the event.
- D. Participants are responsible for bringing all of the necessary supplies to the semifinalist demonstration. National TSA will not provide any supplies.

EVALUATION

PRELIMINARY ROUND

- A. The averaged team's test score

SEMIFINAL ROUND

- A. The demonstration/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Crime scene investigator
- Forensic anthropologist
- Forensic pathologist
- Forensic engineering scientist

FORENSIC TECHNOLOGY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Team is present for demonstration

☐ ENTRY NOT EVALUATED

TEST SCORES (50 points)

Record the average score of the two (2) team members in the box below. Record the team average in the column space to the right.

Average score of the two (2) team members

TEST SUBTOTAL (50 points)

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (50 points)

SKILLS DEMONSTRATION (80 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	The team is unprepared and unorganized for the presentation.	Team members have a general understanding of the forensic science skill and answer questions adequately.	There is clear evidence that team members have a thorough understanding of the forensic science skill; they answer questions well and confidently.	
Articulation (X1)	The team's demonstration is not logical or articulate; leadership and/or 21 st century skills are not evident.	The team's demonstration is logical and generally articulate; leadership and/or 21 st century skills are somewhat evident.	The team provides a concise, logical, and clear explanation of the skill they are demonstrating; leadership and/or 21 st century skills are clearly evident.	

SKILLS DEMONSTRATION (80 points) – continued			
Team Participation (X1)	Only one team member communicates with judges; there is no participation from the other team members.	Team members participate equally, but only one member seems to fully understand the concepts.	Team members seem to fully understand the concepts and share an equal role in the demonstration.
Skill Demonstration (X2)	Team members are unable to successfully demonstrate the designated skill.	Team members are able to partially demonstrate the designated skill.	Team members are successful and effective in their skill demonstration.
Technique (X2)	No indication of proper technique is used during the demonstration	Some indication of proper technique is used during demonstration.	Proper technique is used during the demonstration.
SKILLS DEMONSTRATION SUBTOTAL (80 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
SEMIFINAL SUBTOTAL (80 points)			
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (130 points)			

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

FORENSIC TECHNOLOGY

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Proctors
 - 1. Preliminary round, three (3) or more
- C. Judges
 - 1. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Stopwatches for timekeepers, one (1) per room
- C. Tables and chairs in the analysis room
- D. Copies of the selected skill, one (1) per team and one (1) per judge

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the [National TSA Conference Competition Requirements](#)).
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. All participants and judges should be in the room at this time.
- F. In order to compete, participants must be on the registration list or must have approval of the CRC.
- G. Monitor the one (1)-hour test.
- H. Upon completion of the test, students will receive a copy of the selected skill to be demonstrated during the semi-finals.
- I. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- J. Create a semifinalist sign-up sheet with fifteen (15) minute intervals.

SEMIFINAL ROUND

- A. Facilitate semifinalist sign-up times at the designated location.
- B. When each team enters the demonstration room, guide them to the demonstration area.
- C. Notify students when time begins. Allow teams five (5) minutes to set up and prepare for the demonstration.

- D. Allow ten (10) minutes for the students to demonstrate the selected skill, seven (7) minutes for the demonstration, and three (3) minutes for the interview.
- E. At the end of the ten (10)-minute period, allow students three (3) minutes to clean up the work space and then escort the students out of the competition room.
- F. Collect all materials, including any notes, prior to dismissing the participants.
- G. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- H. Manage security and removal of all materials from the competition area.

INVENTIONS AND INNOVATIONS



OVERVIEW

Applying leadership and 21st century skills, participants research and brainstorm the need for an invention or innovation of a device, system, or process that can be produced using recycled materials. Team entries must include documentation of the team's work through a static display and a model/prototype. Semifinalists give a "sales pitch" presentation to a panel of judges (who act as venture capital investors) to persuade the panel to invest in their invention/innovation.

ELIGIBILITY

PRELIMINARY ROUND

One (1) team of two (2) to four (4) individuals per chapter may participate.

TIME LIMITS

Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:

- A. One (1) minute for set-up
- B. Seven (7) minutes for the "sales pitch" presentation
- C. Two (2) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants concentrate their efforts in researching and identifying a need that has the potential for the invention/innovation of a device, system, or process.
- C. Participants design a scaled model/prototype using a majority of recycled materials, which may be a conceptual model or a working model of a device, system, or process.
- D. Participants document the design through the creation of a static display.

- E. Participants prepare a "sales pitch" presentation that further explains the invention/innovation to a panel of judges acting as venture capitalists.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in:
 - 1. The static display
 - 2. The model/prototype
- B. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:
 - 1. Judges score the static display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted. Only the information provided on the display board will be judged during the preliminary round.
 - 2. Judges score the model/prototype criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalists.
- C. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

- A. One (1) team representative reports at the time and place stated in the conference program to sign up for a "sales pitch" presentation/interview time.
- B. Two (2) to four (4) team representatives report at least ten (10) minutes prior to their assigned time to the designated place for their "sales pitch" presentation/interview.
- C. Semifinalist team representatives make a "sales pitch" presentation to the judges (venture capitalists).
- D. The "sales pitch" presentation beings on the timekeepers signal.
- E. The team answers questions from the judges.
- F. Team representatives pick-up their entry from the display area at the time and place stated in the conference program.
- G. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. The invention/innovation entry (of a device, system, or process) must be the result of an identified need.
- B. Static display:
 1. The size of the display (including the model/prototype) for the invention/innovation may not exceed 15" deep x 30" wide x 48" high.
 2. Content:
 - a. Invention name
 - b. Description of the invention/innovation
 - c. Explanation of the necessity
 - d. Description of brainstorming process and how recycled items were utilized in the final product
 - e. Drawings or illustrations of the invention/innovation. If the entry contains images of people, proof of consent but be included on the back of the display. Images of minors require parental consent (see 2026 & 2027 Forms Appendix)
 - f. Process of building the model/prototype
 3. A/C electricity may not be used.
 4. If the display and/or model/prototype requires power, they must be powered by dry-cell batteries or photo-voltaic cells.
 - a. The power supply must physically fit within the display dimensions.
 - b. All power must be switched off once the team has completed set-up.
 - c. If teams want judges to activate any electronic device in their model/prototype/display, complete instructions must be provided to judges on how to power up the model/display.
 5. The display must be presented as if it were in a product show case geared toward an audience of venture capitalists.
- C. Model/Prototype:
 1. The model/prototype must be placed with the display and not exceed the display size limits.
 2. The model/prototype may be a scaled version of the invention/innovation idea.
 3. The idea should be realistic and have the potential to be workable.
 4. The following options may be used to provide direct current electrical power for the model/prototype when it is demonstrated during the semifinalist presentation. No other electrical source may be used.
 - a. Up to 4 "C" or "D" batteries, OR
 - b. Up to 8 "AA" or "AAA" batteries
- D. Once the display set-up time frame has closed, participants may not re-enter the event area.
- E. Violation of the following will result in disqualification:
 1. No liquids, viruses, live plants, or animals may be used as part of the display.
 2. No harmful or illegal substances may be displayed.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in:
 1. The static display
 2. The model/prototype
- B. Entries are independently reviewed by judges.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. Two (2) to four (4) representatives from each team report at the time and place stated in the conference program for the "sales pitch" presentation/interview.
- B. Using leadership and/or 21st century skills, the goal of the team is to convince the judges that the invention/innovation is needed and has real-world potential.
- C. Representatives may reference their display and documentation during the presentation and interview.

D. Teams may bring and use audio/visual materials or a laptop for their presentation. AC power will not be provided or available. Projectors and screens are not permitted.

E. Time limits:

1. One (1) minute for set-up
2. Seven (7) minutes for “sales pitch” presentation
3. Two (2) minutes to respond to questions from judges
4. A five-(5) point deduction will be incurred for any time infraction.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. The static display

Tier 2

B. The model/prototype

SEMIFINAL ROUND

A. The “sales pitch” presentation and interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Commercial and industrial design
- Engineer
- Entrepreneur
- Marketing strategist
- Product designer
- Small business owner

INVENTIONS AND INNOVATIONS

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Display is present
- ☐ TIER 2 – Model/prototype is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – STATIC DISPLAY (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Display (X2)	Display is not complete; the information is irrelevant to the invention/innovation, and the display is lacking in aesthetic quality.	Display includes most components; the idea featured is adequately described; the quality is aesthetically pleasing.	Display is complete; information presented promotes the invention/innovation exceptionally well, and the quality is aesthetically pleasing; the display incorporates the required interactive components.	
Need and Description (X2)	Description of need for invention/innovation is provided, but is unclear and unconvincing; there is not enough detail; invention/innovation is not sensible, practical, or rational in nature.	Description of need for invention/innovation is generally convincing, with some detail; invention/innovation is realistic and generally meets the need as defined	Description of need for invention/innovation is fully explained, defined, and detailed precisely; invention/innovation accurately and convincingly meets the need as defined.	
Brainstorming Process (X1)	Description of the brainstorming process provides little or no details.	Description of the brainstorming process is provided, and includes some detail about the process; includes little detail about recycled materials used.	There is clear evidence that brainstorming served as a key component of the process; there is detailed descriptions and explanation of recycled materials used in product.	
Illustration/ Drawings (X1)	Only one or two illustrations/ drawings of the model/prototype are displayed, with little or no evidence that a model/prototype was well-developed or utilized.	Adequate illustrations/drawings of the model/prototype are included and are generally representative of the concept.	High quality illustrations/drawings of the model/prototype are provided and documented completely.	
TIER 1 – STATIC DISPLAY SUBTOTAL (60 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				

TIER 1 – SUBTOTAL (60 points)**TIER 2 – MODEL/PROTOTYPE (60 points)**

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Craftsmanship (X2)	The model/prototype shows minimal craftsmanship in design and creation.	The model/prototype shows adequate craftsmanship in design and creation.	The model/prototype shows exceptional craftsmanship in design and creation.	
Feasibility (X1)	The model/prototype is not realistic and does not have the potential to be workable.	The model/prototype is mostly realistic, but would need some changes to be fully workable.	The model/prototype is very realistic and is completely workable in it's current iteration.	
Recycled Materials (X1)	Little or no recycled materials are present in the model/prototype.	The implementation of recycled materials in the model/prototype is adequate.	The model/prototype shows exceptional use of recycled materials.	
Product Overall Marketability (X2)	The model/prototype does not represent a marketable product and would not be of interest to venture capitalists.	The model/prototype is adequately marketable and would be of slight interest to venture capitalists.	The model/prototype is high marketable and would be of great interest to venture capitalists.	
TIER 2 – MODEL/PROTOTYPE SUBTOTAL (60 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (120 points)**SEMIFINAL SALES PITCH PRESENTATION/INTERVIEW (60 points)**

CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participants seem unorganized and unprepared for the presentation; illogical explanation of the need and solution is presented.	Participants are generally prepared for the presentation; explanation of the need and solution are communicated and generally organized.	The presentation is logical, well organized, and easy to follow; the need and solution are communicated in a concise manner.	
Knowledge (X1)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit a basic understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the concepts in their project.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	

SEMIFINAL SALES PITCH PRESENTATION/INTERVIEW (60 points) – continued			
Creativity of Presentation (X2)	Presentation lacks imagination, originality, and detail; there is limited use of audio/visual materials.	Presentation is generally effective, innovative, and convincing; use of audio/video materials provides information about the invention/innovation.	Presentation is inspiring, inventive, resourceful, and completely convincing; use of audio/visual materials is exceptional in providing information about the invention/innovation.
Team Participation (X1)	The majority of the delivery is made by one member of the team; the partner(s) are disengaged from the presentation.	Team members are generally engaged in the process, though one member may take on more responsibility than the other(s).	Team members are actively involved in the presentation and responses to the interview questions; there is shared responsibility among the team members.
SEMIFINAL SALES PITCH PRESENTATION/INTERVIEW SUBTOTAL (60 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
TIME DEDUCTIONS			
A five-(5) point deduction will be incurred for any time infraction.			
SEMIFINAL SUBTOTAL (60 points)			
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.			TOTAL (180 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

INVENTIONS AND INNOVATIONS

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 1. Preliminary round, two (2) or more
 2. Semifinal round, two (2) or more
- C. Assistants, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 1. Event guidelines, one (1) copy for the coordinator and for each judge.
 2. TSA Event Coordinator Report
 3. Stick-on labels for identifying entries
- B. Display tables for entries
- C. Table and chairs for team presentations, as needed
- D. Stopwatch
- E. Measuring tape
- F. Table and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

CHECK-IN

- A. Check-in the entries at the time stated in the conference program.
- B. No more than two (2) team representatives may check-in the:
 1. Static display
 2. Model/prototype
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- F. Each entry must include the team's identification number in the upper right-hand corner of the display and somewhere visible on the model/ prototype.

PRELIMINARY ROUND

- A. Judges independently review the entries.
 1. The initial round of judging scores the static display entries to determine the top twenty-four (24) participants.
 2. The second round of judging scores the model/ prototype of the twenty-four (24) identified participants based on the initial round of judging to determine the twelve (12) semifinalists.
- B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 1. To deduct twenty percent (20%) of the total possible points in this round
 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- C. Create a sign-up sheet for semifinalist presentations and communicate any instructions to the semifinalist participants.

SEMIFINAL ROUND

- A. Semifinalists report to sign-up for the presentation at the time and place stated in the conference program.
- B. Check in semifinalists and confirm with the teams their order of presentation and the procedure. Team representatives should report at least ten (10) minutes prior to their assigned time to the designated place for their presentation.
- C. Manage the presentations and interviews.
- D. Discuss rule violations and have all relevant parties initial the rating form.
- E. Judges independently assess the semifinal presentation and interview.
- F. If necessary, manage security and the removal of materials from the event area.

LEADERSHIP STRATEGIES



OVERVIEW

Applying leadership and 21st century skills, teams prepare a presentation based on a selected challenge that TSA chapter officers might encounter, within a specified, limited amount of time.

ELIGIBILITY

Three (3) teams of three (3) individuals per state may participate.

TIME LIMITS

- A. Presentations must be between three and five (3-5) minutes.
- B. Teams will be penalized one (1) point per ten (10) seconds for speaking over five (5) minutes or under three (3) minutes.
- C. Time commences when the first team member begins talking and concludes at the end of the presentation.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Team members report to the event area at the time and place stated in the conference program to sign up for a presentation time.
- B. Participants report at the time and place stated in the conference program, five (5) minutes prior to the assigned presentation time with three (3) writing pens (one [1] per team member).
- C. At the team's assigned time, one (1) team member draws three (3) cards – each containing one (1) topic – from a box and selects one (1) topic from the three (3) on which the team presents. The cards with the unused topics are returned to the box.

- D. After selecting a topic, the first team enters a preparation room separate from the presentation room and is given fifteen (15) minutes to prepare the team presentation. Multiple teams may be in the preparation room together during the rotation period due to conference logistics.
- E. Teams may not use their cell phones or any other electronic technology while in the preparation or presentation rooms.
- F. The event coordinator introduces each team by team identification number only in the order of the sign-up time.
- G. As participants present, the timekeeper visually notifies the team when it has reached the three minutes minimum time by holding up a 5" x 7" notecard with "3:00" displayed.
- H. After speaking, the team returns the topic card to the judges.
- I. Judges independently score each presentation according to the criteria on the official rating form.
- J. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

- A. The semifinal round will follow the same procedures as the preliminary round and will use a new blank rubric.
- B. Participants report at the time and place stated in the conference program to sign up for a presentation time.
- C. Participants report at their assigned time and place for the same procedure used in the preliminary round, but with a different set of topics.
- D. Ten (10) finalists are announced during the conference award ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

- A. No reference may be made concerning the name of the team, the team members, or their school.
 - B. Each presentation must be the result of the team's own effort.
 - C. Each team must bring three (3) writing pens (one [1] per team member) to the preparation room.
 - D. No reference materials or devices may be brought to the preparation room.
 - E. Any notes for the presentation must be written during the fifteen (15)-minute preparation period.
 - 1. Each team is provided a maximum of three (3) 3" x 5" blank notecards.
 - 2. Although teams are permitted to use notes when speaking, it should be noted that deductions in scoring may be made for this practice if it detracts from the effectiveness of the presentation.
 - F. No observers are allowed in the event or preparation rooms during heats, though they may be present during the semifinals.
 - 1. No talking or gesturing is permitted.
 - 2. Observers are NOT allowed to enter or leave during a presentation.
 - 3. THERE IS NO APPLAUSE UNTIL THE PRESENTATION HAS CONCLUDED.
 - G. Teams are penalized one (1) point per ten (10) seconds for speaking over five (5) minutes or under three (3) minutes.
 - H. Each member of a team must participate in the presentation.
- 3. Plans to upgrade athletic facilities within the school district severely limit previously approved plans to fund technology education with needed maintenance and equipment expenditures. What will you do in an attempt to resolve this dilemma?
 - 4. A school policy that pertains to fundraising activity now requires such funds to become part of a general pool to be distributed at the discretion of the school administrator regardless of who or how they were raised. What will be your plan of action to reverse this decision?
 - 5. Conflict exists within the local chapter related to members who are not engaged within their committee assignments. If the problems are not resolved, the health of the chapter will deteriorate. How do you plan to solve this problem?
 - 6. It is assumed that our school's technology and engineering program will be strengthened through the support of local businesses and industries. Currently, there is little participation of these groups. How might we change this?

EVALUATION

- A. The presentation

Note: Scores are reset for the semifinal round and are not added to the preliminary scores.

Refer to the official rating form for more information.

NOTE

Participants are encouraged to rely on an outline format in presenting their plans, using short sentences and bullet points as appropriate.

Sample Topic Questions:

Topics that might be encountered by teams in this event may include but are not limited to:

- 1. A newly created school ruling does not allow for school related travel out of state. How will your chapter approach a solution to the problem?
- 2. A mandated curriculum severely limits opportunities for students to select and schedule elective courses such as technology education. What can be done to resolve this issue?

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the career areas featured in the TSA AND CAREERS section of this guide. Careers will vary based on the student's area of interest.

LEADERSHIP STRATEGIES

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ A team of three individuals is present

☐ ENTRY NOT EVALUATED

PRESENTATION (100 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Communication of Problem (X1)	It is difficult to understand the problem being communicated.	The problem is communicated in a somewhat organized and/or concise manner.	Communication of the problem is presented in an organized, clear, and concise manner.	
Communication of Solution (X2)	It is difficult to understand the solution being communicated.	The solution is communicated in a somewhat organized and/or concise manner.	Communication of the solution is presented in an organized, clear, and concise manner.	
Impact (X2)	The presentation is unconvincing, uninteresting, and/or lacks compelling and attention-holding ideas.	The presentation is somewhat convincing, with generally interesting ideas.	The presentation is convincing, with compelling and attention-holding ideas.	
Team Participation (X2)	The majority of the delivery is made by one member of the team; the partner(s) are disengaged from the presentation; leadership and/or 21 st century skills are not evident.	Team members generally are engaged in the process, though one member may take on more responsibility than the other(s); leadership and/or 21 st century skills are somewhat evident.	All team members are equally and actively involved in the presentation; leadership and/or 21 st century skills are clearly evident.	
Team Presence (X1)	The team's appearance is unprofessional, sloppy, and inappropriate.	The team's appearance is adequate, appropriate, and professional.	The team's appearance is appropriate, professional, and polished.	
Delivery (X2)	The team is verbose and/or uncertain in its presentation; participants' posture, gestures, and lack of eye contact diminish the presentation.	The team is somewhat well-spoken and distinct in its presentation; participants' posture, gestures, and eye contact are acceptable in the presentation.	The team is well-spoken and distinct in its presentation; participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation.	
PRESENTATION SUBTOTAL (100 points)				

TIME DEDUCTIONS		Record scores in the column spaces below.
One (1) point per ten (10)-second interval is to be deducted for speaking under the three (3) minutes or over the five (5) minutes allotted for the presentation. Presentation time commences when the first presenter begins speaking.		
PRESENTATION DELIVERY TIME		
TOTAL TIME DEDUCTION POINTS		
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____		
SUBTOTAL (100 points)		
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.		TOTAL (100 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

LEADERSHIP STRATEGIES

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more, for the first round of presentations
 - 2. Semifinal round, two (2) or more for semifinalist presentations
- C. Timekeepers for recording presentation start/stop times, one (1) per event room
- D. Monitors, one (1) per event room

MATERIALS

- A. Coordinator's packet containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Speaker's stand/podium
- C. Stopwatches for timekeepers, one (1) per heat and two (2) per preparation room
- D. One (1) 5" x 7" notecard with "3:00 minutes" displayed (see Procedure F)
- E. 3" x 5" blank notecards, for participants to outline their presentation
- F. 3" x 5" topic cards – a minimum of five (5) different topics from which to select
- G. Tables and chairs in the preparation room
- H. Chairs for the audience, for semifinals only
- I. Table and chairs for the event coordinator, judges, and timekeeper

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.

- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Manage the smooth flow of participants according to these procedures:
 - 1. After selecting a topic from the three (3) randomly drawn topics, the first team enters a preparation room separate from the presentation delivery room and is given fifteen (15) minutes to prepare a presentation.
 - 2. Seven (7) minutes after the first team enters the preparation room, the second team enters the preparation room, goes to a different section of the room, and begins its presentation preparation, again with fifteen (15) minutes allowed for preparation; and so on for all the teams.
 - 3. Each team in turn is allowed to enter the presentation room at seven (7)-minute intervals, thus enabling a constant flow of participants to speak before the evaluators in a timely fashion. (This allows for one [1] minute to enter the presentation room and announce the entry number, up to five [5] minutes for the presentation, and one [1] minute to exit the room.)
 - 4. The event coordinator introduces each team by registration number only in the order of the sign-up time.
- B. After speaking, the team returns the topic card to the judges so that it can be returned to the topics box.
- C. Judges independently rate each presentation according to the criteria on the official rating form.

- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. The timekeepers notify judges of any time under three (3) minutes or over five (5) minutes, for which deductions should be made.
- F. When teams have finished, judges determine the semifinalists, consulting the timekeeper's record.
- G. Create a semifinal sign-up sheet for presentations.

SEMIFINAL ROUND

- A. Repeat the preliminary round processes to determine the finalists.
- B. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
- C. If necessary, manage security and the removal of materials from the area.

MASS PRODUCTION



OVERVIEW

Applying leadership and 21st century skills, participants manufacture a marketable product addressing the annual theme, noted on the [TSA website](#) under *Themes & Problems*. Teams create three (3) identical copies of a product to demonstrate their skill and knowledge pertaining to the mass production process, however, the official entry is comprised of a documentation portfolio and only one (1) prototype. Semifinalists present their product and participate in an interview.

ELIGIBILITY

One (1) team of at least three individuals per chapter may participate.

TIME LIMITS

Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:

- A. One (1) minute for set-up
- B. Seven (7) minutes for the presentation
- C. Two (2) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme on the [TSA website](#) under *Themes & Problems*.
- C. Participants research designs for products and develop a full-size prototype that can be replicated for three (3) identical copies of the product. Only one (1) prototype will be submitted on-site as a part of the entry.
- D. Create working drawings and devise production plan flow chart(s).
- E. Tool up for production.

- F. Conduct a trial run and evaluate the process for effectiveness and efficiency.
- G. Manufacture several products using line production techniques.
- H. Document the process with photographic images of the additional copies of the prototype as part of the portfolio submission.
- I. Document the team project with a photo timeline.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in:
 - 1. One (1) full-size prototype
 - 2. The documentation portfolio
- B. Entries are reviewed by judges with neither students nor advisors present.
 - 1. Judges score the Prototype criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the Documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. One (1) team representative reports at the time and place stated in the conference program to sign-up for a presentation time.
- B. Up to three (3) team representatives report at least ten (10) minutes prior to their assigned time to the designated place for their presentation.
- C. Semifinalist team representatives participate in the on-site presentation and interview.
- D. Team representatives pick-up their entry from the display area at the time and place stated in the conference program.
- E. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

A. Documentation Portfolio:

1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order.
 - a. Title page with the event title, the product name, the conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Description of the product: a written description of the product, instructions for its use, the overall advantages and usefulness of the product, its audience, and related safety considerations; one (1) page
 - d. Design efforts: sketches, pictures, magazine clippings, and other graphic design elements used in the development of the final design; maximum three (3) pages
 - e. Drawings:
 - i. An orthographic drawing in three (3) views with dimensions to aid production; one (1) page
 - ii. An assembly drawing or a pictorial drawing with labels; one (1) page
 - f. Materials list: a list of materials (including sizes and market value) used to fabricate the product; each item or sub-assembly should be identified as student produced, standard stock item, or purchased subassembly; one (1) page
 - g. Tools and machines list: a list of any hand, power, and stationary tools and/or machines used to fabricate the product; one (1) page
 - h. Production plan: a production outline or flow chart; pages as needed.
 - i. Photo/Film/Video Consent and Release Forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed
 - j. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
 - k. Photographic Verification – photographic or digital images that verify the mass production of at least three (3) identical copies of the product including the prototype and photographic or digital images that document the team project with a photo timeline; maximum five (5) pages.
2. All documentation must be contained in the portfolio.
3. Tabs or dividers may be used between the sections of the portfolio and are not counted as pages.
4. The documentation portfolio will be collected by the coordinator during event submission and placed with the prototype following all submissions.

B. Prototype of Product:

1. The full-size prototype must be displayed in such a way that it does not exceed 15" deep x 24" wide x 24" high. The documentation portfolio will be collected by the coordinator and placed with the prototype following all submissions.
2. Craftsmanship:
 - a. The product must display good craftsmanship.
 - b. The product should show creative thinking by the students.
3. Appropriate materials: The product must use the materials in a manner that adds value to the product.
4. Efficiency of design: The product must address the identified consumer need, and use the materials effectively.
5. Aesthetics: The product must be pleasing to view.

6. Ergonomics: The product must be easy to use.
7. Appropriate solution: The product must function in a manner that solves the identified problem.
8. Creativity: The product must display an original solution to the identified consumer need.
9. ONLY the documentation contained within the portfolio and the prototype may be submitted for judging.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in:
 1. One (1) prototype
 2. The documentation portfolio
- B. Entries are independently reviewed by judges with neither students nor advisors present.
- C. Once the display set-up time frame has closed, participants may not re-enter the event area.

SEMIFINAL ROUND

- A. Semifinalist team representative report at the time and place stated in the conference program to sign up for a presentation/interview.
- B. Semifinalist teams report at the time and location stated in the conference program for the presentation/interview.
- C. Using leadership and/or 21st century skills, the goal of the team is to demonstrate their skill and knowledge pertaining to the mass production process.
- D. The presentation is limited to three (3) team representatives. Each member should be an active participant in the presentation/interview.
- E. Representatives may reference their documentation portfolio and prototype during the presentation and interview.
- F. Teams may bring and use audio/visual materials or a laptop for their presentation. AC power will not be provided or available. Projectors and screens are not permitted.

G. Time limits:

1. One (1) minute for set-up
2. Seven (7) minutes for presentation
3. Two (2) minutes to respond to questions from judges
4. A five-(5) point deduction will be incurred for any time infraction.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. The prototype

Tier 2

- B. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Electromechanical engineer
- Mechanical drafter
- Production planner
- Standards engineer

MASS PRODUCTION

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – One (1) copy of the prototype is present and does not exceed 15" x 24" x 24"
- ☐ TIER 2 – Documentation portfolio is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – PROTOTYPE (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Product (X2)	Product is not built to detailed standards; it is poorly constructed and finished; it lacks creativity and imagination.	Product is somewhat built to detailed standards; the design is satisfactory and works; creativity or uniqueness are somewhat apparent.	Product is built to detailed standards, and is of a quality that could be purchased by a consumer; the design is unique and demonstrates creativity.	
Product Functionality (X1)	Little specific functionality per the original specification is demonstrated.	The product meets some of the functionality per the original specification.	The end product exhibits functionality as per the original specifications.	
Tolerance of Example (X1)	Materials are not joined cleanly and are not consistent with working drawings.	Materials are somewhat joined cleanly and are fairly consistent with working drawings.	Materials are joined cleanly and are consistent with working drawings.	
TIER 1 – PROTOTYPE SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
TIER 1 – SUBTOTAL (40 points)				

TIER 2 – DOCUMENTATION PORTFOLIO (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is missing several components and/or is unorganized; it is messy and lacking quality.	Most components are included; portfolio is generally organized and displays some quality.	All components are included in order; effort and quality of work is evident.	
Description of Product Instructions for use (X1)	Description of the product and instruction for its use are unclear.	Description of the product and instructions for its use are somewhat defined and explained.	Description of the product and instructions for its use are defined and explained precisely.	
Design Efforts (X1)	Most design effort components are missing, and/or they are unorganized; they are messy and lack quality.	Some design effort components are included; components are generally organized and display overall quality.	Design effort components are included; effort and high quality of work is evident.	
Working Drawings (X1)	Working drawings are sloppy and disorganized; they do not demonstrate labeling and dimensioning.	Working drawings are of sufficient quality; most are labeled and dimensioned.	Working drawings are of excellent quality and are correctly labeled and dimensioned.	
Materials List Tools and Machines List (X1)	Lists are missing several components, and/or they are unorganized.	Most components are included; the lists are generally organized and complete.	All components are included and organized in the lists.	
Production Plan (X1)	Plan is missing several components, and/or it is unorganized.	Components of the plan are mostly included, and the plan is generally organized.	All components are included and well-organized in the plan.	
Photographic Verification (X1)	Photographic verification is not complete, and/or images are missing.	Photographic verification is complete, and the quality of images is adequate.	Photographic verification is clear and supports all aspects of the process.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (70 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
PRELIMINARY SUBTOTAL (110 points)				

SEMIFINAL PRESENTATION/INTERVIEW (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization Use of audio/visual materials (X1)	Participants seem unorganized and unprepared for the presentation; illogical explanation of the product is presented; team exhibits difficulty with the use of audio/visual materials.	Participants are generally prepared for the presentation; explanation of the product is communicated and generally organized; audio/visual materials are used somewhat effectively.	The presentation is logical, well-organized, and easy to follow; explanation of the product is communicated in an organized and concise manner; the use of audio/visual materials is effective, organized, and logical.	

SEMIFINAL PRESENTATION/INTERVIEW (70 points) – continued				
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; the presentation does not clearly define the product; answers to questions may be vague.	Participants exhibit an understanding of the concepts in their project; the presentation is, for the most part, logical and clear.	Participants show clear evidence of a thorough understanding of the project; the presentation is concise and logical.	
Articulation (X1)	Communication of the project is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the project is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the project is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Integration of Theme (X2)	The current theme is not demonstrated in the product, and/or the product is not realistic and functional.	The current theme is adequately demonstrated in the product, and the product is fairly realistic and functional.	The current theme is exceptionally well demonstrated in the product, and the product is extremely realistic and functional.	
Delivery (X1)	The team is verbose and/or uncertain in its presentation/interview; participants' posture, gestures, and lack of eye contact diminish the presentation/interview.	The team is somewhat well-spoken and clear in its presentation/interview; participants' posture, gestures, and eye contact are acceptable in the presentation/interview.	The team is well-spoken and distinct in its presentation/interview; participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation/interview.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (70 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIME DEDUCTIONS	
A five-(5) point deduction will be incurred for any time infraction.	

SEMIFINAL SUBTOTAL (70 points)	
---------------------------------------	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (180 points)
---	---------------------------

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

MASS PRODUCTION

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Display tables for entries
- C. Table and chairs for team presentations, as needed
- D. Stopwatch
- E. Measuring tape
- F. Table and chairs for event coordinators and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

CHECK-IN

- A. Check in the entries at the time stated in the conference program.
- B. No more than two (2) team representatives may check in:
 - 1. One (1) prototype
 - 2. The documentation portfolio
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- F. Each entry must include the team's identification number in the upper right-hand corner of the documentation portfolio and somewhere visible on the model/prototype.

PRELIMINARY ROUND

- A. Judges independently review the entries based on the following criteria:

Tier 1

Judges score the prototype criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

Judges score the documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

- B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- C. Create a sign-up sheet for semifinalist presentations.

SEMIFINAL ROUND

- A. Prepare a list of five to ten (5-10) standard interview questions to be asked of all participants.
- B. Semifinalists report to sign-up for the presentation at the time and place stated in the conference program.
- C. Check in semifinalists and confirm with the teams their order of presentation and the procedure. Team representatives should report at least ten (10) minutes prior to their assigned time to the designated place for their presentation.
- D. Manage the presentations and interviews.
- E. If judges ask questions, they should be the same questions for each team for the purpose of comparing one team to another.
- F. Discuss rule violations and have all relevant parties initial the rating form.
- G. Judges independently assess the semifinal presentation and interview.
- H. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skill, participants design and build a mechanical device to solve the problem statement for the identified theme. Through device functionality and documentation, team members demonstrate their knowledge of mechanical engineering and the application of their solution. The challenge is to design and make a working catapult, within the specified guidelines. The current year's problem statement will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

TIME LIMITS

- A. Catapults must be started and completed during the current school year.
- B. The catapult and portfolio must be checked in at the time and place stated in the conference program. At check-in, one team member will sign up for a time slot to return and evaluate the solution.

PRELIMINARY ROUND

- A. Three (3) minutes of practice time.
- B. Two (2) minutes to score shooting eight (8) cornhole bags.

SEMIFINAL ROUND

- A. One (1) minute of warm-up before each round.
- B. One (1) minute shootout of four (4) cornhole bags.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Teams must identify and understand the use of subsystems with a larger system

- C. Teams must research and identify an engineering design progress that influenced the design of the catapult.
- D. Participants review and implement the annual problem statement posted on the [TSA website](#) under *Themes & Problems*.

PRELIMINARY ROUND

- A. Participants check in their entries at the time and place stated in the conference program:
 - 1. One (1) Catapult
 - 2. Documentation Portfolio
- B. Catapults are inspected by evaluators to determine among other things, safety. Catapults that meet all compliance regulations will be approved for the performance stage of the event. Any unsafe devices will be disqualified. (Unsafe catapults include those with parts that detach during operation or those with a dangerous rotation or throwing motion, either of which could cause harm or damage.) Judges will make a final determination about the operational safety of a catapult.
- C. If an entry's catapult is approved for the performance stage of the event and the team qualifies for the semifinal round, the entry's design portfolio will be evaluated.
- D. A time sheet will be provided for sign-up at check-in.
- E. Both students must be present for the performance stage of the event.
- F. Teams will receive eight (8) cornhole bags to score points in a two (2)-minute scoring stage. Cornhole bags are square bags that measure 6" x 6" (15.24 cm x 15.24 cm) with a thickness between 1.125" - 1.5" (28.5-38.1 mm). The weight of cornhole bags is regulated at between 14-16.5 oz (440-470 g).
- G. Students must bring and wear safety glasses for this stage of the event.
- H. One (1) team member will use a tape measure for measuring and calibrating the distance from the front of the catapult to the front of the cornhole board. The tape measure will be provided by the team.
- I. The team will be given three (3) minutes of practice to adjust catapult for accuracy to the required distance.

- J. Teams will position their catapult on the “firing range” and wait for the command to “FIRE”.
- K. When teams receive their bags and the fire command is given, they will have two (2) minutes to launch up to eight (8) bags to accumulate as many points as possible on/in the cornhole board. Each team must cease firing at two (2) minutes. No shots taken after time has been called will count. The two (2)-minute timer begins with the “FIRE” command.
- L. Teams are allowed to repair a catapult malfunction, but the timer does not pause.
- M. Each team must cease firing at two (2) minutes. No shots taken after time has been called will count. A shot fired before the two (2) minutes and lands after the two (2) minutes will still count. The amount of time left is recorded if teams finish before the two (2) minute times.
- N. Scoring is as follows: in the hole, five (5) points; on the board, three (3) points; off the board in the marked area, one (1) point. A bag on the line will count as “in the marked area.” (A vertical plane extends upward from the outside of the line and counts a one [1] point).
- O. Scores are not tabulated until the two (2)-minute timer has expired. Cornhole bags are scored where they rest at the end of two (2) minutes or the team declares they are finished with their turn.
- P. Ties will be broken as follows.
 1. The team with the highest score and the fewest number of bags scored.
 2. The team with the greatest time remaining.
- Q. Team members must collect all eight (8) cornhole bags once judges complete recording points and before leaving the event area and returning them to the firing range.
- R. Preliminary ranking will be determined from points earned.
- S. Lack of catapult compliance may result in disqualification.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 1. Title Page with the event title, the conference city and state, the year, and the team/chapter ID number; one (1) page
 2. Table of contents; pages as necessary
 3. Identification and explanation of the engineering design process used to produce the catapult; maximum six (6) pages
 4. Bill of materials; one (1) page
 5. Design sketches and detailed drawings; maximum two (2) pages
 6. References and resources; one (1) page
 7. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
- B. Participants must bring and wear safety goggles during the performance stage of the event.
- C. Teams must provide their own tape measure (at least 25' length).
- D. The catapult may be no larger than 24" tall x 24" long x 18" wide while in the loaded position. The launch arm may extend beyond the size parameters of the catapult only while launching.
- E. The base of the catapult should accommodate the provided ballast. The ballast will be two (2) 25lbs Olympic plate weights, provided by TSA on site.
- F. The catapult must operate completely within the given area; the launch arm may extend beyond the front of the catapult only while launching.
- G. The catapult may have any type of spring mechanism to power the arm, but all parts must be contained within the 24" tall x 24" long x 18" wide maximum footprint prior to launch.

- H. The catapult's total weight must not exceed fifteen (15) pounds.
- I. All parts of the catapult must initiate behind the launch line, but parts may extend over the line during and after the last launch.
- J. The catapult cannot have wheels.
- K. The catapult must be made entirely from PVC pipe, with the exception of the launch mechanism, firing mechanism, fasteners, and safety items. These items may be wood or metal and must be constructed in a safe way, so as not to damage the device, the testing area, or cause harm to others.
- L. The following may not be used:
 - 1. Glass
 - 2. Flammable, corrosive, or explosive materials
 - 3. Compounds that produce odors or gases
- M. The catapult must have at least a five (5)-foot pull cord to launch from a safe distance.
- N. When the catapult is on display or not in the performance stage, it must be fully disabled and unable to be readied for firing.

SEMIFINAL ROUND

- A. The portfolios of the twelve (12) semifinalist teams are independently reviewed by judges, with neither students or advisors present.
- B. The top twelve (12) catapults report to the firing range at the posted time.
- C. The top twelve (12) compete in a head-to-head single elimination bracket to earn point for the finals.
- D. Teams will face off head-to-head in a one (1)-minute shootout of four (4) bags on their designated cornhole board. Teams will receive a one (1)-minute warm-up before each round.
- E. The team with the highest score or tie breaker will advance to the next round of the bracket.
- F. Semifinal ties are broken as follows:
 - 1. The team with the highest score and fewest number of bags scored.
 - 2. The team with the shortest time recorded to score the most points.

- 3. Sudden death shootout (one [1] launch each with the higher score for the launch determining the team to advance – repeating as necessary).

- G. Following the completion of the bracket, participants pick up their entries from the display area at the time and place stated in the conference program.
- H. Preliminary and portfolio scores are combined with bracket points to determine the final standings.
- I. Ten (10) finalists are announced during the conference award ceremony.

EVALUATION

PRELIMINARY ROUND

- A. Preliminary scores

SEMIFINAL ROUND

- A. Portfolio
- B. Score from head-to-head bracket

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Engineer
- Multimedia designer
- Product designer
- Small business owner

MECHANICAL ENGINEERING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Entry passes compliance checklist

☐ ENTRY NOT EVALUATED

PRELIMINARY ROUND – SCORING (40 points)

1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	6th: 35 Points	7th: 34 Points
8th: 33 Points	9th: 32 Points	10th: 31 Points	11th: 30 Points	12th: 29 Points	13th or greater: 28 Points	
PRELIMINARY ROUND – SCORING SUBTOTAL (40 points)						

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (40 points)

SEMIFINAL – DOCUMENTATION PORTFOLIO (60 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
Portfolio (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.
Engineering Design Process (X1)	The description of the engineering design process is unclear, unconvincing, and lacks project details.	The description of the engineering design process includes some project details and explanation of steps in the design process.	The description of the engineering design process includes a full explanation of each step in the process; the description is defined, and detailed
Bill of Materials (X1)	Bill of materials is included, but more than one (1) material is missing.	Bill of materials is included with one (1) material missing; List is generally organized.	Bill of materials is included; it includes all materials, components, quantities, and is well organized.

Record scores
in the column
spaces below.

SEMIFINAL – DOCUMENTATION PORTFOLIO (60 points) – continued					
Design Sketches and Drawings (X1)	Only one (1) drawing/sketch of the model is included with little to no labeling.	Two (2) or more drawings/sketches of the model are included and are labeled.	Detailed drawings/sketches of the entire device/model are provided and appropriately labeled		
Work Log (X1)	The Work Log lacks major elements of the project.	The Work Log is somewhat complete, and generally reflects the time and work necessary for the project	The Work Log completely and accurately reflects the time and work necessary for the projects.		
References and Resources (X1)	There is little to no effort to provide references and resources.	References and resources are included and are somewhat credible.	There is clear evidence of the appropriate use of applicable references and resources.		
SEMIFINAL – DOCUMENTATION PORTFOLIO SUBTOTAL (60 points)					

SEMIFINAL ROUND – HEAD-TO-HEAD SCORE (20 points)						
1st: 20 Points	2nd: 18 Points	3rd: 16 Points	4th: 14 Points	5th-8th: 12 Points	9th-12th: 10 Points	
SEMIFINAL ROUND – HEAD-TO-HEAD SCORE SUBTOTAL (20 points)						

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
---	--

SEMIFINAL SUBTOTAL (80 points)	
---------------------------------------	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (120 points)	
---	---------------------------	--

Comments: I certify these results to be true and accurate to the best of my knowledge. JUDGE Printed name: _____ Signature: _____

CATAPULT SCORE SHEET

A shot attempt points may be modified if it is affected by a later shot

Participant/Team ID# _____

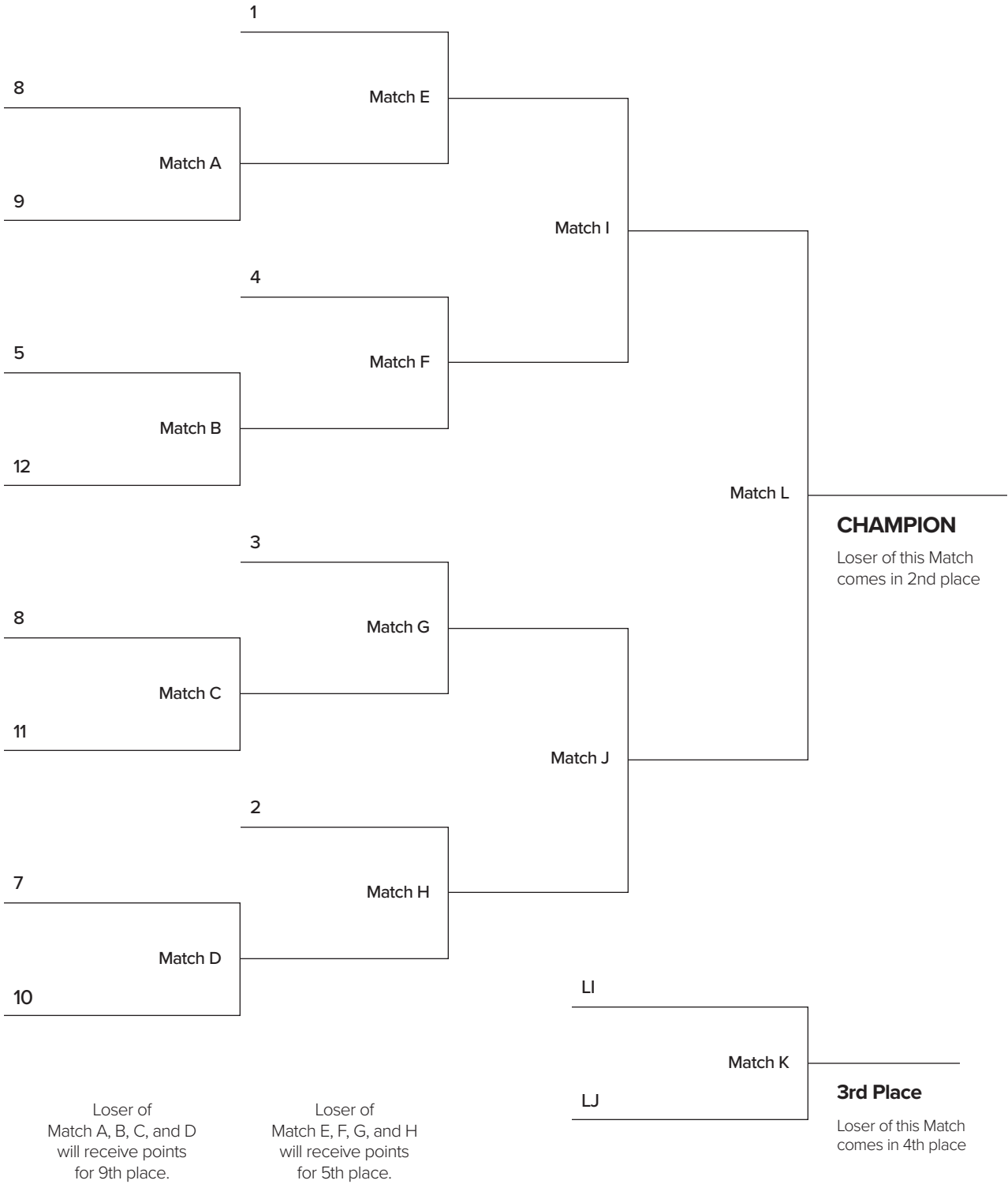
CATAPULT COMPLIANCE CHECKLIST		YES	NO
Does the team have safety goggles?			
Can the catapult be weighed down with two (2) Olympic weight plates (two [2] 25lbs)?			
Is the catapult within the size specifications?			
Is the catapult built with the correct materials?			
Does the catapult launch with a pull cord with at least five (5) feet of safety distance?			
Does the catapult have a safe launching mechanism?			
Is the catapult safe to operate?			
Please mark an "X" for each criteria. A "No" to any of the requirements will not advance.			

Shot Attempt #	Missed Shot 0 Points	Floor Zone 1 Point	On the Board 3 Points	In the Hole 5 Points	Modified Points
#1					
#2					
#3					
#4					
#5					
#6					
#7					
#8					

Total Points Scored:	
Time Remaining: (MM:SS)	

HEAD-TO-HEAD BRACKET

Tournament Bracket Seeded 12 Team Field



MECHANICAL ENGINEERING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, portfolio judging, two (2) or more
- C. Assistants
 - 1. One (1) per evaluation area
 - 2. One (1) to manage the evaluation area.
 - 3. One (1) to collect the score sheets from the judges and sort from highest to lowest totals.

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries (two [2] per team)
- B. Two (2) cornhole boards
- C. Four (4) sets (four [4] bags make a set) of different colored cornhole bags
- D. Four (4) 25lb. Olympic weight plates
- E. Tape (masking) to mark the firing range and 1' floor scoring area
- F. Two (2) tape measures for evaluators
- G. Bathroom weight scale
- H. Tables for displaying and evaluating catapults
- I. Tables and chairs for judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.

- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Tape down the necessary scoring lines needed for scoring and the launching distance for the two evaluation areas. Prepare the area to allow the head-to-head semifinal round to take place.
- E. At least one (1) hour before the event is to begin meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system

CHECK-IN

- A. Check in the entries at the time stated in the conference program.
- B. Check the entry for safety only.
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- F. Secure the entries in the designated area.
- G. Each entry must include the participant's identification number in the upper right-hand corner of the entry.

PRELIMINARY ROUND

- A. Meet with judges to review time limits, procedures, and regulations. If questions arise that cannot be answered, speak to the event manager before the event begins.
- B. Conduct the evaluations of each entry. The judges will enter the scores.

- C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- D. Determine the top twelve (12) and post the semifinalist list.

SEMIFINAL ROUND

- A. Judges independently assess the portfolio entries of the twelve (12) semifinalists and enter scores in the respective rubrics
- B. Report at the time and location provided in the conference program for the head-to-head challenge.
- C. Manage the head-to-head bracket to determine the bracket points.
- D. Enter the placement points from the head-to-head bracket.
- E. Manage security and student pick up of all entries at the assigned time and the removal of materials from the area.

OVERVIEW

Applying leadership and 21st century skills, participants conduct research on a contemporary medical technology issue, related to the current year's theme, noted on the [TSA website](#) under *Themes & Problems*, and present their research through a static display and multimedia video. The entry may include student research, or a re-creation or simulation of research performed by the scientific community. A student-designed informational pamphlet will also be included as part of the display. Semifinalist teams participate in an on-site interview.

ELIGIBILITY

Three (3) teams per state; a minimum of two (2) members and a maximum of three (3) members per team.

TIME LIMITS

Five (5) minutes are allowed for the semifinal interview broken down as follows:

- A. One (1) minute for set-up
- B. Four (4) minutes to respond to questions from judges.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants research the issue using credible sources.
- D. Participants prepare their multimedia video, display, and informational pamphlet according to the regulations.

PRELIMINARY ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to check in the display and the informational pamphlet.
- B. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:
 - 1. Judges score the static display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the informational pamphlet and multimedia video criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalists.
- C. A list of twelve (12) semifinalist teams (in random order) is posted.

SEMIFINAL ROUND

- A. One (1) team representative reports at the time and place stated in the conference program to sign-up for an interview time.
- B. Up to three (3) team representatives report at least ten (10) minutes prior to their assigned time to the designated place for their interview.
- C. Semifinalist team representatives participate in the on-site interview.
- D. Team representatives pick-up their entry from the display area at the time and place stated in the conference program.
- E. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

A. Students must understand the fundamental concepts and principles of the contemporary issue and medical technology they research. Research about the issue should focus on significant impacts (opportunities and risks) on the environment, economy, and society, as well as any important ethical considerations.

B. Display Guidelines:

1. The size of the display may not exceed 15" deep x 30" wide x 48" high.
2. The display should include information about the disease, condition, or problem addressed by the medical technology. Participants should prepare the display as if they were delivering a presentation on the particular issue at a medical conference. Information about the solution/medical technology should be reserved for the informational pamphlet and multimedia video.
3. Supporting materials, such as research, references, and resources, should be exhibited in a creative and effective manner on a stand-alone display and should include at a minimum:
 - a. Team identification number
 - b. Overview of the issue researched, including its prevalence in the United States, how it is diagnosed, and the science behind the disease state or condition.
 - c. Societal impact of the problem, including economic, social, environmental, and ethical considerations
 - d. Supporting information such as logs, graphs, sketches, drawings, illustrations, and photographs
 - e. The display must incorporate the use of design principles. Participants highlight the display with creative visual elements (e.g., photos, titles) to actively engage the audience. Participants may use images "labeled for "re-use," but must properly cite the source.

f. Copyright: Citation of all ideas, fonts, and images from sources other than the designer, and/or that are copyrighted (most fonts and images found on the web are copyrighted material unless purchased or offered as free-domain) and **MUST** be included on the back of the display. Clip art must be documented.

g. Photo/Film/Video Consent and Release Forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person and included on the back of the display (see 2026 & 2027 Forms Appendix).

4. A/C electricity may not be used.

5. If the display requires power, they must be powered by dry-cell batteries or photo-voltaic cells.

a. The power supply must physically fit within the display dimensions.

b. All power must be switched off once the team has completed set-up.

c. If teams want judges to activate any electronic device in their model/prototype/display, complete instructions must be provided to judges on how to power up the model/display.

C. Informational Pamphlet

1. Participants must create an informational pamphlet detailing the medical technology solution chosen
2. Physicians often give pamphlets to patients and their families as a means to disperse more information about a suggested treatment or procedure. Participants should design their informational pamphlet to be used in this setting.
3. The pamphlet should be folded and composed of one (1) sheet of paper not to exceed the size 8.5" x 11".

4. The pamphlet should include information about the chosen medical technology/solution, including how it works, its cost, its availability, and its overall impact on society.
5. A QR code linking the multimedia video should be included somewhere on the pamphlet. Judges should be able to scan the QR code with a smartphone.
6. Resources used by participants in the informational pamphlet must be cited using the same guidelines outlined for the display and included on the citation list adhered to the back of the display.

D. Multimedia Video

1. The participants should describe the medical technology/solution in the multimedia video as if they were describing it to a patient, who is considering this technology, or their family.
2. The video should include what to expect if the patient moves forward with the procedure/technology/solution, including associated risks, and any other treatment that will be required to supplement the technology.
3. Teams must demonstrate strong science communication skills by delivering the information in an accessible manner so that a general audience could understand it.
4. The length of the video should be no less than two (2) minutes and not exceed four (4) minutes.
5. Teams may use any editing software of their choosing along with features such as sounds and effects the software offers. The software and any effects/sounds/graphics used not belonging to the team must be appropriately cited according to procedures outlined for the display and included in the citation list adhered to the back of the display.
6. The multimedia video should be linked to a QR code, which should be included in the informational pamphlet.
7. The multimedia video accessed using the QR code cannot require any authorization, download, or password to view. If required, the multimedia video will not be judged. Tier 2 will not be judged without the video component.

- E. Once the display set-up time frame has closed, participants may not re-enter the event area.
- F. Violation of the following will result in disqualification:
 1. No liquids, viruses, live plants, or animals may be used as a part of the display.
 2. No harmful or illegal substances may be displayed.

SEMIFINAL ROUND

- A. The interview is limited to three (3) team representatives. Each member should be an active participant in the interview.
- B. Representatives may reference their display and informational pamphlet during the interview.
- C. Time limits:
 1. One (1) minute for set-up
 2. Four (4) minutes to respond to questions from judges
 3. A five-(5) point deduction will be incurred for any time infraction.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. The display

Tier 2

- B. The informational pamphlet/multimedia video

SEMIFINAL ROUND

- A. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Dietitian
- Doctor
- Medical technologist
- Nurse
- Pharmacist
- Prosthetics practitioner

MEDICAL TECHNOLOGY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Display is present
- ☐ TIER 2 – Informational pamphlet is present and video is accessible through QR code
- ☐ ENTRY NOT EVALUATED

TIER 1 – DISPLAY (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Explanation of Impacts (X2)	Explanation is missing a discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	Explanation addresses some of the issue's relevance to environmental, economic, social, and/or ethical considerations.	Explanation includes a full discussion of the issue's relevance to environmental, economic, social, and/or ethical considerations.	
Supporting Information (X1)	Supporting information does not help to clarify the issue and/or it is of little significance to the issue.	Supporting information is somewhat appropriate and helps provide some clarity to the issue.	Supporting information is highly effective and of excellent quality.	
Research, References, and Resources (X1)	Research is inadequate, and/or very few credible sources are referenced.	Research is conducted appropriately, with some adequate credible sources.	A comprehensive research base that includes credible sources is evident.	
Communication of Issue (X2)	It is difficult to understand the issue being communicated; an illogical explanation is presented.	The issue is communicated and thoughts are somewhat organized.	The issue is communicated in an organized, clear, and concise manner.	
Creativity (X1)	The display lacks creativity; no, or very few, design principles are integrated in the display.	Some elements of creativity exist in the display, and essential design principles are generally evident.	The display exudes creativity; essential design principles and elements are well integrated.	
Aesthetics and Artanship (X1)	Work is unorganized and sloppy; display seems to be an afterthought or thrown together.	Display generally depicts the issue.	Display is exemplary in logically communicating the issue.	
TIER 1 – DISPLAY SUBTOTAL (80 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
TIER 1 – SUBTOTAL (80 points)				

TIER 2 – INFORMATIONAL PAMPHLET (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Informational Pamphlet (X1)	Informational pamphlet is present but not complete, or is lacking in overall quality.	Informational pamphlet is good quality but missing detail(s) to adequately portray the project's focus.	Exemplary informational pamphlet is present and shows great detail in portraying the project's focus.	
Communication of Solution (X1)	It is difficult to understand the solution/technology being communicated; an illogical explanation is presented.	The solution/technology is communicated and thoughts are somewhat organized.	The solution/technology is communicated in an organized, clear, and concise manner.	
Creativity and Originality (X1)	There is little original thought or creativity in the information presented and how it is arranged.	The pamphlet demonstrates some original thought and creativity in the information that is presented and how it is arranged.	Originality and creativity are at the forefront of the pamphlet, both in content and style.	
Aesthetics and Artanship (X1)	Work is sloppy; informational pamphlet seems to be an afterthought or thrown together.	Informational pamphlet utilizes good design techniques for a visually appealing and somewhat informational resource.	Informational pamphlet uses exceptional design principles to produce a resource that is both visually appealing and informational.	
TIER 2 – INFORMATIONAL PAMPHLET SUBTOTAL (40 points)				

TIER 2 – MULTIMEDIA VIDEO (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Audio (X1)	The audio quality is poor, a result of primary use of the on-camera microphone for recording.	The audio quality is clear and reflects an appropriate use of microphones and audio techniques.	The audio quality is excellent, with use of additional audio clips/cues that enhance the video production.	
Continuity and Pacing (X1)	The sequencing of the multimedia video is confusing or incomprehensible; transitions are distracting from the flow of the multimedia video.	The pace and timing are generally structured; there is a good use of transitions, allowing for an adequate flow of the multimedia video.	The pace and timing are exceptional; transitions provide a seamless flow for the multimedia video and give directionality to the work.	
Lighting (X1)	The video reflects poor ambient lighting choices and/or the use of heavy backlighting.	The video reflects adequate lighting on subjects and the proper use of lighting techniques.	The video reflects and excellent use of lighting, which contributes to the effectiveness of the multimedia video.	
Video Effectiveness (X1)	The work does not meet the project goals, has an unclear message, and reflects sloppy work.	The topic is presented with some insight, and the video meets most project goals.	The video is focused, with a clear message and a rich variety of supporting material.	
TIER 2 – MULTIMEDIA VIDEO SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
--	--

PRELIMINARY SUBTOTAL (160 points)	
--	--

SEMIFINAL INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Knowledge (X1)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit a basic understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the project.	
Articulation (X1)	Communication of the solution is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the solution is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the solution is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Team Participation (X1)	Majority of the interview is made by one member of the team; the partners may be disengaged from the presentation.	Team members generally are engaged in the interview, though one member may take on more responsibility than the other(s).	All team members are actively involved in the interview; there is shared responsibility among team members.	
Delivery (X1)	The team is verbose and/or uncertain in its interview; participants' posture, gestures, and lack of eye contact diminish the presentation.	The team is somewhat well-spoken and clear in its interview; participants' posture, gestures, and eye contact are acceptable in the presentation.	The team is well-spoken and distinct in its interview; participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation.	
SEMIFINAL INTERVIEW SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIME DEDUCTIONS

A five-(5) point deduction will be incurred for any time infraction.

SEMIFINAL SUBTOTAL (40 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL (200 points)**

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

MEDICAL TECHNOLOGY EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator, one (1)
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants for check-in, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Display tables for entries
- C. Table and chairs for team presentations, as needed
- D. Stopwatch
- E. Measuring tape
- F. Table and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Check in the entries at the time stated in the conference program.
- B. No more than two (2) team representatives set up and check in:
 - 1. The display
 - 2. The informational pamphlet
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Each entry must include the team's identification number in the upper right-hand corner of the display and somewhere visible on the informational pamphlet.

PRELIMINARY ROUND

- A. Independently score the display, the informational pamphlet, and the multimedia video.
- B. Judges independently review the entries based on the following criteria:

Tier 1

Judges score the display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

Judges score the Informational Pamphlet/Multimedia Video criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

- C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- D. Create a sign-up sheet for semifinalist interviews.

SEMIFINAL ROUND

- A. Prepare a list of five to ten (5-10) standard interview questions to be asked of all participants.
- B. Semifinalists report to sign-up for the interview at the time and place stated in the conference program.
- C. Check in semifinalists and confirm with the teams their order of interview and the procedure. Team representatives should report at least ten (10) minutes prior to their assigned time to the designated place for their interview.
- D. Manage the interviews.
- E. Discuss rule violations and have all relevant parties initial the rating form.
- F. Judges independently assess the semifinal interview.
- G. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants design a working digital device (product) with real-world applications. Through product demonstration and documentation, the team demonstrates in detail its knowledge of microcontroller programming, simple circuitry, and product design and marketing. The project should have educational and social value, and conform to the theme for the year. The theme will be posted on the [TSA website](#) under *Themes & Problems*. Teams demonstrate and promote their work in a timed presentation.

ELIGIBILITY

One (1) team per chapter may participate; individual entries are permitted

TIME LIMITS

Seven (7) minutes are allowed for the presentation, broken down as follows:

- A. Two (2) minutes to set up for the presentation.
- B. Three (3) minutes for the presentation.
- C. Two (2) minutes to return the device to the storage location.
- D. A deduction of five (5) points will be incurred for exceeding the three (3)-minute presentation time limit.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants concentrate their efforts in the design and development of a working digital device (product).
- C. Participants create a portfolio, documenting their projects and process.
- D. Participants prepare an oral presentation.

ON-SITE CHALLENGE

- A. No more than three (3) team representatives report at the time and place stated in the conference program with:
 - 1. The device (product)
 - 2. The documentation portfolio
- B. Participants sign up for a presentation time upon check-in.
- C. One (1) to three (3) team representatives present the following at their assigned time:
 - 1. Research findings
 - 2. An explanation of the value of their product
 - 3. A demonstration of the product's functionality
- D. The device will be returned by the team to the storage location at the completion of the presentation and then will be picked up by the team at the time and location posted in the conference program.
- E. Judges score the entries and presentation.
- F. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Documentation Portfolio:
 - 1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team identification number; one (1) page
 - b. Table of contents; pages as needed
 - c. A description of the team's project, including an explanation of the theme; pages as needed

- d. Work Log (see 2026 & 2027 Forms Appendix); pages as needed
- e. Source code; pages as needed
- f. Materials list; one (1) page
- g. Team's evaluation of its work; one (1) page
- h. References and resources list in a professional citation style of the competitors choosing; pages as needed. Failure to use a professional citation style will result in a rules violation of 20% (twenty percent). Some examples of professional citation styles include MLA, APA, Chicago, and IEEE.

B. The Device (product):

- 1. The device (product) must include a programmed microcontroller that controls the device functionality.
- 2. Aesthetics: The product must be well-designed, design elements are incorporated, and show good craftsmanship.
- 3. Functionality: The product must operate as intended; remote control technology may be used to operate the device.
- 4. The product must be relevant to the given theme.
- 5. Coding and Circuit Design: The product must show proper and effective use of coding and circuit methods.
- 6. AC power and/or a dry cell battery may be used.
- 7. The device may be no larger than 18" x 12" x 12".

ON-SITE CHALLENGE

A. The Presentation:

- 1. Participants are given two (2) minutes to set up their device and three (3) minutes to explain the problem and demonstrate the functionality of the device.
- 2. A deduction of five (5) points will be incurred for exceeding the five (5)-minute presentation time limit.
- 3. Participants are allowed two (2) minutes to return the device to the storage tables following the presentation. Any other items must be taken from the space.

- B. The device and portfolio will be picked up by the team at the time and location posted in the conference program.

EVALUATION

- A. The documentation
- B. The device/product
- C. The presentation

Refer to the official rating form for more information.

RESOURCES

www.pictutorials.com/what_is_microcontroller.htm

www.newbiehack.com/MicrocontrollerTutorial.aspx

people.ece.cornell.edu/land/courses/ece4760/FinalProjects

www.circuitstoday.com/8051-projects-and-circuits

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Manufacturing
- Software engineer
- Technical writer

MICROCONTROLLER DESIGN

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ The documentation portfolio is present
- ☐ The device (product) is present
- ☐ ENTRY NOT EVALUATED

DOCUMENTATION PORTFOLIO (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Project Documentation (X1)	The project documentation is inadequate, and/or very few credible sources are referenced.	The project documentation is adequate, and mostly credible sources are included.	The project documentation is comprehensive, and credible resources are included.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
DOCUMENTATION PORTFOLIO SUBTOTAL (30 points)				

DEVICE/PRODUCT (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Complexity (X2)	Product lacks complexity; it includes little or no code, and/or circuit design and control technology complexity.	The product exhibits some degree of complexity; it includes code, circuit design, and control technology complexity.	The product is complex and highly functional; it includes code, circuit design, and control technology complexity.	
Creativity (X2)	The product lacks creativity; very little original thought in developing the project is evident.	Some elements of creativity are expressed; the product is somewhat original.	The work exudes creativity; the product is highly original.	
Technical Skill (X2)	Little technical skill is exhibited in the code and circuit design.	A beyond-basic degree of technical skill is exhibited in the code and circuit design.	A level of mastery of coding and circuit design is exhibited.	

DEVICE/PRODUCT (80 points) – continued				
Connection to the Theme (X1)	Product does not have any, or has very little, connection to the theme.	The product has adequate connection to the theme.	The product has high connection to the theme.	
Design Principles (X1)	Product demonstrates little to no use of design principles.	Product demonstrates adequate use of design principles.	Product demonstrates exceptional use of design principles.	
DEVICE/PRODUCT SUBTOTAL (80 points)				

PRESENTATION (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participant(s) seem unorganized and unprepared for the presentation; the presentation is illogical.	Participant(s) are generally prepared for the presentation; explanation of the product is communicated and generally organized.	The presentation is logical, well organized, and easy to follow; explanation of the product is communicated in an organized and concise manner.	
Articulation (X1)	Communication of the product functionality and design process is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the product functionality and design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the product functionality and design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Product Demonstration (X2)	Participant(s) is/are unable to successfully demonstrate the product, and/or the product does not work, or barely works, as intended.	Participant(s) is/are able to partially demonstrate the functionality of the product; the product somewhat works as intended.	Participant(s) is/are successful and effective in the product demonstration; the product works exactly as intended.	
PRESENTATION SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	
Time violation (a deduction of five [5] points total will be incurred for exceeding the five [5]-minute presentation time limit). Record the deduction.	

PRELIMINARY SUBTOTAL (150 points)	
--	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (150 points)	
--	---------------------------	--

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____



MICROCONTROLLER DESIGN

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 1. Portfolio and Device/Product, two (2) or more
 2. Presentation, two (2) or more (can be the same judge as #1)

MATERIALS

- A. Coordinator's packet, containing:
 1. Event guidelines, one (1) copy for the coordinator and for each judge
 2. TSA Event Coordinator Report
 3. Stick-on labels for identifying entries
- B. Two (2) tables and chairs for presentations, as needed
- C. Measuring tape
- D. Display tables for entries
- E. Stopwatch for timing presentations
- F. Access to a AC outlet during presentations

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with your judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

ON-SITE CHALLENGE

- A. No more than three (3) team representatives report at the time and place stated in the conference program with:
 1. The device (product)
 2. The documentation portfolio
- B. Each entry must include the participant's identification number in the upper right-hand corner of the entry.
- C. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Assign students a five (5)-minute time frame for their presentation at check-in. As one team is returning their device to the storage table, the next team can be setting up.
- F. Judges circulate to review documentation and entry prior to the presentation.
- G. Presentations:
 1. Inspect the area in which the presentations are to be held.
 2. There must be seating for at least five (5) people at a table with space for a computer and device (product).
 3. Conduct presentations.
 4. A deduction of five (5) points will be incurred for exceeding the five (5)-minute time limit.
- H. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 1. To deduct twenty percent (20%) of the total possible points in this round
 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- I. Manage security and the removal of materials from the area.

OVERVIEW

Throughout the world, people are working to become more self-sustaining when it comes to landscaping and architectural design. Sometimes, the purpose is to live off the grid, and other times it is to create a smaller carbon footprint; yet other times, it is to meet a need in locations where there is not access to power, water, or other basics. There are many options for sustainability throughout the world, but sometimes a location limits or enables those options.

Applying leadership and 21st century skills, participants conduct research on a sustainable energy source or system for a home in a specified country. Teams will create a display, a model, and accompanying documentation. The model can be of the entire home the team designs or of the specific sustainable process or system highlighted by the team.

Semifinalists demonstrate competency by presenting and responding to questions about their design. The design brief for this competition will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) teams per state may participate; individual entries are permitted

TIME LIMITS

Ten (10) minutes are allowed for the semifinal presentation and interview broken down as follows:

- A. Seven (7) minutes for the presentation
- B. Three (3) minutes to respond to questions from judges

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

- B. Participants access the design brief posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants develop a solution to address the challenge using sustainable, off-the-grid practices based on the given country's climate conditions, infrastructure, construction methods, and a given budget.
- D. Participants prepare their documentation, display, and model according to the regulations.
- E. Participants prepare a presentation and be prepared to respond to questions from judges.

PRELIMINARY ROUND

- A. No more than two (2) team representative report at the time and place stated in the conference program to check in:
 - 1. The display
 - 2. The model
 - 3. The documentation portfolio
- B. Entries are evaluated by judges independently, with neither students nor advisors present based on the following criteria:
 - 1. Judges score the Display and Model criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
 - 2. Judges score the Documentation Portfolio criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.
- C. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

- A. No more than two (2) team representatives report at the time and place stated in the conference program to sign up for a presentation/interview time.
- B. One to three (1-3) team representatives report at the assigned time and place stated in the conference program to participate in the presentation/interview.
- C. Judges score the presentation/interviews.
- D. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

A. Participants must understand the fundamental concepts and principles of the sustainable architecture that they are implementing. Research should focus on the technology and processes used to optimize sustainability and promote off-the-grid living for the scenario described in the annual design brief.

B. Display and Model:

1. The display must include:
 - a. Model of the solution or model of a system used for the design
 - b. Name of the given location, including city/town and country/province.
 - c. A world map clearly indicating the location of the given country
 - d. A description of the challenge or problem being addressed
 - e. An overview of the team's solution, including how it works and how it will address the given issue.
 - f. Supporting information that conveys the utility of the given solution (i.e. graphs, charts, schematics, figures), as needed.
2. The size of the display and model may not exceed 15" deep x 30" wide x 48" tall. The documentation portfolio must be placed in the display with the model and remain within the dimensions.
3. A/C electricity may not be used.
4. If the display and/or model requires power, they must be powered by dry-cell batteries or photo-voltaic cells.
 - a. The power supply must physically fit within the display dimensions.
 - b. All power must be switched off once the team has completed set-up.

- c. If teams want judges to activate any electronic device in their model/display, complete instructions must be provided to judges on how to power up the model/display.

5. Once the display set-up time frame has closed, participants may not re-enter the event area.

C. Documentation Portfolio:

1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team ID number; one (1) page
 - b. Table of contents; one (1) page
 - c. A description of how the team interpreted the design challenge and an explanation of the style and merits of the solution; maximum two (2) pages
 - d. A description of the country's environment, climate, natural resources, and economy; maximum two (2) pages
 - e. Bill of materials comprised of items used to construct and implement the solution, including project costs of each line item, which is not to exceed the given budget from the annual design brief; pages as needed
 - f. Description of alternative solutions and a discussion of why the chosen solution was utilized instead; maximum two (2) pages
 - g. CAD drawings (each drawing to be submitted on maximum drawing sheets cut size B [11" x 17"] with appropriate scale size noted on the drawing); pages as needed
 - i. Schematics showing the incorporation of the system into existing infrastructure in the given design brief
 - ii. Diagrams explaining design elements
 - h. References and resources page. Participants must cite a minimum of three (3) credible resources; pages as needed
 - i. Work Log (see 2026 & 2027 Forms Appendix); pages as needed

PRELIMINARY ROUND

- A. No more than two (2) team representatives check in the entry and set up the display, observing the following:
1. Models must fit within the allotted display space
 2. A/C electricity is prohibited
 3. Dry cell or photo-voltaic cells may be used for power, if desired
 4. The power source used must fit within the maximum display area
 5. All power must be turned off once setup is complete.
 6. If operating instructions or login information are necessary, they must be clearly displayed
- B. Violation of the following will result in disqualification:
1. No liquids, viruses, live plants, or animals may be used as a part of the display.
 2. No harmful or illegal substances may be displayed.

SEMIFINAL ROUND

- A. Participants may reference the display, model, and portfolio. No additional materials or devices are allowed for the presentation.

EVALUATION**PRELIMINARY ROUND****Tier 1**

- A. The display
B. The model

Tier 2

- C. The documentation portfolio

SEMIFINAL ROUND

- A. The presentation/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Appraiser
- Architect
- Construction manager
- Environmental engineer
- Interior designer
- Urban and regional planner

OFF THE GRID

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Display and model are present
- ☐ TIER 2 – Documentation portfolio is present
- ☐ ENTRY NOT EVALUATED

TIER 1 – DISPLAY (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Display Components (X1)	Display is unorganized and/or is missing three (3) or more components.	Display has most components and is generally organized; it has sufficient content.	All components are included in the display content and the organization is excellent.	
Communication of Solution (X3)	The description of the country and design is unclear or vague; there is little to no mention of the sustainable design features.	The description of the country and design of the style are included; however, some sustainable design features are missing.	The description of the country and design of the style are clear, effective, and convincing.	
Creativity (X1)	Display lacks originality; none or very few design principles are integrated in the display.	Some resourcefulness and ingenuity are evident in the display; some essential design principles are generally used.	There is clear evidence of an inventive, unique, and creative display; essential design principles and elements are integrated.	
TIER 1 – DISPLAY SUBTOTAL (50 points)				

TIER 1 – MODEL DESIGN (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Effectiveness of Design (X1)	The design is ineffective in meeting the needs of the challenge.	The design is somewhat effective in meeting the needs of the challenge.	The design is clearly effective in meeting the needs of the challenge.	
Aesthetic Appeal and Functionality (X1)	There is little to no evidence of consideration of aesthetics and curb appeal in the design; systems in the model do not match or work with the design.	There is some evidence that aesthetics and curb appeal have been considered in the design; most systems in the model match or work with the design.	There is clear evidence that aesthetics and curb appeal are fully and effectively integrated into the design; all systems in the model work with the design.	
Creativity and Innovation (X1)	The design lacks originality and exhibits few, if any, creative and/or innovative applications.	Some unique, innovative, and creative concepts are incorporated in the overall design.	Unique, creative, and innovative approaches are fully incorporated into the design.	

TIER 1 – MODEL DESIGN (40 points) – continued				
Model Design (X1)	The model is not informative and does not effectively nor accurately represent the design.	Most of the model is informative and accurately represents the design.	The model effectively displays the solution to the challenge and accurately represents the design.	
TIER 1 – MODEL DESIGN SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
---	--

TIER 1 – SUBTOTAL (90 points)	
--------------------------------------	--

TIER 2 – DOCUMENTATION PORTFOLIO (80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Description of Design Interpretation (X1)	The description of the design and style is unclear or vague; includes little or no mention of the sustainable design features.	The description of the design and explanation of the style are included; they are adequately presented.	The descriptions and merits of the design, and an explanation of the style, are clear, effective, and convincing.	
Description of the Chosen Country (X1)	There is little or no evidence of research done; the country's environment, climate, natural resources, and economy were not addressed clearly or at all.	Most, but not all, of the required areas of research are present and documented; some aspects are missing.	All of the research areas are clearly documented and well presented.	
Description of Alternative Solutions (X1)	The team fails to articulate alternative solutions and/or why the chosen solution was utilized.	The team makes an attempt at outlining other potential solutions and somewhat explains the choice of the solution presented.	The team highlights alternatives to the proposed solution and is able to logically convey why the chosen solution was selected.	
Bill of Materials (X1)	Many elements are missing or incomplete.	Most, but not all, elements are included.	All elements are included, and are detailed and explained clearly.	
Drawings (X1)	A few of the required drawings are present, but they are lacking in quality.	Most, but not all, of the required drawings are included and are in the proper format.	All required drawings are included and in the proper format.	
Resources/References (X1)	There is little to no effort to provide resources or references.	Resources and references are included and are somewhat credible.	There is clear evidence of the appropriate use of applicable resources and references.	
Work Log (X1)	The Work Log lacks major elements of documentation.	The Work Log is somewhat complete, and generally reflects the time and work necessary for the project.	The Work Log completely and accurately reflects the time and work necessary for the project.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (80 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
---	--

PRELIMINARY SUBTOTAL (170 points)**SEMIFINAL PRESENTATION/INTERVIEW (50 points)**

CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
Organization (X1)	Participant(s) seem unorganized and unprepared for the presentation; illogical explanation of the problem and solution is presented.	Participant(s) is/are generally prepared for the presentation; explanation of the problem and solution are generally communicated.	Presentation is logical and easy to follow; the problem and solution are communicated in a concise manner.
Articulation (X1)	Communication of the design project is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design project is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design project is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.
Delivery (X1)	The participant(s) is/are verbose and/or uncertain in its presentation/ interview; participants' posture, gestures, and lack of eye contact diminish the presentation/interview.	The participant(s) is/are somewhat well-spoken and clear in its presentation/interview; participants' posture, gestures, and eye contact are acceptable in the presentation/ interview.	The participant(s) is/are well-spoken and distinct in its presentation/ interview; participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation/interview.
Knowledge (X2)	Participant(s) exhibit little understanding of the concepts in their project; answers to questions may be vague.	Participant(s) exhibit a general understanding of the concepts in their project.	Participant(s) show clear evidence of a thorough understanding of the project.

Record scores
in the column
spaces below.**SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)**

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (50 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.

TOTAL (220 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

OFF THE GRID

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants for check-in, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for entries, as needed
- B. Tables for entries
- C. Tables and chairs for event coordinator and judges

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Check in participants at the time and place stated in the conference program.
- B. No more than two (2) participants check in:
 - 1. The model
 - 2. The display
 - 3. The documentation portfolio
- C. Ensure the model and display does not exceed: 15" deep x 30" wide x 48" high. Measurements need to be taken immediately after the model and display have been set up by the competitors.
- D. Check to make sure the entry does NOT include:
 - 1. Liquids, viruses, live plants, or animals.
 - 2. Harmful or illegal substances.
- E. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- F. In order to compete, participants must be on the entry list or must have approval of the CRC.
- G. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- H. Place a team identification number stick-on label in the lower right-hand corner of each display.
- I. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Judges independently review the entries based on the following criteria:

Tier 1

- 1. Judges score the display and model criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.

Tier 2

- 2. Judges score the documentation criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

C. Create a presentation/interview sign-up sheet.

SEMIFINAL ROUND

- A. One (1) team representative reports at the assigned time and place to sign up for the presentation/interview.
- B. One to three (1-3) team representatives report at the assigned time and place for the presentation/interview.
- C. Participants may reference the display and the portfolio. No additional materials or devices are allowed for the presentation.
- D. Judges review the presentation and ask pertinent interview questions.
- E. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- F. Manage security and removal of all materials from the area.

PREPARED SPEECH



OVERVIEW

Participant delivers a speech that reflects the theme of the current National TSA Conference. See the [TSA website](#) under *Themes & Problems* for the current conference theme.

ELIGIBILITY

Three (3) individuals per state may participate.

TIME LIMITS

- A. Each speech should be no less than three (3) minutes and no more than five (5) minutes.
- B. A one (1)-point deduction will be incurred for each ten (10)-second interval under or over the allotted time for speaking.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants report to the event area at the time and place stated in the conference program to receive an assigned presentation time.
- B. The event coordinator introduces each student by identification number only and in the order of scheduled times.
- C. Participants deliver their speech in the allotted time limit.
- D. A list of twelve (12) semifinalists (in random order) will be posted.

SEMIFINAL ROUND

- A. Semifinalists report to the event area at the time and place stated in the conference program to receive an assigned presentation time.
- B. Semifinalist speeches follow the same procedure as in the preliminary round.
- C. Ten (10) finalists will be announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Each speech must be the result of the participant's own efforts.
- B. The topic for the Prepared Speech event is the published theme of the current year's conference.
- C. Information about technology and TSA is appropriate as long as it relates to the published theme.
- D. Participants are not permitted to use any type of props, computers, display boards, etc.
- E. Costumes are not permitted.
- F. Participants may use note cards during the speech.
- G. Participants may not reveal their school, chapter name, or city, but the appearance of a state name on an official TSA badge is acceptable.
- H. TSA will provide a podium in the event room.
- I. Participant scores are penalized one (1) point per ten (10)-second interval for speaking over or under the allotted time.
- J. Time commences when the speech begins.
- K. Participants are not allowed to hear the speech of any other participant.
- L. No observers are allowed in the event room during heats.

SEMIFINAL ROUND

- A. All regulations and requirements from the preliminary round apply to the semifinal round.
- B. Observers are allowed to sit in the audience during the semifinals.
- C. Observers may not enter or leave during a speech.
- D. No talking, gesturing, or interaction with the audience is permitted.
- E. There should be no applause until a speech has concluded.
- F. No audio or visual recording devices (including cell phones, digital cameras, etc.) by any observer are permitted.

EVALUATION

- A. The quality, content, and effectiveness of the speech
- B. The speaker's stage presence

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Entertainment/television broadcaster
- Lawyer
- Politician
- Speech writer
- Teacher/trainer

PREPARED SPEECH

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Participant is present
☐ ENTRY NOT EVALUATED

CONTENT (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Introduction (X1)	Introduction is weak, with little effort made to highlight the theme and/or to generate interest and enthusiasm for the topic.	Effort is evident; introduction creates a moderate level of interest.	Introduction is effective, stimulating, and engaging; a strong effort is made to highlight the theme and/or to generate interest and enthusiasm for the topic.	
Body (X1)	Body of speech is poorly organized; content does not properly cover or represent the concepts and theme being presented.	Body of speech is adequately presented and is somewhat interesting.	Body of speech is clearly and effectively presented in an exceptionally interesting manner; the speech is memorable and conveys the theme.	
Conclusion (X1)	Conclusion fails to summarize or clarify the information presented in the speech.	Conclusion generally summarizes the content and theme of the speech.	The conclusion is effective, interesting, and memorable; the summary fully brings finality to the speech and the theme.	
CONTENT SUBTOTAL (30 points)				

STAGE PRESENCE (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Appearance (X1)	Participant's appearance is unprofessional, sloppy, and/or inappropriate.	Participant's appearance is adequate and appropriate.	Participant's appearance is clearly exceptional, appropriate, and professional.	
Confidence (X1)	Participant appears nervous during speech; poor posture, poor eye contact, and lack of confidence are evident; participant reads speech from note cards.	Participant is generally poised, displays eye contact, and is confident, with little sign of nervousness; participant's use of note cards is minimal.	Participant "commands" the room, and is exceptionally poised, confident, and positive; participant does not use note cards for the speech, or note cards do not detract from the overall speech.	

PREPARED SPEECH

STAGE PRESENCE (30 points) – continued				
Articulation (X1)	Participant conveys an inconsistent use of proper grammar, word pronunciation, acceptable pitch, and tone.	Participant generally uses proper grammar and pronunciation, and varies the tone and pitch in the delivery.	Smooth and effective articulation, proper grammar, correct pronunciation, and varied tone and pitch are used throughout the speech.	
STAGE PRESENCE SUBTOTAL (30 points)				

ORGANIZATION (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Effectiveness and Quality of Presentation (X2)	Speech is poorly prepared, not interesting, and not representative of the stated theme.	Speech is adequate in most areas but exceptional in none of them.	Speech is exceptional and memorable; the observer can easily understand and relate to the speech.	
Knowledge of Material (X1)	Minimal factual support is evident in the speech; the content of the speech does not relate to the theme; the participant does not convey understanding of the theme.	Factual support is provided in the speech, but the support does not always relate to the theme or match the content of the speech.	Factual support is provided in the speech; it has a strong relationship to the theme and enhances the content of the speech.	
Organization (X1)	The speech is difficult to follow or understand.	The speech is adequately organized and delivered.	The speech is clearly organized and easy to follow; the delivery is exceptional.	
ORGANIZATION SUBTOTAL (40 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
--	--

TIME DEDUCTIONS		Record scores in the column spaces below.
One (1) point per ten (10)-second interval is deducted for speaking under three (3) minutes or over five (5) minutes. Time commences when the competitor begins speaking.		
TOTAL TIME FOR SPEECH		
TOTAL TIME DEDUCTIONS		

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (100 points)
--	---------------------------

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

PREPARED SPEECH EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) per heat room
 - 2. Semifinal round, two (2) or more
- C. Timekeeper, one (1)
- D. Coordinator assistant (to serve as an escort from the holding area to the heat rooms), one (1)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. One (1) stopwatch for each event room
- C. Podium
- D. Chairs for audience
- E. Tables and chairs for two (2) or more judges
- F. Prepared sign-up list indicating eight to ten (8-10) minute intervals for each heat to accommodate all registered participants.

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory coordinator's meeting.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, and regulations. If questions arise that cannot be answered, speak to the event manager before the event begins.

- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Begin the check-in and sign-up presentation time at the time and location provided in the conference program.
- B. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
- C. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.

PRELIMINARY ROUND

- A. The event coordinator takes the first participant to the event room.
- B. The event coordinator or assistant introduces the participant by entry number only. No name tags or clothing identifying the hometown, school, or chapter of the participant are allowed. A state name on a TSA patch is acceptable.
- C. Approximately every eight to ten (8-10) minutes, the event coordinator or the coordinator assistant will escort another participant to the event room.
- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. Judges determine the twelve (12) semifinalists.
- F. Create a sign-up sheet for the semifinal round.

SEMIFINAL ROUND

- A. Using the same official rating form for semifinalists, judges assess the semifinalist speeches and determine the ten (10) finalists.
- B. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- C. If necessary, manage security and the removal of materials from the event area.

PROBLEM SOLVING



OVERVIEW

Participants apply leadership and 21st century skills in problem solving to develop a finite solution to a stated problem provided on-site. Participants work as a team to provide the best solution, which is measured objectively.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

TIME LIMITS

Ninety (90) minutes for the design and construction of the solution are permitted.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

ON-SITE CHALLENGE

- A. Participants report to the event area at the time and place stated in the conference program with their team tool box and safety eye glasses.
- B. The problem, the evaluation criteria, and the materials are distributed.
- C. Teams will be given a two (2) hour window to begin construction of a solution
- D. Teams are allowed ninety (90) minutes to design and construct a solution.
- E. Each solution is tested as soon as possible after the construction phase is completed. Some problems may require teams to be present for testing.
- F. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

- A. All work must be completed in the event area during the time specified for the event.
- B. Specific materials related to the on-site problem will be provided by TSA.
- C. Only the materials issued to each team by the event coordinator may be used in the development of the solution. Exceptions are adhesives (glue) and masking tape from each team's toolbox.
- D. Participants are required to provide their own tool box/container, which must:
 - 1. Include identification (school name, address, and advisor cell phone number).
 - 2. Not exceed twenty (20) inches (508 mm) length x ten (10) inches (254 mm) width x ten (10) inches (254 mm) height.
 - 3. Contain all tools and materials needed to fabricate the solution. The following is a suggested list:
 - a. Cutting devices; NONE may be electric
 - b. Adhesives:
 - i. Aerosol and electric applicators are not allowed
 - ii. A bottle of Uncure or Debonder is recommended
 - c. Temporary fastening devices
 - i. Straight pins
 - ii. Clamps
 - iii. Tape (only masking tape may be used as a construction material of the final solution)
 - d. A cutting surface that prevents table-top marring (required)

- e. Rulers, straightedges, and/or measuring scales
 - f. Marking devices (pens, pencils, etc.) and sharpener
 - g. Sheet of wax paper, (up to 9 square feet)
 - h. Pliers, wrenches, nut drivers, as needed
 - i. Safety glasses and side shields, as required
 - j. One (1) roll masking tape no wider than $\frac{3}{4}$ "
- E. Participants without a tool box will not be allowed to compete.
- F. As teams enter the competition area, each will be given a copy of the Verifications Sheet. The Verifications Sheet is a list of tool box contents as listed in Rules and Regulations D3 a-k.
- G. Sharing tools between teams is not permitted.
- H. Participants are required to provide and wear safety-approved eyewear for this event.
1. Safety eyewear shall be worn at event check-in.
 2. Prescription eyewear will need to have side shields to be considered safety eyewear.
 3. Should a team member remove his/her eyewear, he/she will be reminded once to replace it.
 4. Sunglasses are not suitable eyewear.

EVALUATION

- A. Each team's solution is evaluated objectively.
- B. A finite measure, such as elapsed time, horizontal or vertical distance, and/or strength, is used to determine the best solution.
- C. Ties shall be broken according to the entry with the earlier testing time given the advantage

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer network specialist
- Detective
- Mechanical engineer
- Nurse
- Project manager

PROBLEM SOLVING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Tool box is present and meets size requirements
- ☐ Safety glasses are worn by each team member
- ☐ ENTRY NOT EVALUATED

TESTING OF SOLUTION (60 points)					
Evaluation: A finite unit of measure, such as elapsed time, linear distance, and/or strength, etc. is used to determine ranking for this event.					
1st: 60 Points	2nd: 55 Points	3rd: 50 Points	4th: 45 Points	5th: 40 Points	6th: 35 Points
7th: 30 Points	8th: 25 Points	9th: 20 Points	10th: 15 Points	11th: 10 Points	12th: 5 Points
TESTING OF SOLUTION SUBTOTAL (60 points)					
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.					
Indicate the rule violated: _____					
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.					TOTAL (60 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

PROBLEM SOLVING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
- C. Timekeeper/monitors, one (1) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Stopwatch for timekeeper
- C. A well-written, technologically appropriate problem that can be objectively measured; one (1) for each team
- D. Adequate conditions (inside or outside), on-site problem materials, monitoring, and testing devices for the designated problem
- E. Tables and chairs for participants
- F. Tables and chairs for event coordinator and judges
- G. Tool Box Verifications Sheet, one (1) copy per team

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

ON-SITE CHALLENGE

- A. Distribute materials as appropriate, prior to the start of the event.
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. All participants and judges should be in the room at this time.
- D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. Each team will submit their tool box to the coordinator and judges for size verification.
- G. Once teams are seated (checked against the entry list) and general announcements have been made, the event problem will be distributed, reviewed, and time will be started.
- H. Judges and monitors observe the entire construction phase, with judges measuring solutions as soon as appropriate.
- I. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- J. If necessary, manage security and the removal of materials from the event area.

PROMOTIONAL MARKETING



OVERVIEW

Applying leadership and 21st century skills, participants create a marketing portfolio observing the theme and required elements, which will be electronically submitted pre-conference. The theme will be posted on the [TSA website](#) under *Themes & Problems*. Semifinalists design a solution to an on-site challenge.

ELIGIBILITY

One (1) individual per chapter may participate.

TIME LIMITS

PRE-CONFERENCE

- A. All components of the chapter's entry (marketing portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.
- B. Entries received, or changes made to submitted entries after this deadline will not be judged.

SEMIFINAL ROUND

- A. Fifteen (15) minutes is allowed to set up.
- B. One (1) hour is allotted to complete, save, and submit the on-site challenge.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme and on the [TSA website](#) under *Themes & Problems*.
- C. Participants concentrate their efforts in creating a three (3)-part marketing portfolio containing:
 - 1. A printable advertisement
 - 2. A wearable design
 - 3. Digital signage

- D. Participants submit the printable advertisement, wearable design, and digital marketing portfolio as a multi-page PDF in the following order:

1. Title page
2. Printable advertisement
3. Supporting documentation for the printable advertisement
4. Wearable design
5. Supporting documentation for the wearable design
6. Digital Signage URL
7. Supporting documentation for the digital signage.
8. Copyright Information

- E. All components of the chapter's entry (marketing portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.

- F. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalists (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program with a laptop computer.
- B. Semifinalists complete the on-site layout and design challenge within the one (1)-hour time limit.
- C. Once semi-finalists complete their solution to the the design challenge, they will save their final entry as a PDF and submit their design solution using the submission procedures provided by the event coordinator.
- D. Once the participant has submitted the solution, the participant may leave and take their computer out of the room.
- E. Judges independently assess the entries without the student member present.
- F. Ten (10) finalists are announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. The required elements are posted on the [TSA website](#) under *Themes & Problems*.
- B. Use of the TSA logo is permitted as long as it adheres to the current TSA Branding Guide.
- C. Pre-conference, participants design the following three (3) components as part of the marketing portfolio based on the theme and requirements posted on the [TSA website](#) under *Themes & Problems*:
 1. Printable signage must be viewable and saved as a PDF.
 2. Participants design a wearable design. This design must be viewable and saved as a PDF.
 3. Participants create digital signage using presentation software. This multimedia signage would be scrolling throughout an event.
 - a. A URL link must be provided for submission, which points directly to the team's entry.
 - b. The URL submission of the digital signage must point directly to the individual's entry. Entries that require a request for access be granted will not be judged.
- D. Supporting documentation must be included in the Promotional Marketing portfolio for each of the three (3) required items. Supporting documentation is limited to three (3) pages per item and must include the following information:
 1. Explanation of how the design meets the challenge criteria
 2. Notation of type of software program(s) used for design and layout
 3. Explanation of the process used to create the item
 4. Notation of all fonts, images, and ideas that are completely original for that design component
 5. Citation of all ideas, fonts, and images from sources other than the designer, and/or that are copyrighted.
- E. Copyright information
 1. Student Copyright Checklist (see 2026 & 2027 Forms Appendix); one (1) page.
 2. Photo/Film/Video Consent and Release forms. If the entry contains images of people (minors require parental consent), proof of consent must be included for each person (see 2026 & 2027 Forms Appendix); pages as needed
- F. The Promotional Marketing solution must be submitted as a single, multi-page PDF file with the file name "Marketing Portfolio" and the individual student ID number. The portfolio should consist of the following components:
 1. Title Page – with the event title, the conference city and state, the individual ID number, and the year; one (1) page
 2. Printable signage design; one (1) page
 3. Printable signage documentation; maximum three (3) pages
 4. Wearable design; one (1) page
 5. Wearable design documentation; maximum three (3) pages
 6. Digital Signage URL; one (1) page
 7. Digital signage documentation; maximum three (3) pages
 8. Student Copyright Checklist (see 2026 & 2027 Forms Appendix); one (1) page
 9. Photo/Film/Video Consent and Release forms, if needed (see 2026 & 2027 Forms Appendix); pages as needed
- G. All components of the chapter's entry must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.

SEMIFINAL ROUND

- A. Semifinalists report at the time and location provided in the conference program to complete the on-site layout and design challenge with-in the one (1) hour time limit.
- B. Participants are required to bring:
 1. One (1) laptop. Laptop must be fully charged with enough battery for the duration of the semifinal round.
 2. One (1) computer mouse (if desired).
 3. Pencils and/or Pens.
 4. Notebook paper or copier paper.
 5. Portable power bank (if desired).
- C. Participants do NOT have access to electrical power/ outlets during the event; however, they may choose to bring a portable power bank.
- D. Clip art may be used for the on-site challenge, but the use of a template is not permitted. Students are responsible for providing their own graphic library.
- E. The on-site work must be an original creation.
- F. Internet access is permitted, but it will not be provided by TSA. TSA cannot guarantee the availability of cellular and/or Wi-Fi signals in the competition room.
- G. Students save the design solution for the on-site problem as a PDF and submit their design solution using the submission procedures provided by the event coordinator.
- H. Participants may leave the room only with permission from the event coordinator.

EVALUATION

PRELIMINARY ROUND

- A. The marketing portfolio

SEMIFINAL ROUND

- A. The design challenge

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Ad copy writer
- Telecommunications manager
- Internal communications manager
- Volunteer manager
- Public affairs specialist

PROMOTIONAL MARKETING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ The PDF of the Marketing Portfolio was submitted
- ☐ ENTRY NOT EVALUATED

PRINT DESIGN (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Layout and Design (X1)	Layout/design does not resemble a promotional printed design, is not the correct size, and/or it is missing essential elements.	Most elements of design are followed; the design is the correct size, and few mistakes are made in the layout.	Printed design encompasses all standardized layout practices, and creativity is at the forefront of the design.	
Content (X1)	Printed design is missing three (3) or more of the following elements: Conference city/state and year, TSA logo, time, place, date and cost of the event.	Printed design is missing one to two (1-2) elements listed in the minimal criteria.	All elements are included in the printed design.	
Effectiveness (X1)	Printed design does not convey intended message, and/or it contains unrelated text/graphics.	Printed design generally conveys the intended message, and contains topic-related text/graphics.	Printed design message is easily understood and interpreted, with exceptional use of related graphics and text.	
Incorporation of Graphic Design Principles (X2)	Design principles (alignment, consistency, contrast, unity, white space) are not incorporated adequately into the printed design, and/or they are considered as an afterthought.	Printed design exhibits incorporation of most design principles (alignment, consistency, contrast, unity, white space), and the overall layout is aesthetically pleasing.	Printed design is aesthetically pleasing, and all design principles are well incorporated into the design and layout.	
Print Design Documentation (X1)	The documentation minimally explains how the design meets the challenge criteria and is missing several of the required components of the design documentation.	The documentation includes and a somewhat explains how the design meets the challenge criteria, and includes most of the following: notation of the software program(s) used, explanation of the process for creating the item, and notation/citation of ideas, fonts, and images.	The documentation includes and a thorough explanation of how the design meets the challenge criteria, notes the software program(s) used, explains the process for creating the item, and notes/cites all ideas, fonts, and images.	
PRINT DESIGN SUBTOTAL (60 points)				

WEARABLE DESIGN (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
First Impression (X1)	The design is sloppy, and/or it is difficult to see; there is poor choice of colors; the artwork is not suited for a wearable item, and/or it leaves an unfavorable impression.	The design has good points, but some details may detract from the overall quality.	The design is eye catching; attention to detail is obvious.	
Dominance (X1)	Eyes are drawn away from what should have been the focal point by some other component of the graphic.	A general attempt is made to use a graphic component that draws attention to the design's main idea.	The design's main components draw eyes to the appropriate location and/or focal point of the graphic.	
Use of Fonts/Words in Design (X1)	Fonts/words are not readable; location or size are not appropriate for the design.	Fonts/words are mostly appropriate, but there is room for improvement.	Fonts/words, their size, and their location are clearly appropriate for the design.	
Wearable Design Documentation (X1)	The documentation minimally explains how the design meets the challenge criteria and is missing several of the required components of the design documentation.	The documentation includes and a somewhat explains how the design meets the challenge criteria, and includes most of the following: notation of the software program(s) used, explanation of the process for creating the item, and notation/citation of ideas, fonts, and images .	The documentation includes and a thorough explanation of how the design meets the challenge criteria, notes the software program(s) used, explains the process for creating the item, and notes/cites all ideas, fonts, and images.	
Incorporation of Graphic Design Principles (X2)	Design principles (alignment, consistency, contrast, unity, white space), are not incorporated adequately into the graphic, and/or they are considered as an afterthought.	Graphic incorporates most design principles (alignment, consistency, contrast, unity, white space), and the overall layout is somewhat aesthetically pleasing.	Graphic is aesthetically pleasing, and all design principles are well incorporated into the design and layout.	
WEARABLE DESIGN SUBTOTAL (60 points)				

DIGITAL SIGNAGE (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Layout and Design (X1)	Signage lacks the use of design principles (alignment, consistency, contrast, unity, white space, color scheme) and presentation formatting (animations, transitions, and timings).	Signage contains some design principles and presentation formatting.	Excellent use of formatting to develop the design and layout of the presentation is evident.	
Audience (X1)	Audience is not considered in the development of the signage; inadequate language is used.	Tone and language are of average quality for the audience.	The signage is written specifically for an audience, with professional tone and language.	
Sentence Structure (X1)	Simple sentence structure is used throughout the signage, and there are multiple grammatical errors.	Writing is generally engaging and informative; only a few grammatical errors are evident.	Signage is well-written with little to no grammatical errors evident.	
Content (X1)	Signage is missing three or more of the required elements.	Signage is missing one or two of the required elements.	All elements are included in the signage.	
Effectiveness (X1)	Signage does not convey intended message appropriately, and/or it contains unrelated text/graphics.	Signage conveys overall intended message, but it contains some inadequate and/or unrelated text/graphics.	The message of the signage is easily understood and interpreted, with exceptional use of related text/graphics.	

DIGITAL SIGNAGE (60 points) – continued				
Digital Signage (X1)	The documentation minimally explains how the design meets the challenge criteria and is missing several of the required components of the design documentation.	The documentation includes and a somewhat explains how the design meets the challenge criteria, and includes most of the following: notation of the software program(s) used, explanation of the process for creating the item, and notation/citation of ideas, fonts, and images.	The documentation includes and a thorough explanation of how the design meets the challenge criteria, notes the software program(s) used, explains the process for creating the item, and notes/cites all ideas, fonts, and images.	
DIGITAL SIGNAGE SUBTOTAL (60 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
---	--

PRELIMINARY SUBTOTAL (180 points)	
-----------------------------------	--

SEMIFINAL CHALLENGE (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Layout and Design (X1)	Layout/design does not incorporate or consider three (3) or more of the following design principles: alignment, consistency, contrast, unity, white space.	Layout/design includes most design principles; overall layout is somewhat aesthetically pleasing.	Aesthetically pleasing design is evident, and all design principles are incorporated into the design and layout.	
Solution to Project (X2)	Project is missing three (3) or more attributes of the solution's criteria.	Most attributes of the solution's criteria are included.	All attributes of the solution's criteria are included.	
Effectiveness (X1)	Project does not convey intended message appropriately, and/or it contains unrelated text/graphics.	Project delivers the overall intended message, and it contains basic graphics.	Project message is easily understood and interpreted, with exceptional use of related graphics and text.	
Originality (X1)	Project does not incorporate or consider four (4) or more of the following principles of creativity: freshness, idea cultivation, realness, bravery, momentum, visual signaling; leadership and/or 21 st century skills are not evident.	Project incorporates most creative principles, and results in an adequate/average presentation; leadership and/or 21 st century skills are somewhat evident.	Project is a truly unique presentation; it includes most of the applicable principles of creativity; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL CHALLENGE SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
---	--

SEMIFINAL SUBTOTAL (50 points)	
--------------------------------	--

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.

TOTAL (230 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

PROMOTIONAL MARKETING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. On-site problem for semifinalists, twelve (12) copies
- C. Tables and chairs for the judges and coordinator
- D. Tables, one (1) per participant
- E. Chairs, one (1) per participant

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges
- B. Review entries as they are submitted to the designated online storage utility
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Results are posted on-site at the national conference on the first full day.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.

- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. On the first full day of the conference, post a list of the twelve (12) semifinalists in random order.

SEMIFINAL ROUND

- A. At the designated time, check-in the semifinalists and monitor them as they set up their work area.
- B. Provide the on-site problem.
- C. Supervise the one (1) hour on-site layout and design problem.
- D. Manage the evaluation process of the on-site problem.
- E. Judges independently review each entry with neither students nor advisors present.
- F. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- G. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants design, build, assemble, document, and test an open-source vehicle according to stated specifications and to meet the challenge of the yearly theme/problem. The annual theme will be posted on the [TSA website](#) under *Themes & Problems*.

ELIGIBILITY

Three (3) teams of two to six (2-6) members per state may participate.

TIME LIMITS

- A. Ten (10) minutes prior to assigned times teams can set up their assigned pit area.
- B. Thirty (30) minutes session to test and correct any problems. During this time judges will also perform a safety check.
- C. Ten (10) minutes running clock to complete the challenge.
- D. Ten (10) minutes for semifinalist interview

ATTIRE

TSA competition attire, safety glasses, and safety vests are required for this event.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme on the [TSA website](#) under *Themes & Problems*.
- C. Participants prepare their design, build, and test their robots.
- D. Drones are not permitted in this event.

PRELIMINARY ROUND

- A. Participants will sign up for a setup and testing time at the time and place stated in the conference program.
- B. While wearing safety glasses and safety vests, participants attend a fifteen (15)-minute orientation meeting at the time and place stated in the conference program.
- C. Students will arrive at the assigned place and time to:
 - 1. Set up their pit areas
 - 2. Set up their robot
- D. Entries are reviewed by judges to determine safety.
- E. Safe robots will be given an opportunity to test.
- F. Top sixteen (16) scores on robot testing will have a semifinalist interview
- G. A list of sixteen (16) semifinalists (in random order) is posted.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program to sign up for an interview time.
- B. Participants report at the assigned time and place for the ten (10) minutes interview.
- C. Challenge, and interviews scores are combined with testing points to determine the final standings.
- D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. Participants must check for the current year's design challenge specifications on the [TSA website](#) under *Themes & Problems*.
- B. Participants must wear safety glasses and safety vests at all times during the event.

Robotics Pit and Safety Procedures

- A. Pit Area Assignment. The Event Coordinator will provide a designated area for robotics teams to work on and prepare their robots for the challenge.
 1. If the team brings any type of powered tool or soldering iron, the teams are required to bring two (2) welding blankets (4' x 6' minimum) to cover the table and floor in their pit area. In this case, failure to bring the two (2) welding blankets will result in an automatic disqualification.
 2. Teams bring to pit area, for inspection, robot, radio controller(s), chargers, batteries, tools box, power strip, 3-prong electrical extension cord, replacement parts, spare parts, and tools.
 3. All necessary computers and associated software for the competition.
 4. All equipment, tools, chargers, and computers are to be arranged for inspection and safety check. The use of tools with combustible fuel sources is prohibited.
 5. In the pit area, battery chargers and batteries, as they are being charged, must be placed on the fireproof welding blanket in the pit charging area.
- B. When the Robot is out of the competition area, all batteries must be removed. NO EXCEPTIONS.
- C. When a team member enters the competition tent field, only at the direction of the event coordinator may the team members attach the battery cable and turn on their robot. When a robot is outside of the competition tent area, all batteries must be unplugged. NO EXCEPTIONS.
- D. The judge will inspect the robot to ensure safe operation.
- E. When the competition is taking place and when a practice session is under way with a robot in the competition area, all robots in the pit area or outside the pit area must be POWERED OFF. This is an automatic ten (10) point deduction if this occurs.
- F. All batteries will be inspected prior to practice and the competition.
- G. All robots must be driven ONLY in the Competition field.

Robot Specifications

- A. The robot must be assembled from open-sourced parts. The robot can be purchased as a kit that can be built, reconfigured, changed, and modified with different components.
- B. The robot structure can be made from plastic, metal, wood, 3D printed materials (carbon fiber, PLA plastic, ABS plastic, resin, metal combined plastic or resin). Parts can be purchased commercially and modified.
- C. Battery packs must only be commercially available lithium-ion or lithium-ion polymer batteries that are purchased from open-sourced 3rd parties (i.e., Amazon, hobby shops, etc.).
- D. The robot will have to navigate through a level course. Depending on the location the surface may change so designs must take that into consideration. Teams may choose to use legs, tracks, wheels, or other methods developed by the team.
- E. At the start of the challenge robots must fit in a 16" x 16" x 16" cube. Once the challenge has begun the robot can exceed that size.
- F. Robots must have removable part/coverings for internal inspection.
- G. A camera may be incorporated into the design if needed to help complete the assigned task.

EVALUATION

PRELIMINARY ROUND

- A. Robot testing

SEMINFINAL ROUND

- A. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability.

ROBOTICS

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Robot is present
☐ ENTRY NOT EVALUATED

PRELIMINARY TESTING OF ROBOT (60 points)

Evaluation: A finite unit of measure, such as elapsed time, linear distance, and/or strength, etc., is used to determine ranking.

1st: 60 Points	2nd: 55 Points	3rd: 50 Points	4th: 45 Points	5th: 40 Points	6th: 35 Points
7th: 30 Points	8th: 25 Points	9th: 20 Points	10th: 15 Points	11th: 10 Points	12th-16th: 5 Points

PRELIMINARY TESTING OF ROBOT SUBTOTAL (60 points)

Rules violations (a deduction of 20% of the total possible points in the semifinalist section) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (60 points)

SEMIFINAL INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit a general understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the concepts in their project.	
Articulation (X1)	Communication of the project is unclear, unorganized, and/or illogical; leadership and/or 21 st century skills are not evident.	Communication of the project is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the project is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Team Participation (X1)	The majority of the delivery is made by one (1) member of the team; the partners may be disengaged from the Interview.	Team members are generally engaged in the process, though one member may take on more responsibility than the others.	Team members are actively involved in the Interview and responses to interview questions; there is shared responsibility on the part of team members.	
SEMIFINAL INTERVIEW SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points in the semifinalist section) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (40 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (100 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ROBOTICS

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more
- C. Pit Area Judge/Inspector (1)
- D. Assistants, two (2)

MATERIALS

- A. Coordinator's packet and box, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for entries, as needed
- B. Time trial record sheet
- C. Qualifier interview time slot sheet
- D. Interview questions
- E. Testing Arena- 1" PVC pipe setup to the year's theme
- F. Course materials based on theme
- G. Table with power for the Pit Area for the competitors
- H. Table for inspection and tabulation
- I. 2-Step ladder with platform for judging
- J. Stations or caution tape to separate spectators and participants
- K. Safety glasses and vests for judges and coordinators
- L. Blue and red tamper proof tape for indicating frequency of controllers
- M. Fire extinguisher

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.

- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Participants report to the time and place stated in the conference program and check in and sign up for setup/testing time
- B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
- C. In order to compete, participants must be on the entry list or must have approval of the CRC.
- D. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
- E. Secure the entries in the designated area.

PRELIMINARY ROUND

- A. Assist judges with the check in/setup of pit area.
- B. Assist judges with the Robot of the event.
- C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

- D. Begin the time trials at the scheduled time.
 - 1. Every Robot that is compliant with rules should have the opportunity to be tested.
 - 2. Public viewing is allowed.
 - 3. Announce at starting time: The challenges must be completed in the allotted ten (10)-minute window. If necessary, repairs or adjustments to the robot or its components can be made, but the clock will not stop and no additional time will be awarded to complete the challenges.
- E. Position a judge on either side of the testing area to view the trial.
- F. If a time is not properly recorded, a rerun may be ordered at the discretion of the event coordinator.
- G. Record preliminary times on a time trial record sheet.
- H. The testing of the robot times will determine the top sixteen (16) results.
- I. Create the semifinalist sign-up sheet.

SEMIFINAL ROUND

- A. Post the top sixteen (16) teams with interviews times.
- B. Robot builders report to the designated area posted time for a ten (10)-minute Robot Team interview.
- C. Conduct interviews with the qualifying top sixteen (16) Robot Teams.
- D. Begin the semifinals at the scheduled time.
- E. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
- F. If necessary, manage the security and removal of materials from the event area.

OVERVIEW

Solar Racer provides a hands-on opportunity for students to apply science, technology, engineering, and mathematics (STEM) concepts, along with leadership and 21st century skills such as creativity, teamwork, and problem-solving skills, as they design, construct, and race a solar-powered car.

ELIGIBILITY

One (1) team of two to four (2-4) individuals per chapter may participate; one (1) entry per team.

TIME LIMITS

PRELIMINARY ROUND

- A. All components of the chapter's entry turned in at the place and time stated in the conference program.
- B. All models meeting safety and performance criteria are given up to two (2) time trials.

SEMIFINAL ROUND

- A. Five (5) minutes are allowed for the semifinal interview.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants design and create their solar-powered car while working within the required specifications.
- C. Participants prepare and record their design processes on a showcase display that includes their solar car model.
- D. The chapter will create a car name (the name cannot identify the school or chapter). The car name will be incorporated into the theme of the showcase display.

- E. The team will create video evidence of the solution using solar power, but the preliminary round will take place indoors during the conference.

PRELIMINARY ROUND

- A. Participants report to the time and place stated in the conference program and check in:
 - 1. A solar-powered model car
 - 2. The showcase display
- B. Entries are reviewed by judges to determine specification adherence and safety on the track.
- C. Solutions that are disqualified for any reason will not be permitted to participate in time trials or the preliminary round head-to-head races.
- D. All models meeting safety and performance criteria will be given up to two (2) time trials.
- E. During the preliminary round, the following procedures apply.
 - 1. The fastest times from the time trials will determine the top twenty-four (24) teams. A list of the twenty-four (24) teams is posted on-site.
 - 2. Judges score the top twenty-four (24) showcase displays and models. The top twenty-four (24) teams will complete a single elimination head-to-head race bracket to score points.
- F. The results of the showcase display and model are combined with scores from the single elimination head-to-head race to determine the semifinalists.
- G. A list of twelve (12) semifinalists will be posted.

SEMIFINAL ROUND

- A. The semifinalist interview must include a minimum of two (2) team members.
- B. Five (5) minutes are allowed for the semifinal interview.
- C. The combined score of the showcase display, model, head-to-head bracket, and the interview will determine the top ten (10) finalists.
- D. Ten (10) finalists will be announced during the conference award ceremony.

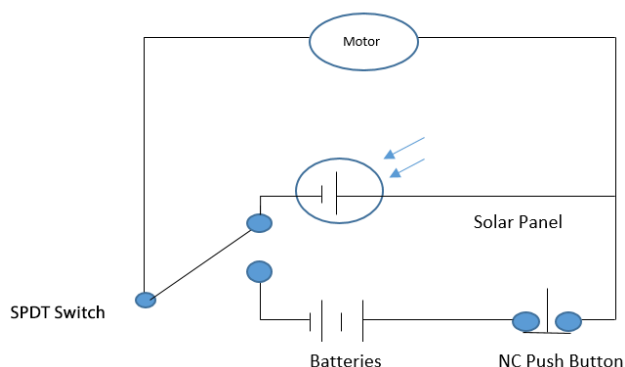
REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. The team must create a showcase display.
- B. The size of the showcase display (including the solar car model) may not exceed 15" deep x 30" wide x 48" high.
- C. The showcase display must include the following components:
 1. Car name (the name cannot identify the school or chapter).
 2. Engineering Design Process. Each step clearly identified. Design process description, including pre-testing notes of various configurations of the model and revision notes about the model design throughout the process.
 3. Design drawings:
 - a. The drawings must show the model with a minimum of two (2) views.
 - b. The drawings must be developed using standard engineering practices and procedures (including measurements/ dimensions).
 - c. The drawings may be produced using traditional drafting methods or CAD.
 4. Components list (see Supplied Components List worksheet).
 5. Run Log (see Run Log form) with times using solar power or battery power on a 20-meter track. The minimum number of runs is five (5).
 6. QR code with a link to three (3) videos of your car using solely solar power to travel 20 meters. Must show that batteries are not in place for the three (3) runs. The maximum length of each video is ninety (90) seconds. The judges should be able to scan the QR code with a smartphone. Viewing cannot require any downloads or require any permissions.
- D. Solar-Powered Model Car:
 1. The materials used to construct the model car must cost less than \$50. The \$50 does not include the cost of the panel.
 2. Original receipts for all materials purchased must be recorded in the Supplied Components List.
 3. If using recycled materials, documentation must show how these items were obtained.
 4. Recycled materials are not included in the \$50 maximum.
 5. Model cars that exceed the \$50 construction cost limit will be disqualified from the competition.
 6. The vehicle must be structurally sound without the solar panel attached.
- E. Solar Paneling:
 1. One (1) solar panel (limited to a maximum output of 3.2 W), and one (1) motor (limited to a maximum 3.0 VDC) are allowed per car
 2. Solar panels cannot be shaved, drilled, or delaminated.
 3. Only the motor supplied in the kit can be used.
 4. Motors cannot be re-wound or disassembled.
 5. If an evaluation group convened by the event coordinator determines that the solar panel and/or motor have been modified, the car and team will be disqualified from the competition.
 6. The solar panel cannot be used as the chassis, or body, of the car.
 7. The axles and wheels cannot be directly attached to the solar panel.
 8. Reflectors, supports, and power leads can be added to these components as needed, but they must fit within the required dimensions.
 9. The model car must, with the solar panel attached, not exceed the following dimensions:
 - a. 60 cm (23 $\frac{5}{8}$ inches) length
 - b. 30 cm (11 $\frac{13}{16}$ inches) width
 - c. 30 cm (11 $\frac{3}{4}$ inches) height (as measured from the surface the car is resting upon to the highest point of the car, with all its components attached)

Wiring Diagram



10. Each vehicle must include a mounted battery holder that is capable of holding two AA batteries. This system needs a switch and button as illustrated. This is to be used at the starting line as follows:
 - a. on the “READY” command the button is depressed and held, this opens the circuit.
 - b. on the “SET” command the switch is flipped, to connect the batteries to the motor.
 - c. on the “GO” command the button is released, thereby closing the circuit, sending power to the motor, causing the vehicle to move.
 11. The team is encouraged to decorate the body of the car, but a clearly visible 3cm square space must be available on the car to display the team ID#.
 - a. All onsite racing will be completed indoors using two (2) provided AA batteries
- F. A student-designed attachment device must be part of the car to accommodate the easy attachment and removal from a guide wire for steering. A purchased screw eye or eye bolt is not considered a student designed attachment device.
1. A guide wire, such as fishing line, will be no more than 1.5 cm ($\frac{5}{8}$ ") from the surface of the track.
 2. The wire will go through the attachment device on the car and serve as a steering mechanism to keep the car in its lane. **This must be done without disconnecting the guide wire.**

3. Both ends of the guide wire will be fixed to the track. This is the only allowable method of steering the car.
4. No radio control is permitted in the car.
5. Lane changing or lane crossing will result in a Did Not Finish (DNF) standing.
6. A car's race that is impacted by an out-of-control vehicle will be allowed an opportunity to run the race again.
7. A car that lacks steering control and interferes with other cars in other lanes will not be allowed to race again.

- G. If a car is deemed unsafe, it will not be allowed to run in the time trials or the semifinalist races.
- H. If the model is safe, but does not meet the required specifications, it will be allowed to run in the time trials but not the semifinalist races.
- I. The remainder of the vehicle can be innovative in design and materials.

PRELIMINARY ROUND

Time Trials

- A. The race lane must be 60cm wide and 20m long.
- B. The track will be a hard flat surface indoors such as convention center concrete. The floor surface for the national TSA conference will be posted on the [TSA website](#) under *Themes & Problems*.
- C. All on-site racing will be conducted using provided batteries however your solar panel must still be in place.
- D. The time trial/race specifications are as follows:
 1. Tables will be set up for teams to make adjustments and minor repairs to cars prior to each time trial and the semifinalist heats.
 - a. Teams that are “next up” to be timed or raced are given priority to use the tables.
 - b. Teams must supply their own tools.

2. Time trials and semifinalist races will not be delayed to permit adjustments or repairs to cars. If a repair is needed during time trials, a three (3)-minute time limit for repairs will be permitted.
3. Prior to semifinals, teams will have an opportunity to perform up to two (2) trial races during a practice run session.
4. At race time, each car will be placed with the most forward part of the vehicle set even with the starting line and all of its wheels in contact with the ground.
5. No more than two (2) team representatives will be allowed in the race area; one at the starting line, one to catch.
6. All cars will be started when the official signal is given.
 - a. Each car will have up to two (2) time trials, unless otherwise determined by the event coordinator.
 - b. The fastest time recorded will determine the twenty four (24) cars to race in the head-to-head single elimination tournament and display judging portion.
 - c. If, for any reason, a car is not able to participate in the time trials, or race at its scheduled time, it may be disqualified.
7. The judges will note the official time for each time trial.
 - a. At the time designated, if a car does not start the time trial, OR if during the time trial it does not finish, it will be noted as a Did Not Finish (DNF).
 - b. If a car has a false start, the entry will be given one (1) more opportunity to race.
8. One (1) team member must wait at the finish line to catch the vehicle for each timed trial. Team members are responsible for finding someone to catch their vehicle if another team member is unavailable.
9. After each timed trial or race, the vehicle and team member must remain at the finish line until the time is recorded for the vehicle.
10. No one, including team members and spectators, may accompany or touch the vehicle on the track during a timed trial or semifinalist race.
 - a. Vehicles stalled on the track can be retrieved after the end of the trial or the race has been declared by the lead judge.
 - b. A violation of this rule will result in disqualification of the offending team.
11. Challenges must be made before the next timed trial or race begins.
 - a. Any challenges must come from team members who are actively competing, not the coach/advisor, parent, or coordinator.
 - b. Any challenges need to be directed to the lead judge.
 - c. The decisions of the judges regarding challenges are final.
12. Only competing students and race officials may be in the race area.
 - a. Spectators, including coaches/advisors, parents, coordinators, and non-competing students, must remain in the designated spectator area throughout the duration of races.
 - b. Teams will be disqualified if a spectator, including a coach/advisor or parent, interferes with a race. This includes a coach/advisor or parent helping team members get their car on/off the guide wire.
13. Judges may inspect cars at any time before, during, and after timed trials or semifinalist races.
14. Any additional rules, regulations, or guidelines established by the event coordinator must be followed.

SEMIFINAL ROUND

- A. A team representative will report at the time and location printed in the conference program for an interview time.
- B. A minimum of two (2) team members must be present for the interview.
- C. Participants will participate in a five (5) minute interview to answer questions related to solar power and the manufacturing process of the solar car model.
- D. All teams will report at the time and location printed in the conference program to pick up the entries.

EVALUATION

PRELIMINARY ROUND

Tier 1

- A. Time trial results

Tier 2

- B. Showcase display and model
- C. Head-to-head race results

SEMIFINAL ROUND

- A. Semifinalist interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability.

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Energy efficiency technician
- Mechanical engineer
- Solar engineer
- Solar panel installer
- Solar sales consultant

SUPPLIED COMPONENTS LIST

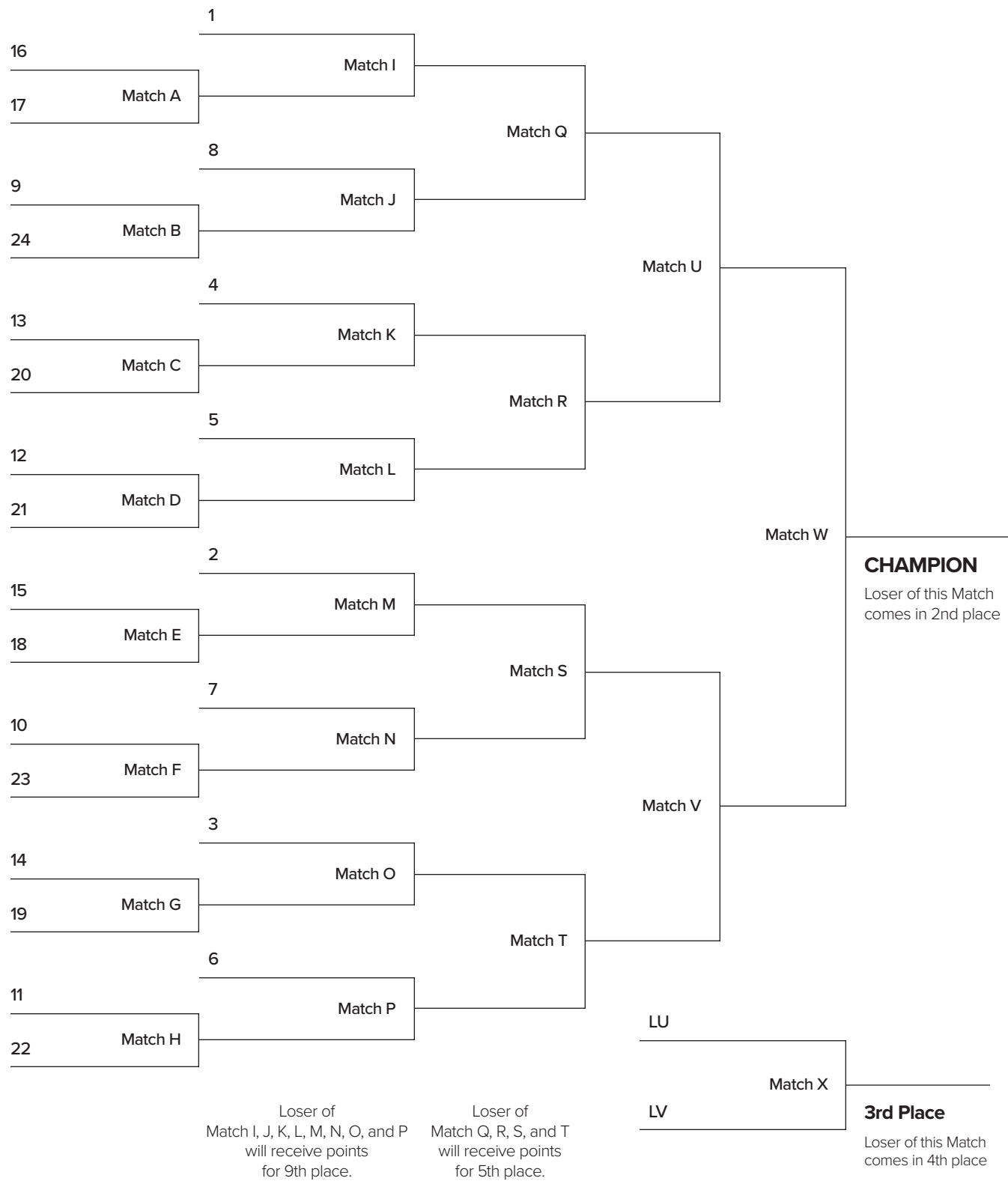
[illegible]

RUN LOG

RACE CAR NAME:			
Race #	Time	Battery or Solar	Notes
#1			
#2			
#3			
#4			
#5			
#6			
#7			
#8			
#9			
#10			

HEAD-TO-HEAD BRACKET

Tournament Bracket Seeded 24 Team Field



SOLAR RACER

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Solar Car Model Present with student made attachment device
- ☐ TIER 2– Showcase display and QR Code with operational video link
- ☐ ENTRY NOT EVALUATED

TIER 1 – TIME TRIALS		
Fastest Duration Time	Seconds	
TIER 1 – TIME TRIALS TOTAL (Seconds)		

Rules violations (a deduction of 20% of the total combined time trials for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (Seconds)	
-----------------------------	--

TIER 2 – SHOWCASE DISPLAY AND MODEL (130 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Display Components (X1)	A number of display components are missing.	Most of the display components are included, but the quality is minimal.	The display includes all required components in order; it is neat and properly organized; effort and quality are evident.	
Engineering Design Process (X2)	The Engineering Design Process is not defined or is lacking significant portions; it is messy and demonstrates lack of effort.	The Engineering Design Process is acceptable, with most information included.	The Engineering Design Process is complete and accurate; the information is neat and orderly; a great deal of effort is evident.	
Design Drawings (X1)	Some drawings are missing and/or drawings are of poor quality.	Drawings are acceptable; all required views are shown.	Drawings are accurate and complete; all required views are present; rough sketches are included.	

TIER 2 – SHOWCASE DISPLAY AND MODEL (130 points) – continued					
Design Details/ Components List (X1)	Several details of the model, such as model size, wheel size, and gear ratio are missing and/or are poor; the components list is very limited.	Most details of the model, such as model size, wheel size, and gear ratio are included; most components are included.	All details of the model, such as model size, wheel size, and gear ratio are present; all components are included.		
Run Log (X1)	No Run Log created or less than five (5) runs recorded.	Run Log created with multiple runs but little detail related to performance and adjustments.	Run Log included with multiple runs and detailed information about performance, variables and adjustments.		
Videos using QR Code (X1)	The QR code is not included or the videos are not accessible.	The QR code is present and one (1) to two (2) videos are provided as evidence of model car running on solar power alone.	The QR code is present and three (3) videos are provided as evidence of model car running on solar power alone.		
Model Design (X2)	The design of the solar model is poor and shows little effort.	The design of the solar model is adequate but not of exceptional quality.	The design of the solar model exhibits exceptional quality.		
Model Creativity/ Originality (X2)	The solar model car design lacks creativity and originality; little effort is apparent; car is an exact, or nearly an exact replica of purchased kit.	The solar model car design demonstrates an adequate level of creativity and originality; at least one (1) modification has been made to the car.	The solar model car design shows exceptional creativity, originality, artisanship, and engineering.		
Model Construction (X2)	The solar model car lacks quality of construction.	The solar model car demonstrates adequate quality of construction.	The solar model car demonstrates exceptional quality of construction.		
TIER 2 – SHOWCASE DISPLAY AND MODEL SUBTOTAL (130 points)					

TIER 2 – HEAD-TO-HEAD RACE (80 points)							
Points are assigned based on results from double elimination bracket.							
1st	2nd	3rd & 4th	5th – 8th	9th – 16th	17th – 24th		
80 Points	75 Points	70 Points	60 Points	50 Points	40 Points		
SEMIFINAL RACE SUBTOTAL (80 points)							

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

PRELIMINARY TOTAL (210 points)	
---------------------------------------	--

SEMIFINAL CAR BUILDER INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Car Builder Interview (X2)	The team shows very limited knowledge of (and has difficulty articulating) how the car was produced or decisions made during the production; the student exhibits a basic understanding of design elements and functionality, and the rationale is inconsistent or absent.	The team demonstrates some knowledge of the vehicle production and has adequate knowledge of some processes or reasoning behind the vehicle design.	The team shows competence and knowledge related to the design and production of the vehicle; the student is able to articulate "reasoning" behind the decisions made.	
Articulation (X2)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL CAR BUILDER INTERVIEW SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL TOTAL (40 points)				
To arrive at the FINAL TOTAL score, subtract rules violation points, as necessary.				
TOTAL (250 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

SOLAR RACER

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more for preliminary round
- C. Judges, two (2) or more for semifinal round (can be the same judges as preliminary round)
- D. Assistants, six (6) or more

MATERIALS

- A. Coordinator's packet containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick-on labels for identifying entries
- B. Race bracket form
- C. Batteries (AA 1.5 V), two (2) per entry plus spares on-site
- D. Braided fishing line for the track:
 - 1. Four (4) pre-tied
 - 2. Two (2) on track
- E. Race track set, including a starting gate and finish gate with digital timer
- F. Spare stopwatches for back-ups
- G. Padding for the finish gate
- H. Tables for the display and evaluation of entries (cars and portfolios)
- I. Lane Assignment Board to be used for a display of semifinals racing
- J. Tables and chairs for event coordinator, judges, and official assistants
- K. A large display for the final twenty-four (24) bracket
- L. A gauge to measure line height at the beginning and end of the line

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event will be held for appropriate set-up, including location for displays and the evaluation of portfolios, racing site, chairs, tables, outlets, etc.
- E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Participants report to the time and place stated in the conference program and check in:
 - 1. The solar-powered model car and decorated shoebox
- B. Secure the entries in the designated area.
- C. Requirements for attire do NOT apply during check-in, only on the first day of the conference.
- D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. Position the displays and models for viewing by the judges, and assist them as necessary during judging.
- G. Set up the race track prior to the time trials. Make necessary adjustments.
- H. Permit all vehicles (that can be safely operated) to participate in time trials.

- I. Vehicles that are disqualified will NOT be permitted to participate in the time trials or head-to-head races.
- J. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- K. Judges evaluate the showcase display and models of the top twenty-four (24) entries.
- L. Complete the head-to-head races using the single elimination bracket at the time and location printed in the conference program.
- M. Only the twenty four (24) qualifying cars will race.
- N. Public viewing is allowed.
- O. If necessary, manage the security and removal of materials from the event area.

SEMIFINAL ROUND

- A. Semifinalists report to sign-up for the interview at the time and place stated in the conference program. Instruct the teams to arrive at least ten (10) minutes prior to their interview time.
- B. Check in semifinalists and confirm with the teams their order of interviews and the procedure. Team representatives should report at least ten (10) minutes prior to their assigned time to the designated place for their interview.
- C. Manage the interviews.
- D. Discuss rule violations and have all relevant parties initial the rating form.
- E. Judges independently assess the semifinal interviews.
- F. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants use computer graphics tools and design processes (e.g., animation) to communicate, inform, analyze and/or illustrate a topic, idea, subject, or concept that focuses on one (1) or more of the STEM (science, technology, engineering, or mathematics) areas. Sound may accompany graphic images. Participants will find the current year's theme posted on the [TSA website](#) under *Themes & Problems* for this information. Semifinalists make a presentation.

ELIGIBILITY

Three (3) teams of at least two (2) individuals per state may participate.

TIME LIMITS

PRELIMINARY ROUND

- A. All components of the chapter's entry (documentation portfolio PDF and animation URL) must be finished, submitted, and accessible via the Internet by 11:59 p.m. EST on a designated date in mid-May.
- B. Entries received or changes made to submitted entries after this deadline will not be judged.
- C. The animation is not to exceed three (3) minutes in length. There is no minimum time requirement.
- D. There will be a three (3)-point deduction for each fifteen (15) seconds, or fraction thereof, over the three (3)-minute length.
- E. The animation time is calculated from the start of the first image or sound to the end of the last image or sound.

SEMIFINAL ROUND

- A. Ten (10) minutes is allowed to make a presentation to the judges.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the annual theme posted on the [TSA website](#) under *Themes & Problems*.
- C. Participants design and create a STEM animation video and documentation portfolio.
- D. Participants submit the documentation portfolio PDF and animation URL by 11:59 p.m. EST on a designated date in mid-May.
- E. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalists (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Semifinalist team representative reports at the time and location provided in the conference program to sign-up for a presentation time.
- B. No more than two (2) representatives from each semifinalist team may report to the event area for the presentation at the time and place stated in the conference program.
- C. Each semifinalist team explains its portfolio to the judges, discussing the purpose, value, research and design, and development process of its work.
- D. Ten (10) finalists will be announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND**A. Documentation Portfolio:**

1. Documentation portfolio is required and must be submitted as a single, multi-page PDF document with pages in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team ID number; one (1) page
 - b. Table of contents; pages as needed
 - c. Purpose of animation; one (1) page
 - d. Images of a hand-sketched storyboard (no digital screen captures) that documents the flow and progression of the animation, with written notes; special effects, audio cues, dialogue, transitions, and scene duration should be incorporated into the storyboard; pages as needed
 - e. Written description of what the animation illustrates or demonstrates; one (1) page
 - f. List of references that includes sources for materials, copyrighted and otherwise; the term "Fair Use" and similar terms are not acceptable citations when creating the list of references; pages as needed
 - g. Permission letters for copyrighted material; pages as needed
 - h. List of software and hardware used in the development of the animation; one (1) page
 - i. Work Log that indicates preparation for the event, as noted by date, task, time involved, team member responsible, and comments (see 2026 & 2027 Forms Appendix); pages as needed
 - j. Student Copyright Checklist (see 2026 & 2027 Forms Appendix); one (1) page

B. Animation:

1. The animation must be submitted in a common video format suitable for viewing with a VLC player, utilizing a Microsoft Windows operating system.
2. The URL must point to the main page of the team's entry. Entries that require a request for access to be granted will not be judged.
3. Each animation must advance automatically once it has been opened and started by judges.
4. The animation is not to exceed three (3) minutes in length.
5. There will be a three (3)-point deduction for each fifteen (15) seconds, or fraction thereof, over the three (3)-minute maximum length. For example: An animation that runs 37 seconds beyond the three (3)-minute limit will receive a deduction of nine (9) points.
6. Sound may accompany the animation, but it is not required.
7. All graphics, images (including background), and audio clips must be originally created by the participants.
8. Where applicable, all ideas, text, images, and sound from other sources must be cited.
9. If copyrighted material is used, proper written permission must be included in the documentation portfolio (see Student Copyright Checklist in 2026 & 2027 Forms Appendix).
10. Absolutely no purchased content may be used in any part of the animation. (Purchased content includes, but is not limited to, texture, models, and royalty free music.)
11. Purchased items from sites like PowToons and Animaker are not acceptable for this event.

EVALUATION**PRELIMINARY ROUND**

- A. The quality of the portfolio and the animation

SEMIFINAL ROUND

- A. The presentation

Refer to the official rating form for more information.

RESOURCES

- Khan Academy – Pixar in a Box: Free courses on storytelling, animation basics, and the math / science behind animation. [khanacademy.org](https://www.khanacademy.org)
- AnimateSpace: Free online tutorials for beginners focusing on animation basics. animate.space

In addition, there are multiple free online tools to use to develop your animations, should you choose to use them.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer animator
- Game designer
- Instructional technologist
- Software engineer

STEM ANIMATION

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Portfolio was submitted prior to conference
- ☐ Animation was submitted prior to conference
- ☐ ENTRY NOT EVALUATED

DOCUMENTATION PORTFOLIO (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Digital Portfolio (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Purpose and Description (X1)	The purpose and description of the animation idea are unclear.	The purpose and description of the animation are explained appropriately.	The purpose and description of the animation are clear and concisely written.	
Storyboard (X1)	The hand-sketched storyboard is sloppy, seems to have been thrown together after the creation of the animation, and/or it does not correlate with the animation.	The storyboard is drawn appropriately and largely correlates with the completed animation.	The storyboard is of exceptional aesthetic and artistic value and clearly correlates with the animation.	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
DOCUMENTATION PORTFOLIO SUBTOTAL (40 points)				

ANIMATION (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Communication of Animation (X1)	It is difficult to understand the idea being communicated; an illogical explanation is presented.	The idea is communicated, and thoughts are somewhat organized and/or concise.	The idea is communicated in an organized, clear, and concise manner.	
Creativity (X1)	The animation lacks creativity; no, or very few, design principles are integrated in the animation.	Some elements of creativity are expressed, and essential design principles are used somewhat effectively.	The animation exudes creativity; essential design principles and elements are integrated.	

ANIMATION (50 points) – continued				
Aesthetics and Artisanship (X1)	Unorganized, sloppy work is evident; the animation seems to be an afterthought and/or thrown together.	An organized presentation of essential issues in a logical format is evident.	An exemplary use of layout and design principles to logically communicate important data is evident.	
Graphical Representations (X1)	Graphical representations do not help to clarify the idea, or they are of little significance to the idea.	Graphical representations are appropriate and help supplement the idea by providing some clarity.	Graphical representations are of excellent quality and completely clarify the idea.	
Originality (X1)	The animation lacks imagination, originality, and artistic detail.	The animation is effective and innovative, conveying some depth.	The animation is inspiring, inventive, resourceful, and completely motivating.	
ANIMATION SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
--	--

TIME DEDUCTIONS		
A three (3)-point deduction will be incurred for each fifteen (15) seconds, or fraction thereof, over the three (3)-minute animation maximum length.		Record scores in the column spaces below.
TOTAL ANIMATION TIME		
NUMBER OF TIME INTERVAL DEDUCTIONS		
TOTAL TIME DEDUCTIONS		

PRELIMINARY SUBTOTAL (90 points)	
---	--

SEMIFINAL PRESENTATION (60 points)				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participants seem unorganized and unprepared for the presentation; illogical explanation of the idea is presented.	Participants are generally prepared for the presentation; an explanation of idea is communicated and generally organized.	The presentation is logical, well organized, and easy to follow; the idea is communicated in an organized and concise manner.	
Articulation (X1)	The presentation is full of illogical thoughts that lack clarity, and/or there is insufficient information provided describing the project.	The presentation is somewhat logical, easy to follow, and/or there is sufficient information provided describing the project.	The presentation is clear, concise, and there is ample information provided describing the project.	

SEMIFINAL PRESENTATION (60 points) – continued			
Delivery (X1)	The team is verbose and/or uncertain in its presentation; participants' posture, gestures, and lack of eye contact diminish the presentation.	The team is somewhat well-spoken and clear in its presentation; participants' posture, gestures, and eye contact are acceptable in the presentation.	The team is well-spoken and distinct in its presentation; participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation.
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project.	Participants exhibit an understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of their project.
Team Participation (X1)	The majority of the delivery is made by one member of the team; the partner may be disengaged from the presentation.	Both team members generally are engaged in the process, though one member may take on more responsibility than the other.	Both team members are actively involved in the presentation and responses to any questions; there is shared responsibility between team members.
SEMIFINAL PRESENTATION SUBTOTAL (60 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
SEMIFINAL SUBTOTAL (60 points)			
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (150 points)			

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

STEM ANIMATION

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Tables and chairs for event coordinator, judges, and participants

RESPONSIBILITIES

PRE-CONFERENCE/PRELIMINARY ROUND

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the evaluation of entries is to begin, meet with your judges and check in personnel to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the CRC event manager before the evaluation begins.
- F. Ensure the judges have access to the online judging system.

SEMIFINAL ROUND

- A. Semifinalists report to the event area at the time and place stated in the conference program.
- B. Each semifinalist team signs up for a time to present its presentation.
- C. During the presentation, semifinalist team representatives will explain their work and answer any questions the judges may ask.
- D. Judges independently assess the twelve (12) semifinalist teams.
- E. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- F. If necessary, manage security and the removal of equipment and materials from the area.

OVERVIEW

Participants apply leadership and 21st century skills by collaborating with their team members to build a designated structure. Teams apply the principles of structural design and engineering through research, design, construction, destructive testing, and assessment to determine the design efficiency of a structure. Details about the structure and information related to it will be posted as a design brief on the [TSA website](#) under *Themes & Problems*. The on-site semifinalist problem is a variation of the pre-conference design brief posted on the TSA website.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

SAFETY EYEWEAR

- A. Participants are required to wear safety-approved eyewear during the on-site phase of this event.
- B. Prescription eyewear needs to have side shields to be considered safety eyewear.
- C. Should a team member remove the eyewear and fail to replace it, s/he will be reminded once.
- D. If there is a second infraction, the team will be asked to leave the competition.
- E. Sunglasses are not suitable.

TIME LIMITS

- A. On-site structures (semifinalist teams only) must be started, completed, and checked in during the three (3) hours allowed for design and construction.
- B. Semifinalist participants with time conflicts must present a written explanation of the conflict to the event coordinator at least one (1) hour before the construction time stated in the conference program. Work must begin during the time scheduled for the event.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the design brief on the [TSA website](#) under *Themes & Problems*.
- C. Participants conduct research and apply principles of structural design and engineering to their current structure.
- D. Pre-built structures must be started and completed during the current school year.
- E. All work must be completed by the team members only, and verified by the team's chapter advisor using the Team Verification form on the [TSA website](#) under *Themes & Problems*.
- F. Participants render a full-size, three (3)-view (front, top, and right end) drawing (hand or computer-generated) of their structure.
- G. The team develops a cuts part list of materials.

PRELIMINARY ROUND

On-site Destructive Testing of Pre-Built Structures

- A. Participants report at the time and place stated in the conference program and check-in:
 - 1. Pre-built structure with a label that includes only the team number
 - 2. Documentation portfolio
 - 3. Pre-built structure and documentation portfolio in a plastic storage box. The plastic storage box size cannot exceed 12" tall, 16" wide, 18" long. The plastic box will protect the solution and contain the necessary documentation
- B. Structures are assessed and undergo destructive testing.
- C. Destructive testing of pre-built structures is not open for public viewing.
- D. Destructive testing is completed using structural testing equipment, as designated by TSA.
- E. When the destructive testing is completed, a list of twenty (20) semifinalist teams is posted.

SEMIFINAL ROUND

On-site Construction

- A. The twenty (20) semifinalist teams participate in the on-site problem, which feature the construction and destructive testing of a designated structure to determine the ten (10) finalist teams.
- B. Twenty (20) semifinalist teams report to the event area at the time and place stated in the conference program.
- C. Teams are seated by a monitor.
- D. The design problem is explained and directions for the construction problem is provided.
- E. Teams have a three (3)-hour window when drawing begins and building stops, typically allotted as:
 1. Thirty (30) minutes to review the problem and create a sketch/drawing of their solution.
 2. Two and one-half (2 and ½) hours to review the problem and construct a solution.
- F. During the building of the team's structure, construction regulations must be observed.
- G. All work stops at the coordinator's signal. Teams that fail to comply with coordinator or monitor directions, after one (1) warning, will be issued a penalty of 20% of the team's total score.
- H. Participants may leave early, but they must first complete check-out as directed.
- I. Teams return all supplied items as directed, and clean and clear their work stations. Failure to do so will result in a 20% penalty deduction.
- J. Teams must identify their structure with only their team ID number, using the label provided.
- K. Structures are allowed to dry in a secure area until destructive testing time.

Destructive Testing

- A. Structures are checked for rules violations and weighed before testing.
- B. Destructive testing is completed by evaluators and is open for spectator viewing.

- C. When all testing is completed, the greatest failure weight of all tested structures is recorded on the rating form, the efficiency rating of individual structures is calculated, and ranking is determined.
- D. Subjective criteria are scored only after all destructive testing is completed using the provided rubric guideline. The subjective criteria are the team participation during semifinals, the pre-built drawing in the portfolio, and the portfolio.
- E. The top ten (10) finalist teams are announced at the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

- A. Documentation Portfolio:
 1. Documentation portfolio is required and must be secured in a [clear front report cover](#) with the following single-sided, 8½" x 11" pages, in this order:
 - a. Title page that includes the event title, the conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Team Verification form; one (1) page
 - c. Full-size, three (3)-view drawing of the structure (solution can be on 11" x 17" paper and folded as necessary)
 - d. A cut parts list of the materials used in the construction of the structure.

PRELIMINARY ROUND

- A. Pre-built structures and documentation portfolio must be completed prior to check-in.
- B. The testing of pre-built structures is not open to spectators.

SEMIFINAL ROUND

- A. Participants must provide and wear safety glasses for this portion of the event.
- B. Participants are required to provide their own tool box (with identification [school name, address, and advisor cell phone number]), which should not exceed twenty (20) inches (508 mm) length x ten (10) inches (254 mm) width x ten (10) inches (254 mm) height. The box must contain all items needed to fabricate the solution.
 1. The following is a suggested list:
 - a. Cutting devices; NONE may be electric
 - b. Adhesives
 - i. Aerosol and electric applicators are not allowed
 - ii. A bottle of Uncure or Debonder is recommended
 - c. Temporary fastening devices
 - i. Straight pins
 - ii. Clamps
 - iii. Tape
 - d. A cutting surface that prevents table-top marring (required)
 - e. Rulers, straightedges, and/or measuring scales
 - f. Abrasives sheets, sanding sponges, emery boards
 - g. Marking devices (pens, pencils, etc.) and sharpener
 - h. Sheet of wax paper, minimum 11" x 17"
 - i. Pliers, wrenches, nut drivers, as needed
 - j. Safety glasses and side shields, as required
- C. Planning and fabrication supplies are provided by TSA. Teams are issued a packet of construction materials (such as a specific type of wood) to use for fabrication of the on-site designed structure once the team's drawing of the on-site solution is complete.
 1. Planning and fabrication supplies (these materials may not be part of the structure submitted for testing):
 - a. 11" x 17" paper with ¼" grids for sketching the structure
 - b. Pin board
 - c. A sheet of wax paper
 - d. Structure label
 - D. Teams that fail to comply with coordinator or monitor directions, after one (1) warning, will be issued a penalty of 20% of the team's total score.
 - E. Filming and the taking of photographs is prohibited during the viewing of structure, judging, and testing.
 - F. Subjective criteria is scored during or after semifinal construction.
 - G. In the event of a tie at the conclusion of the semifinal round, the solution that is higher on the 'on-site structure total points' will receive the higher placement on the top ten (10) finalist teams that will be announced at the awards ceremony.

EVALUATION

Evaluation is based on the compliance and design efficiency of a pre-built structure and an on-site structure (semifinalists only), both of which are destructively tested. Subjective criteria will be scored only after all destructive testing is completed.

Please see the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

TSA AND CAREERS

This competition has connections to one (1) or more of the careers below:

- Architect
- Civil engineer
- Engineering technician
- Mathematician
- Structural engineer
- Structural iron and steel work technician

STRUCTURAL ENGINEERING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an “adequate” score of 7 for an X1 criterion = 7 points; an “adequate” score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Team of two is present
- ☐ The structure is present, identified, and in its storage box
- ☐ The Team Portfolio is present and includes:
 - ☐ The Team Verification form (completed)
 - ☐ Three (3)-view drawing of the structure
 - ☐ A complete cut parts list for the structure
- ☐ ENTRY NOT EVALUATED

PRE-BUILT STRUCTURE (On-Site Destructive Testing) – (40 points)

Indicate N for non-compliant or C for compliant, for each regulation in the Construction section. One non-compliant mark will result in the entry not being evaluated. Dimensional criteria will have a tolerance identified in the design brief.

Regulation	Noncompliant	Compliant
Outside Width of Structure	The outside width of the structure is greater than the designated construction width.	The outside width of the structure is within the designated tolerance of the assigned construction width.
Outside Height of Structure	The height of the structure is greater or less than the designated tolerance of the assigned construction height.	The height of the structure is within the designated tolerance of the assigned construction height.
Inside Structure Width	The inside structure space, if provided in the design brief, is less than the required construction space.	The inside structure space, if provided in the design brief, is greater than the required construction space.
Construction Materials	Material other than the correct construction material was used in the construction of the solution.	Only the correct construction materials are used in the construction of the solution.
Substructure	Substructure, if allowed in the design brief, is not applied correctly.	Substructure, if allowed in the design brief, is applied correctly.
Laminations	Laminations used in the construction of the solution fail to comply with the design brief.	Laminations used in the construction of the solution comply with the design brief.
Coating of Materials	Coating of the construction materials with glue is present.	No coating of the construction materials with glue is present.
Testing Rod and Block Clearance	The testing block and rod cannot be placed and passed through the center of the structure to allow for testing.	The testing block and rod pass freely through the center of the structure to allow for testing.
DISQUALIFIED		
PRE-BUILT STRUCTURE APPROVED FOR TESTING		
Record the mass (weight) of the structure (in grams to the nearest tenth of a gram) prior to testing.		

PRE-BUILT STRUCTURE (On-Site Destructive Testing) – continued					
Record the failure weight in pounds to the nearest tenth of a pound.					
Determine the efficiency (shown to three decimal places) by multiplying the failure weight (or adjusted failure weight, as applicable) by 4.54 and then dividing by the mass (weight) of the structure.					
1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	
6th: 35 Points	7th: 34 Points	8th: 33 Points	9th: 32 Points	10th: 31 Points	
11th: 30 Points	12th: 29 Points	13th: 28 Points	14th: 27 Points	15th: 26 Points	
16th: 25 Points	17th: 24 Points	18th: 23 Points	19th: 22 Points	20th: 21 Points	
PRE-BUILT STRUCTURE TOTAL POINTS (40 points)					

PRELIMINARY SUBTOTAL (40 points)	
---	--

ON-SITE STRUCTURE (Qualification) – (40 points)				
For the ON-SITE STRUCTURE: Indicate N for non-compliant or C for compliant, in the Qualification and Construction sections below. In the Qualification section, one non-compliant mark will result in disqualification. In the Construction section, one non-compliant mark will result in the structure not being evaluated. Dimensional height criteria will have a tolerance identified in the design brief.				
Regulation	Noncompliant		Compliant	
Team of Two	Only one (1) team member is present.		Both team members are present.	
Safety Eyewear	Warnings about eyewear are issued.		No warnings about eyewear are issued.	
Structure Identification	The identification sticker is not attached.		The identification sticker is attached.	
Tools and Fabrication Supplies	Inappropriate tools or supplies are brought to the event.		Appropriate tools and supplies are brought to the event.	
ON-SITE STRUCTURE (Construction)				
Outside Width of Structure	The outside width of the structure is greater than the designated construction width.		The outside width of the structure is within the designated tolerance of the assigned construction width.	
Outside Height of Structure	The height of the structure is greater or less than the designated tolerance of the assigned construction height.		The height of the structure is within the designated tolerance of the assigned construction height.	
Inside Structure Width	The inside structure space, if provided in the design brief, is less than the required construction space.		The inside structure space, if provided in the design brief, is greater than the required construction space.	
Construction Materials	Material other than the correct construction material was used in the construction of the solution.		Only the correct construction materials are used in the construction of the solution.	
Substructure	Substructure, if allowed in the design brief, is not applied correctly.		Substructure, if allowed in the design brief, is applied correctly.	
Laminations	Laminations used in the construction of the solution fail to comply with the design brief.		Laminations used in the construction of the solution comply with the design brief.	
Coating of Materials	Coating of the construction materials with glue is present.		No coating of the construction materials with glue is present.	
Testing Rod and Block Clearance	The testing block and rod cannot be placed and passed through the center of the structure to allow for testing.		The testing block and rod pass freely through the center of the structure to allow for testing.	

ON-SITE STRUCTURE (40 points) – continued					
DISQUALIFIED					
ON-SITE STRUCTURE APPROVED FOR TESTING					
Record the mass (weight) of the structure (in grams to the nearest tenth of a gram) prior to testing.					
Record the failure weight in pounds to the nearest tenth of a pound.					
Determine the efficiency (shown to three decimal places) by multiplying the failure weight (or adjusted failure weight, as applicable) by 4.54 and then dividing by the mass (weight) of the structure.					
1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	
6th: 35 Points	7th: 34 Points	8th: 33 Points	9th: 32 Points	10th: 31 Points	
11th: 30 Points	12th: 29 Points	13th: 28 Points	14th: 27 Points	15th: 26 Points	
16th: 25 Points	17th: 24 Points	18th: 23 Points	19th: 22 Points	20th: 21 Points	
ON-SITE STRUCTURE TOTAL POINTS (40 points)					

SUBJECTIVE CRITERIA (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Team Participation On-Site (X1)	The majority of the construction is completed by one member of the team; the partner may be disengaged.	Both team members generally are engaged in the process, though one member may take on more responsibility than the other.	Both team members are actively involved in the construction; there is shared responsibility between team members.	
Pre-Built Drawing and Parts List (X1)	The submitted drawing was incomplete, not accurate, of proper quality, or was not to scale; a complete parts list was not included.	The submitted drawing was complete but lacked clarity, accuracy, or was of poor quality, the parts diagram was not complete or was incorrect.	The submitted drawing was complete, accurate, and to scale; the parts list was complete and accurate.	
Portfolio (X1)	Portfolio is unorganized and/or missing three (3) or more components; leadership and/or 21 st century skills are not evident.	Portfolio includes most components and is generally organized; leadership and/or 21 st century skills are somewhat evident.	All components of the portfolio are included in order, and content and organization are clearly evident; leadership and/or 21 st century skills are clearly evident.	
SUBJECTIVE CRITERIA SUBTOTAL (30 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL SUBTOTAL (70 points)				
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.				
TOTAL (110 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____



STRUCTURAL ENGINEERING

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges
 - 1. Preliminary round to evaluate pre-built structures, two (2) or more
 - 2. Semifinal round, to qualify structures after construction, two (2) or more
 - 3. Semifinal round, destructive test judges, two (2) or more
 - a. One (1) to weigh the structure, record structure weight, and record failure weight
 - b. One (1) to bring the structure to the testing location, position the structure on the testing device, operate the tester, and then remove and store the structure following testing
- C. Construction monitor, one (1) per twenty teams
- D. Timekeeper, one (1)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
 - 3. Stick on labels for identifying entries
- B. Twenty (20) blank stick on labels for the semifinal round
- C. Testing equipment, provided by TSA
- D. Sample structures for both testing sessions that can be used to demonstrate the testing procedure and to determine that the testing equipment is working properly.
- E. Evaluation and recording equipment
 - 1. Gram scale (3-decimal place calculation)
 - 2. Tape measure or 2' ruler
 - 3. Evaluation gauges (rulers)
- F. Site requirements
 - 1. Construction session
 - a. Tables and chairs suitable for cutting and gluing
 - b. Work area, at least 2' x 3' for each team (suggested space is two (2) teams per 6' x 2' or 8' x 2' area)
 - c. One (1) chair per participant
 - d. Tables for equipment check-out and check-in
 - e. Tables and chairs for evaluators
 - f. Secured area for drying entries and storing supplies
 - 2. Testing session
 - a. Tables for storage of structures
 - b. Table for weighing
 - c. Table for testing
 - d. Table for recording
 - e. Tables for storage of failed structures
 - f. Chairs for spectators
 - g. Barricade to separate testing area from spectators
 - 3. Semifinalist team packets provided by TSA containing construction materials and instructions.
 - a. Construction tools per team, to be used and returned to the event coordinator or helpers after construction:
 - i. Pin board as supplied, but generally a one-foot by two-foot (1' x 2') piece of fiber or foam board
 - ii. Grid paper, 1/4" x 1/4" grid on 11" x 17" paper for structure sketch (to remain with the completed structure when turned in)
 - iii. Wax paper to cover the pin board (to remain with the completed structure when turned in)
 - iv. Label for structure
 - b. Construction materials – balsa as needed for each team
 - c. Instructions

RESPONSIBILITIES

- A. Prepare the structure problem statement (including any necessary related information) for posting on the TSA website.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Check to see that all event equipment and materials have been secured.
- E. One (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

- A. Check in participants at the time and place stated in the conference program.
- B. Participants check in:
 - 1. The pre-built structures
 - 2. The documentation portfolio
- C. Anyone reporting who is not on the entry list may check in only after official notification is received from the CRC.
- D. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
- E. Requirements for attire do NOT apply during check-in, only on the first day of the conference.

PRELIMINARY ROUND/PRE-BUILT STRUCTURE

- A. Coordinate and manage the on-site testing of pre-built structures, the recording of results, and the determination of the twenty (20) semifinalist teams.

- B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

- 1. To deduct 20% of the total possible points or
- 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

- C. Submit semifinalist results to the CRC for posting.
- D. Assemble semifinalist packets of construction materials and directions for the twenty (20) on-site semifinalist teams.

SEMIFINAL ROUND

Team Check-in for On-site Construction

- A. Check-in will begin at the time noted in the conference program and will continue until all teams arriving on time have been checked in and seated. The event begins at the posted time.
- B. Both members of a team must be present during check-in.
- C. No team is allowed to begin late unless its members have complied with the following: Participants with time conflicts must present a written explanation of the conflict to the event coordinator at least one (1) hour before the construction time stated in the conference program.
- D. Work must begin during the timeframe scheduled for this portion of the event.

On-site Construction

- A. Assign team construction locations.
- B. When all teams are seated, distribute instructions and review these, as well as any details for the assigned structure.
- C. Teams are allowed a maximum of three (3) hours to complete their structure:
 - 1. Thirty (30) minutes of this time is allotted for completing the design drawing.
 - 2. Two and one-half (2½) hours, is allotted for actual construction.
- D. When a team notifies a monitor that the required sketch is complete, and the monitor confirms this, the team receives a materials packet and may begin the on-site construction phase of the event.

- E. No additional supplies are provided during the event.
- F. Call time at the end of the allotted three (3)-hour timeframe. All teams must stop working at this point.
- G. All work stops at the coordinator's signal. Failure to comply with instructions will result in a penalty of 20% to the team's total score.

Team Check-out

- A. Establish the procedure for check-in and recording of finished structures.
- B. Designate an area for storage, and allow for the return of construction materials.
- C. Coordinate the return and removal of all supplied items and ensure that teams clean and clear their work stations. Deduct a 20% penalty for teams that do not comply.
- D. Teams check in excess supplies as directed by the monitors.
- E. Ensure that teams identify their structure with only their team ID number, using the label provided.
- F. Teams place their structures in the storage area with the sketch as directed by the monitor. The structure must be identified with the team number only (using the label provided in the materials packet).
- G. Once check-out is complete, all participants leave the competition area. Participants may leave early, but they must complete check-out as directed.
- H. The structures are secured by the monitor and allowed to dry for a minimum of twelve (12) hours.

Destructive Testing

- A. After the structures have dried, judges report at the time and place stated in the conference program.
- B. Judges test each structure and score the results.
- C. Judges score the Subjective Criteria for semifinalists after destructive testing has taken place.

EVALUATION

- A. Check (with assistance from judges) all structures for regulations compliance. Structures that are in compliance are tested without penalty.
 - 1. Weigh all structures before testing and record the weight on the evaluation rubric.
- 2. Use the testing device, designated by TSA, to test each structure. (A specific testing block or attachment for the structure may be necessary for the on-site problem.)
- 3. Apply an increasing load to the structure, via the test block or attachment, until the structure fails.
- 4. Record the greatest failure weight on the rubric. This weight is the greatest weight recorded (of all the tested structures) during testing before they fail.
- 5. Determine each structure's efficiency by the greatest failure weight x 4.54, divided by the weight of the structure in grams; round off the efficiency to three (3) decimal places and record it on the rubric.
- 6. The highest numeric efficiency determines the winner. In the case of an efficiency tie, the greatest weight held by the tied entries determines the winner.
- B. Structures will not be tested if:
 - 1. A non-compliance construction regulation violation was determined before testing.
 - 2. The structure cannot be placed on the tester.
 - 3. The testing attachment cannot be properly placed within or on the structure.
 - 4. Straight pins are left in the structure.
 - 5. There is a failure of a participant to wear safety eyewear and/or to follow safe practices.
 - 6. Laminations fail to comply with the guidelines as specified in the current year's challenge.
 - 7. Failure to use each of the materials specified in the current year's challenge.
- C. Manage, with assistance from judges, the destructive testing of all structures that were not officially tested due to non-compliance.
- D. Discuss rule violations (e.g., 20% deduction, disqualification) and have all relevant parties initial the rating form.
- E. If necessary, manage the security and removal of materials from the event area.
- F. Semifinalist teams may pick up their structures at a time determined by the event coordinator.

OVERVIEW

Applying leadership and 21st century skills, participants use a team approach to develop a computer- controlled model solution to a given problem, typically one based on an industrial setting. Teams analyze the problem, build a computer-controlled mechanical model, program the model, explain the program and mechanical features of the model-solution, and leave instructions for judges to operate the device.

ELIGIBILITY

One (1) team of three (3) individuals per state may participate.

TIME LIMITS

The preliminary round consists of three (3) phases:

- A. Thirty (30) minutes for set up (team captain)
- B. Fifteen (15) minutes for analysis (team)
- C. Two and a half (2 ½) hours for problem solution (team)
- D. All students participate in an interview at the conclusion of the programming

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Each team selects a team captain prior to the orientation meeting.
- B. A team orientation meeting will take place at the beginning of the event at the conference.
- C. The captain checks-in for the team during the set-up time by submitting his/her student identification number and the team's identification number for the written and model portions of the event.

- D. The problem and the Inventor's Log are presented to teams at the beginning of the thirty (30) minute problem analysis session prior to model-building.
- E. Teams must complete their description or interpretation of the problem during this time.
- F. Each team is given a maximum of two and one-half (2½) hours to:
 - 1. Construct a model that simulates realistic industrial processes
 - 2. Program the model
 - 3. Test the solution
 - 4. Describe the program and mechanical features of the model-solution
 - 5. Complete directions for judges to use to activate the model
- G. When finished, teams save their programs and leave them on-screen in operable form with the ability to be reset.
 - 1. Before leaving the event room, teams demonstrate the operation of the model with judges present.
 - 2. Judges may ask questions during the demonstration.
 - 3. After judges have observed the operation of a team's model, the team leaves the room.
 - 4. The coordinator determines the amount of time permitted for the team's demonstration based on the number of teams and the complexity of the problem.
 - 5. Evaluation of the solutions takes place without the teams present.
- H. Team members report to the event area at the time and place stated by the event coordinator to pick up their equipment.
- I. Ten (10) finalists will be announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. No reference materials or building cards are allowed.
- B. Participants provide their own laptop computer with hardware and software systems. All equipment must be labeled with the team's identification number, advisor name, and advisor contact information.
- C. Access to AC power will be provided. One electrical outlet per team. Each team should bring a power strip.
- D. Each team provides pencils and scrap paper along with its own materials kit, which must be appropriate to build a system that can identify, secure, and move objects and that has light and/or sound outputs.
- E. A problem will be developed by the coordinator based upon the assumption that every materials kit will contain at least:
 - 1. Two (2) optical sensors
 - 2. Two (2) touch sensors
 - 3. Two (2) motors
 - 4. Two (2) audio and two (2) light outputs
 - 5. Gears, wheels, and axles appropriate to build a motorized vehicle and/or conveyor belt
 - 6. Balls, blocks, and pegs that can be used as objects to be moved and manipulated
 - 7. Velcro, tape, clamps, and other materials to secure or move the above objects (balls, blocks, and pegs)
 - 8. No cutting devices may be used during the on-site challenge; materials must retain the original form in which they were brought to the competition.
 - 9. Power tools may not be used.
- F. The following definitions are an integral part of the event regulations:
 - 1. Repeatability – the device is programmed to reset automatically.
 - 2. Functional control – the device must accomplish the task in an efficient manner and be user friendly.
 - 3. Model-solution – the physical device must simulate the realistic processes used in the industry.

- 4. Conservation of materials – the model reflects the best use of materials to solve the problem, without being overbuilt.

- G. Programs must be written completely on-site.
- H. Use or modification of any programs written prior to the competition result in disqualification.
- I. An example of a problem for this event is provided below to help students understand and interpret a typical issue common to business and industry that might be used at a national TSA conference.

Johnson Recycling Center needs an automatic system for separating its two primary types of recycling material products. If type A material is detected in the product, the system should move the product to the right side of a sorting line. If type B material is detected on the assembly line, the product should be delivered to the left side. Whenever a product is delivered, a light or buzzer should activate for 10 seconds to alert workers that a product is available on one or the other of the sorting lines. The system should then reset to separate and deliver a new product. Design a prototype that can automatically deliver at least four products without any user intervention, then reset.

EVALUATION

- A. Inventor's Log
- B. Model solution
- C. Programming structure and efficiency
- D. Interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- CNC programmer
- Computer programmer
- Robotics engineer

SYSTEM CONTROL TECHNOLOGY INVENTOR'S LOG

Participant/Team ID# _____

Use only the space provided. This section must be completed DURING the process of problem analysis.

1. Description or interpretation of the given problem:

The two parts below must be completed AFTER the problem analysis session.

2. Description of the team solution (explain the unique features of the program and model):

3. Directions to evaluators to start the system:

SYSTEM CONTROL TECHNOLOGY

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Laptop computer is present
- ☐ Materials kit is present
- ☐ ENTRY NOT EVALUATED

INVENTOR'S LOG (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Description of Problem (X1)	The written description is incomplete, and/or it is illogical and unorganized; the description is simply a restatement of the problem's guidelines.	The written description includes a logical and general understanding of the problem's guidelines.	An organized, logical, and concisely written description of the problem is provided; it includes all major aspects of the problem's guidelines, as well as original thoughts.	
Description of Solution and Activation Instructions (X1)	The team's written solution does not correlate with the final system creation; the solution is illogical related to the problem's guidelines, and/or directions to activate the solution are included but incomplete.	The team's written solution correlates generally with the final system creation; adequate directions to activate the solution are included.	A strong correlation between the team's written solution and final system creation is provided; the solution is written clearly and concisely; activation instructions are included and written concisely.	
INVENTOR'S LOG SUBTOTAL (20 points)				

SOLUTION TO PROBLEM (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Realistic Simulation (X1)	The simulation is not realistic; it has an abstract design that would not work effectively in its intended environment.	The simulation is somewhat realistic and logically designed, and it may work effectively in its intended environment.	The simulation is realistic and similar to a system that would be effective in its intended environment.	
Dependability of Solution (X1)	The solution is not constructed with dependability in mind; when the system is operated, construction pieces fall off, etc.	Most of the solution is well constructed and dependable, with only a few components that are questionable.	Every component of the solution is well constructed and dependable; practical construction techniques have been used.	
Conservation of Materials (X1)	An inefficient use of construction materials is obvious; too many unnecessary materials are incorporated into the design.	Most of the components of the solution are designed with conservation in mind; the construction is generally adequate.	All components of the solution are designed and assembled with conservation of materials in mind; the construction is elegant and not overbuilt.	

SOLUTION TO PROBLEM (60 points) – continued				
Solution to Problem (X2)	The solution is missing three (3) or more required attributes/criteria and several do not function as intended.	The solution includes most of the required attributes/criteria, and they function adequately.	The solution includes all required attributes/criteria listed in the design details, and all attributes function appropriately and correctly.	
Ingenuity and Creativity (X1)	The solution and design are unauthentic, complex, and do not function as a system.	The solution has some original ideas in its design, and its construction is adequate.	The solution is truly unique and authentic; its construction is concise and designed with simplicity.	
SOLUTION TO PROBLEM SUBTOTAL (60 points)				

PROGRAMMING STRUCTURE (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Programming Efficiency (X1)	The software used to program the system is overly complex and inefficient; advanced programming techniques, which would have simplified the programming of specific tasks, are not included.	The programming software is efficient, with some advanced features that simplify the solution's criteria and/or attributes.	A concise and logical programming application is used that incorporates advanced features to simplify the solution's criteria and/or attributes.	
Program Structure (X1)	The programming structure is illogical, unorganized, or overly complicated and/or complex; the program does not reset.	There is evidence of an organized programming structure and adequate use of sub-routines; the program resets.	The programming structure is concise and predictable; there is appropriate use of sub-routines where needed; the program resets.	
PROGRAMMING STRUCTURE SUBTOTAL (20 points)				

DEMONSTRATION/INTERVIEW (10 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
DEMONSTRATION/INTERVIEW SUBTOTAL (10 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (110 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.

TOTAL (110 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

SYSTEM CONTROL TECHNOLOGY EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more
- C. Assistants, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and each judge
 - 2. TSA Event Coordinator Report
- B. Stopwatches
- C. Copies of the written problem
- D. Copies of the Inventor's Log
- E. Large room with sufficient electrical outlets to accommodate a first place team from every state and affiliated country
- F. Power strips with surge protectors and extension cords
- G. One (1) table and three (3) chairs per team

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Begin the event at the scheduled time by closing the doors and checking the entry list.
- B. All participants and judges should be in the room at this time.
- C. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- D. In order to compete, participants must be on the entry list or must have approval of the CRC.
- E. Secure teams' equipment in the area designated.
- F. At the orientation meeting, obtain identification numbers for each team captain and the captain's respective team.
 - 1. Judges must be present at the orientation meeting.
 - 2. Review the time limits, procedure, and regulations with team captains.
- G. Distribute the problem and Inventor's Log to teams at the beginning of the event.
- H. Teams have fifteen (15) minutes to complete their interpretation of the problem, using the Inventor's Log.
- I. Each team is given two and one-half (2 ½) hours to complete the remaining portion of the event.
- J. Prior to leaving the event room, teams must demonstrate that their device/model is operable and has the ability to reset.
 - 1. Judges must observe this demonstration and may ask a few questions.
 - 2. Judges also may take notes, but evaluation of a team's work occurs only after the team has left the event room.

- K. Judges independently evaluate the entries.
- L. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges and CRC manager must all initial either of the violations on the rating form.
- M. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Participants demonstrate their knowledge of TSA and concepts addressed in the technology content standards by completing an objective test. Applying leadership and 21st century skills, semifinalist teams participate in a question/response, head-to-head competition.

ELIGIBILITY

One (1) team of three (3) individuals per chapter may participate. Teams that take the test and advance to the semifinalist portion of the event must be comprised of the same three (3) members.

TIME LIMITS

PRELIMINARY ROUND

- A. One (1) hour is allowed for the test. Three (3) members must be present at the same time and take the test individually.

SEMIFINAL ROUND

- A. Teams selected as semifinalists must be available as scheduled for oral competition.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Participants report to the event area at the time and place stated in the conference program for the test.
- B. Participants follow the specific regulations and adhere to the directions provided on-site by the event coordinator.
- C. All team members take the test individually.
- D. The sixteen (16) top-scoring teams qualify as semifinalists.

- E. The preliminary round score will be the average test score for the three (3) members. The sixteen (16) top scoring teams qualify for the semifinal round as semifinalists.
- F. A semifinalist list (in random order) is posted.

SEMIFINAL ROUND

- A. Semifinalist team members (and their advisor, if available) report to the oral event area holding room at the time and place stated in the conference program.
- B. After a short briefing, advisors leave and the teams remain in the holding room until they are called for competition.
- C. Teams are paired using the semifinalist teams' bracket.
- D. Each team will be listed on the bracket by their chapter ID.
- E. When instructed to do so, two (2) teams enter the event area and are seated according to instructions.
- F. Questions are drawn from a card file resource bank.
- G. If equipment malfunctions, a question that is being considered at that time automatically is eliminated. If equipment malfunctions three (3) times, time is called by the event coordinator to set up back-up equipment. After equipment has been set up and tested, the event continues from the point where it stopped.
- H. Once a team is eliminated, the team is out of the oral competition except for the round in which the third and fourth positions are determined.
- I. The top ten (10) finalist teams will be announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND/WRITTEN EXAM

- A. Team members take the test individually.
- B. Tests will be administered online only. Written (paper) tests will not be used unless deemed necessary prior to, or on-site at the national conference. Online testing will require each member to provide a laptop, and all members must take the test online. Please review the [National TSA Conference Competition Requirements](#).
- C. Participants are responsible for bringing two (2) sharpened No.2 pencils to the test site.
- D. In the event the test must be administered using a hard copy and scantron:
 - 1. Student identification numbers must be entered on the scantron form in the space indicated.
 - 2. Participants must stop work immediately when time is called.
 - 3. Should a participant complete the test before the time allocated is over, the participant will submit the test and scantron form to the coordinator without any form of communication with any other member. Failure to do so results in disqualification of the participant.
 - 4. All tests and scantrons must be turned in before leaving the test area.
- E. The average of the test scores of all three (3) team members determines team ranking.
- F. These same three (3) team members compete in the semifinal round, should the team qualify.
- F. Teams that leave the holding room before being called for competition are eliminated.
- G. Teams may visit with other teams in the holding room.
- H. No advisors or visitors may enter the holding room once the semifinal round begins.
- I. Team members may not enter the oral event area as spectators until after their team has been entirely eliminated from competition.
- J. Questions, to include the bonus question, may not be discussed by teams. Team members may only discuss the additional question.
- K. Ten (10) teams are announced as finalists during the awards ceremony. The highest test scores of the teams that were eliminated in the initial round will receive 9th and 10th place. The highest test scores of the teams that were eliminated in the second round will be used to determine 5th-8th place.
- L. The procedures for reading questions and “buzzing in” are as follows:
 - 1. The team member who buzzes in to answer a question has five (5) seconds to answer the question without discussion.
 - 2. After a full question is read, competing teams have ten (10) seconds to answer without discussion. If neither team buzzes in, the reader moves to the next question.
 - 3. If a team member buzzes in before a question is finished being read, the reader ceases reading and the team member must give the exact answer as printed with the question without discussion. If the answer is incorrect, the reader reads the entire question for the opposing team.

SEMIFINAL ROUND/ORAL COMPETITION

- A. Sixteen (16) teams, based upon the test results, are selected as semifinalists.
- B. All three (3) members of a semifinalist team must be available to participate at the scheduled time for the oral competition portion of the event.
- C. If a team or member is late for participation, that team forfeits and is eliminated from competition.
- D. No transmitting or recording devices are permitted to assist in answering a question in the event area.
- E. No prompting is permitted.
- M. A team’s score is derived from the total number of correct answers to the questions asked:
 - 1. Twelve (12) questions and an additional question are asked per round; no questions are repeated in another round.
 - 2. For questions 1-11, a correct answer gives the team ten (10) points, and an incorrect answer results in a loss of five (5) points.

3. The 12th question is the bonus question and is worth fifteen (15) points; there is no penalty for an incorrect answer.
4. If the bonus question is not answered correctly, participants are not given an additional question.
5. If a team answers the bonus question correctly, the team is given an additional question to answer. There is no penalty for an incorrect answer. The team may discuss this question. A correct answer for the additional question is worth five (5) points.
6. In case of a tie, three (3) additional questions are asked. This procedure continues until the tie is broken.

EVALUATION

PRELIMINARY ROUND

- A. Each team's average written test score is used to determine the sixteen (16) semifinalist teams.

SEMIFINAL ROUND

- A. Semifinalists will be evaluated on their performance during the oral competition.

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer technician
- Construction analyst
- Engineer
- Entrepreneur
- Technology education instructor

TECH BOWL

2026 & 2027 OFFICIAL SCORE SHEET

MIDDLE SCHOOL

Competition Round ID# _____

Team # _____ (A) Team # _____ (B)

Scorekeeper's Signature _____

SEMIFINAL ROUND – ORAL COMPETITION		TEAM A	TEAM B
Mark an X in the box beside the team that gives the correct response to the question and an O beside the team that gives an incorrect response. Record the scores for each response in the column to the right.			
Question #	Points		
1.	+10 for correct, -5 for incorrect response		
2.	+10 for correct, -5 for incorrect response		
3.	+10 for correct, -5 for incorrect response		
4.	+10 for correct, -5 for incorrect response		
5.	+10 for correct, -5 for incorrect response		
6.	+10 for correct, -5 for incorrect response		
7.	+10 for correct, -5 for incorrect response		
8.	+10 for correct, -5 for incorrect response		
9.	+10 for correct, -5 for incorrect response		
10.	+10 for correct, -5 for incorrect response		
11.	+10 for correct, -5 for incorrect response		
12. Bonus question	+15 for answering the bonus question correctly; no penalty for answering the bonus question incorrectly		
Additional question	+5 for answering the additional question correctly; no penalty for answering the additional question incorrectly		
ORAL COMPETITION SUBTOTAL (130 points)			

Tie Breaker Questions			
1.	+10 for correct, -5 for incorrect response		
2.	+10 for correct, -5 for incorrect response		
3.	+10 for correct, -5 for incorrect response		
TIE BREAKER QUESTIONS SUBTOTAL (30 points)			

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

--	--

SEMIFINAL SUBTOTAL (160 points)

--	--

To arrive at the TOTAL score, subtract rules violation points, as necessary.

TOTAL (160 points)

--	--

Comments:

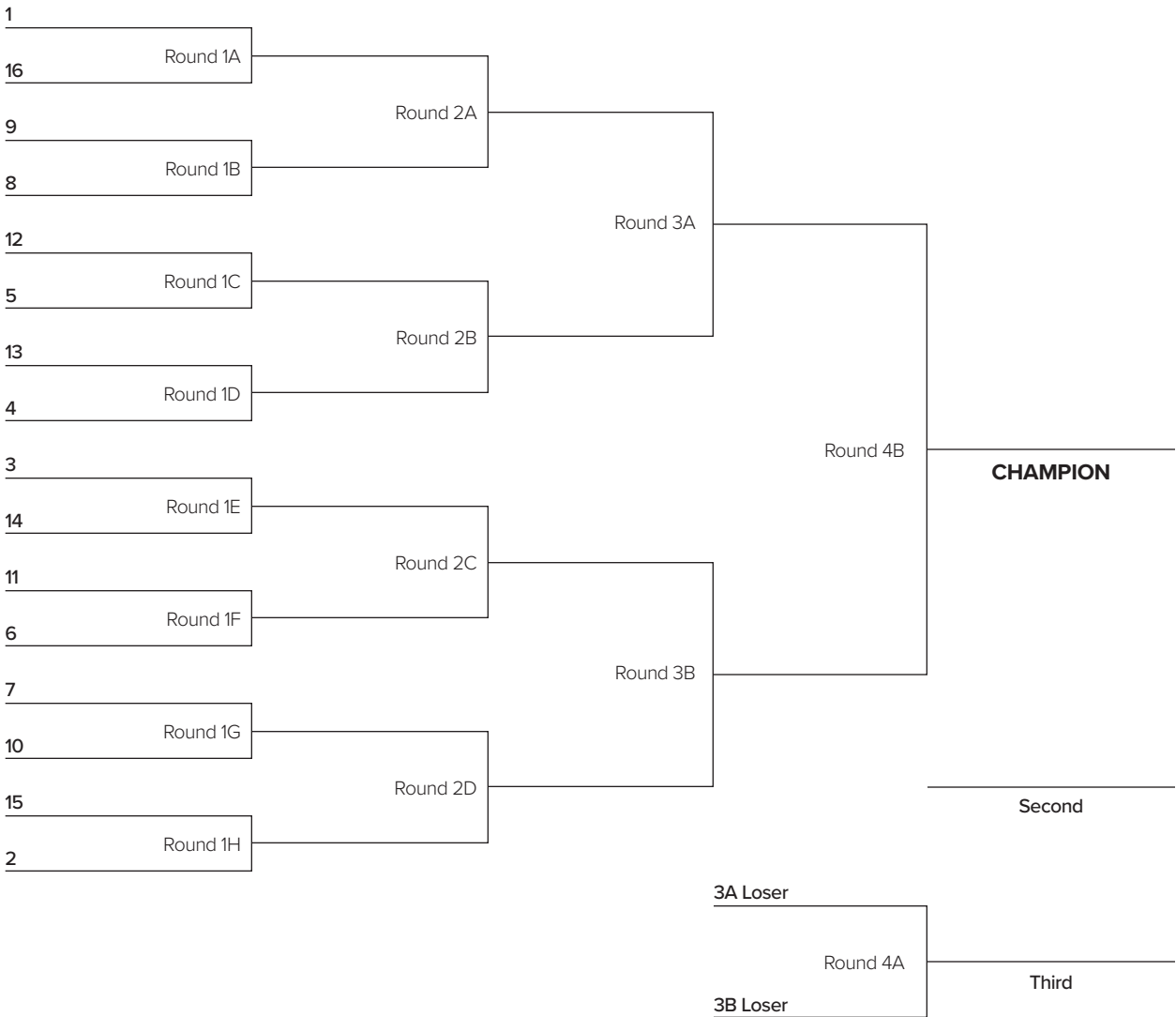
I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

SEMIFINALIST TEAMS BRACKET

MIDDLE SCHOOL



Note to evaluators: This is a single elimination format (semifinalist teams ONLY). Seed is determined by team ranking on written test.			
Seed 1	Team #	Seed 9	Team #
Seed 2	Team #	Seed 10	Team #
Seed 3	Team #	Seed 11	Team #
Seed 4	Team #	Seed 12	Team #
Seed 5	Team #	Seed 13	Team #
Seed 6	Team #	Seed 14	Team #
Seed 7	Team #	Seed 15	Team #
Seed 8	Team #	Seed 16	Team #



TECH BOWL

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Timer for written exam, one (1)
- C. Proctors for written exam, four (4)
- D. Timekeeper for oral competition, one (1)
- E. Scorekeeper for oral competition, one (1)
- F. Moderator for oral competition, one (1)
- G. Assistants for oral competition, two (2)

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. Tables and chairs for the participants and judges
- C. Oral competition
 - 1. Table and chairs for the event judges
 - 2. Two (2) tables and six (6) chairs for the event team, facing the moderator and audience
 - 3. Tech Bowl bracket
 - 4. List of chapters for the event
 - 5. Buzzer system and controls
 - 6. A printed sign (to be placed outside the oral competition room) stating that no filming, taking of photos, or use of any electronic recording devices will be allowed in the competition room
 - 7. Stopwatch for timekeeper
 - 8. 5" x 8" question cards selected from the tech bowl test bank, with questions and the acceptable answer(s) clearly typed

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and any other details associated with the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. Prior to the conference, the event coordinator shall become familiar with the format of the exam (refer to the National TSA Conference Competition Requirements).
- B. Begin the event at the scheduled time by closing the doors and checking the entry list.
- C. All participants and event judges should be in the room at this time.
- D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
- E. In order to compete, participants must be on the entry list or must have approval of the CRC.
- F. All participants and judges should be in the room at this time.
- G. In order to compete, participants must be on the registration list or must have approval of the CRC.
- H. Monitor the one (1)-hour test.

- I. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entryThe event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- J. Determine the sixteen (16) semifinalist teams based on team members' averaged score on the written test.

SEMIFINAL ROUND/ORAL COMPETITION

- A. Run the oral component of the event as described in the Procedure section.
- B. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
- C. Determine the ten (10) finalists. The highest test scores of the teams that were eliminated in the initial round will receive 9th and 10th place. The highest test scores of the teams that were eliminated in the second round will be used to determine 5th-8th place.
- D. If necessary, manage security and the removal of materials from the event area.

TECHNICAL DESIGN



OVERVIEW

Applying leadership and 21st century skills, participants demonstrate their ability to use the technical design process to solve an engineering design problem on-site at the conference.

ELIGIBILITY

One (1) team of two (2) individuals per chapter may participate.

TIME LIMITS

Twenty-four (24) hours is allowed to solve the engineering design problem.

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.

PRELIMINARY ROUND

- A. Teams report to the event area at the time and place stated in the conference program to receive the design brief and instructions about how to submit their solution to the problem as a portfolio PDF the next day. TSA competition attire is required to receive the design brief.
- B. Teams follow the technical design process loop to solve the provided engineering design problem.
- C. All work must be completed solely by the teams entered in this competition. No outside help is permitted.
- D. Teams are responsible for submitting a portfolio PDF of their solution using the submission procedures provided by the event coordinator.
- E. National TSA will not provide wireless Internet.
- F. Entries are reviewed independently by the judges.
- G. The top ten (10) finalists are announced at the awards ceremony.

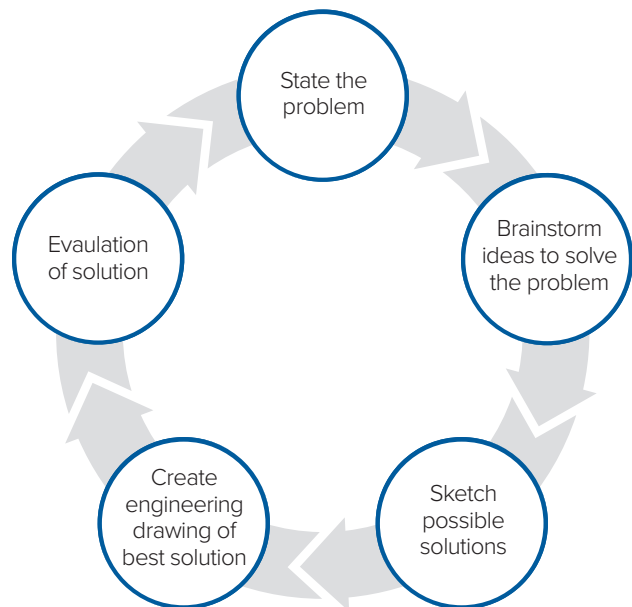


Figure 1: Technical design process loop

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

Design Preparation

- A. Students prepare a portfolio that includes each step of the technical design process loop. (Figure 1).
- B. Students develop a problem statement interpretation of the problem provided.
- C. The portfolio should show a logical progression from one step of the loop to the next.

D. Documentation Portfolio:

1. The documentation portfolio must be submitted as a multi-page PDF document with pages in this order:
 - a. Title page with the event title, conference city and state, the year, and the team/chapter ID number; one (1) page
 - b. Table of contents; one (1) page
 - c. Team's interpretation of the problem, including a list of criteria and constraints set forth in the design brief; one (1) page
 - d. Demonstrated use of a brainstorming technique of the team's choice (mind mapping, reverse engineering, word association, etc.), to develop ideas to solve the problem; brainstorming ideas should be documented; one (1) page
 - e. At least three (3) hand-drawn sketches of different solution options to a given problem. Photo/scan/convert each to a PDF as necessary to insert into the portfolio PDF. One (1) page for each hand-drawn sketch; three (3) pages total
 - i. Each hand-drawn solution must be developed based on the selected brainstorming technique.
 - ii. Each hand-drawn sketch also must include a solution pro/con list written on each sketch to aid in selecting the best design;
 - iii. Label the first solution "Solution Option 1," the second "Solution Option 2," and the third "Solution Option 3."
 - f. Based on the pro/con list for each of the hand-drawn solutions to the problem, select the best solution and create an engineering drawing based on the solution; one (1) page.
 - g. Using the engineering drawing of the final solution, write a paragraph that evaluates the final solution and answers the following question, "Does the final design meet all the elements set forth in the design brief?"; one (1) page.

- E. A sample design brief is provided below to help students understand a typical engineering design problem for this event

Design Brief Sample

(This design brief is ONLY an example of the type of problem that participants may expect at the conference.)

Aircraft carriers are much shorter than a typical airport runway. How do aircraft manage to gain enough speed for takeoff over a much shorter distance? A catapult gives them the extra boost they need to take-off.

An elementary school teacher would like to use the basic principles of this technology to teach his/her students about kinetic and potential energy, but apply the same knowledge to launching a small paper airplane.

Design a hand-held paper airplane launcher that a classroom teacher can incorporate into a classroom lesson and have the students build as a class project.

Your solution must be able to be built by 5th grade students using materials and supplies that they can bring from home and incorporate a rubber band as the power source.

The solutions have a maximum length of 9", width of 8", and a height of 7".

The solution is contained in a documentation portfolio. A prototype or model is not allowed.

EVALUATION

- A. Each element of the portfolio
- B. The overall technical design process

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Designer
- Engineer
- Quality assurance engineer
- Engineering manager
- Creative consultant

TECHNICAL DESIGN

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Portfolio is present
☐ ENTRY NOT EVALUATED

SOLUTION (100 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Interpretation of Problem (X1)	Interpretation of the problem is vague, with few or no criteria/constraints included in the description; statement is difficult to understand.	Interpretation of the problem, criteria, and constraints are included and generally identified.	Interpretation of the problem is well-developed and further investigates the included criteria/constraints.	
Brainstorming Technique (X1)	There is no clear evidence of the use of brainstorming to interpret the design of the problem.	Use of brainstorming (which incorporates the problem statement, criteria, and constraints to solve problem) is apparent.	Exceptional and organized use of brainstorming (which incorporates each element of the design brief) is evident.	
Solution Option 1 (X1)	Sketch is sloppy and poorly constructed, and/or it appears to be included as an afterthought to the design; there is no design pro/con list, or it is incomplete.	Sketch is generally well drawn and includes the pro/con list; evidence of the final design is illustrated in the sketch.	Sketch is of exceptional quality and includes a creative pro/con list; clear transformation from the sketch to the final design is evident.	
Solution Option 2 (X1)	Sketch is sloppy and poorly constructed, and/or it appears to be included as an afterthought to the design; there is no design pro/con list, or it is incomplete.	Sketch is generally well drawn and includes pro/con list; evidence of the final design is illustrated in the sketch.	Sketch is of exceptional quality and includes a creative pro/con list; clear transformation from the sketch to the final design is evident.	
Solution Option 3 (X1)	Sketch is sloppy and poorly constructed, and/or it appears to be included as an afterthought to the design; there is no design pro/con list, or it is incomplete.	Sketch is generally well drawn and includes a pro/con list; evidence of the final design is illustrated in the sketch.	Sketch is of exceptional quality and includes a creative pro/con list; clear transformation from the sketch to the final design is evident.	
Final Solution (X2)	Solution conveys a sloppy design, and/or does not incorporate key elements in the design brief, and/or drafting techniques are not proper.	Solution incorporates most elements laid out in the design brief; drawing uses proper drafting techniques and methods.	Solution exudes creativity and addresses all design brief elements; proper drafting techniques are used in the design.	

SOLUTION (100 points) – continued			
Evaluation of Design (X2)	Evaluation is poorly written; it is a reiteration of the design brief elements, with little or no examination of the finished design.	Evaluation satisfactorily answers the question "Does the final design meet all the elements set forth in the design brief?"	Evaluation response is creative and unbiased; it is well written and answers the posed question completely.
SOLUTION SUBTOTAL (100 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.			TOTAL (100 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

TECHNICAL DESIGN EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. One (1) copy of the technical design problem (in design brief format) for each team and judge
- C. Tables and chairs for the judges
- D. Computer with software capable of viewing PDF files is needed for each judge
- E. Extension cords (25' minimum length), as needed

RESPONSIBILITIES

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. At least one (1) hour before the judging of solutions is scheduled to begin, meet with judges to review the procedures, regulations, evaluation, and any other details associated with the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
- F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN AND DESIGN PROBLEM DISTRIBUTION

- A. Meet with all teams at the time and location scheduled in the conference program.
- B. Distribute a copy of the technical design problem to each team.
- C. Ensure that all participants understand the event requirements, as well as the time and place to submit their entry.
- D. Begin entry check-in at the time and place stated in the conference program.

PRELIMINARY ROUND

- A. Ensure that the judges have access to the portfolio PDFs submitted electronically.
- B. Judges independently review each entry with neither students nor advisors present.
 - 1. The event is judged in sections with two (2) judges per section.
 - 2. The top twelve (12) entries will be reviewed by a second set of two (2) judges using the same rubric, but a blank new rubric. For example, if three (3) sections are used, the top (4) from each section will advance. If six (6) sections are used, the top two (2) from each section will advance.
- C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- D. If necessary, manage security and the removal of materials from the event area.

VIDEO GAME DESIGN



OVERVIEW

Applying leadership and 21st century skills, participants develop a video game that focuses on the subject of their choice. The game must be interesting, exciting, visually appealing, and intellectually challenging. The rating of the game must meet the ESRB rating for E for Everyone.

The game and all required documentation must be submitted on-line, Pre-conference. Semifinalist teams participate in an on-site presentation and interview to demonstrate the knowledge and expertise they gained during the development of the game.

ELIGIBILITY

One (1) team of two (2) to six (6) individuals per chapter may participate.

TIME LIMITS

PRE-CONFERENCE/PRELIMINARY ROUND

- A. All components of the chapter's entry (website address URL and documentation PDF) must be submitted by 11:59 p.m. ET on a designated date in mid-May.
- B. The game submitted for evaluation must be greater than three (3) minutes in length of play and must be interactive.
- C. A deduction of five (5) points total will be incurred for a game that completes in under the three (3)- minute time minimum.
- D. The timing of the game segment starts with the first image or sound presented.
- E. Games must be playable from the deadline until the end of the National TSA Conference.

SEMIFINAL ROUND

- A. Ten (10) minutes are allowed for the on-site presentation and interview broken down as follows:
 - 1. Five (5) minutes for the presentation
 - 2. Five (5) minutes for the interview

ATTIRE

TSA competition attire is required for the semifinal presentation/interview.

PROCEDURE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants design an online game based on the subject of their choice.
- C. Participants record their process through a documentation portfolio.
- D. The game entry (website address URL and documentation portfolio PDF) must be submitted by 11:59 p.m. ET on a designated date in mid-May.
- E. The submission information and deadline will be provided in January on the [TSA website](#) under *Competition Updates*.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

- A. All online game submissions must be a hyperlink to the online game and be accessible for evaluation by the deadline posted on the [TSA website](#) under *Competition Updates*. Participants may choose the hosting site, but the hyperlink must point directly to the entry.
- B. Entries received, or changes made to submitted entries after this deadline will not be judged.
- C. The URL must point to the team's entry. Entries that require a software download, running an executable file (.exe), or request that access be granted will not be judged.

D. Video Game:

1. Must be a hyperlink to the online game.
2. Must be the original work of the team.
3. Work that is not created by the team must have proper documentation, showing copyright permissions and/or license for usage in the game segment.
4. When creating the game, the game must be free of any weapons or violence as stated in the general rules.
5. Game instructions must be clear and understandable.
6. Judges must be able to play the game to the third (3rd) level.
7. The game submitted for evaluation must be greater than three (3) minutes in length of play and must be interactive.
8. A deduction of five (5) points total will be incurred for a game that completes under the three(3)-minute time minimum.
9. The timing of the game segment starts with the first image or sound presented.
10. Games must be playable from the submission deadline until the end of the National TSA Conference.
11. Bonus points may be awarded for exceptional game features or content.

E. The Documentation Portfolio

1. The portfolio must include the following pages in a multi-page PDF document with pages in this order:
 - a. Title page with the event title, the conference city and state, the year, and the team's identification number; one (1) page
 - b. Table of contents; pages as needed
 - c. Purpose and description of the game, the target audience, including market research, explanation of how the game is unique, and a detailed explanation of how to play the game, including a list of control functions; two (2) pages

- d. A hand-drawn or digitally sketched storyboard that depicts the design concept of the video game (No screenshots of the game will be permitted as part of the storyboard); pages as needed
- e. A link to a video (a URL to a video hosted on a service of the team's choice that does not require any download or authorization), maximum two (2) minutes in length, of one (1) student demonstrating how to play the game with appropriate voiceover to include which buttons to press and actions to complete to make each move necessary to complete at least three (3) levels; one (1) page
- f. Student Copyright Checklist (see 2026 & 2027 Forms Appendix); pages as needed
- g. Permission letters for the use of copyrighted material, if applicable; pages as needed.
- h. Work Log (see 2026 & 2027 Forms Appendix); pages as needed

F. Judges score the entries.

- G. A list of twelve (12) semifinalist teams (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Participants from each semifinalist team report at the time and place stated in the conference program to sign up for a presentation/interview time.
- B. Semifinalist teams report to the assigned time and place for the presentation/interview.
- C. The team may bring one (1) device (laptop preferred) to be used by the team during the presentation. The laptop must be fully charged as no access to electricity will be provided. TSA will not provide access to WiFi or guarantee that a team provided hotspot will operate.
- D. No projection devices are permitted during the semifinal round.
- E. Participants make a presentation to the judges about their video game and respond to questions.

- F. Semifinalists do NOT have access to electrical power/ outlets during the event.
- G. The presentation shall not last longer than five (5) minutes.
- H. Judges will ask questions up to five (5) minutes following the presentation.
- I. The top ten (10) finalists are announced at the TSA conference awards ceremony.

EVALUATION

PRELIMINARY ROUND

- A. The first three (3) levels of the game
- B. The documentation portfolio
- C. Up to ten (10) bonus points may be added by the judges for exceptional game features, or for content showing exemplary education and social value.

SEMIFINAL ROUND

- A. The presentation
- B. The interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Animator
- Audio Engineer
- Computer Programmer
- Game Designer
- Game Play Tester
- Professional Gamer
- Writer

VIDEO GAME DESIGN

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ PDF of the documentation portfolio was submitted
- ☐ Video game is accessible and playable online
- ☐ ENTRY NOT EVALUATED

VIDEO GAME DESIGN (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Creativity and Artwork (X1)	The game lacks creativity and originality; little effort is evident in the quality of the artwork.	The game is original and exhibits adequate creativity and artistic quality.	The game is original, highly creative, and well crafted; artwork and graphics contribute to the enjoyability of the game.	
Technical Skill (X1)	The game lacks evidence of programming skills.	The game shows some evidence of programming skills.	The game shows strong evidence of programming skills.	
Storyline/Flow of the Game (X1)	The game follows little or no storyline; there is little to no logical flow to the game.	The game follows a storyline and flows adequately from one (1) scene/level to another.	The game is well-organized and flows smoothly from one (1) scene/level to the next, enhancing the game play.	
Game Mechanics (X1)	The game lacks appropriate use of game mechanics; little feedback is provided to the player, the controls are difficult to use, and/or the objectives of the game are unclear.	The game has adequate use of game mechanics and there is some evidence care in the design of player feedback, objectives, or player controls.	The game has exemplary use of game mechanics and there is evidence of significant consideration placed on the design of player feedback, objectives, and controls.	
Overall Appeal (X2)	Playing the game is not enjoyable, dull, and/or monotonous; interacting in game play is a struggle.	The game is somewhat interesting, easy, and enjoyable to play; most game design concepts are incorporated.	The game is innovative and entertaining; design principles are incorporated, which make playing the game easy and enjoyable.	
VIDEO GAME DESIGN SUBTOTAL (60 points)				

DOCUMENTATION PORTFOLIO (70 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.	
Game Directions, Control Functions, and Market Research (X1)	The game explanation is difficult to follow; no market research is provided; functions provided are illogical or incorrect.	The game directions can be followed and generally sync with overall workings of the game; some market research is provided; most control functions match the functions of the game.	The game explanation is easy to follow, market research is thorough, and control functions clearly match the game functions.	
Storyboard (X2)	The storyboard is sloppy, disorganized, and incomplete and/or does not follow overall flow of the game design.	The storyboard is generally organized and includes aspects and overall scenes of the game.	The storyboard is complete, concise, neat, and follows the overall flow of the game.	
Video (X2)	The video is missing details and content and appears to be lacking information; minimum effort appears to have been given.	The video provides adequate information and includes some of the necessary information for completing the three levels.	The video is clear and concise and includes all of the necessary information for completing the three (3) levels	
Work Log (X1)	Log is poorly organized and/or incomplete.	Log is adequately detailed and organized and contains most of the required components.	Log is well documented and contains all the required components.	
DOCUMENTATION PORTFOLIO SUBTOTAL (70 points)				

TIME DEDUCTIONS	
A deduction of five (5) points total will be incurred for a game that completes under the three (3) minute time minimum.	
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

PRELIMINARY SUBTOTAL (130 points)	
--	--

SEMIFINAL PRESENTATION/INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Participants seem unorganized and unprepared for the presentation; illogical explanation of the game is presented.	Participants are generally prepared for the presentation; explanation of the game is communicated and generally organized.	The presentation is logical, well organized, and easy to follow; the game explanation is communicated in an organized and concise manner.	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit an understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of their project.	

SEMIFINAL PRESENTATION/INTERVIEW (50 points) – continued			
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.
Team Participation (X1)	The majority of the delivery is made by one (1) member of the team; the partner(s) may be disengaged in the presentation.	Team members are generally engaged in the presentation, though one (1) member may take on more responsibility than the other(s).	All team members are actively involved in the presentation and responses to questions; there is shared responsibility among team members.
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)			
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____			
SEMIFINAL SUBTOTAL (50 points)			
BONUS (10 points)			
CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
Bonus Points Unique and exceptional features (X1)	The game demonstrates some unique and exceptional features and/or exemplary educational value.	The game is very good but limited in uniqueness.	The game is outstanding and unique.
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. TOTAL (180 points)			

Record scores
in the column
spaces below.

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

VIDEO GAME DESIGN

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more (preferably the same judges from the preliminary round).

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. One (1) extension cord for the semifinalist evaluation team
- C. One (1) power bar with surge protection for semifinalist evaluation team, as needed
- D. Laptop computer with high speed Internet capability
- E. Tables and chairs for event coordinator, semifinalist judges, and participants

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results will be shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Judges determine the twelve (12) semifinalists.
- E. Create a sign-up sheet for semifinalists.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. On the first full day of the conference, post a list of the twelve (12) semifinalists in random order.

SEMIFINAL ROUND

- A. At least one (1) hour before the event is scheduled to begin, meet with judges, and review time limits, procedures, regulations, evaluation, and all other details related to the event.
- B. No more than two (2) semifinalists representatives report at the time and place stated in the conference program to sign up and participate in the on-site presentation/interview.
- C. Up to six (6) representatives report at the assigned time and place for the presentation/interview.
- D. Manage completion of the presentations/interviews.
- E. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- F. If necessary, manage security and the removal of materials from the event area.

OVERVIEW

Applying leadership and 21st century skills, participants are required to design, build, and launch a website that features the team's ability to incorporate the elements of website design, graphic layout, and proper coding techniques. The design brief for this event will be posted on the [TSA website](#) under *Themes & Problems*. Semifinalists participate in a presentation/interview, with an emphasis on web design as it pertains to their solution, to demonstrate the knowledge and expertise gained during the design process.

ELIGIBILITY

One (1) team of at least three (3) and a maximum of six (6) individuals per chapter may participate.

TIME LIMITS

PRE-CONFERENCE/PRELIMINARY ROUND

- A. All components of the chapter's entry, including the website address (URL) for the entry, must be finished, submitted, and accessible via the Internet by 11:59 p.m. ET on a designated date in mid-May.

SEMIFINAL ROUND

- A. Ten (10) minutes are allowed for the on-site presentation and interview broken down as follows:
 1. Five (5) minutes for the presentation
 2. Five (5) minutes for the interview

ATTIRE

TSA competition attire is required.

PROCEDURE

PRE-CONFERENCE

- A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules and listed in the individual competitive event rules.
- B. Participants access the middle school event design brief from the [TSA website](#) under *Themes & Problems*.
- C. Participants design a website while observing the theme and design requirements.

- D. The website address URL (which includes the documentation) must be submitted by 11:59 p.m. ET on a designated date in mid-May.
- E. Submission information will be provided on the [TSA website](#) under *Competition Updates*.

PRELIMINARY ROUND

- A. A list of twelve (12) semifinalist teams (in random order) is posted on the first full day of conference.

SEMIFINAL ROUND

- A. Participants report at the time and place stated in the conference program to sign-up for a presentation/interview.
- B. Participants report at the assigned time and place for the presentation/interview.
- C. Judges score the interviews/presentations.
- D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE/PRELIMINARY ROUND

- A. Participants must launch their entry on a web server that can be accessed via the Internet twenty-four (24) hours a day, seven (7) days a week, fifty-two (52) weeks per year.
- B. Each entry must consist of web pages that specifically display the chapter's solution to the middle school design brief.
- C. The URL must point to the main page of the team's entry. Entries that are password protected or require a request for access be granted will not be judged.
- D. All web pages must be completed during the current school year.

- E. The solution to the design brief is developed as a series of web pages with a minimum of three (3) pages and no maximum of pages linked under the main design brief solution web page.
1. One (1) of the pages must list all sources of information used to create the website and be labeled as “Reference” on the website.
 2. If copyrighted material, such as text, images, or sound from other sources is used, proper written permission must be included/documented on the “Reference” page.
 3. Participants must submit a completed Student Copyright Checklist (see 2026 & 2027 Forms Appendix) in PDF format as a link on the “Reference” page; one (1) page
 4. Participants also must include a completed Work Log (see 2026 & 2027 Forms Appendix) in PDF format as a link on the “Reference” page; pages as needed
- F. All entries must be compatible using the latest versions of Microsoft Edge, Firefox, Chrome, etc. on a desktop/laptop computer.
- G. In addition to basic HTML code, the website may contain HTML5 and other state-of-the-art web-based technologies (e.g. JavaScript, jQuery, Angular, etc.).
- H. Framework systems, such as Drupal, Joomla, Wordpress, Bootstrap, or other current technologies may be used; however, pre-built templates and themes for these sites are not permissible. If a framework system is used, a statement affirming that the template or theme used on the framework was built by the team must be posted on the “Reference” page.
- I. The website must be finished, submitted, and accessible via the Internet by 11:59 p.m. EST on a designated date in mid-May.
- J. Changes made after submission will result in disqualification from the event.

SEMIFINAL ROUND

- A. Up to six (6) team representatives can participate in the presentation/interview.
- B. The presentation is limited to five (5) minutes.

- C. The team may bring one device (laptop preferred) to be used by the team during the presentation. The laptop must be fully charged as no access to electricity will be provided. TSA will not provide access to WiFi or guarantee that a team provided hotspot will operate.
- D. No projection devices are permitted during the semifinal round.
- E. The interview is limited to five (5) minutes.
- F. During the interview, the team must be prepared to discuss at a minimum:
 1. Overall website design and originality
 2. Design brief
 3. The appropriate use of new Internet and web-based applications (different browsers, monitor resolution, plug-ins, etc.).

EVALUATION

PRELIMINARY ROUND

- A. The website

SEMIFINAL ROUND

- A. The presentation/interview

Refer to the official rating form for more information.

TSA HONOR STATEMENT

All work must be created and completed by individual competitors or teams. Plagiarism, the use of Generative Artificial Intelligence (GenAI) software, copyright violation, cheating, and falsification of information are prohibited. Participants may NOT use any generative artificial intelligence (GenAI) tools (e.g. ChatGPT, Google Gemini, GitHub Copilot, etc.). Any attempt to gain an unfair advantage will not be tolerated. Competitors at any level of TSA competition understand and agree to abide by the TSA Honor Statement.

If it is determined that a student violated the TSA Honor Statement, a rules violation of twenty percent (20%) will be incurred.

STEM INTEGRATION

This event has connections to the STEM areas of Science, Technology, Engineering, and Mathematics.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer engineer
- Computer programmer
- Researcher
- Software designer
- Webmaster

WEBSITE DESIGN

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ Website URL that is functional on a desktop/laptop
- ☐ Design brief solution with no copyright or plagiarism issues
- ☐ ENTRY NOT EVALUATED

PRELIMINARY ROUND – WEBSITE (130 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Theme (X2)	The annual theme is not addressed.	The annual theme is somewhat addressed, but the supporting pages do not adequately support or contribute to the overall design.	The annual theme is addressed and is reflected in the supporting pages.	
Challenge (X3)	The challenge is not addressed, or addressed but not in detail; it is generally ineffective, and/or missing many parts of the required research and presentation; unrelated or distracting elements are present.	The challenge solution is generally well presented; it addresses most major parts of the required research and presentation; there are few or no unrelated or distracting elements.	The design brief solution is well presented, well researched, and highly effective; all expected components are present, and additional related elements that enhance the final product are incorporated; there are no unrelated or distracting elements.	
Content (X2)	The content lacks originality and does not contribute to the overall design of the webpage; the content does not align with the purpose of the website.	Very basic information is presented; the content aligns somewhat with the purpose of the website; some pages are irrelevant.	The content aligns well with the purpose of the website and adds to its effectiveness.	
Layout and Navigation (X2)	The web pages are cluttered and confusing; it is often difficult to locate important elements; the navigation structure is unclear, unintuitive, and ineffective in getting users to relevant information.	The web pages have a reasonably usable layout, and all major elements can be found; the design is generally pleasing to view; the navigation structure is generally effective and intuitive, and provides reasonable ability to navigate the website.	The layout is exceptionally user-friendly; the relationship of elements and content are effective and attractive to the viewer; the navigation structure is highly intuitive, and provides efficient access to all pertinent information on the website.	
Graphics and Color Scheme (X2)	Graphic content is nonexistent or of low quality and questionable relation to the topic; colors are of poor contrast and detract from the user experience.	Graphic content effectively relates to the purpose of the site, provides enhancement to the user experience, and is of acceptable quality; the color scheme is effective and does not detract from the viewer's experience.	Graphics are well-used, of high quality, and clearly enhance the user experience; interactive elements effectively engage the user; the color scheme is attractive, appropriate, and clearly enhances the viewing experience.	

PRELIMINARY ROUND – WEBSITE (130 points) – continued					
Function and Compatibility (X1)	There are several broken links and images, and/or the website does not render properly on multiple browsers.	There are no broken images, and/or few, if any, broken links; the website renders properly on most major browsers.	There are no broken images or links; the web site renders properly on most major browsers and is usable on mobile devices.		
Spelling and Grammar (X1)	There are numerous spelling and grammatical errors.	There are only a few spelling and/or grammatical errors.	There are few, if any, spelling and grammatical errors.		
PRELIMINARY ROUND – WEBSITE SUBTOTAL (130 points)					
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____					

PRELIMINARY SUBTOTAL (130 points)					
--	--	--	--	--	--

SEMIFINAL PRESENTATION/INTERVIEW (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Presentation (X1)	Participants seem unorganized and unprepared for the presentation.	Participants are generally prepared and are somewhat organized for the presentation.	Participants' presentation is organized, logical, and easy to follow.	
Knowledge (X2)	Participants seem to have little understanding of the concepts in their project; answers to questions may be vague.	Participants exhibit an basic understanding of the concepts in their project.	Participants show clear evidence of a thorough understanding of the project.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
Team Participation (X1)	The majority of the delivery is made by one (1) member of the team; the partner(s) may be disengaged from the presentation/interview.	Team members are generally engaged in the process, though one (1) member may take on more responsibility than the other(s).	All team members are actively involved in the presentation/ interview and responses to questions; there is shared responsibility among team members.	
Delivery (X1)	The team is verbose and/or uncertain in its presentation/ interview; participants' posture, gestures, and lack of eye contact diminish the presentation/interview.	The team is somewhat well spoken and clear in its presentation/ interview; participants' posture, gestures, and eye contact are acceptable in the presentation/ interview.	The team is well-spoken and distinct in its presentation/interview; the participants' posture, gestures, and eye contact result in a polished, natural, and effective presentation/ interview.	
SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (60 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (60 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.

TOTAL (190 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

WEBSITE DESIGN

EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

- A. Event coordinator, one (1)
- B. Judges:
 - 1. Preliminary round, two (2) or more
 - 2. Semifinal round, two (2) or more

MATERIALS

- A. Coordinator's packet, containing:
 - 1. Event guidelines, one (1) copy for the coordinator and for each judge
 - 2. TSA Event Coordinator Report
- B. The latest version of Microsoft Edge, Firefox, Chrome, etc.
- C. List of questions for on-site interviews
- D. Laptop computer with high speed Internet capability

RESPONSIBILITIES

PRE-CONFERENCE

- A. National TSA will collect entries until 11:59 p.m. ET on a designated date in mid-May. The results are shared with the CRC manager, event coordinator, and assigned judges.
- B. Review entries as they are submitted to the designated online storage utility.
- C. Manage communication and pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with the Judge Manager.
- D. Judges determine the twelve (12) semifinalists.

AT THE CONFERENCE

- A. Attend the mandatory event coordinator's meeting at the designated time and location.
- B. Report to the CRC room and check the contents of the coordinator's packet.
- C. Review the event guidelines and check to see that enough personnel have been scheduled.
- D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
- E. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

- A. On the first full day of the conference, post a list of the twelve (12) semifinalists in random order.
- B. Create a semifinalist sign-up sheet.

SEMIFINAL ROUND

- A. Semifinalists report at the time and place stated in the conference program to sign-up for a presentation/interview.
- B. Semifinalists report at the assigned time and place for the presentation/interview.
- C. Manage completion of the on-site presentation/interviews.
- D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:
 - 1. To deduct twenty percent (20%) of the total possible points in this round
 - 2. To disqualify the entry
 The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.
- E. If necessary, manage security and the removal of materials from the event area.

2026 & 2027 FORMS APPENDIX

Downloadable forms are available in
TSA Membership System/Total TSA/
Competition Forms and Resources

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time involved	Team member responsible (student initials)	Comments
1.				
2.				
3.				
4.				
5.				
6.				



STUDENT COPYRIGHT CHECKLIST

(for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☐ NO

If NO, go to question 2.

If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO

If YES, move to question 1A. If NO, move to question 1B.

- 1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.
- 1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☐ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☐ NO

If YES, move to question 2A. If NO, move to question 2B.

- 2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.
- 2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☐ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☐ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, _____ (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Initials: _____

Chapter Advisor Signature: _____

PHOTO/FILM/VIDEO CONSENT AND RELEASE

I hereby give permission for images of my child or myself (as applicable), captured during Technology Student Association (TSA) activities through film, photo or digital camera, to be used solely for the purposes of TSA promotional materials and publications, including the TSA website, and I waive any rights of compensation or ownership thereto.

Name of Minor in Images (please print)

Name of Minor's Parent/Guardian (please print)

Name of Adult in Images (please print)

Parent/Guardian or Adult's Signature (as applicable)

Date

NEW COMPETITIVE EVENT PROPOSAL

New proposals may only be submitted by a chapter or state advisor or TSA alumni and must be submitted by September 1 of the prior conference year in order to be considered for the next guide. Please attach any additional pages as necessary.

Name of Competitive Event: _____

Level: ☐ High School ☐ Middle School

Overview (description of the event and participant expectations):

Eligibility for entry (how many teams/individuals can participate):

Limitations (such as time or entry submission requirements):

Resources (i.e. are the resources a limiting factor, or are they affordable/readily available to all populations? Can this be executed at the national level?):

Specific regulations:

Required personnel:

Alignment with STEM standards (how does this align with STEM standards?):

What are the societal benefits for learning this information? How can this be applied in a real world context?

Do you know of a TSA Chapter, at the regional or state level, that executes this event at conferences? If so, whom?

Name Date

Email Phone Number

How are you affiliated with TSA? ☐ Chapter Advisor ☐ Alumni ☐ Other: _____

Mail to: CRC, c/o National TSA, 1904 Association Drive, Reston, VA 20191-1540; Email to: general@tsaweb.org



EVENT REVISION SUGGESTION

As TSA expands its membership and participation in competitive events increases, competitive events may require revision. TSA consistently tracks and monitors misinterpretations and strives to revise the guide to improve clarity. TSA encourages input so that competitive events continue to improve. Use this form to note how outcomes for competitive events may be improved.

Competitive Event: _____

Level: ☐ High School ☐ Middle School

Note a reference to the exact section and page number (if applicable): _____

Specifically state the suggestion. List exactly what should be deleted, replaced, and/or added to the event rule or procedure.

Provide a rationale and list the pros and cons of this proposed update.

In your opinion, will the update to this event change the space requirements at the conference? ☐ YES ☐ NO
If yes, provide your rationale.

In your opinion, will the update to this event require additional resources? ☐ YES ☐ NO
If yes, provide your rationale.

Enter any additional comments

Print Name Signature Date

State Advisor's Name Signature Date

Contact Email Contact Phone

Mail to: CRC, c/o National TSA, 1904 Association Drive, Reston, VA 20191-1540; Email to: general@tsaweb.org



RULES INTERPRETATION PANEL GRIEVANCE

Site of National TSA Conference _____

Advisor's Name _____

Advisor's Cell: _____

School Name _____

State _____

State Advisor's Cell: _____

Competitive Event (including level) _____

Student or Team Identification Number _____

STATEMENT OF CONCERN (Please print or type.)

Signature of Advisor Date

Signature of State Advisor Date

The decisions of the Rules Interpretation Panel (RIP) at the National Conference are final. There is no appeal process.

RULES INTERPRETATION PANEL RESPONSE TO GRIEVANCE

PANEL MEMBERS

Signature

Date

Signature

Date

Signature

Date

Site of National TSA Conference

Date

Competitive Event (including level)

Student or Team Identification Number

Advisor's Name

STATEMENT OF RESPONSE

The decisions of the Rules Interpretation Panel (RIP) at the National Conference are final. There is no appeal process.

